

ALL ABOUT
MINIATURE PLANTS AND GARDENS
INDOORS AND OUT

BERNICE BRILMAYER

The book cover has a blue textured background, possibly a cloth or paper with a fine, woven pattern. The text is centered and printed in a black, serif, all-caps font. The title is split into three lines: 'ALL ABOUT' at the top, 'MINIATURE PLANTS AND GARDENS' in the middle, and 'INDOORS AND OUT' at the bottom of the title block. The author's name is centered below the title.

ALL ABOUT
MINIATURE PLANTS AND GARDENS
INDOORS AND OUT

BERNICE BRILMAYER

The cover image was created by the transcriber and is placed in the public domain.

The Project Gutenberg eBook of All about miniature plants and gardens indoors and out

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: All about miniature plants and gardens indoors and out

Author: Bernice Brilmayer

Illustrator: Kathleen Bourke
Fritz Schaefer

Release date: January 12, 2023 [eBook #69767]
Most recently updated: October 19, 2024

Language: English

Original publication: United States: Doubleday and Company, 1963

Other information and formats: www.gutenberg.org/ebooks/69767

Credits: Stephen Hutcheson, Karin Spence and the Online Distributed Proofreading Team at <https://www.pgdp.net>

***** START OF THE PROJECT GUTENBERG EBOOK ALL ABOUT
MINIATURE PLANTS AND GARDENS INDOORS AND OUT *****

By the same author:

ALL ABOUT BEGONIAS

ALL ABOUT VINES AND HANGING PLANTS

BERNICE BRILMAYER

ALL ABOUT
MINIATURE PLANTS
AND GARDENS
INDOORS AND OUT

*Sketches and Landscape Designs
by Fritz Schaefer*

*Additional Art Work
by Kathleen Bourke*

DOUBLEDAY & COMPANY, INC., GARDEN CITY, NEW YORK
1963

*For my family, again and again.
The future is coming closer.*

*Library of Congress Catalog Card Number 63-18225
Copyright © 1963 by Doubleday & Company, Inc.
All Rights Reserved
Printed in the United States of America
First Edition*

CONTENTS

	AUTHOR'S NOTE	<u>9</u>
	ACKNOWLEDGMENTS	<u>13</u>
<i>Chapter 1</i>	MINIATURE WINDOW GARDENS	<u>17</u>
<i>Chapter 2</i>	MINIATURE GARDENS WITH ARTIFICIAL LIGHT	<u>23</u>
<i>Chapter 3</i>	MINIATURE GARDENS IN CONTAINERS	<u>38</u>
<i>Chapter 4</i>	MINIATURE GARDENS IN GLASS	<u>53</u>
<i>Chapter 5</i>	MINIATURE GREENHOUSE GARDENS	<u>61</u>
<i>Chapter 6</i>	MINIATURE HOUSE AND GREENHOUSE PLANTS	<u>74</u>
<i>Chapter 7</i>	MINIATURE ROSES, INDOORS AND OUT	<u>137</u>
<i>Chapter 8</i>	MINIATURE SINK GARDENS	<u>150</u>
<i>Chapter 9</i>	MINIATURE PLANTS, BONSAI-STYLE	<u>159</u>
<i>Chapter 10</i>	MINIATURE GARDENS IN THE LANDSCAPE	<u>177</u>
<i>Chapter 11</i>	MINIATURE ROCK AND WALL GARDENS	<u>183</u>
<i>Chapter 12</i>	MINIATURE POOLS AND WATER PLANTS	<u>199</u>
<i>Chapter 13</i>	MINIATURE WOODLAND GARDENS AND PLANTS	<u>211</u>
<i>Chapter 14</i>	MINIATURE TREES AND SHRUBS	<u>226</u>
<i>Chapter 15</i>	MINIATURE PERENNIALS AND ROCK PLANTS	<u>251</u>
<i>Chapter 16</i>	MINIATURE ANNUALS	<u>277</u>
<i>Chapter 17</i>	MINIATURE GARDEN BULBS	<u>288</u>
	EPILOGUE	<u>299</u>
	BIBLIOGRAPHY	<u>300</u>
	WHERE TO BUY MINIATURE PLANTS AND SUPPLIES	<u>301</u>
	INDEX	<u>307</u>

List of Illustrations

COLOR

By the author except as noted

	BETWEEN PAGES
Formal garden in a wash-boiler lid	<u>32–33</u>
Tiny tropical garden with pool	<u>64–65</u>
Achimenes, a beautiful gesneriad	<u>96–97</u>
Twenty-three varieties of miniature and dwarf geraniums	<u>128–129</u>
Streptocarpus, Weismoor hybrid	<u>160–161</u>
Rose and miniature rose	<u>192–193</u>
Garden in the landscape	<u>224–225</u>
Rock garden effectively composed	<u>256–257</u>

BLACK AND WHITE

By the author except as noted

Miniature geraniums in uniform rows	<u>20</u>
Mexican motif with cacti in window box	<u>21</u>
Child's cactus garden over radiator	<u>21</u>
Small plants in a lighted bookcase	<u>25</u>
An indoor "jungle garden"	<u>26</u>
Light-case planted with various small plants	<u>28</u>
Light shelves with begonias	<u>29</u>
Kenilworth ivy in gnome strawberry jar	<u>40</u>
Pawnbroker's planter with ivy	<u>41</u>
Pruning a dish garden	<u>43</u>
Apple-tree root with pocket for plants	<u>45</u>
Miniature plants in fish bowls <i>(Industrial Photographic Specialists)</i>	<u>54</u>
Converting a cider jug into a terrarium	<u>56–57</u>

Commercially produced terrarium (<i>Russ Stone</i>)	<u>65</u>
Author's succulent garden	<u>68</u>
Rampant greenhouse	<u>69</u>
Unusual propagation box	<u>87</u>
Sprouted stem cuttings of dwarf geraniums	<u>88</u>
'Spaulding,' bushy dwarf begonia	<u>98</u>
Group of dwarf begonias	<u>99</u>
<i>Caladium humboldti</i>	<u>108</u>
Miniature climber, <i>Cissus striata</i>	<u>112</u>
Ivy meagheri	<u>119</u>
<i>Helxine soleioli</i>	<u>121</u>
<i>Leuconeura massangeana</i>	<u>123</u>
<i>Oxalis hedysaroides rubra</i> (<i>Merry Gardens</i>)	<u>125</u>
Three dwarf geraniums (<i>Merry Gardens</i>)	<u>127</u>
Dwarf geranium, 'Robin Hood' (<i>Merry Gardens</i>)	<u>127</u>
Creeping <i>Pilea depressa</i>	<u>130</u>
Hardy <i>Saxifraga sarmentosa</i>	<u>133</u>
<i>Sinningia pusilla</i> , miniature of miniatures	<u>135</u>
A miniature rose grows	<u>146–147</u>
Miniature garden of dwarf evergreens	<u>152</u>
Rock garden in a wash-boiler lid	<u>157</u>
Variety of bonsai trees	<u>162</u>
Bonsai in citrus	<u>163</u>
White poppies in a tiny garden	<u>178</u>
Raised flower bed	<u>186</u>
Saxifraga seedlings	<u>188</u>
Trimmed lemon thyme and ivy	<u>194</u>
Constructing a no-cost pool	<u>204–205</u>
Wild garden in New York City	<u>213</u>
Bloodroot	<u>214</u>

Juniper with pansies	<u>245</u>
Planted cold frame	<u>257</u>
Author in her \$00.00 greenhouse	<u>279</u>

DRAWINGS

Dream greenhouse (<i>Kathleen Bourke</i>)	<u>66</u>
A fancy to build on (<i>Kathleen Bourke</i>)	<u>67</u>
Foliage details of five popular miniature trees and shrubs (<i>Fritz Schaefer</i>)	<u>237</u>
Construction diagram for low-cost greenhouse (<i>Hal Gearhardt</i>)	<u>280</u>

AUTHOR'S NOTE

Naturally, the children's welfare was the compelling reason for moving our family out of New York and into Connecticut. But we can't deny that we also had visions of more expansive gardening. So we set out to find an old (meaning dilapidated—not antique), spacious, window-rich house with acres of neglected land where we could indulge our yen for flower borders with delphiniums by the dozens, sweeping green expanses of lawn, even obese bullfrogs on lily pads in a modest lake.

These naïve notions were quickly canceled by the orbital prices of Connecticut real estate. In order to achieve our principal purpose, we had to make concessions to the second. The house we settled for is small; its windows are few and runty; and it has less than an acre of cultivatable land. It is one hundred feet at its widest, nearly six hundred feet long, and less than a hundred feet level in any one expanse. In other words, we got split-level land instead of a split-level house. But it is charming. Neighbors with great expanses of gardens and lawns actually envy us for our “natural setting.”

Actually, my favorite landscape architect, who happens to be my husband, Bob, would be lost if given a perfectly flat piece of land of equal length and width. He would have no contours to follow and would probably go fishing. As it is, both of us have plenty of challenges and the fun of running up and down ridges in our plantings. The acreage is ample for two persons who have little more than so-called “spare” time.

From this quick summation of facts, it is obvious why we gave up our grandiose ideas of immense perennial beds, a half-acre vegetable plot, naturalized bulbs by the thousands. Instead, we've learned how to tuck little gardens into odd corners; to compensate for limited space with intimate miniature perfection; to hunt for and find the small plants that are in sympathy and in scale with our small house and landscape. Cramped growing quarters indoors have even led us to collect miniature house plants. And when, some sweet day, we have our own personal greenhouse on the place, it's bound to be in scale with the rest of it.

Fortunately, we are by no means a minority. More small homes than large are being built today, and on more small lots. Gardeners are intensifying their demands for small plants of all sorts; and hybridists and suppliers are working nobly at filling the need. We now have four-inch ‘Wee Willie’ sweet William, tiny Twinkle Phlox, other dwarf annuals and perennials. Some nurseries are beginning to feature dwarf trees and shrubs. Florists and greenhouses are giving us minuscule house plants such as *Sinningia pusilla* and orchids with one-inch flowers. The charm and intimacy of the miniature is replacing the magnificence (and oppressive maintenance) of the massive.

There you have the beginning of this book and the reason why it contains many quite new projects. They would be illustrated as “before and after,” except that the “after” is yet to be written. Regardless of how long miniature gardening has been practiced, we feel the greatest developments are yet to come. Small houses and small plots of land force us to this conclusion.

Admittedly many of our personal opinions are based on experience and observations in Northeastern gardens. However, whenever possible we have included reliable information for other climates. You will, of course, make your own interpretations and adaptations. This a reader must always do, no matter where an author lives and gardens. And there is always your county agent to consult or your local garden-supply florist with whom to discuss your particular situation. Always an added pleasure.

PLANTS INCLUDED

As the author, I have used two criteria for including or omitting plants at the time of writing. I am concerned with those that are readily available from florists, nurseries, and the suppliers listed in the Appendix; and those that in my opinion are suitable for miniature gardens.

Except for the specific art of bonsai, I have not included plants that are unnaturally dwarfed by pruning or other means. I have omitted plants that look like miniatures when they are young, grow slowly, but eventually get out of miniature proportions if given time. I have not attempted to differentiate between miniatures and dwarfs, nor have I set up restrictive dimensions. Sizes vary with types of plants. A miniature orchid may be three inches high, a miniature shrub three feet or more.

PLANT NAMES

This book has been written by an amateur gardener for other amateurs; and I have made it as readable and enjoyable as I could. But in the interests of clarity and accuracy, Latin botanical names are used in preference to the vernacular. This is the only way to be sure plants are correctly identified. Popular names are confusing. Kenilworth ivy, grape ivy, and English ivy certainly sound as if they were related in some way; but when you use botanical names (*Cymbalaria muralis*, *Cissus striata*, and *Hedera helix*, you know they are not. By using the botanical names you are more likely to find the ivy you want in a reference book or catalogue.

For most plants, *Hortus Second* has been used as the authority for identification and spelling of names; but in the interests of readability, the double *ii* ending has been reduced to a single *i*. For a number of plants that have become available since *Hortus* was last revised (1941), I have referred to *Exotica II*, by A. B. Graf.

Unless a plant name is complete (genus plus species—plus variety, if any), it is neither capitalized nor italicized. (The caladium is a favorite foliage plant.) Complete botanical names are italicized, but only the generic name has an initial capital letter, even when the specific name has been derived from the proper name of some person or place. (The diminutive *Caladium humboldti* needs humidity.) When you see a plant name in italics, you will know that this is a recognized botanical species or one of its varieties, and not a man-made hybrid.

The names of recognized hybrids, seedlings, and mutations of either or both are not italicized, but are capitalized and enclosed in single quotation marks (caladium ‘Little Rascal’). Common or popular names are set in regular type with initial capital letters only for proper nouns, when they appear in text. In separate listings each word is capitalized.

BOTANICAL TERMS

In order to make a gardening book completely accurate and understandable, it is almost mandatory to use some so-called “scientific” terms which should really be as much a part of a gardener’s vocabulary as “annual” or “evergreen.” The following words are used in their technical sense:

Genus (plural, *genera*)—A group of plants related to each other by botanical characteristics. The name of the genus is like a human family's surname, Smith, but it is written first instead of last. *Oncidium* is a genus of orchids.

Species (plural, *species*)—A plant that differs from others within a genus, usually occurring in a natural state and capable of reproducing itself in identical form. The name of a species is like a person's first name, Alice, but is written last. *Oncidium pusillum* is one of several species in a genus of orchids.

Hybrid—Generally the result of fertilizing the flowers of one plant with the pollen of another; the resulting seedlings are hybrids.

Mutation or *sport*—A variation in any part of a plant that remains constant when that part is severed and propagated.

The word *variety*, however—although it has a strict botanical application—has been used more loosely and may often be defined here simply as “variation.”

ACKNOWLEDGMENTS

I wonder if anyone ever wrote a book without being indebted to many persons for some sort of help or inspiration. Certainly, I couldn't do it. Subtract the encouragement and time-consuming assistance of my family, friends, and horticultural acquaintances, and this would be less a book.

I am deeply grateful to: Fritz Schaefer for landscape designs and drawings of rare delicacy, and for letting me benefit by his wide horticultural training and talents; to Kari Berggrav for her enthusiastic contributions to the manuscript and for all sorts of help with plants and photographs; to Mrs. John Lee and to F. H. Michaud of Alpenglöw Gardens for their help and the use of their artistic photographs; to Adolph Adukas of the Julius Roehrs Company for his talented arrangements of dish gardens; to Kathleen Bourke for her fanciful drawings and to Elvin McDonald of McDonald and Bourke for his assistance and advice; to *Flower and Garden* for allowing me to adapt material that had appeared in that magazine; to Mary Ellen Ross of Merry Gardens for her assistance and the photographs of miniature plants she allowed me to use; and to all the friends and tolerant gardeners who allowed me to put my camera tripod in the midst of their plants—Mr. and Mrs. H. Lincoln Foster, Mr. and Mrs. Alex O'Hare, Mr. and Mrs. Norman Cherry, and our neighbors, Mr. and Mrs. Henry Fuller. To Ernesta Ballard and Peggie Schulz, well-known garden writers, and Mrs. N. E. Dilliard of Tropical Gardens, my gratitude for your assistance. I thank my mother, Alice Gaines, and her keen eye for catching my witless errors.

ALL ABOUT
MINIATURE PLANTS AND GARDENS,
INDOORS AND OUT

CHAPTER 1

MINIATURE WINDOW GARDENS

In a living room so small that two dogs asleep before the fire must be roused to let you pass through, monstrous cut-leaf monstera would be out of place—literally and most certainly no asset. In our house, to be truthful, anything larger than a three-inch pot begins to get out of proportion. When we were buying the place, we called it “quaint” and “cozy.” But when we moved in our favorite house plants, it was just too crowded for words.

This was the origin of our intense interest in miniature house plants. But limited space is by no means the only reason why these little fellows are such cheerful and desirable indoor decorators.

First, of course, there’s the charm of the diminutive—the same lure that leads some people to collect figurines or doll’s furniture. But plants are alive and growing; you can pore over each leaf and flower as it matures to small-scale perfection.

Because miniature plants occupy little space, you can grow more of them, and in greater variety. Three dwarf geraniums will bloom their heads off where a single large one might be crowded. Modern, narrow window sills are adequately spacious for a dozen or so two-inch pots of colorful cacti. One cattleya orchid can be replaced by several equally exotic, and much more personable, dwarf “botanical” orchids in delightful variety. Where full-sized narcissus and “daffy’s” that have been forced often seem to be just that, “forced,” miniatures fit in, add gaiety and color, along with naturalness.

Most important, miniature plants and gardens are thoroughly in tune with today’s decorating trends. They’re in scale with small rooms and low ceilings, in harmony with the spirit of suburban homes, mobile enough to facilitate change and rearrangement, even functional because they’re more carefree. And they certainly go along as we leave last year’s stark, bare, uncluttered look behind and move toward the warmer, more personal décor that once more allows us to display snapshots of the children on the mantel.

Miniature plants are often less costly than large specimens, and require less care. They grow slowly, require fertilizing and repotting less frequently, don't outgrow bounds, and seldom need to be renewed or replaced.

When I first started to collect miniature house plants, I had no idea how many were available, or in what delightful and wide varieties. There are miniatures in almost all of our best-known plant families, and there are some groups that have almost nothing but miniatures to offer. There are small-scale trailers, climbers, creepers; leaf rosettes or bushlets; tropical plants and mountain-dwellers; those with striking foliage, spectacular foliage, or both. Once you discover the wealth of Lilliputian plants you can grow in your home, I warn you, your will power had better be strong, else you never will stop following this fascinating hobby of raising the little fellows. It will run away with you before you know it.

WINDOW GARDENS

The window is the place most naturally suited for a living garden. It is nearest to the fresh out of doors and brings the plants closer to the environment where they are at home. By creating a transition, the plants in turn seem to bring the outdoors inside. A window is often, also, the only place where indoor plants can get the daylight and sunlight they need to keep in good condition.

But a real window garden is not a motley assortment of plants in pots, haphazardly arranged (or not arranged at all) or lined up in precise, military rows. It is an artistic composition, a grouping of plants with some sound design in mind—an arrangement of plants and their containers for pretty and refreshing effect. The more natural the plants look, the less obvious or contrived the lines of the design, the more decorative the result. This principle is, of course, integral to all kinds of gardens, indoors and out; but it is particularly vital in a window where our eyes stray a dozen times a day.

With miniature plants I find it easier to achieve good composition—much easier than with large ones. There are more elements with which to work; there is more opportunity to rearrange, a wider choice of colors, textures, and forms—the possibility of blending or playing them against each other. I recall a small window in an old country house, deeply recessed by the width of the thick stone wall. Three or four large or medium-sized plants might have stood on the two-foot-deep sill. But there were a dozen or

so dwarfs and miniatures all blended and accented by two small baskets of miniature ivies. The display was so lovingly arranged and cared for, the effect was more of a garden than an obvious decoration.

(In a rich selection such as this one, there is a natural danger of “too-muchness.” Don’t crowd these plants. Just the addition of one extra pot can spoil the effect of a perfect garden. Miniatures are not meant to be massed. When crammed close together they can look like a weedy, unmown lawn. Give each plant enough space to set off its modest charm, then you’ll find each one doubly charming in its space.)

And so an assortment of small potted plants can be arranged as effectively in a window as perennials can be in a flower border. There should be a careful selection and placing of colors for both contrast and harmony; the interplay of foliage forms and textures; the blending of plants into one design with eye-catching accents where accent is needed. For a container, use a shallow galvanized metal tray made to fit the window sill and painted a matching white. It should hold about an inch of water with a layer of pebbles thick enough to keep the pots above the water. The evaporating moisture humidifies the air. Use miniature plants of several families but all needing approximately the same amount of light and sun. For color, there are the flowers of begonias and impatiens; for foliage contrast, peperomias; for accent, taller plants; with *Ficus pumila* ascending the window frame and small-leaved creepers dangling over the edge to soften harsh lines and blend the garden into the room.

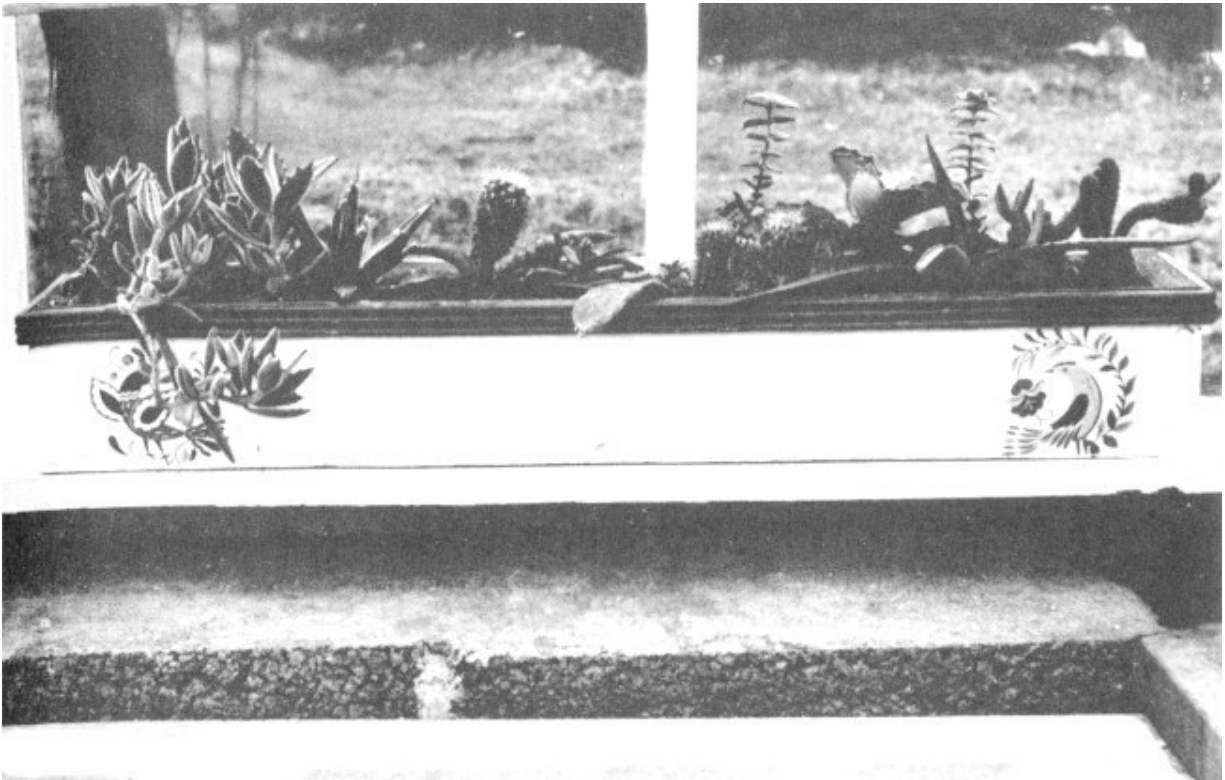
There is equal charm in a collection of miniature plants of the same general type and of nearly the same size. Neat rows of cacti and other succulents in small pots look gay and colorful lined up on the sill and on glass shelves in the window above it—glass, of course, to permit all possible sun to reach the plants. Between the pots, at irregular intervals, set a collection of crystal wine glasses or figurines. Or line up impudent miniature geraniums as in the photograph. Here, the pleasure comes, not from the artistic composition, but rather in the uniformity of the rows of small-scale pots and plants.



Miniature geraniums arranged in uniform rows

For an indoor version of the outdoor window box, use a box made to fit on the sill, gaily painted and decorated in the Mexican spirit of the cacti growing in it. It should be deep enough (about four inches) for healthy root growth. The cacti are not potted, but planted in the sandy soil in the box. These indoor window boxes can be of all sizes and shapes—large enough to cover the sill of a big window plus the radiator under it; triangular, to fit in corner windows; suitable for the top of a child's play table in a sunny bedroom or playroom.

All of these gardens are planned for windows with full sun, or nearly so. With less sun the choice of plants changes. For example, miniature gesneriads (African violets, streptocarpus, episcias) might be combined with ferns and other foliage plants; a selection of the widely varying types of peperomias would be effective where sun is very scarce indeed.



Mexican motif with cacti in a homemade window box



Child's cactus garden over a radiator—fine for a playroom

WINDOW GREENHOUSE

Available light, or sunlight, is the first consideration in selecting plants for a window area, or in selecting the window for the plants you have or want. Light can be brighter (it even comes from overhead) inside a greenhouse that extends out from the window. You can buy these in all combinations of measurements, ready-made and assembled, or ready to be assembled. Or you can make them, or have them made, from the materials sold in most hardware stores for those who build their own screens and storm windows.

The greenhouse fits flush to the outside of the window frame and is sealed with a calking-gun after it has been screwed firmly in place. It may rest on the outside of the sill, or be supported by metal or wooden brackets on the underside. The top lifts open for ventilation, and the opening is

covered with a screen. Glass shelves permit light to penetrate fully. A tray at the bottom holds moist vermiculite to humidify the air.

The window sash can be removed or not, as you wish. You can install an inexpensive, thermostat-controlled heater for extra warmth in winter.

If the light is right, and if humidity can be kept high enough, an installation such as this can contain not only all sorts of window-garden plants, but also many of those recommended for the greenhouse in Chapter 6.

A window greenhouse filled with growing, blooming plants is an attractive outdoor decoration on almost any house. Its effect indoors is always cheerful and refreshing. And it is especially suited for miniatures. Numerous small plants make a better decorative effect than a few large ones.

(For suitable plants, please refer to list at end of Chapter 6.)

CHAPTER 2

MINIATURE GARDENS WITH ARTIFICIAL LIGHT

The three tiny rooms of the Greenwich Village apartment possess a total of two narrow, old-fashioned windows; yet in its darkest corners bloom some of the most gorgeous gesneriads I've ever seen. In similar fourth-floor quarters on New York's dreary 41st Street, miniature orchids and other tropicals make a flamboyant jungle. In an attic in Levittown, a cellar in Bayside, a heated garage in Westchester, plants make it look like July in January, living their life cycles over and over again without ever seeing the sun. The life they must have for existence is supplied by electricity.

Time was, when windows were the only place in the house where plants could be grown. But since government scientists first grew corn to maturity under artificial light at Beltsville, Maryland (back when I had more interest in boys and dating than in gardening), that picture has certainly changed. Now, all sorts of plants can flourish in the most unlikely places. Home decorators can use plants ornamentally wherever they look best, and create the conditions in which they grow best. The hobbyist who can't afford a greenhouse can have a most satisfactory and inexpensive substitute in unused places in the house. And a greenhouse owner can double his growing space without adding another section of glass.

Naturally enough, scientific research in this field has been aimed at helping florists, farmers, and others to whom plants are a business; but amateurs have benefited, too. The principle of photoperiodism—that some plants set buds and flower only when nights are long, some others only when nights are short—led to delaying the flowering of commercial chrysanthemums by interrupting the long night with a period of light. Amateurs have used the same principle to force tuberous begonias to flower in winter by lengthening the day with several hours of artificial light.

The discovery and isolation of a light-sensitive enzyme, photochrome, has been applied to cyclic lighting—a less costly method of regulating flowering by flashing lights on and off at intervals. Probing the mysteries of photochrome has also given orchid fanciers a better understanding of their

plants' blooming habits and has even made it possible, with some species, to have flowers twice or three times a year, rather than just once.

If I may be permitted a slight prejudice, it's these amateur benefits that make me happiest. I love plants; and I think millions of other people do. From the windows of my commuting train I see New York tenement tenants wistfully watering morning glories that pathetically climb fire-escape trellises. More prosperous Manhattanites spend small fortunes on florists' plants to bring the breath of green life into their sterile apartments; and their disappointment, if the plants die, is pitiable. Suburbanites have a yen to make a hobby of collecting plants. And now they can. I know, because I did.

In our roomy, old-fashioned cellar in Bayside we had triple-decker shelves fitted with fluorescent lights where we grew everything from begonias (finally, a collection of more than 350 varieties) to annuals for the gardens out of doors. That was some years ago. The information about lighting was sparse, inconclusive, and often confusing. Our light intensity was inadequate, and there were other deficiencies which we would correct were we setting up that cellar greenhouse today. But our successes were fascinating, our failures a challenge. And the hours we spent working with those plants in the cellar often were our only moments of refreshment and relaxation.

The hobbyist, with his dividends of fun, is not the only one who benefits from this new concept of light and plants. There is the home decorator, the woman of the house, who finds in plants the sort of ornament the entire family enjoys. She'd like the graceful lines of a vine tumbling down from the mantel, jewel-like flowering plants on the shelf of a corner cupboard, a garden of green atop the room divider between the living and dining areas. Frustratingly, she discovers that where the plants are most effective, too often they won't grow and flourish. It is usually because there is insufficient light for their life processes. But now, she can set up a light on the mantel, install fluorescent tubes beneath cupboard shelves, or let ceiling lights flood the plants above the room dividers. Such lighting has a double effect, it enables the plants to flourish, and it gives a dramatic accent to the décor of the house.



Interesting combination of bookcase and lights for African violets and begonias of several varieties

Artificial lighting is a help even for the casual grower—one who has only a few plants, whether by happenstance, for the fun of it, or simply because “a house is not a home” without a plant or two. Table, desk, and floor lamps can be used to supplement the natural light from windows. Too often windows are shielded by trees or the house next door, or perhaps it is winter and there isn’t enough light to keep most plants in a thriving condition. Just turning on a lamp so that the rays fall on a plant can lengthen the hours of light enough to bring out bloom that might otherwise be impossible.



Tropical plants with controlled light, heat, and moisture make a “jungle garden”

Miniature plants and gardens are, of course, shining prospects for growth under artificial light. They take so little space, and since there is a limit to the height, width, and depth a single installation will illuminate, you can make the most use of it if you are growing the little fellows.

Here’s how the “jungle garden” came to be our source of continual refreshment and pleasure. Our living and dining rooms, both rather small, are separated partially by deep shelves. The previous owners of the place, devout music-lovers, used the shelves for their hi-fi set and stacks of phonograph records. Our record player—pardon me, our stereophonic hi-

fidelity music box—has its own cabinet, and that left a gap in the divider between the two rooms. We naturally thought of plants, particularly the tender tropical miniatures I collect. Since we still hope to do extensive remodeling, the garden was not built permanently into the shelves, but was constructed as a separate case.

We are fortunate in having a generous friend who loves to work with fine wood, and can make cabinets with the precision of the real professional. The case he turned out is a beauty. It measures eighteen inches by twenty-four inches inside. The top rests on strong metal rods at the corners. Window glass slides horizontally in the grooves cut in the top and bottom, enabling us to open or close the case as need be. The inside of the top is painted white, thus reflecting the light from the lamps downward on the plants. We use both fluorescent and incandescent lights which are mounted on the underside of the top. The bottom of the cabinet is lined with the heaviest plastic we could find.

At first the case was used as an indoor greenhouse for many potted plants that need protective warmth and humidity. Several inches of vermiculite in the plastic lining were kept moist constantly, with the sides being opened or closed for ventilation.

Later, we filled the bottom with rich potting soil and put the plants' roots right in it—climbers, creepers, tiny bush-shapes and trees. This turned out to be more of a “jungle” than we expected. Some notably delicate residents seeded themselves and started families. A dainty cissus strung itself languorously from one end to the other. The creeping fig nearly strangled the frail, whiskery bertolonia. But the planting was a source of delightful surprises—a bud here, a flower there, increasing colonies of some delicacies we hadn't been able to grow at all, before.

Several years ago a bookcase which I set up in my office as a garden was the object of considerable attention—how much I never realized until I dismantled it and gave away the plants. Then, I was bombarded with questions—and even some complaints that I had taken away this spot of greenery. From the night watchman up to the president of the company, people missed those plants. Some even thought I must have been fired.

There is a little house in Levittown, one which I always enjoyed visiting. The second floor has two finished rooms, one of which then was the office-

den of the hard-working Elvin McDonald of *Flower and Garden*. (He has since moved to Kansas City.) His tiered plant table with fluorescent lights was there for a functional reason, but it had a decorative value as well. In other homes I've seen plants growing by hundreds under lights in unused bedrooms, single specimens displayed in shadow boxes with circular fluorescent tubes, decorative gardens thriving in all sorts of dark corners. With artificial lighting taking care of the space problem, just about anyone can grow plants.



The author's New York office light-case planted with gesneriads, begonias, and other plants

However, before your enthusiasm flies too high, consider this sobering caution. Like anything else, artificial lighting works best only when it is properly planned and executed. Light must have the quality, intensity, and timing that plants need. Specific, accurate, up-to-date information is not always easy to find. Despite many fascinating discoveries and developments, this is still a relatively new horticultural principle, and there is still much more to be learned. Before he begins, the newcomer should locate the very latest and most reliable information; and the experienced grower should keep posted on the constantly changing rules. It has been my pleasant discovery that the big power-and-light companies, ever alert to develop new outlets for their product, are keenly aware of the possibilities of artificial-light plant propagation. Many of them are setting up departments to help horticulturalists. If you are puzzled, try your light company for information. It may take a few phone calls and letters, but eventually I know you will find some likeable chap wanting to help you.



Light shelves of medium height with begonias of many sizes and varieties (note miniatures down front center)

Although it is not necessary to become a botanist, I feel it is urgent to have a clear conception of how plants grow, and particularly how they use light. While we can't all be electrical engineers, it is also helpful to have some basic facts about electric lights and how they relate to plant growth. But if it were possible, I think I'd consider writing the facts I have with invisible ink. Who knows but what today's list of rules will be obsolete, and outmoded by new discoveries, before this book can be published?

Botanical Principles

For normal growth and flowering, plants must have light of the proper sort, intensity, and duration. Thus the leaves can perform their function of making starch, then sugar—the mysterious process called photosynthesis.

Besides normal growth, plants require an extra supply of sugar and starch for producing flowers. True, plants need light, but they also need dark to convert food into energy and growth. And this means complete dark. It has been shown that if light falls on so much as a single leaf, the entire plant continues to operate as if it were day.

For normal growth and flowers, plants require a certain balance of the red and blue rays of the spectrum. In general terms, blue rays are especially effective in developing leaves, stems, and other vegetative growth, and often in greater proportions for seedlings as compared with mature plants. In general, the red rays keep plant growth sturdy, regulate the development of buds and flowers, affect the germination of seeds and the rooting of cuttings.

For normal growth and flowers, different sorts of plants need light of different intensities—depending usually on available light in their natural habitat. Again in a general sense, light of more intensity is needed for flowering as contrasted with the needs for healthy foliage. But light intensity requirements vary with various types of plants.

For normal growth, and flowers, some plants need dark periods of greater duration. This is the principle called photoperiodism. By now a good many plants have been classified as to this requirement, but there are many others whose needs are yet to be determined. Chrysanthemums, poinsettias, and Christmas cactus, for example, will set buds and flowers only when there are more hours of dark and fewer hours of light. These are called *long-night plants*. Tuberous begonias, and other summer-flowering types, come into flower when nights are of short duration, and are called *short-night plants*. Those plants that don't seem to care one way or another are called *day-neutral*. For the sake of consistency you might even call them *night-neutral*. It is also thought that there is some relation between the duration of light and dark periods and temperature. Thus it can be seen how much research is yet to be done. A challenge of course, but that is what makes our scientists great.

Electrical Principles

Artificial light is not the same as daylight—it doesn't have to be. It needs only to supply the right kind of light (blue and red rays) of suitable duration

and intensity. Because it is constant, and consistent, the intensity (as measured in foot-candles) does not have to equal the brightness of a sunny day at high noon. Daylight waxes and wanes from dawn to dark every day, and may be very dim on cloudy and rainy days. Artificial light, coming from generators, is not dimmed by clouds or other external conditions. Duration is controlled by a light switch, or a time clock.

Incandescent bulbs are an adequate source of red rays for plants, but give little blue. They get burning hot, are comparatively expensive, and actually are inefficient to operate. Incandescents are also a source of far-red rays that delay flowering on long-night plants and operate in reverse for short-night plants. According to U. S. Department of Agriculture scientists, incandescent light used as a supplement to fluorescent light “improves the growth habits of many kinds of plants, but is seemingly not required by others.”

Until the introduction of the new Gro-Lux tubes in 1961, fluorescent lamps have given light with more blue than red, and in varying proportions according to the types of lamps. Fluorescent tubes do not get burning hot, and they are comparatively inexpensive to operate, and also efficient. In using the older types, those created especially for illumination, it is important to come as close as possible to the proper balance of the red and blue rays needed by plants. For some plants it has been sufficient to use only fluorescent tubes. For some of the other types many growers use 10 per cent of the wattage in incandescent bulbs.

But the new Gro-Lux fluorescent tubes, developed by Sylvania Electric Products, Inc., are especially for plants and not for illumination. They give a lavender-looking light made up of red and blue rays which are carefully balanced to suit plant needs. Growers who have used them report a spectacular improvement in plant appearance, in plant health, in faster rooting of cuttings, and in increased flowering. If demand warrants it, no doubt other electrical manufacturers will introduce their own brands of fluorescent tubes for plants.

Obviously, in growing plants under artificial light there are so many variable elements it is impossible—and extremely unwise—to set down hard-and-fast rules. The types of plants to be grown, whether the installation is primarily decorative or functional, and the possibility of

continuing research outdating your work, all should be taken into consideration when any installation is set up and put into operation.

CONSTRUCTION AND INSTALLATION

Again, I must write in general terms. I have neither the knowledge nor the experience to explain the intricacies of wiring, ballasts, circuits, and the like. This technical information is available from your electrical supplier and from equipment manufacturers, and often is on the cartons in which the parts are packed. Our installation was so outrageously large we had to hunt up a friendly contractor for help. He was a sympathetic man who loved plants and was fascinated by the idea of growing them under lights. Also, he was a cautious person, mindful of the fact that our electrical system was about twenty-five years old. And that stamped it as being an antique (as your light-and-power men will tell you). Since our basement floor was likely to be damp at times, heavy waterproof cables with special plugs and outlets were used, and grounded to prevent shocks, etc. Be careful about your electrical system, especially if you are going to go into anything as elaborate as our first enthusiasms. Don't build a firetrap for yourself. It's hard on the plants, not to mention the old homestead.

Whether your plants are to be grown in a garden, or in pots on benches, on shelves, or in a greenhouse-like case, the lineal proportions will be determined pretty much by the space that is available in your house, basement, greenhouse, or perhaps, as was in my case, your office. In small decorative planters twenty-five-watt fluorescent tubes (two feet long) are used most frequently. However, it is important to use enough of them, lined up closely to each other, to give a light of sufficient intensity. In fluorescent tubes the light is most intense in the middle and tapers off sharply at the ends. Since short tubes have more end—and less middle—they give off less light. The “shorties” are less efficient, as your plants will tell you.



Miniature roses, begonias, a birdbath, and ground cover made this charming little formal garden.

The distance between the tubes and your plants also affects intensity. The closer they are, the stronger the light. If possible, hang your fixtures on chains so that they can be raised or lowered. Adjust them to accommodate the taller plants and then raise your “little fellers” on upended pots, bricks, or boards so they will not be cheated of their share of light. Please remember, the greater the distance between light and plant, the more tubes you will need. Distance determines the number of tubes!

For greater intensity, and efficiency, forty-watt tubes (four feet long), or even larger, are usually recommended. If these are to be hung from the top of a case or cabinet, the simple strip fixtures are sufficient. If there is to be no “ceiling” directly above the lights, or if it is a decorative arrangement where glare might hurt the eyes of those who see it, use the industrial

fixtures with shield-like reflectors. (In planning your light-garden, please don't forget that the fixtures are a few inches longer than the actual tubes.)

If the case which you may be planning can be enclosed, at least on three sides, it will be easier to maintain the needed humidity. If the enclosing sides are opaque, they—and the “ceiling” above the lights—should be treated so the light rays are bounced back and the plants receive the extra benefit. In our cabinets we usually applied several coats of flat white paint on the inner surfaces. But once, under the blandishments of the aluminum industry, I lined a cabinet with their foil. It was plain foil, not the crinkled sort, so I did my own crinkling. Then I smoothed it out and fastened it in place with a staple gun. Plain foil, like high-gloss white enamel, seems to reflect the light every place except where it should be, on the plants.

In one of the installations we had at our place on Long Island I found it impossible to put in enough fluorescent tubes for the plants we wished to grow. Since they were day-neutral varieties, we made up for the lack of intensity by increasing the length of time the lights were used. Up to a point, increasing the light-hours will help to compensate for the lack of intensity—just to a point, however, and then the old law of diminishing returns takes over. Plants must not be under light so long that they fail to get their necessary periods of darkness. It is as essential as sleep is to a human being—perhaps more so.

OPERATION

In planning a light installation try to squeeze out a few extra dollars for an automatic timer. It will help to guarantee success for the operation. You'll have a certain peace of mind if you tend to be absent-minded. No more will you fret through a P.T.A. meeting, a movie, or a concert wondering if you turned off the lights on your plants. The timer will have done it for you. If you happen to have an enclosed case—one tight enough to conserve the humidity—you can very easily go away on a short trip (a day or two at most) and feel confident your pets will not suffer. If you have postponed buying a timer—actually, they are not expensive—and have to leave your plants for a day or so, it is better to turn off the lights completely. They'll suffer less than if the lights are going full blast. But for peace of mind, particularly that of the plants, we've always used automatic timers. At one time we had three of them. When I was ordering one from a mail-

order company, my husband was buying me one as a birthday gift. And at the very same time the electrical contractor who redid out light system donated one in the interests of our begonias. We had them popping on and off at all hours of the day and night. We even hooked a percolator into one for the morning coffee.

As I look back over our experiments of a few years ago, I find there are more plants which are day-neutral (night-neutral if you prefer) than plants which are short-night or long-night. For these day-neutrals, fourteen to sixteen hours of fluorescent light (of sufficient intensity) every day, all year round, will keep them happy and thriving. They won't know the difference between winter and summer, spring and autumn, Florida or Long Island. That has been our experience, but now I find opinions vary on whether hours of light should be lengthened or shortened in spring and autumn for these seasonal changes. (There is still plenty of room for experimentation. For instance, the light requirements for many plants are still to be worked out—even for closely related plants within various types.)

Some growers, those who specialize in plants for which they know the light requirements, turn on the lights at dawn and turn them off at nightfall. This is a year-around schedule. Others who have plants of assorted types, or of undetermined light requirements, maintain a constant fourteen-hour growing day. And they are often surprised by even second, or third, bursts of bloom. A nice surprise, if you ask me.

INTENSITY

Here again we find the needs of plants vary and fluorescent-light setups vary accordingly. If possible, measure the light in your growing area. The readings of a photographic light meter—the same instrument you employ in your photography—can be translated into foot-candles. Or you can get a meter that registers foot-candles. For advice, consult your camera dealer, or check with your local power-and-light company. Here in Redding we find the Connecticut Power and Light Company vitally interested in artificial-light plant propagation.

Again “in general,” house plants that require “full sun” when grown in a window need 1200 to 1500 foot-candles of artificial light, and for fourteen hours a day. Foliage plants will get by with 500 to 600 foot-candles. At

about 1000 to 1200 foot-candles many plants, and I'm thinking of begonias and gesneriads in particular, will be robust and floriferous.

Should you find it difficult to figure light intensity as suggested above, you might follow the formula worked out by an old friend on Long Island, Elaine Cherry (Mrs. Norman Cherry, the wife of one of our engineering friends). Her formula is easy to follow. "A single forty-watt tube will serve a space approximately four feet long by six inches wide." Small plants that need intense light can be set up close to the tubes.

Here is a tip—ever notice how your television picture is dim but brightens appreciably when you take a dust rag to the surface of the glass? The same is true of your light fixtures. Wipe them off now and then. Clean tubes give more light than dusty ones, and new tubes give more light than old ones. When a tube darkens at the ends, that means it has seen better days and should be replaced. According to Mrs. Cherry, it is a good policy to replace tubes after five thousand hours of service and not wait for the dwindling light to curtail the rays your plants need. While you are at it, it's smart to insert new starters.

TYPES OF TUBES

Until the introduction of the Gro-Lux lamps, we had to choose types designed primarily for illumination. And there were as many choices and combinations as there were tube types. In a private and somewhat limited survey, I've found that when only one type of tube was used, cool white was to be preferred. In combinations of equal or two-to-one proportions, some growers use daylight and natural tubes; others prefer daylight and de-luxe warm white. And there are those who go for cool white and de-luxe warm white. Those who supplement their lights with 10 per cent incandescent light seem to favor all daylight fluorescent tubes.

The object of all these different combinations is to get the most favorable balance of red and blue rays. If you are a hobbyist who grows plants for the love of them, and not necessarily for their value in interior decoration, the new Gro-Lux tubes are less complex and less troublesome. You don't have to be a light expert to get results and have fun with your light-garden.

CARE OF PLANTS

Temperature, humidity, soil, fertilizing, potting—almost without exception, plants growing under artificial light need the same care as window-garden plants. But since the light is an artificial substitute for natural sun and light, watch for signs that the plants are not entirely satisfied with it. When they stretch out, get long and lanky, or the foliage has a weak, wan color, set the plant up closer to the tubes, or over toward the center where the light is strongest. You might do well to make room by shifting some of the plants that have been in the center. Sometimes when a plant has too much light it will become stunted. Until a more exact rule book is written, you will have to use your own good common sense.

Here is the big worry many growers have; the failure of their pets to flower. More often than not that means insufficient light, insufficient red light, or perhaps both.

As of this date it is probably ten years since we first started toying with plants under artificial lights. I say “toying” because it was just that—purely for fun. We kept no records. When frost was in the air we dug up flowers and brought them indoors. My husband even brought in eleven goldfish which he feared would be glacéed in an outdoor pool. We put everything under lights with the fish in terrariums. Eventually he spent thirty dollars for a pool in an untidy corner of the living room. Thirty dollars, not counting the electric bill, I felt was a little expensive for a dollar’s worth of goldfish. I sold twenty dollars worth of photographs of that pool and then included one of them in my book *All About Vines and Hanging Plants*. Eventually he allowed me, very grudgingly, to place episcias around the pool. Mites moved in on them. He sprayed for the mites and killed all of the fish. He replaced the fish with eleven others. Thus the cycle continued.

All the time we had those indoor plantings our neighbors kept asking us what plants were good for lights and what lights were good for plants. Frankly, we couldn’t answer. Ten years ago that book hadn’t been written.

We tried just about everything less than five feet tall. We had wonderful results with African violets, begonias, orchids, and gesneriads. We even had a morning glory which singed itself on a steam pipe. All of them loved the kilowatts.

(In Chapter 6 I have indicated certain plants which are suitable for propagation under artificial lights.)

MINIATURE GARDENS IN CONTAINERS

DECORATIVE CONTAINERS, DISH GARDENS, AND MODEL LANDSCAPES

House plants are usually considered more or less lasting indoor decorations. But they can also be used the same as cut flowers for temporary and changeable displays, and then, like cut flowers, can be discarded when they begin to fade. They cost less and last much longer than bouquets, but because they're temporary decorations, they cause less worry and require less care than the permanent inhabitants of window sills or artificially lighted gardens.

That sounds rather heartless, I know. But it's a defense I've built up—and a perfectly logical one—against the wails of those who take beautiful florists' plants, place them on dark mantels, or in other thoroughly unsuitable growing areas, neglect them wholeheartedly, and then "can't make them grow." How many people do you know who buy lovely Christmas begonias, poinsettias, or cyclamen for the holidays and expect them to bloom the following season?

Honestly, I can't see any reason why plants must be immortal, why they can't refresh and beautify the home as long as they remain healthy and attractive, and not one minute longer, and then be discarded. I do object to stringy, leafless stems of expiring philodendron, dried-up dish gardens, or any plant or combination of plants that has become undecorative because it is dying. Actually, some florists' plants, such as greenhouse primulas and calceolarias, are annuals that come into full bloom only once, and having had their big moment are supposed to die peacefully afterward.

Do I treat my plants in the house so very cruelly? Well, no ... not exactly. My budget includes no allotment for florists' fripperies. I have a different system, and I have a constant supply of healthy plants to use for indoor decoration. My plants spend most of their lives in growing quarters where cultural conditions are good—in bright windows, in the window greenhouse, or on our small sun porch. They are brought in for a few days (never more than a week), then quickly returned to their more healthy, healthful homes. Having done their duty, they go back to grow and prosper.

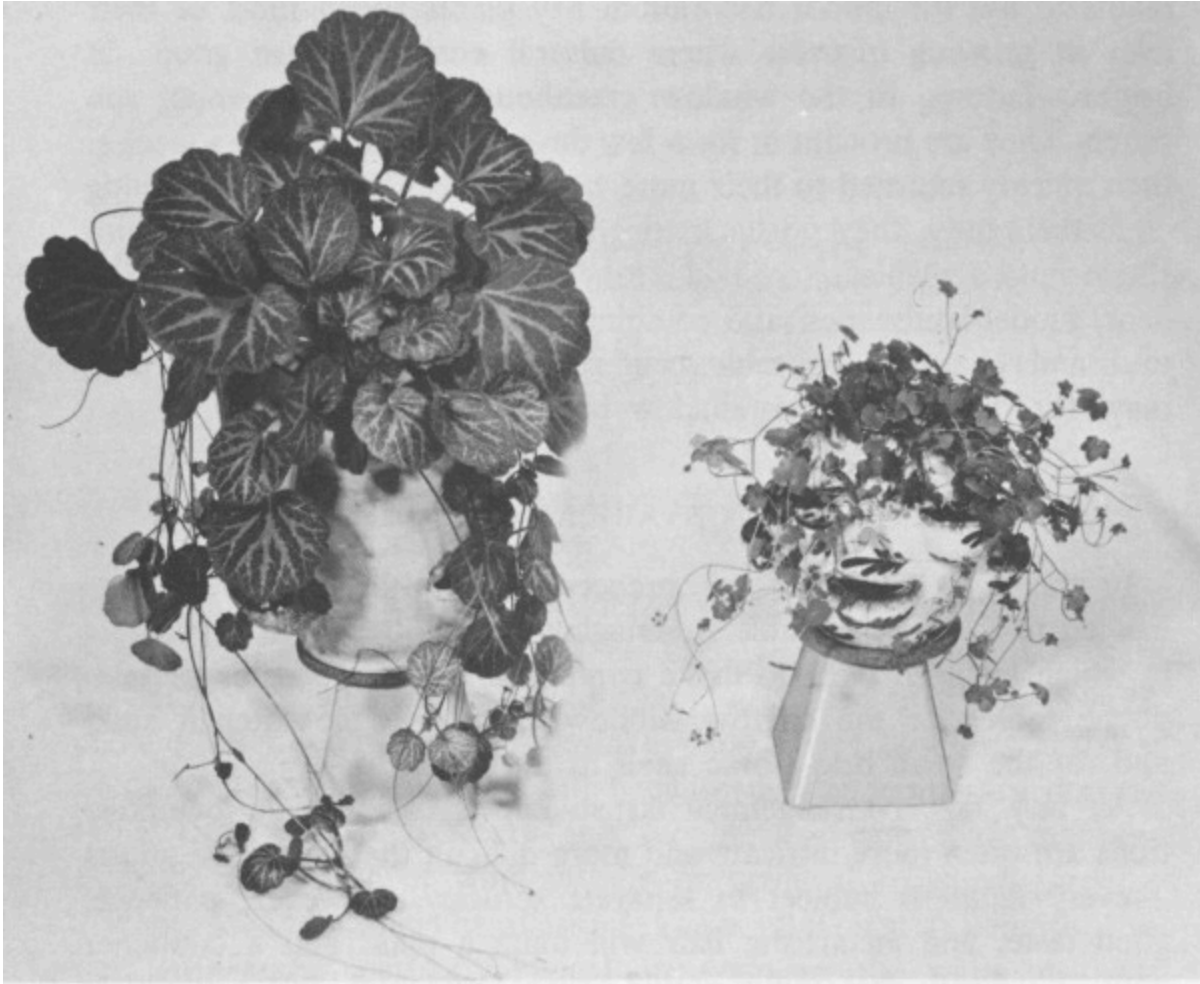
I do this with single potted plants, placed in attractive containers, with dish gardens, model landscapes, and combinations of plants. They are beautiful and charming as table centerpieces, mantel ornaments, displays for the coffee table, shadow box, or bookcase shelf.

PLANT-AND-CONTAINER COMPOSITIONS

In the past few years my preoccupation with miniature plants has led to some pleasurable rummaging and shopping for containers in which to place them to make compositions for a bedside or telephone table, for the narrow window sill above the kitchen sink, and for the small bric-a-brac shelf in the foyer.

As any flower-arrangement artist knows, small-scale compositions are often more intricate and more difficult than full-scale affairs—every detail is subject to separate scrutiny. However, patience, good taste, and an artistic flair will unite a plant and a container with an affinity that looks casual, even accidental, but actually is cunningly contrived. Container and plant become one picture—neither outshining the other—the container setting off the plant, and not sacrificing its own importance.

People who are intrigued with these miniature compositions usually collect containers in wide variety. Some of them are even made for the express purpose of holding plants—from wood, bronze, copper, all sorts of chinaware, glass, and ceramics. But the containers that give the most fun are those made for entirely different purposes. I've seen tiny bird cages, little woven baskets, glass lamp shades, odd-ball ash trays, punch cups, unusual tea or coffee cups, soup tureens, and even an ancient Buick hub cap which a little girl "borrowed" from her father's collection of automobile antiques. Some gourds are just the right size and shape, and with a nice wartiness, to lend enchantment for growing plants. Our cat keeps us well supplied with the tins in which his food is sold—spray them with paint and they are ideal for many plants. Some cocktail or champagne crystal looks precious with miniature vines drooping over the side.



Strawberry jar resembling gnome, planted with Kenilworth ivy

Once for our P.T.A. fair I collected a dozen or so unmatched liquor glasses, put a half-inch of soil in the bottoms, and planted tiny *Sinningia pusilla*. They sold immediately, with people wanting more. A plant sale at such an affair is a rather convincing test of popularity, and whether you have created a good arrangement.

Another favorite I have discovered for unusual containers is *Cymbalaria muralis*, the nostalgic Kenilworth ivy. I planted some in a small strawberry jar. Look at the jar from the right angle and it resembles a round-cheeked dwarf with a sparse green wig. I wish I could remember where I bought that jar—so many friends have wanted one. The “pawnbroker’s” planter cost five cents in a local junk shop. I also planted it with ivy.

Inexpensive hanging containers and wall brackets for miniatures are available in a wide variety at five-and-dime stores. But hanging baskets are not so easy to handle, as they must be suspended from wire or screwed to the wall. I've seen a doll's hat used delightfully, and also some nice little woven baskets. Or try anything of metal or ceramic if it has a lip to hold a wire or chain—or a two-handled consommé bowl; or a soup ladle with its handle fastened to the wall. You can easily punch holes in most plastic containers—and without cracking—by using a red-hot awl or old-fashioned ice pick.



Pawnbroker's planter set with ivy

Occasionally I have seen props or accessories used in these miniature plant-and-container compositions that were successful, but only occasionally were they in perfect scale and harmony. More frequently, the

silk, wood, or ceramic butterfly, bee, or bird is an unnatural and disturbing intrusion.

Be careful when you water plants in decorative containers. If possible keep the plant in its original pot so it can be lifted from the container and taken to the sink, where excess water will drain away. Otherwise, hold off on your watering until you are positive the plant won't wait any longer; then stop before the soil gets soggy and wet. Excess water, trapped by a container, can cause roots to rot, in fact will promote rot in most cases.

Be daring, be creative, be artistic when planning container projects and arrangements. If a fat little fern looks right for a teacup, let the cup be squat and fat; or let it be fluted gracefully and flared up to the delicate frond-fans. If a miniature orchid looks like a gem without a case, set it on pebbles in a clear crystal bowl; or perhaps invert a dome-shaped watch glass over it. If a succulent makes you think of a tough little gnome, for goodness sake don't plant it in one of those grotesqueries which is the hump of a camel's back or a cavity along the spinal column of a ceramic cat. (Remember how ridiculous a Venus stomach clock looks.) Use a little imagination. Perhaps you have something at hand—a droll bucket, a miniature fishing creel, a butter tub. Interesting containers make interesting compositions if you use good taste and imagination. Try to achieve the quality and feeling that the plant and container were “made for each other.”

DISH GARDENS

A dish garden is the combination of a group of living plants and the container holding them. It should be designed and planted with artistry and originality, but without artificiality. Each dish garden should look distinctive—certainly without any resemblance to the ones which florists seem to make by formula. It should be neither crowded with too many plants, nor cluttered with accessories or small ornaments. It should be eye-catching but not brazen, harmonious but not dull, unusual in some manner and yet comfortably natural.

Like cut-flower compositions, dish gardens are arranged so that plant and container together complete an artistic design. And like any artistic design, these gardens follow (or have a good reason for not following) certain basic principles:

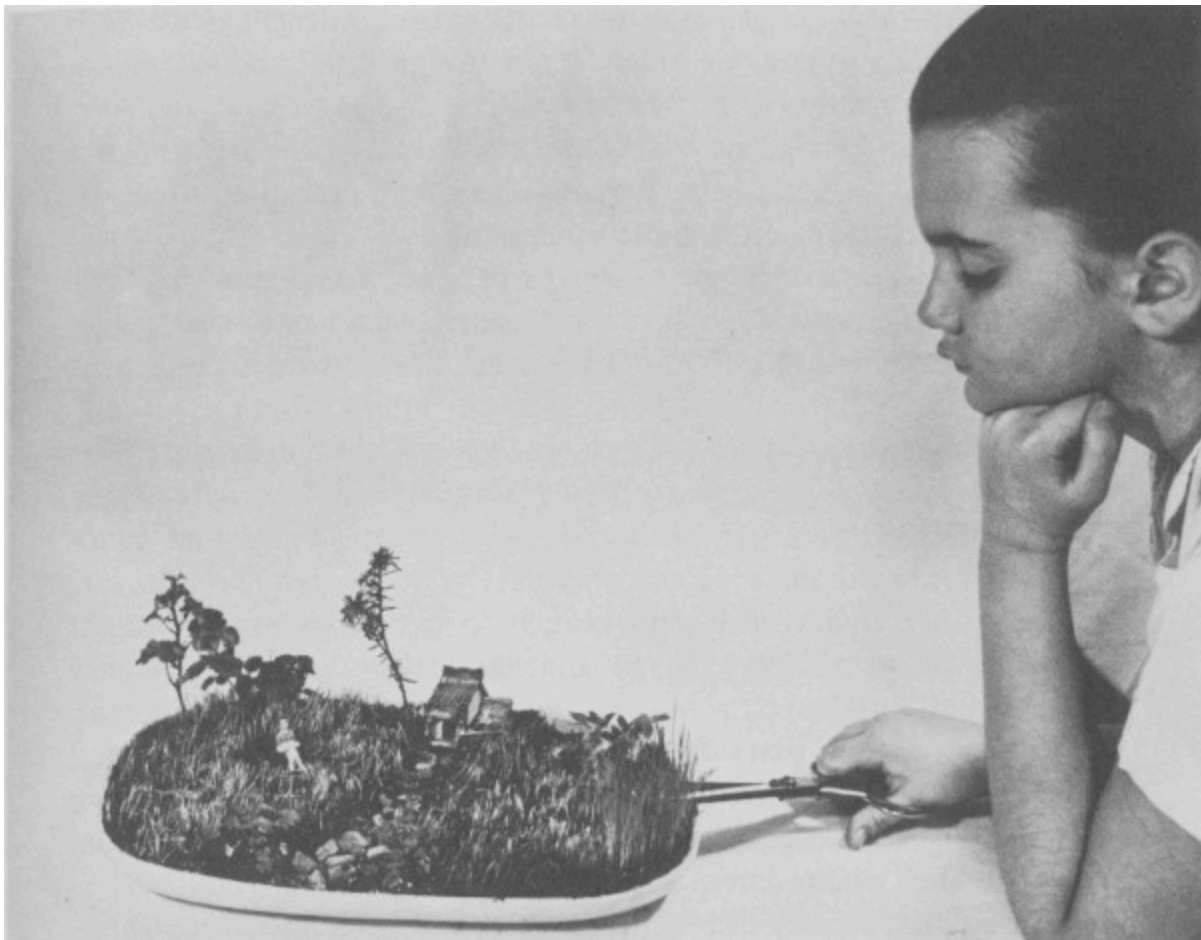
Plants and container blend into one pleasing picture.

Elements of the design interlock, overlap, or otherwise hang together.

The number of elements is limited by restraint and good taste.

All parts of the design are in pleasing relative proportion.

There is one focal point, or center of interest.



Pruning a dish garden to keep elements in size and proportion

If the design has formal balance, the focal point is in the center, with elements of equal weight at the sides.

For informal balance, the focal point is off-center, with heavier elements to balance it.

A design becomes fluid, rhythmic, with the dynamic use of line, and with pleasing contrast of colors, textures, and structural forms.

Of first importance, of course, is the container. It should be of proper size, shape, texture, color, and mood for the plants that will fill it. Rustic pottery is suitable for desert cacti and other succulents; glazed white, or lightly tinted, pottery for dainty flowering plants; copper, pewter, wooden bowls for an arrangement of heavy, masculine-looking foliage plants.

Containers can be of any shape—round, square, rectangle, triangle, ellipse, irregular. If possible they should be at least three inches deep so there is space in which to pack the roots of your plants. And they should not make themselves conspicuous with bold ornament, texture, or color. Plain design and subdued colors bring out the beauty of the plants.

Very few artificial accessories look well in a dish garden; but natural garden or landscape features such as interesting rocks or bits of old wood are often quite successful.

Before you begin to plant a dish garden, set the plants (in their pots) in the container, and then shift them around until they begin to look right. This will give you a rough idea of how an arrangement will turn out. For formal balance, set the tallest or most striking plant in the center, with some low ones nestled around its base. For informal balance, set the accent plant in one corner of a rectangle and let a large expanse of unadorned sand, gravel, or ground cover spread out toward the diagonal corner.

Turn a sharply curved leaf or branch so it falls against a straight up-and-down plant. Play rough foliage against smooth; feathery against solid; bright colors against dull; pattern against plain leaf. Try lifting out one plant to see if the effect is cleaner. To blend plants with the container, let a creeping or hanging plant fall down over the edges. These beforehand experiments will help you avoid having to shift plants later, during the actual planting.

Although not strictly dish gardens, there are some attractive variations that can be composed without benefit of soil, or of a dish to hold it. In the

pockets of a small piece of smooth, silky old root, or driftwood, tuck osmunda fiber (orchid-potting material) for the roots of epiphytic (air growing) plants—most are bromeliads. Terrestrial (soil growing) plants, such as the miniature begonia, are best in sphagnum moss. Or try tiny orchids; some will creep slowly over the surface of the wood. Fasten the plants firmly in place with inconspicuous fine florists' wire. This will hold the plants until their roots penetrate the fiber and attach themselves to the soft wood. If you supply liquid fertilizer at regular intervals, the plants will grow normally. Water by dunking plants and log in a pan or the sink. Feed by adding soluble fertilizer to the water.

Plants will often grow from cavities and crevices in rocks. If the rock is "limy," stick to lime-tolerant plants. Tufa, if you can find it, is especially malleable for gardens like these. It is soft and porous, easily cut and shaped, and with ready-made cavities to hold roots and small amounts of soil or moss. It is perfectly acceptable to acid-loving plants.

Conch shells, and another large shell of a similar type which we used to find on the beach—the sort kiddies hold to their ears when playing the game of "listening-to-the-sea"—offer interesting possibilities. Pack the cavity with moist sphagnum moss and plant with several smallish plants. Water with extreme care, and fertilize only slightly. Almost any moisture-compatible foliage plant that is available will live and grow this way for months.



Root from an apple tree, with a pocket for osmunda and a bromeliad

MODEL LANDSCAPES

Although these indoor gardens also follow the rules of good design, the result is a different effect. Montague Free once called them “an idealization in miniature of an outdoor scene.” They are not arranged to give an artistic impression, but to re-create some part of the out-of-doors on a small scale. Their charm lies in their diminutiveness, intricate detail and, often, in their whimsy.

The elements are: container; tiny plants (for the purist, all must be living) to represent trees, shrubs, grass, and flowers; and props Or accessories such as miniature pools, fences, and other landscape or architectural features. I suppose rocks would be called accessories, too.

Each garden should have a theme, and all elements should be in harmony with the theme and help to carry it out. For example, it’s difficult to combine buoyant hybrid pansies with shy wild flowers. A contemporary garden is best in a container with clean lines, but an old-fashioned garden is fine in a platter with high fluted edges. A desert scene calls for a container

that's bare and stark. A white plastic trellis doesn't belong in a woodland scene. And please, no green bath towels for grass.

Visualize your garden first—sketch the plan on paper. If you can draw it to scale, it will help in the selection of container, plants, and props. It is crucial that each element should be in proper proportion to all others. One element not in scale can ruin the entire effect.

In some gardens a plant or small group of plants will be the object of interest; in others it may be a particularly charming and important feature such as a rustic bridge or a shrine. In gardens of moderate size or less, one feature is usually sufficient, and not more than two in larger ones. Select your main feature first, place it, and make sure all other elements are in scale. For example, a fence should not be more than one and a half inches high under a tree of six inches.

The variety of plants, props, and containers from which you can select can be as wide as your enthusiasm and ingenuity want to make it. Here are a few suggestions.

Tree

Upright plant with a single stem-trunk, foliage at the top, usually taller than it is wide. If the tree is to be the object of interest, look for plants with character rather than symmetry—bent, twisted, gnarled trunk; interesting, lopsided shape; especially lacy foliage; tipsy tendency to lean. There are a number of useful house and greenhouse plants, and more to be found in the woods and fields. For deciduous trees, it is often permissible to use twiggy branches stuck in the soil. I find leafless pieces of mountain laurel very effective.

Shrubs

Upright plants of bushy habit and branching. You'll find many suitable house plants and some in the wild.

Hedge

Tiny-leaved, bushy plants that can be set close together and clipped to shape. The tiniest boxwoods will also do if they are carefully thinned and

each extra leaf is removed separately.

Flowering and Foliage Plants

Miniature house plants are best for these indoor gardens, although you can achieve temporary success with some annuals like alyssum.

Climbing and Trailing Plants

These are needed for training over walls, but even more necessary for planting at the container's edge so they will fall over and softly blend the garden and the container.

Ground Cover

A cover for bare spots in the garden—get sheet moss from the woods. Or plant grass seed and keep it mowed with sharp scissors. Use your own ingenuity. You may very likely come up with something more appropriate.

Urns

Use thimbles, thumb-pots, miniature vases.

Pools

These can be built with Sakrete or plaster of Paris. Or sink a sardine can—painted blue-green—an ash tray, soap dish, or plastic cheese container.

Paths

A path should always be going somewhere, preferably to the point of interest. Make paths with sand, fine gravel, small pebbles, perlite. If your garden is a formal one, make cement sidewalks with Sakrete. (Please, we have no financial interest in Sakrete—don't even know who makes it—but have always found it a most useful material around our gardens for patching, fixing, and repairing.)

Bridges, Fences, and Gates

Here is another chance for your personal ingenuity—and the more ingenuity you use the greater will be your pride when the job is done. Use

matchsticks, toothpicks, balsa wood (it is available in hobby shops, but you can very likely snatch a few pieces from some model airplane the kiddies are making). In my office I get coffee from the corner drugstore, each container having a stirring stick. I save those sticks. It is wonderful what one can do with them—picket fences and the like. A little whittling is all that is necessary.

Rocks

Please, don't use chunks of broken concrete. Hunt around for smooth, interesting specimens, eroded and rounded stones of the correct size. If you happen to come upon one with a lichen, you have a real prize.

There are as many themes for these gardens in miniature as there are outdoor scenes—cultivated or natural—in the world. The only necessity is, once you have decided on a plan, stay with it. See that every plant and prop you use is in harmony. See that every plant has the same cultural requirements—especially if your garden is to be a lasting thing. Here are some general ideas:

Formal Garden

This is probably the easiest to execute, chiefly because it is based on perfectly mathematical balance. The plan is basically geometric—a rectangle with a birdbath in the exact center; walks straight and precise; pairs or quadruplets of plants that are identical in size and shape; hedges that are neatly trimmed. How about trying something different?—an Old World herb garden; perhaps a scene from Colonial Williamsburg; or something from the Elizabethan age.

Informal Garden

Re-create your own garden, or something you hope to have around your house and grounds. It will help you to visualize it in advance. Get a container the shape and proportions of your lot—do a planting with the lawn you want, build up patios and terraces. Build a model of your house

and duplicate the plantings you want on a miniature scale. This sort of garden will give you a real thrill.

Old-fashioned Garden

I wonder if you ever had a wonderful grandfather and grandmother—I wonder if they had a trim house with a picket fence—white of course. If you did, how about trying to duplicate it. If you didn't, do a little dreaming. Dream about what you would like to see—picket fence, billows of bloom from flower beds, climbing things on the walls and fences. Please, let yourself go and improvise à la dream. Next to your own home, I can think of nothing more satisfying than trying to duplicate an old-fashioned garden in the manner of that wonderful past generation. Use your imagination. You'll be happy that you did.

Contemporary Garden

The central figure could be a miniature vase, to represent an urn, at the edge of a square or rectangular pool. Small boxes can be made like redwood planters. To be purely functional, use gravel or paving instead of grass. Plant sparsely and with an eye for modern design.

Oriental Garden

Here is a garden that can fool you with its simplicity. It calls for fewer plants, more minutely perfect props, figurines, stones, and moss. It may be built around a pool with a Japanese bridge. Outwardly, it looks so easy and simple, but it isn't. Just get one feature out of proportion and you will be unhappy. Remember, the Oriental artist is a person of great perfection, one with thousands of years of artistry behind him. Before attempting an Oriental garden, better get some good photographs or drawings. It will help you achieve a good picture and you will have a lasting satisfaction. Good luck.

Tropical Garden

This one should be lush with tropical creepers and climbing tropical trees, as pictured in the color section of this book. The container is a bowl from an overhead light fixture—the sort that used to hang above the dining-

room table. (It cost ten cents in a junk shop.) The back is a masonry wall, made of pebbles and Sakrete, as is the irregular pool. Paint your pool blue-green. Since your plants will very likely require acid soil, separate the construction material from the soil by strong plastic.

Desert Garden

Little but cacti and kindred succulents can grow here, and sparsely at that. *Sedum multiceps*, little Joshua tree, has a picturesque tree-like character. Use a suitable soil mixture completely covered with a layer of desert sand, or very fine gravel. Build a dune perhaps. Or make an oasis with a few palms around a pool—an irregularly shaped pool like one might see in a mirage. How about a few strands of grass—maybe not quite in tune with the setting but it might be considered as bamboo. A little faking is permissible.

Rock Garden

This usually calls for building up a rocky slope supported by hardware cloth in the rear and lined with moss to keep the soil from falling through. Follow good rock-gardening rules—rocks of the same kind but of varying shapes, with their layers, or strata, running horizontal. At the base of the slope you might contrive a small pool overflowing into a plastic-lined stream. Make a rustic gate and bridges with evergreen twigs wired and glued together.

Woodland Garden

Naturalistic arrangements of woodsy plants, rocks, moss, fallen logs. Seedling evergreens are fine. Artificial props are out.

Meadow Garden

A gate might open through a split-rail fence to a winding, foot-trodden path through a field of waving grass and flowers. At the back leafy trees line the edge of the imagined cow pasture.

Most containers for dish gardens and model landscapes are watertight. That is wonderful for any furniture on which they might be placed, but not so good for the plants. There is that eternal danger of overwatering. Roots rot when they stand in mud or water. In tight-bottomed containers it is wise to start with a thick drainage layer—pieces of broken flower pots, pebbles, brick, coarse sand, or even small pieces of charcoal. That gives the excess water a place to go. Cover this bottom layer with burlap or moss to keep the soil from sifting down.

The soil mixture should be suitable for the type of plant which is going to live in it—acid or alkaline, sandy or humus-rich—and should be moist—not muddy—at planting time. One at a time take your plants from their individual pots, set them in place, and make the soil firm enough to support them. Add dangling-edgers and ground cover last. Mist the finished garden with a fine spray of water, thus washing off any dirt and refreshing the foliage. Set the garden in a shaded, protected spot until the plants have recovered from transplanting shock.

Watering these gardens can be tricky. The soil may feel dry on the surface and yet be boggy underneath. Find a small bare spot where you can insert the handle of a spoon or a fork. Dig down to the bottom to make sure that water is really needed. And water with the greatest of care—enough to moisten the soil, but not enough to leave water standing in the bottom. No puddles, please.

Now supposing your hand has slipped—the hand holding the watering-pot—and you have overdone it. If the planting will allow, put the container on its side for a half-hour or so. But, please be careful—actually, I shudder to give you this piece of advice. I’m afraid you might find your creation out of its container and a muddy mess in the kitchen sink. All right, here is something else you can do; dig a hole in a bare spot—a small hole the size of a pencil and in the deepest part of your garden. Suck up the extra moisture with a pipette until the hole is dry. What, no pipette in your garden kit, then try a medicine dropper. No medicine dropper either—try a soda straw, but you had better be nimble or you will get a taste of dish garden. They don’t taste as good as they look.

If your garden is only a temporary decoration, you have given it your all and that is all the care it needs. But I feel you are going to love it so much

you'll want to keep it growing as long as possible. That changes the rules considerably. Place it, not on the coffee table, but in a window where it will get the light and sun the plants need, and where the temperature and humidity are to their liking. (Specific recommendations and plant preferences will be given in Chapters 6 and 16.) Hardy outdoor plants should be kept as cool as possible. You might set them in a cool room, or on an unheated porch, at night and bring them in only for the day. Fertilizing is usually not necessary, except when roots are severely crowded or you are trying to force a plant to bloom.

Keep the garden immaculately clean and neat. Remove faded flowers and tired leaves. Trim those plants that have a tendency to grow too large or straggly. It might be smart to remove any that refuse to stay within proper size. Train your climbers and creepers as you want them to grow. Keep your pools filled with clean fresh water. Mist foliage daily to keep it fresh and dust-free.

The dish gardens and model landscapes you plant this way are easy to care for, but those ones from a florist may present some problems. Now let's be fair to florists—their gardens and landscapes are turned out on a commercial basis in order that they may make money. (Outside of a few fancy floral outfits, none of them gets rich, particularly when one considers the long hard hours they spend on the job.) In the interest of economy they often combine plants of complete cultural incompatibility—dry-growing succulents with moisture-loving aroids; African violets that need sun for flowering with ferns that scorch in it. Too often these dish gardens are crammed with too many plants for the amount of soil; and the roots have been bruised and broken in handling. The florist knows that two-thirds of the customers who buy his product are going to abuse it anyhow. So he takes a “what-the-dickens” attitude. Make it pretty for the moment, for tomorrow it is going to die anyhow. One more word in praise of my many florist friends—just let the man with the green paper, the ribbons, and the carnations sense that you love plants, understand them, and care for them, and he will go to bat for you. He will help you in every possible way. I've never known it to fail. Actually, they are a soft-hearted profession.

All right, so you have a typical florist-shop dish garden. Uncle Charlie bought it for you as an anniversary present because it looked cute in the shop window, and he couldn't think of anything else. For all he knows that

green stuff is spinach. As soon as Uncle Charlie has gone home, start remodeling the garden. Check over the plants and remove any that don't agree with the majority of the other plants on light, moisture, and soil consistency. Pot up the good ones that you want to keep and thin out the others. Remember, those plants are going to do a lot of growing and must not be crowded. Use your ingenuity and common sense. Dig in the soil with that silver-spoon handle and see how much moisture is needed. Set the garden where light and humidity are right for the plants. You will have made yourself a new garden. Care for it as though it were your original creation. And when your next anniversary comes around, and Uncle Charlie comes around with another present—most likely a Chinese silent-butler—he will look at the dish garden and praise you for having a green thumb.

(In the plant list in Chapter 6 you will find plants that are suitable for decorative containers, dish gardens, and model landscapes. In Chapter 16, which is devoted to miniature annuals, you will find additional possibilities.)

CHAPTER 4

MINIATURE GARDENS IN GLASS

From a terraced garden in a huge brandy snifter, or perhaps in a fish bowl, to a single orchid under an upturned tumbler, there are many kinds of glassed-in gardens to be planted and enjoyed—terrariums if you like. This includes the old-fashioned partridgeberry bowl, bottle gardens, and today's version of the classic Wardian case. In any glass container a terrarium is literally a miniature greenhouse in which the temperature and humidity can be controlled. Dish gardens and model landscapes can be a great pleasure, so these terrariums add the additional joy of permanence and variety. Most plantings will live and thrive for twice or thrice as long, and you can grow many delicate plants that wouldn't last a day in the dry, drafty air outside.

TERRARIUMS

Most terrariums are small-scale replicas of woodland scenes with native plants, pools, streams, and rocky cliffs. These are charming and refreshing; and if there is good light, small outdoor plants will live through the winter into spring. But there are other possibilities—tiny tropical plants that love steamy warmth; miniature begonias, aroids, and marantas that are compatible under glass—in fact, that compatibility includes two-inch orchids and other dazzling epiphytes that need extra protection and humidity indoors.

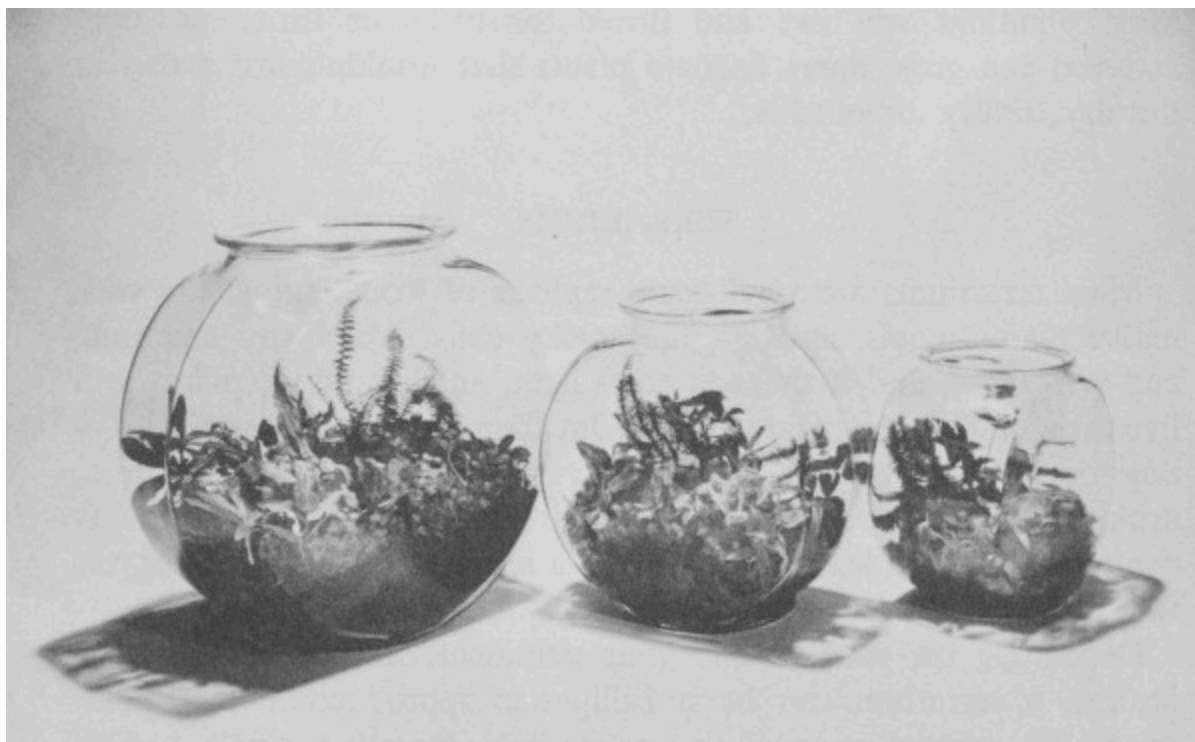
Depending on your plants, your container, and your whims or desires, a terrarium can be a Lilliputian reproduction of a landscape, an artistic composition, or perhaps merely a single perfect plant enhanced by a crystal setting. Since proportions and sizes are more strictly limited as compared with gardens which are minus glass walls, the selection of your container should come first.

CONTAINERS

Ready-made, practical, and not undecorative are the fish bowls and tanks of all sizes. The flat-sided and octagonal bowls are inexpensive, though not always of good glass. Very large, round bowls and brandy snifters of all sizes are slightly more costly, but of much better quality. Rectangular fish

tanks are in good proportion for miniature landscapes, easy to plant, and roomy.

Or perhaps you have an heirloom bell jar; a glass candy dish or bowl; an apothecary jar; a bottle laid on its side (rather tedious to plant. You have to reach through the neck of the bottle with long tweezers and iced-tea spoons, but a good job is very rewarding). You can still buy the traditional round, pear-shaped, and “Ming urn” covered jars for partridgeberries. If your container has no glass cover, cut one from window glass. Buff down the sharp edges. Terrarium glass is best clear, not tinted.



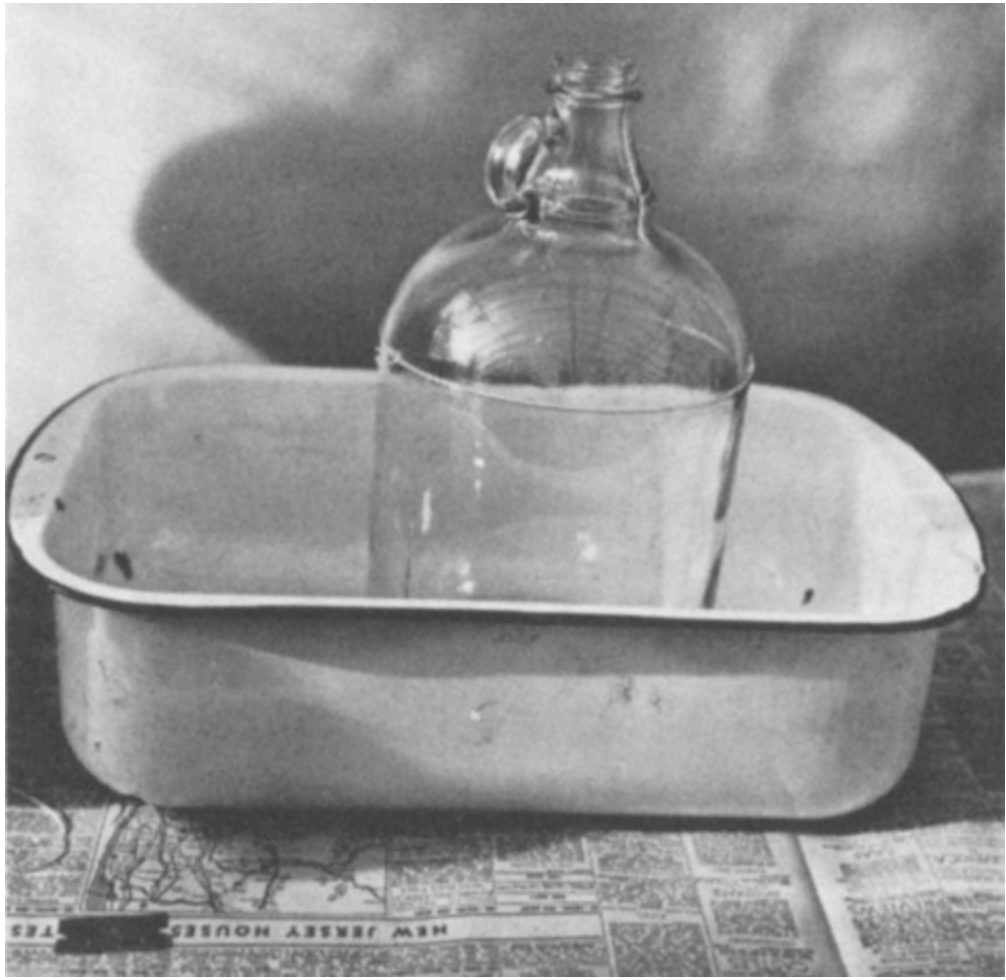
Inexpensive fish bowls planted with ferns and other miniatures

In the do-it-yourself department you can make bell jars out of cider jugs with the burning-string procedure. Soak a piece of string with lighter fluid or gasoline, wrap it around the jug where you want it cut, light the string, and when the flames are gone, dunk the jug in cold water. The first few attempts may not work—I had three tries before I got a nice clean break. But for fifteen cents’ worth of cider jugs I got a nice container. If the edges come out rough and jagged, buff them with an emery cloth or a file. But, please be careful, don’t get splinters of glass in your fingers.

For a miniature-garden-minded child, simply build up a container with window glass and a flat or box. Cut the sides and ends to match the dimensions of the flat or box; then bind the corners with adhesive or masking tape. Fold tape around the top edge and the cover. That will prevent cut fingers. We made one for our children. Just to add interest, we included a turtle and a salamander. They liked their new home so well they dug into the moist soil and hibernated all winter. These tailor-made terrariums can be constructed to fit whatever space is available—long and narrow for a window sill, short and squat for a table top.

History's largest and most elaborate terrarium probably was the Wardian case, invented (or discovered) over a hundred years ago. It is, of course, too monstrous for today's interior décor. But the principle can be applied in designing an indoor greenhouse that is decorative and in good taste—something to fit near a bright window, or to be fitted with fluorescent light if it is to be in a dark corner. It is surprising how a little extra light will often make a deeply shaded spot habitable for plants. In these cases, of course, the plants usually remain in their individual pots. A tray with water and pebbles, or sand, humidifies the air.

Today's adaptation of the Wardian case can be of almost any size or shape. One company manufactures a glass-enclosed box about two feet wide, four feet long, and three feet deep, with detachable furniture legs. I have seen a large coffee table, designed to sit before a picture window no doubt, fitted with plate-glass sides and accommodating a dozen or so medium-sized plants. I've seen several homemade cases as decorative as culturally practical, and sadly I must admit, I have seen some that were neither. I seem to have said before, in plantings of this sort let your plants be the stars.

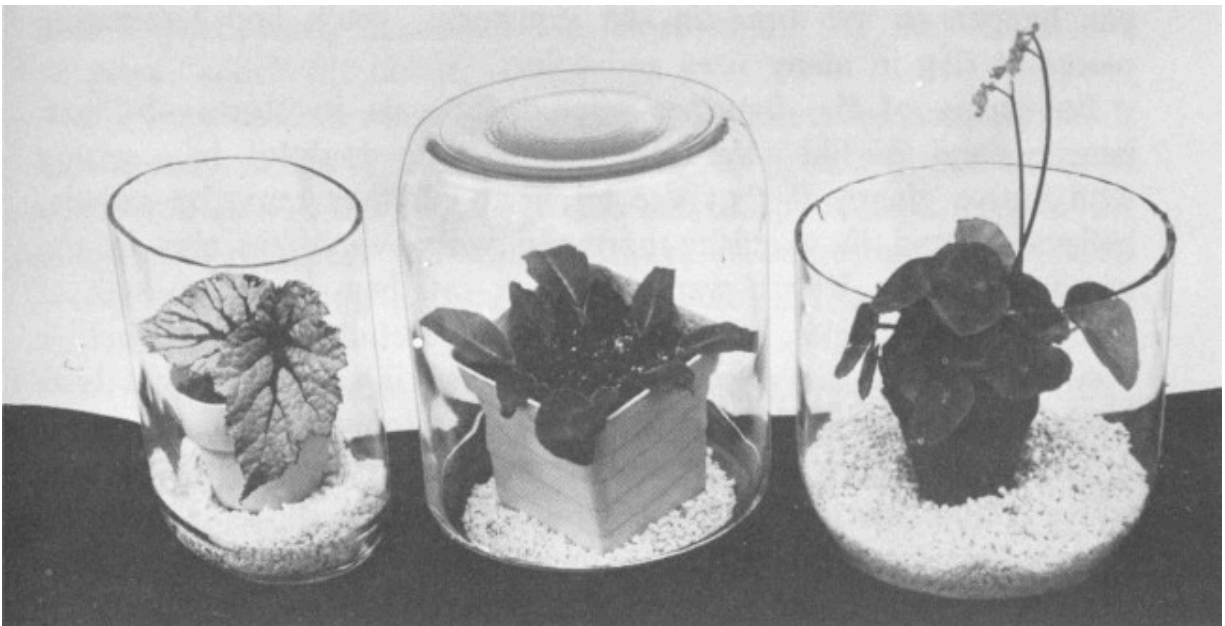




Converting a cider jug into a terrarium: a. Tie string at desired level and soak with lighter fluid b. Light string and immerse jar in cold water when extinguished



c. Lift off top of jug



d. Terrariums in use

AQUATIC GARDENS

Any watertight, glass-sided container can also be used for a fascinating water garden with fish and plants and many types of colorful decorations. To me, the most pleasing are those re-creations of natural underwater scenes with coral reefs, rocks, and the fantastic aquatic plants sold at aquarium supply stores. Quite honestly I'm bothered by the suited underwater diver who brandishes a spear and spouts bubbles at intervals. I'm bothered by the obviously artificial remnants of a wrecked ship, or the opened treasure chest spilling out coins and jewels. It is artificial to the point of being phony.

The planting and care of these underwater gardens has become an intricate science and art and requires knowledge of the delicate balance of plant and animal life, the proper supply of oxygen, and other technical matters beyond my ken. About all I know is, I love a beautifully executed aquatic garden. If you have a yen for one of these gardens, find an expert who knows his subject and ask his advice. If you can't find an expert, there are many excellent books on the subject.

PLANTS AND ACCESSORIES

Rocks, twigs, bits of decaying wood, and other naturally indigenous materials are suitable for gardens in glass. Tufa rock is always good, whether the plants sink their roots into it or not. If you happen to live near an old ironworks, you'll find fascinating pieces of slag in many sizes and colors.

But many of the figurines offered for sale in stores—bridges, benches, and the like—are questionable, even doubtful, in a setting with native plants. If they are to fit at all, they must be artistic, believable, and in exquisite harmony with everything else in the woodland scene. If you want to try it, set them in place, look at them from all angles, and think it over carefully. Decide whether they “belong” or are merely cute. Cuteness is something that lasts for a moment—beauty is something that is forever.

Just as with dish gardens, model landscapes, and other gardens where plants live and grow together, terrarium plants should be chosen first for their cultural compatibility. In fact, the principle is even more vital here because terrarium gardens are usually meant to be lasting. Think for a moment—you wouldn't tuck a dry-growing peperomia in the soil beside a moisture-loving fern, or a cool-growing wild flower beside a tropical selaginella. They just aren't meant to live with each other.

Next, size up your plants as to height—larger and bolder plants for the background or for accent, creepers for blending. In between should come the many flowering and foliage plants that make your garden a thing of beauty.

Many of these plants can be bought at florists' shops or variety stores. Others are available by mail, including the wildlings. The woods are always a source, but conservationists urge all of us not to destroy plants carelessly or move them out of their native haunts to unfavorable locations.

I find that the plants available from mail-order specialists are reasonably priced and are always good sturdy stock. More on this in Chapter 13.

PLANTING AND CARE

Whether your glassed-in garden is an artistic arrangement of plants or an actual garden, give your design a tryout by placing the plants on a temporary basis. Put them in place and appraise the arrangement. Don't rush. You'll save time by not having to do it over later.

In the case of containers rounded on the bottom, line them with thin sheets of moss. Turn the green side out. Next add an adequate layer of pebbles or sand, for drainage. A sprinkling of broken charcoal is particularly desirable. The close humid air may cause soil to grow "sour" without it.

Soil may be acid or not, heavy or light, sandy or humus-rich. It all depends on the type of plant to grow in it, and it will be moist when you use it. Please, make the soil deep enough to give the plants root-room. If you want to make it interesting, build it up gently into slopes or natural mounds. Flat land is dull, not nearly so interesting.

As you put in the plants, add the stones, pebbles, paths, or pools—whatever you have in your design. May I warn you—don't dirty the inside of the glass. Once dirty, it is almost impossible to clean up later. Plant sparsely. Remember, your plants are going to grow. The finishing touch will be the addition of "sod" or moss on the bare areas. Most of the suggestions for dish gardens and model landscapes in earlier chapters apply here. Finally, polish the glass and spray the plants with a fine mist to clean off any soil that may be on leaves and stems. Water very gently.

According to tradition, partridgeberry bowls are arranged differently. The base of the bowl is lined with moss, and covered with drainage layers, charcoal, and then soil. Then the berried plants and any others to be combined with them are placed, one layer at a time, with their faces against the glass. Their roots are toward the center and covered with soil. Finish the top layer facing up; sprinkle any loose berries on top; mist the foliage gently; cover with glass, and deck out with a big bow of Christmas ribbon.

For a few days after planting, set a terrarium where the light is not bright. After that, it needs light or sunlight according to the requirements of the plants. If they grow lank, limp, and leggy, they are usually stretching out for more light. Window-sill plants may lean toward the window—remedy: simply turn the terrarium around, and alternate it every few days so that both sides get an equal share of light.

Temperature also depends on the plants. Hardy outdoor natives are best at 60 degrees or less, even if they are moved to a cool spot at night. Tropical plants, and many others, are happy at the usual living-room temperatures.

Watering can be tricky, but not if you are careful and rational. Obviously, if the top of your container is covered tightly, not much moisture can escape. It will condense inside the glass and drop back into the garden. Water will be needed less frequently. But don't be too happy about this. If there is excess moisture, if too much collects on the glass, if the soil is always soggy, mildew and root-rot may result. Lift off the top for a while and let the inside dry out a little. Keep the soil on the dry side if the cover fits too tightly.

Here is another warning—soil should never become completely dry. (I am afraid I have you walking on a tightrope.) Again, take that handle of a spoon or fork and dig down through a bare spot so you can see how wet the soil is at the bottom. If water is needed, add it very gently. Remember, the small plants must not be deluged and dislodged; use just enough water to moisten the soil, not soak it. Sorry, but it is impossible to write down any definite schedule for watering. Frequency and quantity depend on the size of the terrarium and the size of the plants, on the outside temperature, and on other environmental factors. Even the weather and the season of the year are important. But, I'll bet you need watering much less frequently than you think. You may feel you should fertilize the plants so they will grow more

vigorously. Here is a word of warning on that score—few woodland plants appreciate chemical fertilizers (the powders you buy at the hardware store or in the horticultural supply shops). If you want to feed your plants, the best thing to use is a weak solution of manure water. I am always bundling up manure in gunny sacks, soaking it in a watering pot, and then using the water on my plants.

Glass-covered terrariums usually keep themselves clean inside, because dust just can't get into them. You may need to remove fading flowers and leaves. If you have any plants, particularly creepers, that threaten to take over and smother the others, get a sharp pair of embroidery or manicure scissors and begin pruning. Aliens such as earthworms and slugs are out. Put on your sharpest-pointed heels and step on them.

CHAPTER 5

MINIATURE GREENHOUSE GARDENS

As far as I am concerned, the first part of this chapter is a dream. It is such a precious dream I shouldn't even publish it until it comes true. I am afraid I must confess that parts of it have already been published in *Flower and Garden*. I was so carried away with the idea I rushed it into print, just so I could tell the world about a dream. It is a dream I hope you will dream with me. The magazine has given me permission to repeat the article I wrote for them.

The whole thing started when I began to get bored with prissy rows of pots in my own greenhouse. It is a forty-by-twenty-foot house and really isn't ours. (It belongs to a neighbor who used to use it for the chickens. The chickens didn't like being dispossessed, but they adapted themselves to the horse barn, where they are much happier.) My dream picked up emphasis when I tried a few "potless gardens" with plant roots unconfined and running freely in the soil. I immediately saw how much happier the plants were and how much more naturally they thrive. I am a working girl. I have but a few years to go on my retirement plan, and then I can retire. Hence, I dream about my own personal greenhouse. My dream greenhouse.

I picture it as an extension of our recreation room so I can look out through the large window to the glassed-in garden a dozen times a day, every day of the year. There will be a convenient door for the many times I'll be lured away from my typewriter. Any moment I wish, I can step into my garden to watch a bud unfold, tuck a wandering tendril back around a support, breathe in the fresh, moist fragrance only a greenhouse has.

My garden will be tropical. I love it that way. You, my reader, may wish a desert planted with fabulous cacti, or perhaps a cooler garden for alpine and hardier plants. But I love the exotics; and I will plant each one where it will grow according to its natural whims. Please, no more neat rows of pots on benches. My creepers will be planted where they can clamber over the soil and an occasional rock or rotted log. Climbers will have some picturesque support on which to climb. Epiphytes will have trees to perch on; and the danglers, baskets to hang from.

There will be variations of light, from full sun to patches of dappled or deep shade. If I can manage it, somehow I'd like to have variations in temperature in the different parts of my dream greenhouse. Some plants like the vigor of a cool corner; others revel in tropical warmth. Then there is the matter of moisture. For dry-growing plants there would be rock gardens and raised beds; for moisture-lovers there would be sunken, humus-rich bogs.

But you can't have a greenhouse, dream or otherwise, without having utilities, heating plant, water pipes, and potting places. Those I would conceal under the shade of flowering vines. I think I should allow myself the luxury of a stool—perhaps even a rocking chair—where I can sit down to pot or propagate my plants.

Of course, there would be a pool in my greenhouse garden, with a water lily to flower in it. Around the edge would be arranged rocks to make a home for friendly frogs, salamanders, and turtles. I would teach them to be so friendly I could take them to safe quarters when I had to spray or fumigate. Restful ferns would frame that pool. A fountain would make the musical sound of splashing water. Perhaps the overflow would run into a tiny, winding stream. I'll have to ask Fred, our plumber, about that.

I don't know much about birds, but I'd like to have a canary housed in a cage. He'd be there only when the vents were open. Otherwise he'd be free to fly. I might even have a parakeet for his company. I'd have a radio to play softly—only classical, gentle music. Nothing with brass—mostly violins and soft ones at that. No telephone—never. I'd have a bench for my guests—wrought iron very likely, since I think it would fit the décor, but the sort that is comfortable. Somehow, I'm going to have to figure out a way of labeling my plants so the labels will not show. Perhaps, we can do it electronically. As I have said, I would have a rocking chair, an old and battered one. And the grandchildren—this being a dream I can have grandchildren (our own youngsters are still young). There would be a curiosity corner for them where they could touch the mimosa and see it fold, watch a pitcher plant catch a fly, pick fluffy bolls of cotton, or dig a small peanut crop.

My garden will be full of surprises. Any day of the year a visitor will find something new—a miniature orchid flowering on the branch of the bromeliad tree; iridescent *Selaginella uncinata* stalking on stilt-roots under

a dwarf banana; carved columns, old urns, pieces of ceramic art I expect to pick up in my travels. Here, perhaps an old tree trunk sunk naturally in the soil; there, a log half buried as it would be in the woods; and then some stepping stones, two or three at the most to entice guests to look down on a mound of oxalis in full flower; overhead, a moss-lined basket of flamboyant epiphyllum in spring.

My planting, of course, will be carefully planned, but the plan will not be obvious. The shaggy fishtail palm, *Caryota urens*, would be placed in the perfect spot. The Mexican tree fern would look as if it had lived there always. The bromeliad tree would seem to have lived and died in my indoor jungle, and the plants rooted in its branches would look as if they had planted themselves, as they do in the tropics. Luscious-leaved philodendrons would climb and cover any obvious walls. Vines would be trained to soften sharp corners and provide some shade. In irregular beds I'd plant a natural arrangement of upright flowering and foliage plants—begonias, fuchsias, oleander, all kinds of aroids, a dwarf citrus or two. A walk might be edged with the tiny, delicious Corsican mint, *Mantha requieni*.

In a warm, partly shaded area I'd go crazy with gesneriads—flaming episcias covering the ground, tube-flowered aeschynanthus spilling down from above, fiery-flowered columneas in all their glory. And yes, I'd have African violets—not in pots, but in baskets, in strawberry jars, or sunk in the soil.

If (when?) I have my greenhouse, there are some plants I wouldn't be without. Among vines—silver-leaved *Cissus striata* with its swinging curtain of stringy aerial roots; *Clerodendrum thomsoniae*, its blue-green leaves smothered in red-hearted white flowers in spring; *Passiflora coccinea*, the red passion flower, for glamour. I'd hang baskets of the new soft-hued fuchsia hybrids, and my favorite floriferous begonia, 'Shippy's Garland.'

For fragrance, I'd plant a jasmine, *Stephanotis floribunda*, and *Osmanthus fragrans*, the precious olive. For early spring refreshment, I'd force miniature and dwarf daffodils, bringing in pots of them from the cold frame and sinking them in the soil. I'd want the silky-soft foliage and royal-purple flowers of *Tibouchina semidecandra*, the glory bush; but I'd pinch

and prune it, to keep it fairly low. I'd want a large basket floating airy fronds, one of the davallias, or rabbit's-foot ferns, and a smaller basket for the variegated *Abutilon megapotamicum*, because I love its gold-splashed leaves strung neatly along wiry stems, and its dangling red-and-yellow lantern flowers.

Just one more thing to complete my dream. Near the door of my dream-greenhouse garden, I'd have a special box for my guests, a box filled with small plants from which they could choose a parting gift. (I know if this greenhouse works out as I picture it, I shall have guests. I shall be happy to see them come, and a bit wistful when they leave. I will feel better if they take a small token of my gardening with them as they go.)

Frankly, this idea of a dream greenhouse came upon me quite casually. Much as we love our home here in Redding, Connecticut, we discovered a place which we felt we would like even better—a manor sort of place with stables, pools, formal gardens, a very charming old house, and seemingly endless rock walls. My husband and I fell in love with it—he, I think, because of a quarter-mile of trout streams, I, because it had a most charming greenhouse, the one I have been dreaming about. At this writing the entire project is still very much in the future, but we have hopes.

To be reluctantly honest, my greenhouse garden is not a brand-new idea. Mammoth conservatories are often planted as gardens, and so are the “plant rooms” now built into the more luxurious contemporary homes. Both are often show places, with plants brought in for display at the peak of their flowering beauty and, as they begin to fade, returned to spend the rest of the year in more utilitarian growing quarters. My garden would be a year-round project, the plants allowed to live through their natural cycles of active growth and rest without disturbance. We follow this procedure out of doors when we plant annuals over the spring-flowering bulbs; why not indoors, too?



Tropical garden in early stages with room for creeping ground cover and growing trees.

And, of course, there must be many hobby greenhouses already planted as gardens. I saw one near Boston, a fairly large one set into the side of a steep hill. It was a perfect piece of a desert, with the soil made suitably sandy and the curious cacti and other succulents growing as naturally as though they had never left home. Mrs. Ernesta Ballard has a small tropical greenhouse so realistically planted you feel as if you are in an exotic jungle the minute you step inside the door. But both of these places lack one important asset—room for the rocking chair and the radio with its soft music, things I hope I won't have to be without.



A miniature garden in a glass-covered terrarium

This complete greenhouse garden may not appeal to you, or it may not be practical or possible. In that case, I urge you to try a smaller naturalistic planting of some sort, if only to get that “garden feeling” and see how happy it makes your plants.

As a substitute for that Boston desert garden, I filled a small section of one of our greenhouse benches with a suitable soil mixture and planted it with small cacti and other colorful succulents, with here and there an interesting rock or two. These plants never flowered so freely when confined in pots, never showed off to such advantage. But I made one big mistake. I failed to make allowance for the more lusty growth, and planted the garden much too thickly. In just a few months the garden looked not the

least like a sparse and frugal desert, but more like a menagerie of scrambling, hoydenish pets.



Dream greenhouse, delightful and enchanting

Another garden that is not quite so wild and reckless is planted around a small pool under a greenhouse bench. The pool is actually a large plastic basin that must be emptied and cleaned with fastidious regularity. It would be better, of course, dug out and cement-lined, with a drainage pipe at the bottom. The plants are tropical, because they are in the warmth near the heating pipes; shade-loving, because a bench (even though it is slatted and admits some light) is above them. This is an excellent place for ferns, upright or climbing foliage plants, soft ground covers like selaginella. If I were to install fluorescent lights, I could add a number of flowering varieties.



A fancy to build on in the Oriental
manner



The author's succulent garden

A soil-filled bench is the site of my husband's pet project, a grove of miniature fruit trees. There always seem to be flowers or fruits on the small orange, lemon, kumquat, and pomegranate trees. But again, we erred. We did not realize how vigorously these trees would respond to having their roots free in soil. The dwarf banana and ever-bearing fig now threaten to exceed dwarf proportions at any moment, even to go through the roof.



Rampant greenhouse with citrus trees, banana tree, and birch trunk covered with bromeliads

A naturalistic planting that occupies a minimum of bench space is a bromeliad tree fitted with a sturdy stand to hold it upright. (The stand we are using actually is a Christmas-tree stand.) For the tree itself, we selected the top of a birch tree which blew over in the woods behind our garage. The tropical green contrasts nicely with the white bark of the birch, otherwise our selection wasn't too wise. Birch wood is soft and doesn't last too long, hence we have been propping and wiring branches to keep them in place. Cracks, crotches, and pockets created for the purpose are packed with osmunda fiber tightly wrapped around the base of all kinds of brilliant

bromeliads, some few orchids, a staghorn fern, and other epiphytic plants. The plants need not even be rooted if the osmunda is packed tightly around the base; if they're wired in place so tightly that the spray from the hose can't loosen them, they'll soon be at home on the tree. Their roots will grow through the osmunda and they will attach themselves to the tree. Of course, the osmunda must be kept moist until the roots form. Once rooted they are fed by dusting the outside of the sphagnum with soluble fertilizer and watering it in. As a finishing touch, we threw strands of Spanish moss over the branches. The misty-gray moss thrives, and even flowers.

As further proof that many plants prefer freedom to being confined in pots, try planting just one creeper—an episcia, for example—in the center of a spot filled with suitable soil. See how soon it will garland the top with lush foliage, how freely it flowers. Many plants recommended for hanging baskets are really creepers and are better off when treated this way.

PLANTING AND CARE

Large or small, a greenhouse garden, like any other, should be designed carefully and never planted until you have the plan firmly in mind. Each plant should be given room in relation to all the others, and not merely at planting time but with maturity time in mind. Allow plenty of space for spreaders to spread and creepers to creep; and make sure every climber has its support. And, of course, plants in close proximity should have compatible cultural preferences in soil, moisture, light, and the like.

Light and Sunlight

It shouldn't be difficult, in a complete greenhouse garden, to arrange for areas of full sunlight, partial or dappled sun, and shade, and to do it in a natural manner. Tall plants will get the sun first and for the longest time; but you could also build a slope, a sand dune, or a rock garden that would face the sun and get the most of it. Partial-sun plants could be placed where they are only lightly shaded by other plants or, if necessary, by shadings on the sections of glass above them. Shade-loving plants can grow under small trees and bushes.

Almost any greenhouse, in any area, should have shading applied to the glass in early summer. Several shading compounds are available, in white

or green, and are mixed and applied with either a wide brush or a sprayer. These compounds are constituted so that they slowly weather away during the summer and leave the glass clear by autumn. But if you happen to have plants that need all of the winter sun possible, it is wise also to clean the glass.

Temperature

Greenhouses are generally classed as cool (40–60 degrees), moderate (50–70 degrees), or warm (60–80 degrees). This refers to the minimum night temperatures at which the thermostat is set to regulate the furnace or heater. In warm climates where an air-conditioner is used, it follows the same scale. Greenhouse plants are usually classified similarly. But if other cultural conditions are favorable, many plants will tolerate a much wider range.

In order to grow the widest variety of plants, I set my greenhouse thermostat at 60 degrees in winter. This is a little warm for some cool-growing items such as geraniums, and causes outdoor garden plants (annuals, perennials, and vegetables started from seed) to grow slightly leggy and lank. But they recuperate in a hurry when they are moved outside. This temperature is slightly cool for subtropical plants; some gesneriads merely hold their own through the winter, neither growing nor flowering; some extremely tender begonias are forced into dormancy, from which they awaken in warm spring weather.

Humidity and Ventilation

With such a motley assortment of plants, I try to keep relative humidity at 75 per cent or higher. The minute the gauge dips below that mark, we wet down the walks and benches. Of course, this happens only on bright, sunny days. During dark, gloomy weather we water and mist as little as possible.

Constant circulation of fresh air is extremely important all year long. Our doors and vents are wide open in summer. On quiet winter days we may open doors or vents just a crack for a short time. Automatic ventilation is a marvelous convenience in late winter and early spring when the sun suddenly seems to get very hot and threatens to scorch everything under the glass.

Soil

Whether a greenhouse garden is a small flat, part of a bench, or a large ground bed, soil should be prepared carefully to suit the kind of plant that will grow in it. You'll find "recipes" of all kinds in any authoritative greenhouse garden book. Prepared soil mixtures should be at least a foot deep for beds in which you plan to grow plants of moderate size. For fruit trees and other larger plants, make it even deeper.

Fertilizing

The fertilizing schedule which works so well for my house plants, seems to do equally well for my greenhouse plants. I shall stay with it until I find something better. But it does seem possible that a large greenhouse garden, with large areas of carefully prepared and nutritious soil, would probably need feeding slightly less frequently than plants in pots. I shall have to try this, when my dream greenhouse comes true.

Watering

One of the "musts" in my dream greenhouse will be both hot and cold water, and a provision for mixing them as they flow into the hose. Thus, watering will be like a gentle summer rain. I'll have an extra-fine nozzle to break the stream into a delicate mist. Warmed water in a fine spray surely should not injure tender foliage in any way. And think what fun it would be to water a garden this way, enjoying the slightly musty fragrance of a warm summer night when soil is moistened by rain—and all of this in January when the snow is piled deep in drifts and banks outside the house.

Grooming

These are the little everyday pleasures of gardening in a greenhouse—snipping back a wandering branch, chiding a creeper and pulling it back from crawling over a neighbor, picking off a faded flower, supporting a branch heavily laden with flowers or fruit. It is real fun, and makes the difference between an overgrown mass and a carefully tended garden in which each plant appears at its best and in harmony with the others. Cleanliness—the removal of all organic matter before it rots, the rinsing

away of dust and dirt—is the best protective measure against infiltration of insects and disease.

Insects and Disease

When my retirement time comes, I may have to make a choice. Which do I want most—the convenience of fumigation with greenhouse “bombs,” or the presence of pet frogs, ladybugs, friendly insects, chameleons, and the like among my greenery? I am sure I will choose the latter, because I will have plenty of time to tend my greenhouse garden; and it’s not too much trouble to “spot spray” any plant that shows signs of problems. Anyway, it’s risky to use strong fumigants in a greenhouse that is attached to a dwelling, unless it is tightly closed in and has no cracks or fissures for leakage. And who could be sure of that. Anyhow, if I killed all the insects, what would the frogs find to eat.

GREENHOUSE PROPAGATING

Actually, I have two propagating cases in our large greenhouse. One is completely enclosed in plastic to keep the humid air even more humid. It has three or four inches of perlite on top of an electric coil to give it gentle bottom heat. The cuttings are inserted in rows so they are easier to manage and remove when their time comes. Here I root all cuttings from delicate plants and those with large leaves that need high humidity to keep from wilting and drying while roots are forming. My little plastic boxes of difficult seeds are also set there, where they will keep warm and protected.

The second propagating box is wide open—merely an extra-large flat filled with sharp sand. Here I root the more rugged, or succulent, plants, such as cacti, geraniums, semperflorens begonias, and anything that may rot if kept too moist. Indeed, the sand may dry out occasionally, but they don’t seem to mind. Both cases are shaded by some old bamboo blinds.

(All of the window-garden plants listed in Chapter 6 do very well in greenhouses. See also the forced bulbs in Chapter 13.)

CHAPTER 6

MINIATURE HOUSE AND GREENHOUSE PLANTS

My gardening friends certainly vary as to ability, inclination, talent, and aptitude. At one extreme there is Aunt Minnie, whose green thumb injects magic into every pot and for whom every plant grows to buxom, blooming perfection. At the other end of the scale there is the poor soul who tries so hard but can't keep a plant alive no matter what she does. Why do some plant-lovers never fail, while others never seem to win? There are a number of reasons, including a difference in growing conditions, variations in the types of plants, and even the amount of practical experience or common sense Aunt Minnie is blessed with.

The truth is that there is no single, incontrovertible secret to success, but rather a combination of many factors of equal, or nearly equal, importance. The best soil in the world won't make plants flower unless temperature and sunlight are right. The frequency with which you water plants should be affected by indoor temperature, humidity, and pot size. And don't think it is Aunt Minnie's favorite brand of fertilizer alone that keep her plants in good health.

In our house, and in countless others, it is not possible to make growing conditions as perfect as a plant might wish; but we try to come as close as we can, and find most plants are willing and able to make concessions. One more thing we do. We learn all about each plant's natural home—desert, steamy jungle, Mexican mountainside—so we'll know what combination of conditions it likes best.

CARE OF HOUSE AND GREENHOUSE PLANTS

Light and Sunlight

Plants need daylight to make energy; they can't grow without it, or its equivalent in artificial light. They need some sunlight to set buds, and can't flower without it or a substitute. Intensity and duration of light should vary according to each plant's preferences.

No plant of ours is left to waste its life away on the fireplace mantel, where the light is too dim to read the printing on a book of matches. Granted, we might set it there for a day or so as a decoration, but we would return it to its bright window sill before it began to stretch out weakly in search of light. These are the signs of insufficient light—weak, lopsided growth; leaves that feel limp and look pale; new leaves, if any, growing progressively smaller and smaller; long leaf stems and long internodes.

Our window greenhouse, which faces south, is reserved for flowering plants that need all the light and sun we can give them, particularly in winter. This is Connecticut, remember; the winter sun seldom gets very hot or stays bright very long. And there are days on end when the sun simply does not show its face. California's "shade plants"—begonias and fuchsias, for example—require full sun during a Northern winter.

For plants that are not quite so greedy for sunshine, we have the east-facing playroom picture window, which is lightly shaded by a high-branched deciduous tree in summer. For foliage plants, and a few others that will thrive on little or no sun, there are several other windows around the house where there is plentiful daylight, but the sun is seldom seen.

Your situation may be entirely different—perhaps a glassed-in porch that's brighter than our window sills ever will be; or a contemporary-style glassed-in entry that lets sun stream in all winter long. Keep in mind that various plants have varying needs for light and sunlight; and the only sun that benefits a plant is that which falls directly on it. A south window may actually be shady in summer, and this is truer the farther south you go.

Temperature

Some plants grow their healthiest when the thermometer reads 50 to 55 degrees at night and 5 to 10 degrees higher during the day; some tender tropical *émigrés* suffer a chill when the mercury dips under 60 degrees at night. But the greatest number will tolerate a fairly wide range, and are quite content with whatever the house has to offer. This again assumes that other conditions, such as humidity, are kept up to the plants' requirements.

Temperature tolerances are directly affected by the humidity in the air. Many plants will accept a temperature that is higher than average if the air is moist. Some will accept a temperature that is below average if the air is

not too moist. In air that's too hot and dry, leaf edges will often turn brown and crisp; when it is too cool, a plant may stop growing temporarily; when it is both too cool and too moist, there may be danger of rot. Don't trust the thermostat that governs your house heating system to tell you whether the temperature is right for a plant; measure it on the very spot where the plant is growing.

Humidity

This is often a more crucial factor than temperature, and one that is more troublesome to change. Relative humidity is a measure of the amount of moisture in the air as compared with the maximum amount of moisture the air can hold at a given temperature. It is expressed in percentage figures.

Most plants find it difficult to breathe in dry air; they need some moisture around, on, and in their leaves. (People are better off when they have it, too.) Alone, or teamed up with unsuitable temperature, low humidity can cause leaves to wither, buds to drop before they open, and even an invasion of mites or some other pest.

A humidity gauge is not nearly so familiar a household item as a thermometer, but it is equally useful, usually better looking, and not much more costly. And it can come up with some eye-opening information. The humidity gauge will tell you very quickly, for example, when an air-conditioning system is missing its recommended mark of 50 per cent relative humidity. It will stop you from overwatering in muggy weather. It will show you that humidity can be dangerously low in the window where your plants are growing, even though it is perfectly acceptable three feet away. (Perhaps there is a radiator beneath the window.)

What you do to increase humidity depends on how low it is, and on what type of plant you are growing. At our house we have three systems for the humidity problem—the lazy, the halfway, and the necessary nuisance. The lazy way works fine for dry-air-type plants such as most cacti and other succulents. We simply grow them where the hot blasts from the radiator are strongest. They don't wither, and when on sunny days we water them, we also mist them with a fine water spray. (For an easy misting device, we have an old Windex bottle with a spraying attachment in the cap. It does a

fine job. Of course, there are other commercial products with built-in sprayers. Perhaps you have one in your kitchen.)

To go “halfway,” you set up some simple humidifying device. The containers which you fill with water and hang behind the radiators are effective to some extent. But don’t forget to keep them filled. Pans of water on top of the radiator also help. A steam kettle, such as you use when the kiddies have a chest cold, can be used for a few hours. But remember, that steam is hot, so be careful not to get it too close to your plants. Just keep it in the general vicinity; please, not too close! Here is something else we often use—put some gravel, sand, perlite, or vermiculite in a plate or platter, pour in some water, and put your flower pot in it. The water will evaporate and moisten the air around your plant. If you have a number of plants, get a metal or plastic tray. If you have a decorative garden, you might want to have a tray made to order, to fit your space. Paint it a pleasing color. I have seen such trays decorated with decals, but personally, I’d rather rely on my plants for the necessary decoration. In having a tray made—we’ve had several made by our tinsmith—the larger the tray, the better.

The “nuisance” system is usually necessary only for very delicate tropical plants that flourish in highly humid jungles and rain forests. They will need the platter, or tray of moist sand, plus some enclosure to hold in the moist air. For a single small plant, this can be an inverted bell jar, or a tent made from a polyethylene bag (the sort vegetables are often packed in) and propped over the jar and plant. A larger bag can be used to protect several plants. You can even use kitchen-type plastics such as Saran Wrap. A collection of tropicals almost calls for an indoor greenhouse, like the oldtime Wardian case. In our window greenhouse we can close the inside sash and, with the aid of pans of water, keep the humidity just about where we want it. On cold nights the electric-coil heater isn’t always adequate, so we open the sash to get the benefit of the heat of the room.

The plants that need this extra humidification are definitely in the minority; most are content with simple measures, or none at all.

Soil

When I first became interested in a wide variety of indoor plants, I prepared soil mixtures with the care and precision I had used in making the formulas for our children. It involved trips to the woods for leaf mold, sifting and mixing ingredients, and sterilizing the ordinary soil which we used in order to kill the seeds that are found in almost all soils. In our Long Island home we had an extra stove in the basement which we used for canning and other secondary purposes. I'd fill the oven with old baking pans and other receptacles which contained my potting soil, and light the burner. Frankly, it "reeked" and the reek was all over the house. It was as though we were burning feathers in the fireplace. Now that I am less energetic, and somewhat wiser, I buy prepared soil.

I find that for a reasonable number of plants the prepared soil is not costly and saves a lot of time and preparation. Of course, if you are a professional you may have your own ideas about soil, and your own formulas for a mixture. I'll admit I do a little mixing on my own, but that is because I grow many different kinds of plants in pots, and each has individual preferences. Most cacti and succulents need something sandy that doesn't hold moisture too long, as is natural in their desert homes. Tropical plants need a light, porous medium that is rich in moisture-holding humus, like decayed leaves on the jungle floor. Some plants get nourishment only from acid soil, some need alkaline soil, some like soil nearly neutral. I've found Michigan peat moss a good starting base regardless of requirements of individual plants.

I like to dish it right out of the bag. When done, I always pull the plastic inner liner together to keep it moist. In extra dry weather I often wet a sponge and put it on top of the mixture. Potting is a pleasure, because Michigan peat feels so soft, silky, and clean. At repotting time I discover well-developed, healthy root systems.

For plants that require better-than-average drainage—mostly succulents—I mix peat moss with builders' sand. (Want to know where I got my last batch of builders' sand?—from a contractor who was building a house down the road. He and his wife happen to be begonia lovers and he was charmed to know about my book *All About Begonias*.) The formula for the mixture is two-thirds soil and one-third sand. For even greater aeration, I may throw in a moderate ration of crushed charcoal.

On the other hand, some of my plants insist that their soil always be moist. For them I add coarse vermiculite, perlite, or Pelonex, and in a similar ratio. These soil conditioners have the ability to hold moisture without making the soil muddy or soggy.

There is another item on my potting-soil shelf—a jar of horticultural lime. This is for plants that dislike acidity. I either add it to the potting soil—a scant teaspoon per three-inch pot—or mix it with water, to be applied later. With or without these added ingredients, I feel that a good potting soil has sufficient nutrients for almost any plant for several months; so I seldom mix in fertilizer of any kind.

Potting

When and how to repot a plant should be the least of a gardener's problems. Here is a case where hovering, pampering, and fussing usually does more harm than good. Plants are often better repotted *mañana* than today.

I know many of my plants have benefited from the fact that I have been too busy to repot them whenever the impulse came over me. I hate to think how many I have killed with kindness in my less active years.

Remember my Aunt Minnie? She embarked on a big repotting spree every spring, but not very often in between. Some of her most handsome specimens have lived in coffee tins and large juice cans for years. Instinctively she knows when a plant should be repotted. Don't ask me how she knows it. And when that time comes, she goes about it in a firm, no-nonsense manner, firmly but tenderly.

There is one sure way to tell whether a plant needs a new pot. Turn the old pot upside down, tap the contents loose, and examine the soil ball. If it is completely covered with a network of roots, get busy with a larger pot. If not, don't repot, just slip it back where it was before and give it a loving pat. If you happen to have a seedling you have great hopes of raising to a beautiful maturity, it will have to be repotted more often.

The kind of pot is a matter of personal choice. Plastics are lighter in weight, easier to clean, and capable of keeping soil moist for a longer period of time. This makes them suitable for moisture-loving plants, or for

gardeners who have a tendency to forget the watering pot. Clay pots are porous, and because they let air seep into plants, they dry out faster. This makes them best for dry-growing plants, and for overwaterers (like me).

Actually, the size of the pot is more important than the type. It takes an expert to know how to water a plant when it is overpotted. You are not doing your plant a favor by housing it in a pot several sizes too large; in fact, you may even be signing its death certificate. As a rule, the new pot should be only one size larger than the previous one, thus leaving just enough room around the roots for some fresh soil. Pack the new soil firmly with your finger tips, a pencil, or a slim piece of wood. Be sure there are no empty air-pockets. Water it thoroughly and set it in a light, but not sunny, spot, for about a week. This will give it time to recover from any transplanting “shock.”

This discussion about overpotting applies to plants in general, but it is even more important with miniatures. Smaller pots will keep them down to natural, miniature size.

Fertilizing

Many people who have fewer plants than I—and more time—tend to make a fetish of fertilizing. I am afraid my methods are haphazard, and not to be recommended. Anyway, I will tell you what I do. Follow it if you like, but quite possibly you may have a better method which you prefer.

I keep two kinds of house-plant fertilizer on hand—a reliable brand of soluble commercial chemical plant food (it dissolves in water and has the note “trace elements added” on the label) and an organic food. This organic food is either manure water or fish emulsion. The fish emulsion comes bottled and is diluted with water. As an amusing sidelight, our tomcat goes slightly out of his mind when around a potted plant that has had fish emulsion for its dinner. The cat just can’t find the source of that aroma. Manure water, on the other hand, has no such fascination. It is easy to prepare. Simply wrap a portion of well-rotted manure in a section of cheesecloth or burlap, and steep it in the watering pot long enough to produce a “tea” fluid. Better do this out of doors, in the garage, or in the tool shed. It is usually “olfactorily offensive.”

Once a month I make a solution of the chemical fertilizer, at half the strength recommended on the label of the package, and feed plants as I water them. Two weeks later, and once a week after that if they need it, I feed with the organic solution. This, I think, constitutes a “balanced diet” for most types of plants.

Like overpotting, overfertilizing can lead to lost plants. It is my observation that an underfed plant usually doesn’t die quickly. It simply slows down until you have time to feed it. It is particularly important *not* to fertilize plants that have been repotted recently, plants that are unhealthy or are plagued by insects or disease, plants that are resting right after flowering, plants that are dormant or semidormant, as some of them are at certain times of the year. *Do* fertilize plants that are in active growth, setting buds, or in full bloom, plants that are aglow with good health, plants whose roots have filled, or nearly filled, their pots.

Watering

My children, who are learning to care for plants rather early, have been grounded in one fundamental fact—feel the soil in the pot to see if the plant needs water. If the soil is dry, fetch the watering pot (succulents can be an exception). I have been told my kiddies have had arguments at school about watering the plants which cluster on the sunny window sills. The usual procedure in our Redding school is to assign the care of the plants to a different child each day. He, or she, floods everything with water and then retires to his, or her, desk with the feeling of having done his, or her, good deed for the day. Plants, alas, are not like goldfish. They don’t care for swimming.

This brings up a question I am frequently asked about house plants: “How often should they be watered?” Even an IBM calculator with a thousand cards feeding through its maw couldn’t come up with a better answer than this simple statement: “Water them when they need it.”

At the risk of repeating myself, I will sum up my thoughts on this problem, and then go on to other subjects. Test the soil with your fingers, if it is dry, you can most likely water with impunity. If moist, more water is inviting root rot. When you water, make sure the entire soil ball is so saturated that excess water runs out the drainage hole in the bottom of the

pot. Miniature plants are likely to require more frequent watering, because their pots are small and they are small. Frequency of watering: Here we get into an awful lot of tangibles, and some intangibles. There are such things to consider as the type of plant and its moisture needs, size of the plant and its pot, what sort of soil is in the pot, is the plant root-bound or not, is it in active growth or dormant, what is the weather like out of doors—is it warm or blustery, clear and sunny, dark and humid, or just another day. Remember, on hot dry days the moisture is going to slip away into the atmosphere. On cloudy days it will hang around longer; humidity will take care of that. If you feel your plant needs water and you are afraid of flooding it, you might try another device. Put some moist peat in a larger pot—or you can use vermiculite or perlite—and set the plant and its pot inside this larger pot.

Right now you may be wondering about hanging baskets and those plants that are wrapped in osmunda and mounted on slabs of wood. Give them a good dunking in the kitchen sink but let the excess water drain off before you hang them again, out of deference to your carpeting and furniture. Actually, the plants don't care whether they spot your furnishings or not.

If all of this is confusing, may I sum it up in a phrase: “Just use your common sense” (horse sense may be a better term). I know I have written a few books about plants and gardening, and countless magazine articles, but here is a very candid confession. Until a very few years ago I knew practically nothing about the subject. What I know I learned by reading books and magazine articles, and “doing.” I know I killed a few plants out of ignorance, but I killed more with kindness. Every time a plant withered and died I felt badly. But I considered it a lesson in what to do, or not to do. I tried to find the cause for the demise. Today I have the satisfaction of having raised thousands of plants to beautiful bloomhood, and largely because of the sacrifice of some obscure begonia, or petunia, a few years back. I have been acclaimed as having a “green thumb.” Actually I do not have a green thumb. No one has. I thoroughly dislike the term. I dislike the implication that a person has a God-given quality which makes plants grow. Making plants grow and prosper is largely the mastering of a few rules, and the use of a lot of common sense. I'll grant you, much of your common

sense comes the “hard way.” But it is fun. I bow most respectfully to those plants which I killed with ignorance and kindness.

Grooming

This is the “beauty-parlor” treatment for your plants. Fading flowers and dying leaves are as unattractive on a plant as those unruly ringlets in a woman’s coiffure. In fact, they are even worse. They take strength from a plant and are unsanitary. Remove them as regularly as you can. You’ll be adding to the welfare of your plant. If the plant is of a type that needs to be pruned or pinched to keep it comely and compact, perform the operation before the branches get that gangly, middle-aged spread. Pinching out the new growing tips, either with sharp fingernails or sharper scissors, usually produces two new branches. If you pinch out the tips of these, you will most likely get four new branches and your plant will become pleasingly plump. The exception is the plant that blooms from the ends of new growth. Stop your pinching before the bud-setting season.

Not many miniatures need staking to keep them upright. If stake you must, do it as inconspicuously as possible and tie the stems neatly in their natural growing manner. Train miniature creepers in the direction you want them to grow, and miniature climbers up their supports. Rinse foliage with a fine spray (use that discarded Windex bottle I described earlier) as often as you can to wash away dust and dirt. Please, may I throw in a plea right here—recently I saw a commercial product which some people use to make their plants more green. It is a spray for foliage. For the moment it makes plants look green with its sheen. But it also clogs the pores on the plant and its leaves. For the moment it is beautiful; for permanency it is harmful. Decide, do you want your plant to live for the moment only?

Insects and Disease

Not long ago, when visiting a friend, I was taken out to her sun room to see her plants. She was proud, but I was shocked. What I saw was the most amazingly bug-ridden collection of fuchsias and geraniums I have seen in many a moon, plus a few poor distorted other flowers which were trying to survive. My poor hostess, honestly, I think she had no idea of her problems and the future of her plants. In fact, they had little future. I thought over that

problem for a spell, and then told her the facts of life. Fortunately, she was able to save much of her greenery.

That poor gardener was on one end of the spectrum. On the other end are the overanxious ones who treat their plants with so many sprays and dusts that the leaves never get a chance to breathe pure air. Fortunately, in between there are the sane, sensible growers who are able to cope with the insects and diseases which beset house plants. As for myself, I have been at both extremes.

I now find prevention is the best protection. That is an old saw, but the best one I know. I keep plants and pots clean, remove dead flowers and foliage before they decay, and use germ-free potting soil. People often give me plants. I accept them most politely, and gratefully; but I am cautious about mingling them with my other residents of the greenhouse and window garden. I keep them in a quarantine section for a few weeks. No use asking for trouble, and I trust my friends understand.

For immediate and reliable treatment, I rely on an “all-purpose” house-plant spray or a dip. As long as I am careful to keep it up, I seem to have no great problems. When I get careless, as humans are wont to do, I regret it.

It is very convenient to rely on the new aerosol bombs—those that are specifically marked for house plants, and not those for household insects. There are several reliable brands, each listing on the label what they are to be used for. If there happens to be an added fungicide, so much to the good. That will cover just about everything.

A few words of caution about the use of aerosol bombs. Follow the instructions on the label to the letter. Hold the bomb at least eighteen inches away from your plant, maybe more. Some of those fluids come out of the can at a very low temperature and have the same effect as a blowtorch on foliage. Try to make sure the spray hits all parts of the plant. Remember, leaves also have an underside. Don't let bugs use that underside as a hide-out.

There was a word of caution on each aerosol container I have seen: “Do not throw into an open flame when empty.” My husband wondered what would happen if he did. He tossed an empty can into the trash burner and got behind a tree. The explosion sounded like a hand grenade left over from

World War II and he had to get a new lid for the burner. Please take my word for it and don't take that chance. Let him be crazy if he likes, but not you. I don't like to lose readers so early in my book.

I know I take precautions, but in spite of everything my plants become infected at times. When that happens it's helpful if you can recognize the symptoms, diagnose the ailment, and apply the cure. With your interests in mind, I have put together a list of sorts which includes the most prevalent house-plant insects and diseases, the telltale signs of their arrival, and the insecticides and fungicides which will send them to their happier hunting grounds. If I have skipped anything, my apologies. If you are having special problems, phone your local county agricultural agent. He is a good man to know anyhow. But please, unless you have something that is out of hand, don't bedevil the man. He is probably concerned with ticks on sheep, ringworm, hog cholera, breachy cows, grubworms, and forty people who want to know why their hollyhocks didn't bloom.

PROPAGATING HOUSE PLANTS

It's hard to tell which provides the warmest pleasure, to give a friend or visitor a rooted, potted cutting of some plant she has admired, or to come home with a gift cutting of a new plant you have admired and coveted. There is a thrill also in adding a new plant to your collection by growing seed from a seed house. And I always have another thrill when the church or the school asks me to contribute plants for the annual bazaars. Fortunately, I usually am able to anticipate these requests far enough in advance to have a supply ready and waiting.

PESTS AND TREATMENT

	<i>Symptoms</i>	<i>Treatment</i>
Aphids (Nasty plant lice)	Little louse-like insects. Green in color in our area. Clustered on new leaves, on flowers, and around stems	This is an easy one to get rid of, but likely to return if you don't watch out. Use a solution of Black Leaf 40 (nicotine sulfate) in a combination spray with lindane or pyrethrum and malathion. Or you can use any of them alone
Mealybugs (Don't let them frighten you)	Little white dabs of cotton clinging to tender joints, stems, and the underside of leaves	Dip an artist's paint brush or a cotton-wrapped toothpick in a solution of malathion. Remove the pests one by one

Mites	Buds and new leaves are puckered and distorted. The underside of the leaves may show fine, white silky webs	Dip plants in a solution of malathion, Kelthane, or Tedion according to the directions on the label
Mildew	Silvery dust disfigures the leaves, most likely in dark, muggy summer weather	Dust plants with sulfur, or spray with one of the new fungicides like captan, zineb, etc. Give plants more space and increase air circulation
Scale	Hard brown, or green, scales irregularly on the underside of leaves	Dip plant in a solution of malathion. Repeat according to directions
Slugs	Slimy villains, like shell-less snails, which hide in soil or under pots by day and come out to chew holes in leaves at night	Use special slug-bait containing metaldehyde according to package directions
Thrips	Brownish blotches under leaves. Disfigured flowers	Dip or spray with malathion all-purpose mixture containing lindane, pyrethrum
White Fly	Minute flies swarm in the air about the plant when disturbed	Spray or dip in malathion, or an all-purpose mixture

Since I fashioned my first crude propagating box—a square cake-tin with a rickety frame covered with kitchen plastic—I’ve always had some sort of similar device in operation. They have been of many sizes and shapes, and have included an aquarium vacated by the children’s goldfish, and a fruit crate from the grocery with cut-to-measure glass sides and top (I use masking tape to hold the corners). The one thing common to all of these boxes is the three-inch layer of some moist propagating medium, ready to receive seeds and cuttings.

The latest and most attractive box I’ve used is the Gro-Master—a ready-made plug-in propagator with an electric coil in the bottom to provide gentle bottom heat. Several similar devices are also available. With something like this one can hardly miss germinating even the most difficult house-plant seeds, or rooting even the most delicate cuttings.

The choice of a propagating medium is all yours. I’ve used vermiculite, perlite, and Pelonex with unqualified success. I tried sharp sand but discovered that it dried out too fast for anything but succulents. Peat alone, or mixed with sand, tends to pack and rots rather quickly. I tried a blend of equal parts of peat, vermiculite, and perlite. It held up rather well, but I seldom bother to mix it. Regardless of what you prefer to use, have the

medium moist when you use it, and water just enough to keep it barely moist during the rooting process.

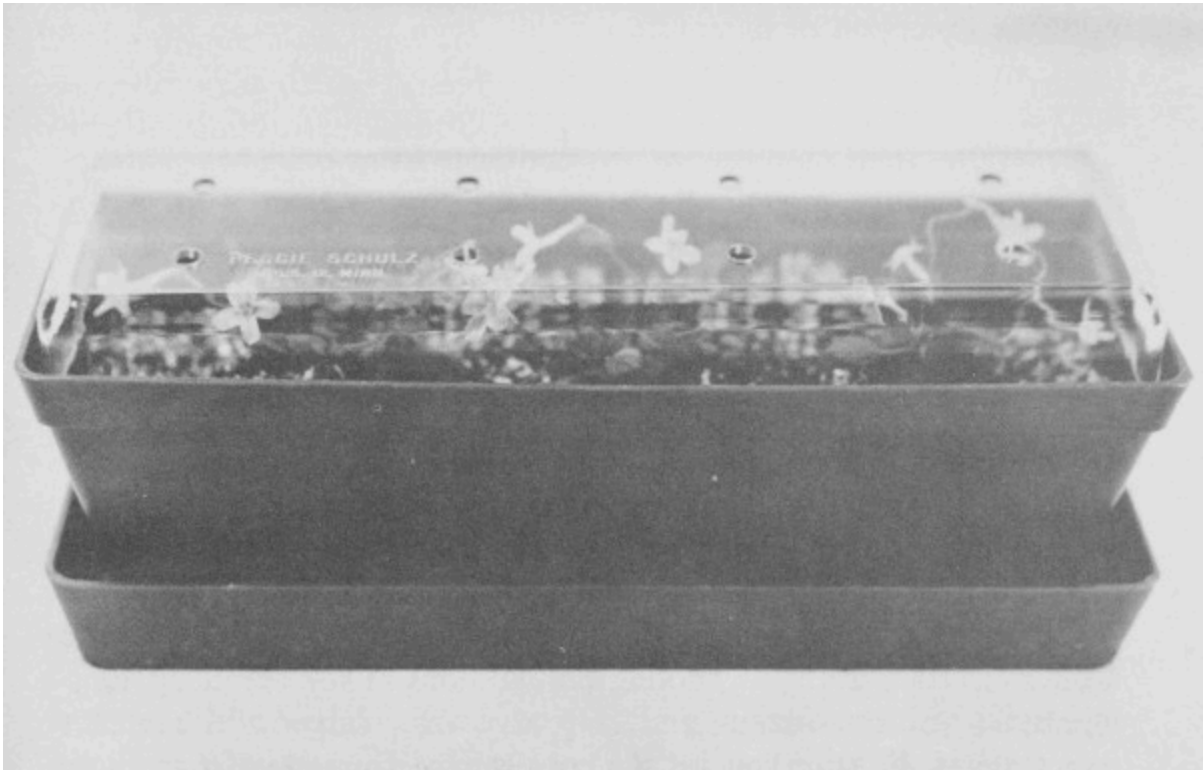
Of course, with many types of plants, you can root “slips” in a glass of water on your kitchen window sill. But in any kind of propagating box, where the air is kept humid and the delicate plants are protected from drafts and drying-out, you have a better chance.

House Plants from Seed

African violets and other gesneriads, all types of begonias and geraniums, and many other indoor plants can be grown quite easily from seed. I like to plant each variety in its own small plastic refrigerator dish with a half-inch or so of soil (moist, sterilized soil that is) in the bottom of the dish. Sometimes I use plastic ice-cube trays. When planting powder-fine seeds, I usually add a thin layer of moist, finely-milled sphagnum moss and make it very smooth. Dust the tiny seeds over this surface, but don’t cover them with soil. Larger seeds should be covered with soil or more moss. Next, cover the container tightly and set it in a warm spot. That is when you will appreciate a propagating box. If the moss and soil begins to dry out, moisten it by gently running drops of water down the sides of the container. Please be gentle and tender. Tiny seeds and seedlings should not be disturbed.

When the seedlings have developed one or two true leaves of fairly substantial size, pick them out very gently and transplant them to another “community” container, or to individual thumb pots. Keep them humid and protected until they are big enough to need transplanting again. After that, they should be nearly ready to become adjusted to grown-up growing conditions in a window, or some other indoor garden.

One of the neatest tricks of the year is the Water ‘N’ Watch indoor garden developed and sold by Peggie Schulz, an attractive plastic planter complete with soil and planted with fourteen types of seeds. You simply follow her instructions, add water, and watch the seedlings germinate and grow. Gradually you remove the plastic dome when the plants are large enough. At transplanting time, remove the seedlings to separate pots, or leave a few to mature in the planter. This device is just about as work-free a method of growing house plants as there is.



Propagation box developed by Peggie Schulz, author and garden columnist



Sprouted stem cuttings of dwarf geraniums

Stem Cuttings

This is the most familiar way to propagate house plants, and the method that works best for most kinds of plants. With the exception of miniature orchids, stem cuttings can be taken from any house plant mentioned in this book, and your chances for success are very good if you have a propagating box for them.

Generally, stem cuttings are the ends cut from branches that are in healthy, active growth. Each should be more than a mere tip, and have at least two nodes, or “joints.” The large, bottom leaves are stripped off very gently and the cut end of the stem is inserted, to about half of its length, in the propagating medium. When you see evidences of new growth, or when roots have developed, dig under it with a spoon and remove it for potting. (Here is the test to determine if roots have developed: Pull on the cutting, very gently of course. If you feel resistance you know it has roots which are holding it down.) Pot your new plant very carefully in soil that is suitable for it; and please, don’t add fertilizer to this soil.

For stem cuttings, and most other methods of propagation, there are now certain hormone rooting preparations. They speed up rooting, make it more certain, and help to develop larger and healthier root systems. Be sure you get the correct type for softwood cuttings, or indoor plants. In using these new hormones, follow the directions on the package carefully.

Leaf Cuttings

It is a well-known fact that African violets will reproduce themselves from a single leaf. However, it is not so well known that there are dozens of other plants that will do the same. For example, more than once I have taken a succulent leaf, laid it out flat on moist sand, and had it grow roots and a small new plant. Leaves of peperomias, and many other plants, can be cut, with or without a piece of stem, and will grow bushy new plantlets. Some kinds of begonias will send out roots from the end of the stem, or new plants from the point above where stem and leaf join.

Except for some succulents, leaf cuttings are usually inserted with their stem ends in a moist propagating medium, with their leaves standing nearly upright above. (Again, you are safer if you have a propagating box, even though it be a crude one.) When they are well rooted and the new plants are

of fair size, the cuttings are potted with the old leaves intact. The parent leaf is removed only when the new plant is strong enough to grow on its own. Of course, in the case of African violets, where more than one plant is produced at the end of the stem, the babies must be separated and each put into its own pot.

Root and Rhizome Cuttings

Sections of roots are sometimes cut to propagate house and greenhouse plants, but the practice is more common with hardy garden plants. But pieces of thickened, stemlike, or rootlike rhizomes can be taken from many types of indoor plants. Rhizomatous begonias, and rex begonias of rhizomatous habit, are familiar examples. Each piece of rhizome should have at least two, but preferably three, “eyes,” or scars from which leafstems have grown. The piece is placed half in and half out of the moist medium in the propagating box. Roots grow from the bottom down into the medium; stems and leaves shoot up from the eyes on top.

Layering

There are two general methods of layering, and in each case the plant is completely rooted before it is cut away from the parent. In “air layering” a thick stem, or cane, is slit, wrapped with moist sphagnum moss, and enclosed in plastic until roots are produced. This method is usually practiced on large, overgrown plants such as dieffenbachias, and truthfully is not necessary, or practical, for most miniatures.

But ground layering is a safe and sure way to propagate almost any plant with lax stems, and is particularly useful on temperamental plants such as some miniature ivies, which are reluctant to strike roots on stem cuttings. It is very simple. Just pin down the stem, not too far from the growing tip, in the soil beside the mature plant. If you like, you can put it in a pot of its own and thus be one jump ahead by having the roots where you want them. I usually steal a hairpin out of my “bun,” but a section of wire bent into a *U* will do. When the stem has rooted, cut it off on the side next to the parent plant. Unless it is already rooted in a pot, dig it out and plant it.

Division of Crowns and Roots

One African-violet plant may grow several crowns, or main stems complete with rosettes. Since single-crown plants usually are more shapely and bloom more abundantly, the extra crowns can be cut off and rooted like ordinary stem cuttings. This is one of the more popular methods of dividing an old plant and getting several new ones from it.

Some plants will eventually make such dense, bushy growth that they not only look obese and unattractive, but are prey to all sorts of rot. Root division not only gives these plants a new lease on life, but also provides several new plants. Remove the overgrown plant from its pot, and very gently, try to split the matted roots and stems into separate sections. Do this with your fingers and not a metal instrument. If you work slowly and with care, several will most likely pull free. If you can't do it with your fingers, you may have to resort to a clean sharp knife, in which case cut down past the stems and through the roots, making several separate sections. Discard any damaged leaves. Pot up the divisions, and keep them shaded and protected for a week or so until they recover from the shock.

Runners, Stolons, and Offsets

Many plants are continually propagating themselves by producing new plants in clusters around the old ones, sometimes at the ends of underground stolons or above-ground runners. *Saxifraga sarmentosa* was named "Strawberry begonia" because it continually produces new plants on runners, strawberry-style. *Episcias* and *chlorophytums* are other examples.

Any of these offsets make new plants easily. Pin them down in soil, give them time to root, and then cut them off and insert them in a propagating box. Next, you pot them. Sometimes you'll find the roots have already formed and the new plant is ready for potting and you didn't even suspect it.

DESCRIPTION OF HOUSE PLANTS

Some pages back I told you that the question asked of me most frequently is "How often should I water plants?" A close runner-up is the question, "Where can I get out-of-the-ordinary plants?" Believe it or not, that question is easier to answer. Local florists and greenhouse people often have plants tucked away—something exotic and special. Most of the people in the industry cherish these plants and are hesitant to part with them to run-

of-the-mine customers. But if the person is a genuine plant-lover, a sale can usually be made, even though it is a reluctant sale. Just demonstrate that you are vitally interested and you will be surprised at what you can buy. Try it.

I have a number of catalogues from mail-order suppliers in all parts of the country who carry unusual plants, and in a variety that increases year by year. Do you know where I came upon their names?—from small advertisements in horticultural magazines. Read those small ads. Often the advertiser is so small, and so specialized, that he can't afford too much space. But he has the treasures you want.

Mail-order plants are naturally young and small. Large plants are too difficult to pack and ship for any distance. Plants that are small usually cost less, and you have an extra dividend in the pleasure of seeing them grow to maturity. You would be amazed at the progress that has been made in packing and shipping in the past few years. Plants that I have ordered recently have come through the mails in from excellent to perfect condition—even plants from the Pacific Northwest. You simply unpack them—carefully, I'll admit—water them if necessary, and keep them out of hot, bright sun and draft for a few days until they are acclimated to their new surroundings. Should any plant seem weak or mildly injured, pop it into the propagating box and usually it will be back on its feet again in a few days. If the damage is too serious, report it to the post office, the express company, or the shipper. I have yet to register one of these complaints.

I have been general in the preceding pages of this chapter. Now I will be more specific and offer a list of plants and their growing requirements. All of the following plants can be grown in the greenhouse. But given proper light, humidity, and/or other special care, they can be grown in the house where you live—even the most delicate types. The information on culture is abbreviated, but consistent use of the same terms, I feel is justified. The following are used in the sections on care under each plant.

Full sun

The sun actually falls on the plant for most of the day.

Partial sun

The sun touches the plants for only a few hours in the early morning or late afternoon in summer, but for longer periods in winter (tropical areas excepted).

Little or no sun

Bright daylight is sufficient to keep some plants healthy.

Temperature

Any figures that are quoted are for daytime. Night temperatures can usually drop about five degrees. (Cool is 40 to 60 degrees; moderate, 50 to 70 degrees; warm, 60 to 80 degrees.)

Humidity

This is noted if the plant needs more than average humidity (50 per cent).

Soil

The plant may prefer a potting mixture that is *humusy*, extra-rich in organic matter; *average*, like the potting soil I have described on other pages; *sandy*, with an extra ration of sand, or a similar substitute. Or it may be *epiphytic* and grow in some suitable medium such as osmunda fiber or sphagnum moss.

Dry

Let the soil dry thoroughly before watering.

Moist

Water when the soil in the top of the pot feels dry to the touch.

Wet

Never allow the soil to dry out.

Methods of propagation are also summarized and, if possible, listed in order of preference. The terms are the same as those used in the sections on propagating house and greenhouse plants.

For most plants, there are short notations on special uses for which they are especially suited, i.e., growing under artificial lights, dish gardens, model landscapes, terrariums and other gardens under glass. Included also are some small tropical trees and shrubs for the small greenhouse. Among the precious miniatures described in Chapter 17, varieties suitable for forcing are so noted. Forcing procedures are also covered.

***Abutilon hybridum savitzi* Malvaceae** Flowering Maple

Delicate tropical shrubs with thin, soft, maple-shaped, dove-gray leaves lavishly splashed with creamy white, and sometimes white all over. (I've never seen it produce the typical dangling lantern flowers.) It grows slowly, keeps plump and bushy, and is content with a small pot for several years.

CARE. Full sun, moderate temperature, and average soil kept constantly moist.

SPECIAL USES. Artificial light, dish gardens, model landscapes, terrariums.

***Achimenes* Gesneriaceae**

A large group of utterly charming plants related to, and grown like, African violets—except that their scaly rhizomes (like minuscule pine cones) are completely dormant in winter. Most have velvet-soft leaves and all produce an overwhelming display of tubular flowers flaring wide-open at the end. None grow to great size. A single rhizome would probably be happy in a two-inch pot; several are usually planted together for a bigger show. The following are among the smallest varieties, upright unless otherwise noted.

‘Charm’—Luscious pink flowers, gold in the center.

coccinea coccinea—Trailer with notched, oval leaves and plentiful flowers of deepest scarlet, half the size of a dime.

‘François Cardinaux’—Flowers in two tones of blue.

‘Little Beauty’—Similar to ‘Charm,’ but in another glowing shade of pink.

‘Masterpiece’—Flowers royal, reddish purple.

‘Miniata’—Ruffled flowers in changeable violet tones.

‘Misera’—Many small white flowers spotted with drops of wine, yellow in the throat.

‘Patens Major’—Orange throat blending into rich plum petals.

‘Peach Blossom’—New dwarf with peach-pink flowers.

‘Violacea Semi-Plena’—The only semidouble-flowering variety in deep, glowing purple.

CARE. Partial sun, warmth, humusy soil kept moist. In late winter, or early spring, pre-root the tiny rhizomes in a light, porous mixture such as vermiculite and peat, barely moist and over bottom heat, or in warmth (65 to 70 degrees). When new growth is about two inches high, pot in light soil enriched with leaf mold or some similar organic material. Pinch out any growing tips or trailers once or twice to encourage branching. In early fall, after flowering has ended, let the pot and soil dry out gradually. Clean the rhizomes, and store them in a polyethylene bag filled with dry peat at a temperature of 50 to 60 degrees. Check the little fellows occasionally. You may be amazed, but sometimes they’ll send out new sprouts in this dark, dry storage, and want to be rooted and begin growth again.

PROPAGATION. Rhizomes will multiply spontaneously. Also use stem cuttings, leaf cuttings, seeds.

SPECIAL USES. Artificial light.

***Acorus gramineus variegatus* Araceae**

Neat, sweet little water-lover, like a miniature grass, with flat fans of slender, four-inch leaves striped lengthwise with sparkling white. It spreads eagerly.

CARE. Partial sun, cool, average soil kept wet.

PROPAGATION. Division of creeping roots.

SPECIAL USES. Artificial light, dish gardens, terrariums. Pretty bog plant for the miniature garden pool.

***Adiantum bellum* Polypodiaceae Bermuda Maidenhair Fern**

Elfin version of the maidenhair fern with ruffly leaflets like upside-down wedges, the points attached to the dark, wiry six-inch stems. The foliage is surprisingly dense for such a delicate, airy effect.

CARE. Little or no sun. Warmth, humidity, humusy soil (lime if acid), kept moist, and wet in winter. The plant seems to rest in summer.

SPECIAL USES. Artificial light, terrariums.

***Allophyton mexicanum* Scrophulariaceae Mexican Foxglove**

Perky combination of oval, dark-green five-inch leaves beneath upstanding stems bearing several long-tubed, flaring half-inch flowers of pastel lavender blending into white, violet in the throat. Blooms in spurts throughout the year.

CARE. Partial sun, moderate warmth, humidity, average soil kept moist.

PROPAGATION. Seeds.

SPECIAL USES. Artificial light, dish gardens.

***Alternanthera* Amaranthaceae Joseph's Coat**

Enthusiastically branching, bushy plants with leaves of many colors, and convolutions, like small, contorted coleus. They're often kept dwarfed by regular shearing, so the white flowers seldom form.

amoena—Bushlet with crisp, oval leaves haphazardly daubed with bright shades of red and orange. It seldom tops four inches.

bettzickiana—Tongue-shaped leaves blotched with cream, yellow, salmon, and red. This is the one that's used for formal carpet bedding. Its green-and-gold variety, *aurea nana*, makes a round three-inch mound.

versicolor—A gnome with gnarled, wrinkled leaves. It is basically green but brightened with shocking pink and white. It can grow six inches high, but stays lower if pinched regularly.

CARE. Full sun (for best color), warmth, average soil kept moist.

PROPAGATION. Stem cuttings, division of roots.

SPECIAL USES. Artificial light, dish gardens, model landscapes, terrariums.

***Anthurium scherzerianum* Araceae Flamingo Flower**

This is a baby in a family predominantly of giants, but it won't outgrow a three-inch pot for years. This exotic tropical plant has leaves like varnished green shields and flowers like a golden Jack preaching from a flamingo-pink pulpit.

CARE. Little or no sun, warmth, humidity, humusy soil kept moist, or even wet.

PROPAGATION. Division of suckers and seeds.

SPECIAL USES. Artificial lights.

Babiana stricta *Iridaceae*

Winter-flowering, South African bulb for forcing in the greenhouse. Grown outdoors only in frost-free climates. Clusters of red or lavender fuchsia-like eight-inch stems above fuzzy, slender leaves indented at the veins. A dwarf variety, 'Blue Gem,' has deep-blue blooms.

CARE. Full sun, moderate warmth, average soil kept moist. Plant in fall for winter flowering.

Begonia *Begoniaceae*

There are enough miniatures in this big happy family of plants to make a sizable collection, and enough variations to keep the collector fascinated. Botanically, begonias are divided into three classes—fibrous-rooted, rhizomatous, and tuberous-rooted. Among the tuberous types, the best-known are the summer-flowering garden beauties—not available, as far as I know, in miniature. Each of the other classes contains miniatures that divide naturally into two general types of begonias with separate personalities, habits, and cultures. You could almost consider each type a completely distinctive group of plants, only technically related to the others.

SEMPERFLORENS BEGONIAS

Fondly known as “wax begonias,” and often called “America’s favorite house plant.” These are bustling, buxom, freely branching plants with watery stems and crisp, nearly round leaves gleaming with a high polish. They cover themselves with continual bursts of white, pink, or red flowers. The furiously flowering singles are the oldest, best-known, and toughest, often used for edging semishady garden beds. The semidoubles (crested or thimble type) have a raspberry-shaped center extending out from a circle of petals. The doubles (rosebud or camellia-flowered) are fluttery, full-petaled spheres. Foliage may be clean green, bronzy, or mahogany.

‘Adeline’ (‘Improved Darling’)—Free-blooming soft, single pink; green leaves.

‘Andy’—Deeper, more luminous pink flowers; green leaves.

‘Little Gem’—Double, rosy-pink flowers; very dark red leaves; small, slow-growing.

‘Pied Piper’—Baby pink, semidouble flowers, the crest sometimes touched with gold; bronzy leaves.

‘Snowdrop’—Smallest I’ve ever seen, has never topped three inches for me, just grows bushier and bushier. Double white flowers like minute snowballs; dark-red foliage.

tausendschoen (‘Thousand Beauties’)—A group of green-leaved, single-flowering dwarfs available in red, pink, or white. Easily grown from seed.

‘Winkie’—Fully double, old rose flowers; masses of dark leaves.

CARE. Partial sun, moderate warmth, average soil kept on the dry side.

PROPAGATION. Stem cuttings (best taken with a branch, so the plants will be self-branching), seeds (singles), division of root and crown.

MISCELLANEOUS FIBROUS-ROOTED BEGONIAS

Some miniatures are of the angel-wing, cane-stemmed type; some are from the hirsute, hairy-leaved group; some can’t be categorized.

albo-picta—Small angel wing with low, arched branches; silver-spotted, sharp-pointed slender leaves; clusters of off-white flowers.

bartonea (‘Winter Jewel’)—This one’s a flirt, flaunting its leaves and tiny pink-tinged flowers one wintry day, collapsing completely the next. It can’t bear dry air or chills but will grow up again cheerfully from the roots. The foliage is finely scalloped on the edges, washed with russet in the center and along the veins, and completely overlaid with silver sheen.

‘Dainty Spray’—Impudent dwarf with little angel-wing leaves, bouquets of face-powder-pink flowers dripping from the drooping stems.

dregei—Maple-leaf begonia with sharply cut, thumbnail-sized leaves bronzy with purple veins, white flowers. This is a semituberous type, the main stem swelling to look like a bulb above the soil at the base. Pinching regularly keeps it fairly small. May be dormant in winter. The variety *macbethi* has smaller leaves completely green, white flowers.

‘Dwarf Houghtoni’—For me, this stays under six inches high and covers itself with pointed leaves upholstered with sheer, tawny-pink velvet. The clusters of large white flowers with pink whiskers are almost too heavy for the branches to hold.

‘Medora’—Miniature angel wing with two-inch, flat silver-spotted leaves and white flowers. Also available as ‘Green Medora,’ with plain green leaves and watermelon-pink flowers.

richardsiana—Another semituberous maple-leaf type, the leaves smallest of all and deeply cut almost into lace. Flowers are faint pink.



Achimenes, one of the most beautiful gesneriads.

CARE. Partial sun, moderate warmth, average soil kept on the dry side. The semituberous types take more moisture during the active growing season.

PROPAGATION. Seeds (for some species), stem cuttings, division of root and crown.

RHIZOMATOUS BEGONIAS

Here's where the hybridists are creating the most intriguing new varieties, because they have more miniature parent species to work with. The leaf and flower stems grow straight up or out from the rhizome, a swollen, scarred rootstock that creeps over the top of the soil, usually sending down roots as it goes. Sprays of trembling flowers stand well above the foliage in late winter or early spring.

aridicaulis—Mounds of tiny, sharp-pointed, lettuce-green leaves seldom over three or four inches high. Small, white, two-petaled flowers like fairy pocketbooks.

boweri—Better known as the “eyelash begonia” because of the black stitching around the edge of the lettuce-green leaves, marked with bristly black hairs. Flowers are small, baby-pink, and plentiful. This is the seed parent of a whole group of popular namesakes; star-leaved ‘Bow-Arriola,’ chocolate-stitched ‘Bow Chance,’ dark-complexioned ‘Bow-Joe,’ bronzy ‘Bow-Nigra.’ All have the eyelash edging and pink flowers, and are happy plants for the window garden or greenhouse.

hydrocotylifolia—“Miniature pond-lily begonia” for the terrarium or shallow basket or pot. Shiny, penny-like leaves overcast with bronze, dark along the veins; pink flowers on six-inch stems.

‘Maphil’ (‘Cleopatra’)—Most famous *boweri* offspring, more dwarf than miniature, just right for small window gardens and baskets. The starlike leaves are satiny, irregularly marked chocolate on chartreuse; the flowers, rich pink.

mazae—Bronzy, half-dollar leaves with light veins that meet at the stem end to make a white eye, wine-red beneath. Pale-pink, red-spotted flowers.

rotundifolia—Very similar to *hydrocotylifolia* except for the bronzy cast and dark veins. May be the smallest of all.

‘Spaulding’—Extra-bushy dwarf with medium-green leaves shading to dark green, edged with whiskers, and oxblood beneath. Showers of pink flowers.

‘Virbob’—Reddish star leaves with yellow-green leaves, bright red beneath. Short stems hold the leaves close to the pot.



‘Spaulding,’ an extra-bushy dwarf begonia

In the past few years, Mrs. H. E. Dillard of Tropical Paradise Greenhouse has introduced a number of delightful new dwarfs and miniatures. Among them:

‘Baby Perfection’—Star leaves splotched mahogany on green.

‘Black Falcon’—Darkest red-brown star leaves, silvery along the veins, whiskery along the edge.

‘Chantilly Lace’—One of my favorites, with cupped, chartreuse leaves stitched with black around the edge.

‘China Doll’—Pointed yellow-green leaves striped brown along the veins.

‘Kathy Diane’—Pointed oval leaves brown splotched with chartreuse.

‘Midget’—Nearly black star leaves silvery green along the veins.

‘Oriental Music’—Dwarf with pebbly apple-green leaves.

‘Persian Brocade’—Green star leaves intricately laced with black along the edge.

‘Raspberry Parfait’—Pointed, velvety, olive-green leaves lighter along the veins; new leaves flushed with bright pink.

CARE. Partial sun, moderate warmth, humusy soil kept moist.

PROPAGATION. Rhizome cuttings, leaf cuttings (slow), seeds (for some species).



Group of dwarf begonias: *lower left*, ‘China Doll’; *upper left*, ‘Silver Jewel’; *upper center*, ‘Bow-Chance’; *upper right*, ‘Bow-Arriola’; *lower right*, ‘Bow-Nigra’; *center*, ‘Chantilly Lace.’

MINIATURE REX BEGONIAS

These begonias are also rhizomatous, but the brilliant patterns of their leaves put them in a class by themselves. Few other foliage plants have

such startling combinations of peacock colors.

‘Baby Rainbow’—Crinkly, jewel-like leaves with bands of royal purple, emerald green, silver, raspberry, amethyst. Grows and shows off best in glass.

‘Dew Drop’—Thin, ivy-shaped leaves completely overlaid with shimmering, lavender-pink mother-of-pearl.

‘It’—Branching, upright rex type with silver-splotched green leaves, multitudes of large pink flowers. Likes some sunlight.

‘Lorraine Closson,’ ‘Louise Closson,’ ‘Lucille Closson,’ ‘Lucy Closson’—A group of aristocratic dwarfs with taffeta-textured leaves in varying patterns of black, purple, red, pink, silvery green.

‘Pansy’—Small, pointed, deep-green leaves with sharp zone of lighter metallic green.

‘Peacock’—Jet-black and scarlet leaves on short stems. Stays small if fed sparingly.

‘Red Berry’—Sheer, shimmering satin leaves of unrelieved claret.

CARE. Little or no sun, warmth, humidity, humusy soil kept moist. Many rex begonias may go partially or completely dormant in winter, dropping some or all of their leaves. Simply keep them warm, with the soil slightly dry, until signs of new growth appear.

PROPAGATION. Rhizome cuttings, leaf cuttings. Seeds will produce an unpredictable mixture, seldom like the parent plant.

Bertonia *Melastomaceae*

The slowly spreading stems look like fat little top-of-the-soil rhizomes; but the foliage is like nothing else on earth. Perfect pointed ovals seem fashioned of sheerest silk; the skeleton of veins is sunken and strikingly marked. Such daintiness, to be endowed with bristly whiskers! Wee flowers cluster at the top of short upright stems. None of the bertonias are very large; these are the available miniatures.

maculata—Sheer leaves deep green shading to light, pencilings of silver along the lengthwise veins, wine red beneath.

pubescens—Many veins pucker the leaves like small-scale seersucker. Colors are copper over green, plum purple down the center.

CARE. Partial sun, warmth, humidity, humusy soil kept moist.

PROPAGATION. Stem cuttings (in warmth), seeds.

SPECIAL USES. Artificial light, terrariums.

Boea hygroskopica *Gesneriaceae*

Six-inch tropical gesneriad with fresh green, quilted leaves and clusters of violet-like flowers campanula-blue, with yellow centers, on willow stems.

CARE. Partial sun, moderate warmth, humidity, humusy soil kept moist.

PROPAGATION. Division of crown, seeds.

***Bouvardia longiflora humboldti* Rubiaceae**

Two-foot tropical shrub to perfume the dream greenhouse in fall and winter. The luxuriant, glossy, evergreen leaves are a fine foil for the celestial white flowers—trumpets with long, slim tubes flaring out into perfect four-pointed stars. Florists grow the larger-flowered variety, 'Albatross,' for cutting.

CARE. Full sun, moderate warmth, humidity, humusy soil kept wet except when the plant is resting after bloom.

PROPAGATION. Stem cuttings of new wood with heel (in warmth), root cuttings in early summer.

SPECIAL USES. Greenhouse shrub.

***Buxus microphylla japonica* Buxaceae** Box, Boxwood

Slow-growing boxwood with small, prim, shiny green leaves filling out the plump shrub shape. It is hardy outdoors, but nice in pots and frequently found at plant counters in variety stores.

CARE. Full sun, cool temperatures, average soil mixture kept moist.

PROPAGATION. Stem cuttings of half-ripe wood.

SPECIAL USES. Artificial light, dish gardens, model landscapes, terrariums, indoor bonsai.

CACTI AND SIMILAR SUCCULENTS

Generally, I prefer to grow a spicy variety of plants. But I am certainly in sympathy with the hobbyists who find enough stimulation in this one group to keep them collecting for a lifetime. Such a weird assortment of shapes, from barrels to humping inchworms. Such unusual patterns and colors of

either leaves or stems that have taken on the shapes and functions of leaves. And such flowers! Some like daisies or water lilies, in incredible neon-bright colors, sometimes three times the size of the plant.

Among cacti and similar succulents are some of our smallest plants, plus plenty more that grow so slowly they're miniature for many years. Here are selected samples, only a small portion of the number available from specialist-growers and other sources.

CARE. Most succulents need full sun, moderate warmth, a sandy soil mixture (not pure sand) kept on the dry side. They need more water and warmth in summer, less in winter when they are resting. If soil is very acid, neutralize with lime.

PROPAGATION. Stem cuttings, leaf cuttings, division of root or crown, seeds.

SPECIAL USES. Dish gardens, model landscapes (with other dry-growing plants), a few for indoor bonsai.

***Aeonium caespitosum spatulatum* Crassulaceae**

Clump-forming succulent with rosettes of leaves like the hen-and-chicks, silvery green sparsely spotted with darker green.

***Agave victoriae-reginae* Amaryllidaceae**

Miniature "century plant" with a stiff rosette of thick, sharp-pointed dull-green leaves with white piping along the edge and streaked with white between. Mature size, six inches.

***Aichryson (Aeonium) domesticum variegatum* Crassulaceae**

I'm not quite sure how high and wide this pretty succulent will grow if left to its own devices. Mine has stayed in a two-inch pot for more than a year, and filled itself out with thin, round green leaves edged with creamy white, blushing faint pink in warm sun. The leaves huddle in tight, overlapping rosettes all around the branching stems.

***Aloe* Liliaceae**

Symmetrical clusters of thick, heavy, sharp-spiny leaves; fall and winter flowers held aloft like a torch. One of the smaller species is *A. brevifolia*, with leaf rosettes about three inches across. *A. variegata* can eventually reach a foot high, but very slowly; and its white-marbled leaves are striking in the meantime.

Aptenia cordifolia (Mesembryanthemum cordifolium) Aizoaceae

Creeping, clustering succulent with thinnish, round-pointed, gray-green leaves in pairs along the stems; brilliant fuchsia-purple daisy-shaped flowers. The variety *variegata* is embellished with creamy-white leaf edgings.

Astrophytum Cactaceae Star Cactus

Thick stem-bodies divided neatly into five sections but still attached together, growing very slowly to four inches across. Outlandishly large, flat, daisy flowers in summer. Try *A. myriostigma*, bishop's cap, or *A. asterias*, sand dollar, both spineless; or silver-dotted *A. ornatum*, with swirling lines and tufts of curved spines.

Cephalocereus senilis Cactaceae Old Man Cactus

A good bet for beginning collectors. This is a columnar cactus covered with a shaggy coat of snow-white hairs, growing up to forty feet high in the desert, but approaching that height at a snail's pace indoors. Flowers are rosy-pink, about two inches across.

Chamaecereus silvestri Cactaceae Peanut Cactus

Gay ground-hugger, sending out in all directions thick green two-inch joints with soft white spines, and keeping its miniature proportions except when it's top-heavy with long-tubed orange-scarlet flowers.

Conophytum Aizoaceae

Very tiny succulents with clusters of plump bodies that are, actually, two leaves joined completely except at the tip. The plant barely reaches one inch high and is content in a three-inch pot for years. Ridiculously large and brilliant flowers pop out through small slits in early fall. Then the leaves

look like little dumplings sitting under a daisy. Of the several species available from specialists, *C. ornianum* is light green with darker freckles, lavender-rose flowers; *C. aureum* has gold flowers.

Coryphantha vivipara *Cactaceae*

Symmetrical two-inch sphere covered with evenly spaced bumps, each like a miniature sun with white rays and a spine sticking up from the center. The fringed rosy or carmine flowers pop out on top, in June; bright-red berries appear in fall. Native to, and hardy in, Manitoba, down to Texas. In time it will form clustered mounds.

Crassula *Crassulaceae*

An oddly assorted group of succulents including many roguish miniatures of fascinating form. *C. cooperi* has tufts of small, pointed leaves with black blotches, little clusters of pale-pink flowers. *C. lycopodioides* mimics the club moss of the woodlands, with slim stems encircled with little needle-like leaves. ‘Morgan’s Pink’ is a variety of dense clusters of spear-shaped leaves, crinkled like gray seersucker, coral flowers. *C. schmidtii* makes a three-inch mat of pointed, pinkish leaves, with generous glowing pink flowers.

Echeveria *Crassulaceae*

Perfect rosettes of succulent leaves in many lustrous colors, some silk-velvety or contrastingly trimmed on the edge. Clusters of brilliant tubular flowers top short stems. The following grow low, with four-inch rosettes.

derenbergii—painted lady—Translucent green leaves with silvery sheen, touched with red at the tip, yellow-orange flowers.

elegans—Mexican snowball—Light blue-green leaves frosted with white, pure white on the edge, coral-pink flowers.

pulvinata—chenille plant—Dusty-green leaves of sheer velvet, trimmed with brick red on the edge, scarlet flowers.

Echinocereus melanocentrus *Cactaceae* Hedgehog Cactus

Small, spiny globes with brilliant carmine flowers from the side of the ball. At its three-inch maturity, the flowers are still larger than the plant.

Echinopsis *Cactaceae* Easter Lily Cactus

Small, round plants with thick ribs and formidable spines, the lily-like flowers usually opening in the evening. A popular species is *E. kermesiana*, with glowing red flowers. *E. grandiflora* is described as only two or three inches across, with five-inch rose-pink flowers.

Euphorbia *Euphorbiaceae*

Two small relatives of the poinsettia that show little family resemblance.

caput-medusae—Medusa’s head—Sneaky, snaky-looking plant with tangles of twisted, gray-green branches, occasionally tipped with small leaves.

splendens bojeri—dwarf crown of thorns—Compared to the sprawly species that grows four feet tall or more, this is really a midget. Mine has kept its six-inch bushy contours for nearly two years. The upright grayish branches are fairly well supplied with roundish, dark-green leaves; the scarlet flower-like bracts are plentiful at intervals all summer and fall.

Faucaria *Aizoaceae* Tiger Jaws

Low, crowded succulents with thick, triangular leaves toothed with spiny hairs, unmistakably resembling an animal’s mouth. In late summer or early fall, golden daisy-like flowers pop up and make fun of the plant’s ferocious appearance.

tigrina—Silvery green leaves flecked with white, two-inch yellow flowers.

tuberculosa—Darker green leaves with little white knobs.

Fenestraria *Aizoaceae* Baby Toes

Clusters of cylindrical leaves, larger at the top, like little flat-tipped baseball bats. The nearly colorless tops feature tiny transparent “windows.”

aurantiaca—Three-inch orange daisy flowers more than twice as wide as the clustered leaf-colony.

rhopalophylla—Leaves more blunt, smaller white flowers.

Gymnocalycium mihanovichi *Cactaceae* Chin Cactus

Just one of many available miniature, globe-shaped cacti with spines on regular shelves, or “chins.” This one produces chartreuse flowers bigger than its body, starts to bloom while quite young.

Haworthia *Liliaceae*

Tight pinwheels of thick, pointed leaves intricately studded with varied patterns of pearly pinheads. The whitish flowers are not a main feature.

fasciata—Zebra-striped succulent often seen in dish gardens.

margaritifera—Slightly larger, dark green with a more scattered pattern of white dots.

Kalanchoe *Crassulaceae*

Among these congenial succulents are several that grow to considerable size in their native homes, but keep pleasantly small in pots or dish gardens. The leaves are fleshy, with indentations along the edge. Lantern-shaped flowers appear in winter.

blossfeldiana—Well-branched bush with overlapping, fresh green leaves, flowering in winter when days are short and nights are long. ‘Tom Thumb’ is a dwarf variety that smothers itself with scarlet blooms for Christmas. Greenhouses grow it from seeds sown in spring, and so can you.

marmorata—penwiper plant—Leaves fold in around the stem and are spattered on both sides with purple blotches.

pumila—Leaves like a doll’s spoon, notched on the edge and sugar-frosted; plum-colored pitcher flowers.

tomentosa—panda plant—Fat leaves covered with white felt, distinctly marked with chocolate at the teeth on the edge.

(*Kitchingia*) *uniflora*—Miniature creeper or dangler for small hanging baskets, with round green leaves marching up and down the stem and rosy or red urn-flowers hanging from thin, short threads.

Kleinia *Compositae*

Curious even among succulents, each of these oddly shaped or strangely decorated plants has a personality of its own and no need for daisy-like flower heads to make it interesting.

pendula—inchworm plant—Weird, round, leafless stems snake up and down over the soil; flowers brilliant red.

repens—Low and somewhat trailing, with thick leaves like long canoes, unbelievable blue.

tomentosa—Cylindrical cocoon-like leaves tapered to sharp points and covered with pure-white down; gold or orange flowers.

Lobivia aurea *Cactaceae* Golden Easter Lily

Squat, round, prickly cactus like a small echinopsis, except that this one opens its friendly water-lily flowers in the daytime. Dozens of different species and varieties are available.

Mammillaria *Cactaceae* Pincushion Cactus

There are many mighty midgets in this group, and in fascinating variety. They're mostly round, from squat to columnar, but all are primly neat. The spines may be soft or not, but are always arranged in a perfect pattern. The flowers are not overlarge, but are arranged in a crown and ripen into attractive, berry-like fruit.

bocasana—powder puff—Soft, white-woolly globes, only one and a half inches across when mature; beige flowers.

elongata—golden lace—Small branching pillar, nicknamed for its tatted pattern of bright-yellow spines.

hahniana—old lady—Fond name for a small, white-haired cushion.

Notocactus *Cactaceae* Ball Cactus

Plump balls with prettily colored spines and large, showy flowers in late spring.

apricus—sun cup—Golden-yellow flowers, Oxford-gray spines.

graessneri—Butter-yellow spines and flowers.

ottonis—Indian head—Reddish spines.

rutilans—Rosy flowers.

Opuntia *Cactaceae*

This is a “crazy, mixed-up” group of cacti. They come in so many sizes, shapes, and forms that any generalized description is impossible. Many optunias are hardy even in Northern gardens.

mamillata—boxing gloves—Resembles a little tree whose branches turn into cockscombs at the tip.

microdasys—bunny ears—Flat, long-oval pads with tiny tufts of soft yellow hairs.

Parodia *Cactaceae*

Fat little balls covered with glistening spines and sending out unbelievably large flowers although the plant measures only an inch across the middle. Even in old age, they're never larger than three inches.

aureispina—Tom Thumb cactus—Gold spines, orange flowers.

mutabilis—Shining yellow flowers.

Portulacaria afra variegata *Portulacaceae* Rainbow Bush

After planting this little tree-like succulent in a dish garden when it was only three inches tall, and finding it less than half an inch taller nearly a year later, I was mildly amazed to learn that it is a version of the twelve-foot elephant bush, or purslane tree, of desert gardens. It's difficult to imagine the fat red stems and fleshy, cream-splashed, red-rimmed leaves ever adorning a plant of such monstrous proportions.

Rebutia *Cactaceae* Crown Cactus

Flat, fat balls with whiskery spines, spreading out into clusters. Each ball, when mature, is circled by large, wide-eyed flowers coming up from the base, often as large as the four-inch plant.

minuscula—Best-known species, with fiery red flowers.

senilis—Orange flowers with turned-back petals.

violaciflora—Rose-pink flowers.

Sedum *Crassulaceae*

Tender relations of the hardy garden sedums, not so numerous but equally varied.

adolphi—Rosettes of fat, yellow-green leaves, white flowers.

hintoni—Oval, grassy-green leaves covered with prickly white hairs, like clumps of tiny porcupine tails; white flowers.

lineare—Many branching, trailing stems covered thickly with needle-shaped leaves. The variety *variegatum* is a gem, each leaf more creamy-white than green.

multiceps—little Joshua tree—Unbelievable bonsai-like plant with trunk, branches, and tufts of needly dark-green leaves like the smallest conifer imaginable.

pachyphyllum—jelly beans—Fat, juicy, berry-like leaves clustering close to the branching stems; yellow flowers in spring.

rubrotinctum (*guatemalense*)—Christmas cheer—Thick layers of small, green, drumstick-shaped leaves turning holiday red in full sun; yellow flowers.

stahli—coral beads, Boston beans—Faintly hairy, beady, reddish leaves strung closely together on branching stems; yellow flowers in summer and fall.

Titanopsis *Aizoaceae*

Stone-mimicking succulents with thickly clustered leaves lavishly spotted with white, wart-like tubercles, and short-lived daisy flowers in fall

or winter.

calcareo—jewel plant—Lustrous gray-green leaves sparkling with white spots, gleaming gold flowers. Leaf rosette only two inches in any direction.

schwantesii—Even smaller cluster of blue-gray, liver-spotted leaves, lemon-yellow flowers.

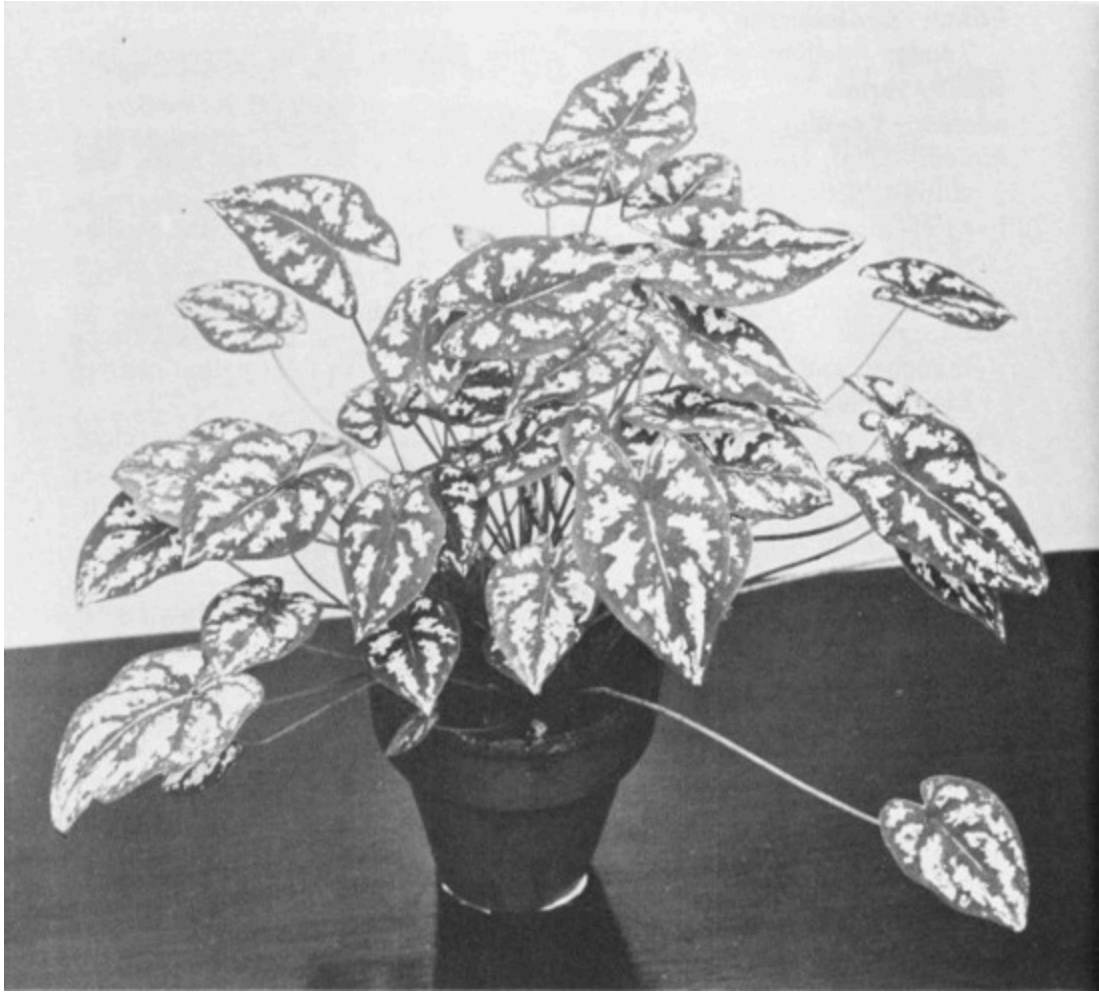
Trichodiadema densum *Aizoaceae* Desert Rose

Picture a bunch of tiny, smooth green pickles, each tipped with a triple crown of ridiculously long, bristly, white hairs. Now, smother this leaf colony under two-inch red daisy flowers. A really outlandish plant!

SUCCULENTS

Caladium *Araceae* Elephant Ears

Bicolor Varieties. Gorgeous arrow-leaved foliage plants for small gardens in the South or sheltered areas where summer temperatures are not likely to drop below 60 degrees for long. The patterns, colors, and contours of the leaves get fancier every year. Many are hybridized for larger and larger size, but specialists also offer a good selection of dwarfs that will stay under eight inches. Among them:



Caladium humboldti—one of the most striking members of the family

‘Candidum, Jr.’—Bushy low version of the standard favorite with green-netted, white leaves.

humboldti—A gemlike species for the connoisseur with willowy stems topped by diminutive deep-green leaves splotted with silvery transparent white between the center veins and the edge. Needs more humidity than the bicolor hybrids.

‘Little Rascal’—Leaf more lance-shaped, wine-red.

‘Miss Marveen’—Pearly-white tinged and blotched with pink over heavy drab green.

‘Mrs. Arno Nehrling’—Bronzy-green, red veins.

‘Pink Radiance’—Small and bushy, leaves centered and veined with shades of pink, pink-flecked on the edge.

‘Sea Gull’—White at the midvein and frosted all over.

‘Twilight’—Glowing flame-pink leaf finely netted with green.

CARE. Partial sun; if in a greenhouse shade against burn. Temperature: warm (75 degrees). Humid. Soil: rich, acid (azalea type). Keep moist.

Fertilizer: feed liquid manure on alternate weeks.

PROPAGATION. Seeds.

SPECIAL USES. Terrariums, window boxes, specimens.

Calathea (Maranta) Marantaceae

There's only one available true miniature in this group of richly colored and patterned tropical foliage plants, but many other species will stay small in a three-inch pot for years. The fact that they're suitable for lush but restrained effects in far Southern gardens is my other excuse for including them.

These plants are maranta-like, the leaf tufts with markings a peacock could crow about, silky-velvet or iridescent sheens that outshine the gowns at a coronation ball. They are reluctant to flower but nobody cares. The following is only a sampling of the available riches:

argyrea—Low and compact, the blotched green leaves overlaid with silver and horizontally level.

insignis—Narrow, permanently waved leaves perfectly patterned with chartreuse, wine red underneath.

louisae—Leaves perpendicular and perky, feather-marked with yellow-green.

micans—Miniature with oval leaves about an inch long (four inches in the tropics), lustrous deep green banded with silver at the center, paler underneath.

picturata vandenheckei—Satiny dark green banded with silver white in center. Wine red on the underside. A new form, 'Wendlinger,' is even more startling. The centers are sterling silver, edged with deep green.

CARE. Provide warmth, humidity. Soil: loam, leaf mold, and sand. Feed heavily for best colors. Shade from direct sun.

PROPAGATION. Divide crowns; tubers and spring cuttings.

SPECIAL USES. Tropical gardens, pots, terrariums, dish gardens.

Callopsiis volkensi Araceae

This is a perfectly proportioned miniature "calla lily" only four inches high, the white porcelain flowers centered with a gold, spear-like spadix, and substantial but small heart-shaped green leaves quaintly crinkled. It grows from an underground rhizome. It branches freely but compactly and is winter-blooming.

CARE. Shade, humidity, humusy soil kept evenly moist.

PROPAGATION. Rhizomes.

SPECIAL USES. Terrariums, dish gardens.

Carex elegantissima (Morrowi variegata) *Cyperaceae*

Lady-like little grass with airy tufts of slimmest leaves, bright green with a white stripe along each edge.

CARE. Loamy garden soil, filtered sun, wet. Intermediate temperature.

PROPAGATION. Seeds.

SPECIAL USES. Terrariums, dish gardens.

Carissa grandiflora nana compacta *Apocynaceae* Natal Plum

When is a dwarf not really a dwarf? In the two years or more I've had this plant it has grown so slowly I have no idea what ultimate height it has in mind, or when it will reach it. It is still about six inches tall, with round, green, overlapping leaves with the shiniest glassy polish of any leaves I have ever seen. Off and on during the year it gives sensual delight with richly fragrant china-white flowers, none of which, so far, have been replaced by scarlet plums. This form is blessedly without spines, always refreshing, and particularly appropriate for Oriental planters and miniature indoor gardens.

CARE. Almost any soil, warmth, light shade, moist, spray leaves. Resents repotting.

PROPAGATION. Cuttings, layers.

SPECIAL USES. Bonsai, planters.

Ceropegia *Asclepiadaceae*

Among the few hanging or climbing vines of suitable size and form for truly miniature baskets, or supports—succulent and easy to grow to boot! Their effect is always dainty, never bold; thin wiry stems may grow long, but never the leaves; waxy tube-like flowers inspire close inspection, but are never showy. Here are four of the daintiest species:

barkleyi—umbrella flower—A climber with pointed wing-shaped leaves feathered with a network of silver; purple-veined greenish flowers like parasols.

caffrorum—Pairs of green heart-leaves march down the threadlike stems; greenish flowers intricately marked with plum.

debilis—Green needle-like leaves penciled silver along the center; green-and-purple flowers.

woodi—rosary vine, string of hearts, hearts entangled—Matching pairs of thick silvery valentines every inch or so along the dangling stem, identically traced with a precisely patterned design. The little pink-lavender flowers are shaped like pixie pipes or urns.

CARE. Soil: loam, leaf mold, sandy and humusy. Intermediate temperature, filtered sun. Moist in summer, on the dry side in winter.

PROPAGATION. Cuttings in spring; tubers, seeds.

SPECIAL USES. Hanging baskets.

Chaenostoma fastigiatum *Scrophulariaceae* Little Stars

Cheerful little tropical herb which keeps its compact contour by branching freely, and covers its stems with myriads of half-inch fragrant leaves. All year it twinkles with a sprinkling of starry white flowers, outdoes itself in the summer months.

CARE. Loamy garden soil, bright light, moist.

PROPAGATION. Seeds, cuttings.

SPECIAL USES. Edgings.

Chamaedorea elegans bella *Palmaceae*

Slow-growing dwarf palm tree, so slow it can be a granddaddy in a six-inch pot and takes eons to top two feet. The reedy green stems cluster in a rosette and curve gracefully at the ends. Leaves are thin, fresh green, airily divided.

CARE. Warm, loamy garden soil, moist, shade.

PROPAGATION. Seeds.

SPECIAL USES. House plant, dish gardens.

Chamaerantherum *Acanthaceae*

A choice, exclusive threesome of tender tropical creepers with small supine leaves embroidered with dainty patterns.

gaudichaudi—Forest-green, long oval leaves of sheer velvet, silver feathers down the center and out along the veins.

(Stenandrium) lindenii—A bolder, golden feather marks the larger, perhaps even sheerer, leaves.

venosum—Symmetrical, oval leaves more broad than slender, dark powder-blue with thin leaves of silver on the network of veins.

CARE. Warm, humid, loamy garden soil, filtered sun, moist.

PROPAGATION. Cuttings.

SPECIAL USES. Terrariums, dish gardens.

Chlorophytum bicheti *Liliaceae*

Spidery little pot plant with tapering, grassy-looking leaves about six inches long, arching somewhat stiffly, with cream-white tidy stripes along the edge and sometimes down the middle. Always looks prim and precise.

CARE. Intermediate temperature, needs fresh air, loamy soil, filtered sun, moist.

PROPAGATION. Division.

SPECIAL USES. Indoor gardens, artificial light.

Cissus striata *Vitaceae* Miniature Grape Ivy

This lacy little climber looks less like grape ivy, more like woodbine, but it certainly does look and act like a miniature. The reddish stems turn and twist at angles, trying to touch the tiny tendrils to a support they can curl around. The one-and-a-half-inch leaves are like five fat separate fingers delicately joined in the center, scalloped or toothed near the rounded tip, burnished green and lined with wine beneath.



Cissus striata—a genuine miniature climber not unlike woodbine

CARE. Intermediate temperature, humid, loamy garden soil, filtered sun, moisture in the pots.

PROPAGATION. Roots and cuttings.

SPECIAL USES. House plant, miniature gardens, artificial light.

Citrus Rutaceae

These are true miniature citrus trees, varieties that are naturally dwarf, or grow so slowly they'll flower and fruit in smallish pots indoors.

aurantifolia—lime—Spiny little tree with small leaves, small tartly acid, thin-skinned, green fruit. The Persian lime is a variety called “one of the most beautiful of all citrus trees.”

limonia ‘*ponderosa*’—ponderosa lemon—Oblong, glassy, evergreen leaves; stiff sharp spines; large waxy, fragrant flowers; heavy (to two and a half pounds) pear-shaped (to five inches long) lemons at intervals through the year, beginning when the tree may not be much more than a foot tall. Supposedly, one lemon “gives enough juice to make a pie.”

mitis—calamondin—Bushy little tree, not prickly, dense with glassy green leaves; flowers fragrant, small and white. May bear fruit all year but usually in winter it produces bright oranges, about one and a half inches in diameter, thin-skinned and somewhat flattened on the end. The lime-like flavor of the juice is reportedly pleasant, but my tree looked so pretty I didn’t pick the fruit. A calamondin can, and often will, fruit in a two-and-a-half-inch pot—a mighty midget.

nobilis deliciosa—tangerine, mandarin orange—Small, nearly thornless tree with willowy leaves; sweet fruit, with loose skin that peels off like a kid glove and segments that separate freely.

taitensis—otaheite orange—Neat little bush for pots, with plenty of two-inch, wavy-edged leaves, few if any spines, and fragrant pink-tinged flowers in January. By Christmas, even if the tree is only eight inches high, the oranges are ripe. They are the size of a plum, shaped like a lemon, tartly flavored like a lime.

CARE. Intermediate temperature, needs fresh air, loamy soil, bright sun, keep on the dry side. For fruit, pollinate. Humidity, not too much fertilizer, not too much water. Sudden temperature changes causes leaves to drop.

PROPAGATION. Seeds, grafting, cuttings (for pot varieties).

SPECIAL USES. Seedlings for miniature gardens and greenhouses.

***Columnea microphylla* Gesneriaceae**

Velvety trailer, suitable for small hanging baskets, with soft little mouse-ear leaves. But wow! the fiery-red flowers never heard of the word *miniature*. Their long slender tubes stand straight up from the dangling stems, flare out to a wide mouth through which you can look to see the yellow throat inside.

CARE. Warm, humid, humusy soil, filtered sun, moist.

PROPAGATION. Seeds, stem and leaf cuttings.

SPECIAL USES. Hanging baskets.

***Cryptanthus* Bromeliaceae Earth Stars**

Not many bromeliads can rightly be called miniatures, and grow either in air or in soil to boot. These are low flat rosettes of stiff painted leaves, often with prickles on the edge and always unusually colored. From the heart,

wee white flowers peep out, usually in summer. Some suppliers offer a grab-bag selection of hybrids, “no two alike.” Or you can buy these smaller species if you wish.

acaulis—Thin gray fuzz over rather mottled green leaves.

bahianus—These leaves are slimmer, curving at the tip and wavy on the edge, flushed or margined brick red, more upright than flat.

beuckeri—Asymmetrical arrangement of ladle-like leaves tapered to a sharp point, dark green marbled over light.

bivittatus minor (*roseus pictus*)—Faintly banded dark-green leaves completely flushed bronzy pink or salmon.

bromelioides tricolor—Slim, pointed leaves with bands of rich cream and green, shaded cherry pink along the edge and in the heart.

lacerdae—‘Silver Star’—Definitely star-shaped and definitely silvery; the only green is in the two stripes down each side of the center.

CARE. Warm, humus, filtered sun, dry side, good drainage.

PROPAGATION. Offsets or cuttings. Peel off one or two basal leaves from stem; pot and root for two to three weeks.

SPECIAL USES. House plant.

Cryptbergia meadi *Bromeliaceae*

Everything said about the cryptanthus applies here. This is the result of crossing a billbergia with a cryptanthus. The leaves are slim spear-shaped and sharply pointed, upright in the center and arching out around the side. The basic green is rather drab. Made more appealing by a bronzy sheen and faint dark mottlings.

Cyanotis *Commelinaceae*

Succulent creeping or trailing cousins of the inch plants, with similar botanical characteristics, but each an individual in its own right:

kewensis—teddy-bear plant—Brown-woolly all over the stem and the tiny pointed-ear leaves.

somaliensis—pussy ears—The fresh green leaves are bent into boat shape, clasping tight to the stem and covered with velvety white fuzz.

veldhoutiana (*Tradescantia villosa*) (*Tradescantia pexata*) (*Tradescantia sillamontana*), and known in the trade as tradescantia ‘White Velvet’ and also ‘White Gossamer’—This tiny plant, more trailing than creeping, has lettuce-green leaves, and stems clothed in long silky silvery-white hairs. It has so many names I fear it would be lost to view if all of its labels were placed around

it. I quote all of them in order to avoid confusion and controversy. But according to the Institute de Biologia in Mexico, the correct name is *Tradescantia sillamontana*.

CARE. Intermediate temperature, needs fresh air, loamy soil, bright light, dry side.

PROPAGATION. Cuttings.

SPECIAL USES. Miniature tropical gardens, greenhouses.

***Dionaea muscipula* Droseraceae** Venus Fly Trap

Never let anyone tell you this little bug-eater will “trap flies on your window sill.” Don’t poke at it to see it close its trap. Avoid disturbing it; let it feed itself naturally. And now that the negatives are covered, do try this braggart elf (give it conditions in which it can grow), for the pleasure in its intricately devised mechanism, one of nature’s wonders. Basically, there’s a five-inch-wide rosette of wide flat stems, a two-part leaf at the end of each making a flat burnished bronze oval with a fringe of thick whiskers. When an insect steps onto this dance floor, the hinges in the center fold the two halves together fast, and that is the end of the bug. White flowers top the tallish stems in spring.

CARE. Cool, humid, sun, soil rich in humus with sphagnum, wet. Dormant in February and March before flowering. Most vigorous growth in spring and summer.

PROPAGATION. By roots or seeds. If seeds are used, start them under a bell jar.

SPECIAL USES. Terrariums.

***Dracaena godseffiana* Liliaceae**

Small tropical foliage shrub, slow-growing and bushy, with glassy, leathery leaves haphazardly splotched with white. The named variety ‘Florida Beauty’ is even more dwarf, takes years to fit a four-inch pot, and boasts golden-yellow splotches.

CARE. Warm, loamy garden soil, filtered sun (more light for color), moist to wet.

PROPAGATION. Pieces of stem with several joints; tip cuttings; air layering.

SPECIAL USES. Miniature gardens.

Drosera rotundifolia *Droseraceae* Sundew

The whole plant can often hide under a half-dollar—the smallest of all the insect-eaters. Flat stems radiate out from a central crown, expand at the ends, become flat spoon-shaped blades covered with reddish hairs that are tipped with drops of glue. Any insect landing on the surface gets his feet stuck while longer hairs on the edge act like tentacles to entangle him even more. Spring flowers top stems that grow straight up in the center.

CARE. Cool, humid, humusy soil made acid with peat moss if necessary, moist but not soggy, good drainage, room temperature is adequate with some direct sunlight but not enough to raise temperature in terrarium to a dangerous level. Transplanting or shipping may retard growth several weeks; don't get soil on leaves. May go dormant after flowering. Do not try to grow in ordinary room without glass cover or a plastic bag. Small amount of plant food (such as Vigoro). Rain water preferred.

PROPAGATION. As in *Dionaea muscipula*.

SPECIAL USES. Terrariums.

Dyckia *Bromeliaceae*

Small, stiff fountains or sprays of spiny-edged leaves, spikes of orange or yellow flowers like a miniature version of the pineapple in various sizes and shapes. Some are not very dwarf. The dwarfs, to the best of my knowledge, are:

altissima—Light-green, tapering leaves armed with brown spines and yellow flowers.

fosteriana—Dense circle of sugary gray leaves with purplish sheen.

rariflora—Slim, sharp-pointed six-inch leaves with minute silver scales, spines soft and black.

CARE. Intermediate temperature, humusy soil, filtered sun, dry side.

PROPAGATION. Easiest from offshoots when suckers are large enough to handle. Root in sphagnum peat. Seeds, germinate on tissue napkins in a shallow tray covered with glass. Keep in light at 65 to 70 degrees.

SPECIAL USES. House plant.

Episcia dianthiflora *Gesneriaceae*

Although none of the episcias grow very high, they do spread out to make wide, handsome pot or hanging-basket plants. This is the only species that can reasonably be called dwarf. Its leaves are the size and shape of my little fingernail, olive green and softly velvet, reddish along the center vein, first forming a tidy rosette. Soon the branches start stretching, rather stiff and woody for an episcia, with new leaf rosettes forming as the branches grow. The silky tube-flowers are pure white and fringed on the edges.

CARE. Warm, humid, humusy soil, filtered sun, moist.

PROPAGATION. Seeds and cuttings.

Exacum affine *Gentianaceae*

This is a compact, free-flowering plant of the gentian family. The flowers are blue with golden stamens and very fragrant. A herbaceous biennial, it can be treated in a greenhouse as a perennial. It has bushy green leaves, ovate and tiny. A dark-lavender form is the variety *atrocaeruleum*.

CARE. Warm, humid, humusy soil, filtered sun, moist.

PROPAGATION. Seeds. Can be planted any month, but a February planting will produce flowering plants for autumn and winter.

SPECIAL USES. Window gardens and greenhouses.

Ficus *Moraceae* Creeping Fig

Here are two delightful foliage creepers that keep their miniature proportions, although the large-leaved form of the first often covers large walls in the South.

pumila minima—Tiny valentine leaves patterned and crinkled with a network of sunken veins, deep green and refreshing.

radicans variegata—Much larger, slim pointed leaves basically silvery green, but marked with creamy white, starting at the edge and blending off irregularly. Like many variegated plants, it's not very vigorous and thus is suitable for terrariums, where it will get the humidity it needs.

CARE. Warm, needs fresh air, loamy soil, filtered sun, moist.

PROPAGATION. Division of the rooting stems.

SPECIAL USES. Terrariums, hanging baskets, outdoors in the South.

Fittonia verschaffelti *Acanthaceae* Nerve or Mosaic Plant

Flat-creeping tropical plants with jewel-like foliage netted with veins of contrasting color. The oval or rounded paper-thin leaves may be large—up to two inches long—but the plants grow slowly and seldom exceed dwarf proportions, particularly if there are any slugs within miles. Fittonias are one of the critters' favorite foods.

You have a choice of three color schemes: The species is dark green with carmine veins; its variety *argyroneura* is emerald with silver; the variety *pearcei*, light green with pink.

CARE. Warm, humid, loamy soil, filtered sun, moist.

PROPAGATION. Stem cuttings.

SPECIAL USES. Foliage plants in greenhouses.

Fortunella *Rutaceae* Kumquat

Smallest, and hardiest, of the evergreen citrus trees or shrubs, with typical glassy, leathery leaves and flowers at intervals during the year, sometimes coincidental with the thin-skinned fruit. The species vary in size, habit, and fruit, some taking a lifetime to reach ten feet. But before they achieve that altitude, they can long pass as dwarfs or miniatures. Who wants to wait a lifetime to watch a kumquat outgrow a three-inch pot?

hindsii—Small spiny tree with small (less than one inch) pea-shaped fruit.

japonica—marumi kumquat—Small tree, well branched and shapely, with perfectly round, somewhat sweet fruit nearly one and a half inches in diameter.

margarita—nagami kumquat—Most frost-resistant and dwarf. Nearly thornless, with sour-sweet, egg-shaped fruit.

CARE. Intermediate temperature (any sudden changes in temperature may cause the leaves to drop), soil on the dry side (not too much fertilizer), bright sun, pollinate if you wish fruit, maintain humidity.

PROPAGATION. Seeds, grafting, cuttings.

SPECIAL USES. Potted plants, seedlings for miniature gardens.

Fuchsia magellanica *Onagraceae*

This “hardy” fuchsia, when given a chance to develop, can hardly be called a miniature. It can cover a wall twenty feet high, I have been told. However, in a climate like that near Philadelphia, it is only “root-hardy” and even then has to have a protective winter mulch. In the spring it has to start new growth all over again. Then it assumes miniature proportions. I’ve seen it in a small hanging basket, the leaves less than one-third the size of the typical hybrid fuchsias, the stems thin and wiry, the red-and-purple flowers quite small. Even more like a miniature is the slower-growing variety *variegata*, with creamy blotches and streaks on the leaf edges. Then there is the slender, more airy and lax variety, *gracilis*, also available in variegated form.

CARE. Cool, needs fresh air, loamy soil, shade, moist.

PROPAGATION. Cuttings of soft green wood.

SPECIAL USES. House plants, hanging baskets.

***Hebe buxifolia variegata* Scrophulariaceae**

This is a charming little plant I have grown in my greenhouse. Although it has some larger relatives, my specimens have been delightfully small. The leaves are about one-half inch long, waxy green with creamy white edges which overlap into a nice pattern. I love the small white flowers which appear frequently.

CARE. Little sun, humusy soil, moderate warmth, moist.

PROPAGATION. Stem cuttings.

SPECIAL USES. House plants.

***Hedera helix* Araliaceas English Ivy**

As long as I’ve grown house plants I’ve always had ivies, sometimes only a few pots of my favorite varieties (when my growing space was limited) but more often a collection of a dozen or more. Once when preparing a magazine article, I had thirty-seven scrambling around the greenhouse! The foliage always is fresh green and glossy, delightful the year round; the climbing or trailing stems are graceful and willing to be trained in just about any pattern. In the tiny-leaved miniatures there is a

particular fascination in the intricate formations of lobes and points, edges which are waved or crinkled, and the markings, which may be dainty, pert, or bold.



(MEAGHERI GREEN FEATHER)

Hedera helix meagheri—a fine small specimen

These variations (actually mutations) of the ubiquitous evergreen ground or wall cover, are less hardy than the species from which they have sprung. Their outdoor use is best limited to moderate or mild climates—but never where it is hot and dry. Some slow creeping types like the variety conglomerata are delightful in not-too-hot rock gardens. Indoors, of course, you can have any or all of them, and you have many to choose from. Here is a representative selection to tempt you:

‘Goldheart’—Deepest green leaves centered with a heart of gold.

‘Itsy Bitsy’—Lacy leaves half the size of a fingernail.

‘Ivalace’—Fresh green with a crocheted edge.

‘Jubilee’—Gray-green spattered with white.

‘Needlepoint’—So sharply cut there’s nearly no leaf at all.

‘Shell’—Waved and crinkled on the veins.

‘Tear Drop’—A baby’s tear at that.

‘Telecurl’—Tight as a brand-new “permanent.”

CARE. Cool, needs fresh air, loamy soil, shade or semishade, moist.

PROPAGATION. Cuttings, layering, seed. (Seeds may take as long as two years to germinate.)

SPECIAL USES. House plants, ground cover, rock gardens, miniature gardens.

Helxine soleioli *Urticaceae* Baby’s Tears

Millions of tiny, button-like leaves hug close to the thready stems of this wee creeper that covers soil, or a pot, in incredible haste. The growth twines and intertwines thickly as if it were weaving a fresh green carpet for a doll’s house.

CARE. Intermediate temperature, needs fresh air, humidity, humusy soil, filtered sun, moist.

PROPAGATION. Cuttings of rooting stems.

SPECIAL USES. Ground cover, rocks and walls. (Outdoors only in mild climates.)

Hoya bella *Asclepiadaceae* Wax Plant

Definitely a dwarf, as wax plants grow, and not climbing, as most others are, little *H. bella* has arching, rather than woody, stems with thick, heavy somber green leaves on either side and clusters of incredible flowers at the tips, porcelain-white, waxy, and fragrant, with a wheel-shaped purple “crown” in the center. Best in a basket because the flowers droop, and the view is prettiest from below.

CARE. Warm, needs fresh air, loamy soil, bright light, dry side.

PROPAGATION. Cuttings in spring.

SPECIAL USES. Good house plant.

Impatiens repens *Balsaminaceae*

This precious creeper bears so little superficial resemblance to the popular “patient Lucy” it’s hard to believe it’s a close kin. I loved it when I first saw it at the Montreal Botanical Gardens, and was delighted when it became available to us here. It is a creeper with red stems, plump button-shaped leaves, and yellow flowers.

CARE. Intermediate temperature, needs fresh air, humusy soil, bright light, moist.

PROPAGATION. Seeds, cuttings.

SPECIAL USES. Good house plant, miniature gardens.

Koellikeria erinoides *Gesneriaceae*

This one is a gem—clusters of pert green-velvet leaves with indented veins and sprinkled with silvery star dust; on short stems, tiny two-lipped bell flowers of creamy white and red, in summer.

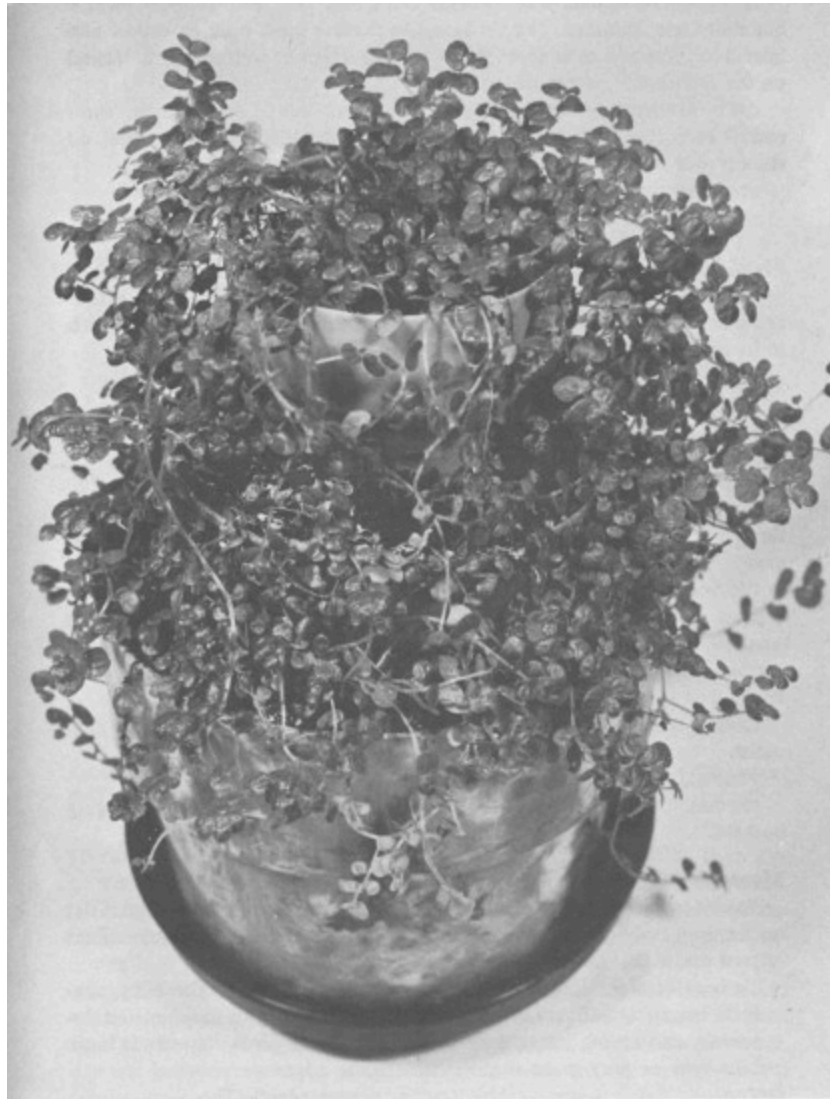
CARE. Warm, humid, humusy soil, filtered sun, moist.

PROPAGATION. Seeds.

SPECIAL USES. House plant.

Lantana camara *Verbenaceae*

Here is a plant I have long enjoyed around the greenhouse. It is a small hairy shrub with angled branches and rough bristly leaves and is sometimes prickly.



Helxine soleioli, baby's tears—a delightful little crawler

Its blooming habits have always delighted me, and it is a prolific bloomer come summer. The verbena-like flowers open pink or yellow and later darken to red or orange. Often several different shades will be found on the same plant at the same time.

CARE. Moderate greenhouse temperatures, or out of doors in the summer. Needs fresh air, loamy garden soil, bright light, and does best on the dry side.

PROPAGATION. Cuttings and seed.

SPECIAL USES. House plant, pots, and containers.

Malpighia coccigera *Malpighiaceae* Miniature Holly

Small glossy leaves sharply toothed like holly, and adorable as a foliage plant, but in addition provides the sweetest white-pink blooms which in its native West Indies produce red, edible fruit rich in vitamin C.

CARE. Partial sun, warmth, humusy soil, moist.

PROPAGATION. Stem cuttings.

SPECIAL USES. House plant, indoor bonsai.

Manettia bicolor *Rubiaceae* Firecracker Vine

Small, sharp-pointed leaves, tube-flowers only a half-inch long, fine thready twining stems—everything about this tropical vine is miniature except the height to which it will climb in the greenhouse, or outdoors in the South. Although its inclination is to climb, it is sold most often as a plant for hanging baskets. Perhaps the lack of a support helps keep it small. The leaves cluster thickly around the intertwining stems, making a massed background for the impertinent flowers, yellow at the tips, fire-engine red at the base.

CARE. Cool greenhouse, needs fresh air, humid, loamy soil, filtered sun, moist.

PROPAGATION. Cuttings over heat, of young growth. Seeds.

SPECIAL USES. Trellis and rafter vines for the greenhouse, hanging baskets.

Maranta *Marantaceae*

Exotic tropical foliage plants, laying their large oval leaves almost flat on the soil, only technically different from the calatheas, and sometimes offered under that name.

(*Calathea*) *bicolor*—Really should be “tricolor,” I think. The silky, six-inch leaves are silvery in the center, feathering out to points at intervals, and fading into a dark-green zone which gives way to the basic blue-gray or gray-green that extends to the edge.

leuconeura kerchoveana—rabbit tracks, prayer plant—There’s a similar grayish feather in the center of the leaf, a lighter silver green to the margin; but in between, mahogany blotches where a “hippity-hopper” might have planted his paws. The leaves fold up in prayer at night.

leuconeura massangeana—A picture is a more vivid description than any word I can call on. The center is silver, the thin curvy lines silvery pink. The basic color is mahogany near the middle,

blending into blue-green. The leaves are tissue-thin with a silky sheen and lined with plum beneath.



Leuconeura massangeana—a most beautiful foliage plant with a silver center and basic mahogany blending into blue-green

CARE. Warmth, minimum night temperature of 60 degrees; humid, and use mist spray; rich soil; use liquid manure (except in winter). Filtered sun (plant will scorch in too direct sunshine), moist. Partial rest in winter, dry out between waterings. Repot in spring, with the crown high in the center; use charcoal in pot.

PROPAGATION. Divide crown.

SPECIAL USES. Foliage plant, artificial light. This plant is prettiest when young. Markings may disappear in older plants.

Myrsine africana *Myrsinaceae* African Boxwood

This is a shrubby, dark-green-leaved plant of the boxwood sort. It is small, neat, and trim and may be shaped into many patterns to resemble the true boxwood.

CARE. Partial sun, moderate temperature, humusy soil, moist.

PROPAGATION. Stem cuttings.

SPECIAL USES. Formal miniature gardens, dish gardens, indoor bonsai.

Myrtus communis *Myrtaceae* Greek Myrtle

This is a fragrant close-leaved evergreen shrub native to the Mediterranean region but raised in most of Europe and this country as a potted plant. It grows out-of-doors in the South. Fragrant white flowers and purple-black berries. In size, Greek Myrtle ranges from dwarfs to ten and fifteen feet in height.

communis microphylla—dwarf myrtle—This compact form, with brown twigs and needle-like dark-green leaves, is a favorite for bridal bouquets, corsages, and decorations. It has a lovely white flower of pleasing fragrance. This sentimental association with the marriage ceremony makes it a favorite house plant.

CARE. Partial sun, cool to moderate temperatures, average soil, keep moist.

PROPAGATION. Stem cuttings.

SPECIAL USES. House plant, indoor bonsai.

Oxalis *Oxalidaceae*

Although most of the familiar varieties with clover-type leaves and fine-petaled buttercup-like flowers stay within our allotted limits for height, they are not miniatures as far as spreading width is concerned. But here are two small indispensables for terrariums and dish gardens.

hedysaroides rubra—firefern—Correctly, a tropical shrublet, but slow-growing and willing to be kept down by pinching. The silky thin leaves are deep wine colored and shrink from your touch. The little flowers, like golden twinkling stars, are frequent and fairly plentiful.

herreriae (*henrei*)—Curious, densely branching succulent for small hanging containers. The sturdy stems are gray-green and swollen, tipped with three thick leaflets. Small yellow flowers in clusters on long stems.

martiana aureo-reticulata—Shamrock leaves of emerald green thickly traced with interlacing veins of gold, easy flowers in loose clusters.

CARE. Cool, loamy garden soil, needs fresh air, bright light, dry side. (*O. hedysariodes rubra* requires more warmth and semishade.)

PROPAGATION. Bulbs, division of roots, seeds.

SPECIAL USES. House plants, hanging containers.



Oxalis hedysaroides rubra

Parochetus communis *Leguminosae* Shamrock Pea, Blue Oxalis

It's neither a shamrock nor an oxalis, although it has three-part leaves resembling both and is marked with a mahogany quarter-moon in the center. But, the blooms are pea-like, all right, with ocean blue blending into pink

on the two side petals. It's a tender tropical trailer never more than three inches tall with creeping stems rooting over the soil.

CARE. Sun, intermediate temperature, average soil.

PROPAGATION. Seeds in March and April. Division in March.

SPECIAL USES. Rock gardens in mild climate, baskets, miniature gardens, edging.

Pelargonium hortorum *Geraniaceae* Geranium

How minute is a miniature? How small is a dwarf? Here is one case where I really don't care. However you classify them, these small-scale geraniums are bewitching. And actually, their stature depends largely on how you grow them.

Their leaves may be as small as your thumbnail, but they are shaped, edged, veined, zoned, and often variegated like their larger relatives. In most varieties the size of the flowers has been cut down proportionately, but sometimes a cluster is as large as the entire plant. With age the stems may become gnarled and twisted, almost bonsai-like.

As a sort of yardstick, if you are very much concerned about height, miniatures are from two to three inches. Anything above that is a semidwarf, usually up to six or seven inches. There are many varieties within these limits, and by the time you read this there will most likely be dozens more. For the moment, here are some varieties you can start on your window sill; they will stay small and meanwhile bloom their heads off.

'Black Vesuvius'—Single, orange-scarlet.

'Dopey'—Large rose-red with white center. Free-blooming. Semidwarf.

'Elf'—Dark leaves, zoned red and black, yellow on edge, single.

'Epsilon'—Large flowers light pink with dark-pink phlox eye, semidwarf.

'Fairyland'—Dull-green leaves with cream on the edge, single, touched with rose-red.

'Fairy Tales'—Flouncy white with lilac at the center.

'Fleurette'—Strong robust free-blooming dark salmon. Semidwarf.

'Goblin'—Outstanding blooms bright red. Free-blooming, dark zoned.

'Imp'—Very miniature salmon-pink. Dark foliage.

'Kleiner Liebling' ('Little Darling')—Variegated, single, pink.

- ‘Minx’—Purple-crimson blend, pretty blackish ruffled leaves.
- ‘Mischief’—Orange-red curled and twisted poinsettia-type petals.
- ‘Perky’—Single, red with bright white center.
- ‘Polaris’—Free-blooming white with pink edge. Single.
- ‘Pride’—Very large full rounded salmon. Robust semidwarf.
- ‘Prince Valiant’—Purple-crimson with orange flush. Semidwarf.
- ‘Robin Hood’—Semidwarf, strong growing, dark red. Good for outdoor planting. Double.
- ‘Rosy Dawn’—A different shade of salmon-orange-red. Dark foliage.
- ‘Ruffles’—Semidouble, small, ruffled, light salmon. Very miniature.
- ‘Salmon Comet’—Somewhat shaggy, single, salmon.
- ‘Saturn’—Bright scarlet, good bloomer, dark foliage.



Three dwarf geraniums: 'Salmon Comet,' 'Pride,' and 'Pygmy'



Dwarf geranium, 'Robin Hood'

'Small Fortune'—Double, pure white, blushing pink in center.

'Sneezy'—Large bright scarlet with white center. Semidwarf.

'Snow-White'—A pure-white single that blooms and blooms.

'Sparkle'—Free-blooming bright rose-red. Semidwarf, dark foliage.

'Sprite'—Small silvery-and-white leaves sometimes pink-tinged, single, salmon.

'Tiny Tim'—Pink or red, tiny blooms. Miniature foliage.

'Twinkle'—Bright rose-pink, double. Dark foliage, semidwarf.

'Variegated Kleiner Liebling'—Small green-and-white leaves. Small single pink blooms.

CARE. Sun (in winter artificial light), fresh air (air-conditioning if possible), cool greenhouse or cool part of dwelling-house, garden soil (lime if soil is extra acid), on the dry side.

PROPAGATION. Cuttings, seeds.

SPECIAL USES. Excellent house plant, artificial-light gardens.

Pellaea rotundifolia *Polypodiaceae* Button Fern

Take a piece of wiry brown yarn and string it with shiny, dark-green, leathery polka dots on either side and you'll have a model of a frond of this tropical fern. The fuzzy stems arch gently but stay close to the soil as the small rhizomes slowly creep around. It's all very un-fernlike but exceedingly sweet.

CARE. Intermediate temperature, humid, loamy soil, shade, moist.

Pellionia *Urticaceae*

Two delicate tropical creepers with attractively colored and patterned leaves lined up and lying flat along the thickish stems. No one cares whether they produce their drab clusters of small greenish flowers or not.

pulchra—Precise oval leaves are dusty blue-gray, with broad lines of black over the network of nerves, violet underneath. Juicy stems look lavender-pink.

repens (daveauana)—Young leaves nearly egg-shaped, metallic copper-green, nearly chartreuse on both sides of the center nerve. Leaves of the mature plants become slimmer and pointed on the end, with more green in the center zone and less copper along the edge.

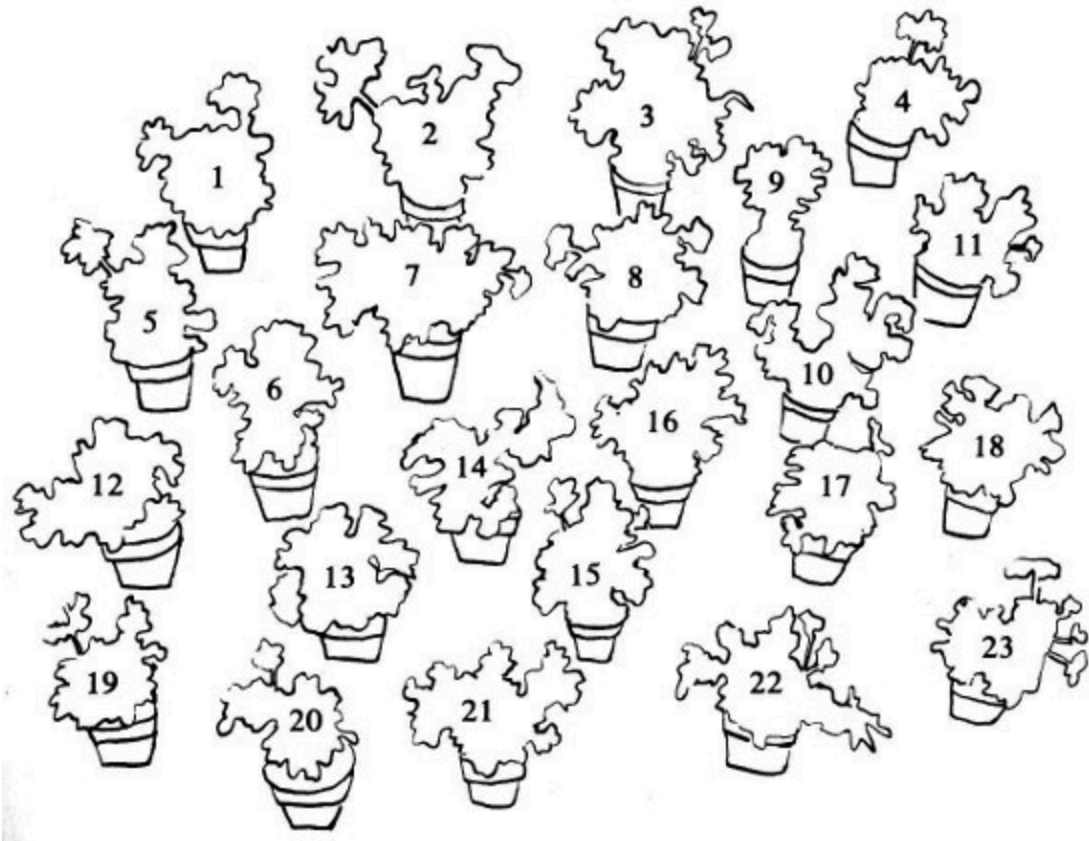
CARE. Filtered sun, warm and humid, loamy soil, moist.

PROPAGATION. Cuttings, division of rooted stems.

SPECIAL USES. Terrariums, artificial light, hanging baskets.



Twenty-three varieties of *Pelargonium hortorum*, miniature and dwarf: 1. Fleurette 2. Snow White 3. Dopey 4. Mischief 5. Robin Hood 6. Minx 7. Epsilon 8. Prince Valiant 9. Polaris 10. Saturn 11. Sparkle 12. Kleiner Liebling 13. Small Fortune 14. Goblin 15. Variegated Kleiner Liebling 16. Fairy Tales 17. Rosy Dawn 18. Pride 19. Sneezy 20. Twinkle 21. Ruffled 22. Imp 23. Salmon Comet



Peperomia *Piperaceae*

Since almost all of these succulents, with their “catkin-like” flower spikes, fit within our miniature measurements, I’ve enjoyed picking a few to serve as a sample of their modest but much varied appeal. I like them, and hope you will also.

‘Astrid’—Layer upon layer of glossy green, pointed oval leaves creased down the center by the main vein, making a dense, symmetrical cone-shaped mound. In fact, there are so many tight branches that the leaves can’t always find room to fill out to full size. Then ‘Astrid’ turns into a version called ‘Pixie.’ To reverse the cycle, root a single stem of ‘Pixie’ and see how it returns to being ‘Astrid’ again.

‘Caperata’ (‘Little Fantasy’)—One perfect little heart leaf, its end pointing down, perched at the top of a sturdy stem. The entire leaf is so deeply wrinkled and crinkled that the basic Kelly-green looks almost black in the indentations, and silvery on the edges. Its big brother, ‘Emerald Ripple,’ is half again as large but otherwise identical.

‘Mamorata’ (‘Silver Heart’)—These heart-leaves taper to a sharper point and are rippled in a much more modest way. There is so much silver on the surface they look plated.

metallica—Upright and branching with slim-oval leaves pointed at both ends and colored chocolate with a coffee luster. There is a wide green stripe straight down the middle.

ornata—Handsome and stately despite its small stature. The cupped, pointed-oval leaves are somber green on top and enlivened with lines of light green along the curved veins. Underneath, the veins become raised ribs and are wine red.

prostrata—Personally, I have my suspicions about this sprightly creeper. When it is young, discontented, or ailing, its fat little button-leaves along the thready stems are a plain light green. It looks so much like the species *rotundiflora* (*nummularifolia*) that you can't tell the two apart. But when *prostrata* is happy and romping around in its favorite environment, the leaves turn blue-green and wear an ornate embroidery of silver over the veins. Botanically, I'm probably wrong. See what happens when you grow it.

rubella—Rosettes of tiny oval moss-green leaves stand out at intervals up and down the straight stems and all the branches. The undersurface is gaudy red, and so are the stems. Pinch often to keep it bushy.

CARE. Warm, loamy garden soil, filtered sun, dry side. Water with care. Drainage to prevent rot.

PROPAGATION. Leaf cuttings, stem cuttings, division of the plant or rooting stem (roots at joints), seeds.

SPECIAL USES. Excellent house plant, dish gardens.

Pilea *Urticaceae*

With one exception, this is a genus of creepers, crisp, fleshy—spreaders, and bushlets for dozens of decorative uses indoors (and outdoors in tropical climates). Their common characteristics are much-branched stems bearing a full measure of variously colored, patterned, and shaped foliage, and very greenish flowers in flat clusters at the branch tips.

cadieri minima—Dwarf form of the popular aluminium plant, or watermelon pilea, eagerly branching into a plump bush. The quilted deep-green leaves are splotted with aluminium-silver. And the splotches are faintly reminiscent of watermelon markings.

depressa—Crowded stems spill over the pot, bearing round sea-green leaves neatly toothed on the edge. Stems root where they touch the soil.



Pilea depressa—a creeping, dish-garden favorite

involutrata—panamiga—Fuzzy pointed oval leaves deeply quilted by a tight network of veins, piled pair on pair along branching stems. Foliage colors darkest green in shade, tinges of bronze in bright light. Flowers pinkish, nestled close to the last pair of leaves.

microphylla (*serpyllifolia*)—artillery plant—Tiny, pointed fresh-green leaves on spreading, somewhat upright branches. The male flowers explode their clouds of pollen when dry.

nummulariaefolia—creeping Charlie—Stem-rooting creeper with crinkled round, or broad heart-shaped, leaves, pale green and fuzzy.

pubescens ('Silver Panamiga')—Oval leaves overlaid with silvery blue, grayish beneath, the veins indented sharply.

repens—blackleaf panamiga—Prostrate and creeping, the branches spreading and bearing nearly round, thin, bronzy leaves lined with violet beneath. This one holds its beady flowers atop tall stems.

seripillacea—Habit like a perfectly proportioned shrub but in most miniature proportions. Small round leaves and plentiful branches, light green and succulent. Flower clusters held out from the foliage by stems.

'Silver Tree'—Copyrighted name for a species with brown-green leaves marked with a broad silver zone on each side of the center vein, corrugated by indented veins.

CARE. Easy, warm, humid, loamy soil with humus, filtered sun, moist.

PROPAGATION. Cuttings, use sharp sand, 65 degrees.

SPECIAL USES. Dish gardens, terrariums.

Polyscias (Aralia) fruticosa *Araliaceae*

Tropical shrub or tree (eight feet or less) better known in Northern greenhouses for its intricately cut, lacy foliage, and growing so slowly it is nearly permanent (and unquestionably picturesque) in a planter or dish garden. The elegant horticultural variety 'Elegans' has leaves cut as fine as a feathery plume. I covet the newly named variety 'Parsley' because it is ruffled, to boot, and because its habit is so restrained, and its form so compact and plump.

CARE. Warm, needs fresh air, loamy soil, bright light, moist.

PROPAGATION. Cuttings of firm wood, eye-cuttings (over bottom heat).

SPECIAL USES. Dish gardens.

Polystichum tsus-simense *Polypodiaceae*

Definitely a dwarf fern, keeping its size moderate in a three-inch pot but not always a subject for terrariums unless they are a larger-than-usual size. It has a tidy habit, always looks clean and refreshing. The small fronds are substantial and taper to a needle-sharp point.

CARE. No sun, cool, humid, loamy soil, moist.

PROPAGATION. Divisions, buds, or offsets.

SPECIAL USES. Hanging baskets. Terrariums.

Sansevieria *Liliaceae* Snake Plant

Here we have some of the most fortuitous "sports" in the history of plants. The tall, stiff, ungainly but omnipresent snake plant has produced spontaneously mutant growth that turned into neat, low, bird's-nest miniatures that are really most attractive. First came the variety named 'Hahni,' a flat spiral of broad leaves tapering to a sharp point, dull green marked crosswise with splashy bands of lighter green. Another sport, 'Silver Hahni,' abandoned most of the crossbands and plated its green with pewter. Even more striking is 'Golden Hahni,' with lavish bands of creamy yellow running lengthwise of the leaf on either side of a center stripe of

mottled green. And to top it all, these picturesque plants are just as easy to grow as the rugged species from which they sprang. Flowers, whitish or yellowish. Slow-growing.

CARE. Filtered sun, warm, loamy soil, slightly moist.

PROPAGATION. Division of clumps. Leaf cuttings (three-inch piece in sand, shade, and 65-degree temperature). Makes stolon-like buds that form the new plants.

SPECIAL USES. Dish gardens. Good house plant.

***Sarcococca ruscifolia* Buxaceae** Sweet Box

Sweet box is a small, free-branching evergreen with attractive leathery leaves, broad at the stem but coming to a point. Has tiny fragrant white flowers.

CARE. Intermediate temperature, fresh air, loamy soil, filtered sun, moist.

PROPAGATION. Cuttings.

SPECIAL USES. Dish gardens, indoor bonsai.

***Saxifraga sarmentosa* Saxifragaceae** Strawberry Begonia or Geranium

One of the few hardy perennials that will thrive indoors the year round. Round, hairy silver-veined leaves grow in a rosette from the crown, which also sends out slender red stems, strawberry style, with new little plants that root and grow wherever they touch soil. In late spring the fall stems are topped by cloudlike soft clusters of small white flowers. 'Maroon Beauty' is slightly darker, and larger.

More miniature, and more tricky, is the variety tricolor, sometimes called 'Magic Carpet,' with smaller, basically gray-green leaves, red-rimmed and variegated with wide areas of creamy white, purple underneath. In cool air and sun, the cream is strongly tinted pink.

CARE. Humid, cool, poor soil, dry.

PROPAGATION. Runners.

SPECIAL USES. Dish and sink gardens.



Saxifraga sarmentosa—a hardy perennial good for indoor miniature gardens

Schizocentron (Heeria, Heterocentron) elegans *Melastomaceae* Spanish Shawl

Mexican creeping perennial with thickly branched stems rooting at the joints, making a plush carpet of tiny, teardrop leaves. In summer the one-inch open-faced, royal-purple flowers seem unbelievably large and rich. This one is very nice in a small hanging basket, but its natural inclination is to creep and it really goes to town when it can cover soil or some sort of porous support such as a moss totem pole. It should be spectacular covering a hanging ball filled with sphagnum moss.

CARE. Intermediate temperature, humid, loamy soil with humus, filtered sun, moist.

PROPAGATION. Cuttings, division of rooted stems.

SPECIAL USES. Small baskets, miniature gardens, ground cover.

Selaginella *Selaginellaceae*

Soft, fluffy foliage plants in all shades of green, including metallic, and with all kinds of growth, low and creepy, upright, even climbing. They bear more resemblance to each other than they do to their cousins the ferns.

emmeliana—sweat plant—As the nickname hints, this ferny plant languishes unless it is “perspiring” in high heat and humidity. In fact, once the fine-lace fronds turn dry and brown, they won’t be fresh and green again.

kraussiana browni—Scotch moss—Soft, symmetrical cushions of bright-green leaves, slowly spreading into larger mounds.

kraussiana (denticulata)—spreading club moss—Branching, rooting, creeping stems thickly set with needlepoint leaves.

lepidophylla—resurrection plant—Antithesis of the sweat plant. When the fan-shaped branches have dried out and curled into a ball, immerse them in water and they will come back fresh and green as ever.

martensi—Young branches stand upright, may drop or creep with age. The variety *variegata* has eye-catching white splashes, splotches, or tips and supports itself on stiff aerial roots from stems to soil.

plumosa—Foamy creeper with short, branching stems overlapped by foliage of fresh woodsy green.

uncinata—Foliage sparse but shimmering peacock-blue in shade. The running, branching stems have a ludicrous way of sending down stilt-like roots into the soil, so they seem to be running above the soil, not in it.

CARE. Fern culture; warm, humid, humus, shade, moist.

PROPAGATION. Cuttings in pots (several pieces per pot), in spring. Put on top of medium and cover with glass at 70 degrees until roots form at joints. Spores, division of rooted stems.

SPECIAL USES. Terrariums, greenhouses, ground cover.

Serissa foetida (japonica) *Rubiaceae*

A boxwood-like plant with tiny white-margined leaves clustered on branchlets. White funnelform flowers to one-half inch long.

CARE. Partial sun, average soil, moderately moist.

PROPAGATION. Cuttings.

SPECIAL USES. Artificial light, dish gardens, indoor bonsai.

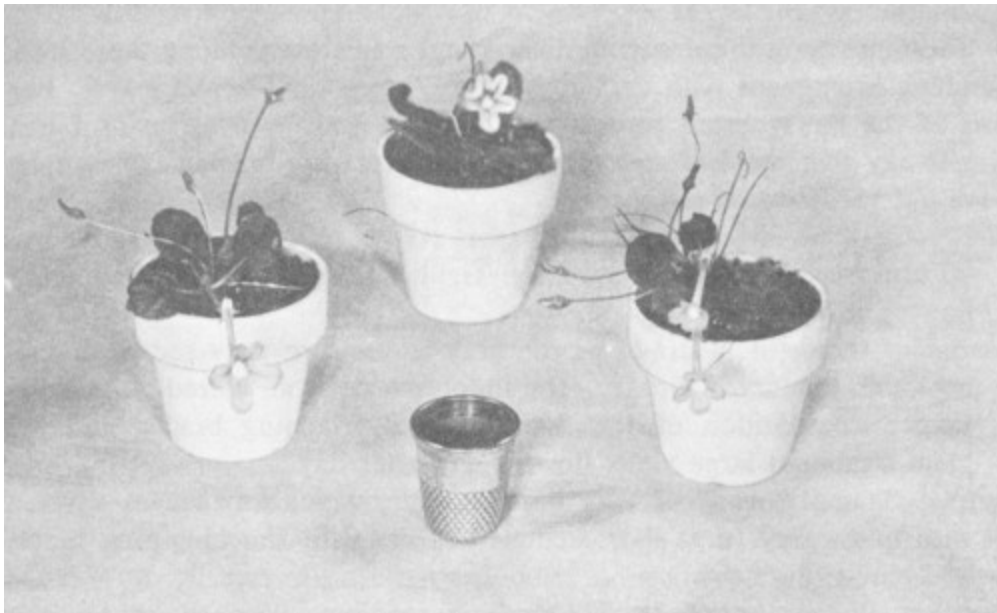
***Sinningia pusilla* Gesneriaceae**

If ever a plant was a miniature, this is it. The tuber is hardly as large as a grain of wheat. The soft round leaves, not even a quarter-inch across, make a flat rosette on the soil in a thimble, or any tiny pot. From the crown arise the thread-thin stems, to the great height of one inch. And from the tip of each stem stands a slim flaring, tube-like flower, pale orchid with violet veins and lemon in the throat. This plant is quite obviously close kin to the familiar gloxinia (*G. speciosa* hybrids), but you almost need a magnifying glass to compare the characteristics.

CARE. Warm, humid, humusy soil, filtered sun, slightly moist.

PROPAGATION. As for the gesneriads (Gesneriaceae).

SPECIAL USES. Terrariums, miniature gardens.



Miniature of miniatures, delightful *Sinningia pusilla*

***Streptocarpus* Gesneriaceae**

When limited growing space frustrates the hobbyist who admires the great, glorious gloxinias (sinningias), here's a selection of pleasing and precious substitutes. In general these plants are distinguished by the fact that the flower stems grow out from the base of the leaf where it joins its

own stem (axil). The flowers are typical gesneriad trumpets, usually nodding; the leaves are mostly rather round and velvety. The following are fibrous-rooted:

hybridus—Botanical name for groups of hybrids called “as complex as the garden geranium,” with quilted light-green leaves and a wide selection of flower colors. One strain of particularly attractive low plants, the German Weismoor hybrids, has fringed and crested flowers up to four inches across, often contrastingly veined or blotched.

rexii—Long-oval, velvety leaves lie very flat. Six-inch stems hold two-inch funnel-flowers, pale orchid with purple throat.

saxorum—Leafy-stemmed species, the stems branching continually, making dense mounds of plump, one-inch oval leaves curled under on the edge and covered with soft silk-velvet. The lavender-flushed white flowers stand out at the end of wiry three-inch stems.

CARE. Cool, humid, loamy soil with humus, filtered sun.

PROPAGATION. Seeds (plant in early spring for flowers fall and winter), leaf cuttings, some offsets.

SPECIAL USES. Artificial light.

Tillandsia Bromeliaceae

There are several entrancing dwarfs and miniatures among these tree-perching bromeliads with curved, leathery, often quill-tipped leaves. For two of the newly listed species (*T. argentea* and *T. tricolepsis*) I can simply say that the leaf rosettes resemble airy, long-bristled cones; they have not yet flowered for me.

circinnata—The silvery leaves, broader at the base, overlap and form an urn-shaped, tuber-like swelling. Tight, flat spikes of glowing-orchid flowers in late winter.

ionantha—Three-inch tuft of pewter-gray leaves turn fiery red at flowering time, January-February. The inflorescence is an incredible paddle-shaped composition of fat, flat, tightly overlapping bracts, and the plant sends out large violet flowers day after day.

stricta—Shaggy “head” of very narrow, every-which-way leaves silvered with finest gray fuzz; short-stemmed spikes with shocking-pink bracts and blue-violet flowers.

CARE. Warm, humid, orchid-growing medium, wire to board with orchid peat. Moist during spring and summer. Dry in dark winter months, when plant is semidormant.

PROPAGATION. Offshoots.

SPECIAL USES. “Log” gardens, tree slabs.

Tradescantia *Commelinaceae* Spiderwort, Inch Plant, Wandering Jew

Most varieties of the inch plants that cheerfully romp all over indoor gardens are, of course, too rambunctious to be called miniature. But there are two species of much more modest proportions and habit.

multiflora—Unmistakably an inch plant, but with stems more threadlike than succulent; small, slim (even quite thin) leaves plain dark green, tinted purple beneath; and clusters of tiny white winking flowers. The effect is, believe it or not, delicate and “ferny.”

navicularis—China plant—Curious succulent creeper with very thick stems threaded through widely spaced pairs of clasping leaves folded tightly down the center; rose-purple flowers.

CARE. Easy culture, intermediate temperature, needs fresh air, loamy soil, filtered sun, dry side.

PROPAGATION. Cuttings of growing shoots, seeds, division.

SPECIAL USES. Baskets.

CHAPTER 7

MINIATURE ROSES, INDOORS AND OUT

Men, women, and children; gardeners, nongardeners, and the family cat—everyone is captivated by a four-inch rosebush with precise little leaves and thorns, studded with button-size buds and flowers, twinkling in its pot on the window sill. On her weekly visit the cleaning woman checks on its health and welfare. The baby-sitter has her boy friend come in to see it. The milkman wants to know where he can get one like it for his green-thumbed wife.

But if I had a dime for every eager buyer who has found these midgets disappointingly difficult to grow indoors, I could start building my dream greenhouse tomorrow. There are simply too many floriferous pictures with thimbles to show how cute the flowers are, and too few responsible growers who give specific cultural directions with every sale.

In the garden there's no problem. These are by nature outdoor plants, mostly sturdier and more winter-hardy than the full-sized hybrid teas and floribundas. For some reason, they even seem to be less subject to the depredations of insects and disease. I can pick handfuls of Japanese beetles from the regulars in the rose garden, but few from the miniatures little more than a hundred feet away.

I'm not implying that miniature roses are impossible indoors. One of the most perfect blooming bushlets I've ever seen came to our flower show from a sparsely windowed, steam-heated Brooklyn apartment. I simply want to spread the gospel that, to avoid risking disappointment, everybody should know what kind of cultural conditions they need.

Potted miniature roses are positively precious in cool, sunny window gardens, with rows of matching pots on glass shelves up and down the window, or singles or small groups in mixed arrangements on the sill or in a window greenhouse. Although they are not the most adaptable subjects for growing under artificial light, I've known several people who have been successful, particularly when the plants were started under lights from seed.

In a harmonious decorative container, a flowering miniature rose makes a small plant-and-container decoration to inspire the prettiest compliments. With suitable environment a tiny bush can be used as a center of interest in an indoor model landscape. A small greenhouse should hardly be without one of these brightly blooming babies.

Miniature roses have many uses in sink or trough gardens, as single specimens, pairs or quadruplets in formal plantings, even hedgerows kept carefully trimmed. I don't know whether anyone has ever tried them for miniature bonsai. This would be a process of dwarfing a dwarf; and my mental picture of the proper plant, artistically trained, is enchanting. I must try this, before long.

Outdoors, miniature roses are delightful in all kinds of containers—tubs, strawberry jars, window boxes, and other planters. They're often used as a low hedge to edge a path, driveway, or the beds of a formal rose garden, or around the base of a birdbath or sundial. In mixed flower borders they're planted singly or in small groups toward the front. In rock gardens they keep most safely cool and moist when planted low, near the base of the garden; and they show off most effectively with something like a dwarf evergreen as background.

Gardens featuring miniature roses are most often formal in design—round, square, rectangular, the beds divided with geometric precision by narrow strips of grass or gravel paths. A small, formal pool or piece of statuary may be the center of interest. In a sunken garden outlined with an eight-inch brick or stone wall, the planting pattern is particularly pleasing. In raised beds each individual shrublet can be enjoyed at eye level. In a single or double row at the base of a low retaining wall, the plants show off to advantage.

Beds of miniature roses can be carefully arranged strips, or groups of separate colors, or mixtures. They can be edged with shrubs such as dwarf box, perennials such as dwarf lavender, dainty annuals such as lobelia or alyssum. In the center a tree or standard is often used as accent. Or a bed may be backed by a wall, fence, trellis, or arch on which miniature climbers are trained. If the soil is slanted slightly up, toward the center, it is easier to see the plump perfection of each little bush.

Dwarf evergreens, particularly junipers, are popular backgrounds for miniature rose gardens. Upright types with symmetrical pyramid, column, or cone shapes are often used as accent—for example, a matching specimen of one of the dwarf varieties of *Juniperus communis* in the exact center of each formal bed.

Miniature or not, rose gardens are most often conceived in formal design. But to me, the cheerful dwarfs are more friendly when planted informally—popping up at the base of a tree stump or boulder, spotted here and there in the rock garden, a few at the top of a flight of small steps.

TYPES OF MINIATURE ROSES

The tight buds may be as big as the eraser on a pencil, or as tiny as a grain of unpolished rice, and the flowers may be single, semidouble, or double. The doubles may be formed like a hybrid tea or be full-petaled and fluffy, in clusters like a rambler rose. Some varieties stay very dwarf and bushy, from four to six inches tall; others are more robust, with larger flowers, and may grow to ten inches.

Climbing miniature roses are usually sports of bush varieties, with supple canes four or five feet long that can be trained on low fences, walls, trellises, or arches. Otherwise, every part is in perfect miniature scale.

All of these types are recognized by fanciers as authentic miniature roses because they grow on their own roots. And so is the rare tree or standard grown with a single trunk-like stem that is kept free of side growth, then pinched at the top to form a crown and symmetrical head. But standards that are budded or grafted onto the stems of other root stocks (which most of our American miniature tree roses are) are excluded by the experts, which is a matter of concern only to the serious collector.

INDOOR PLANTING AND CARE

Since miniature roses seldom spend the summer indoors, they are usually purchased in fall or winter from the local florist or greenhouse, or by mail from house-plant or miniature-rose specialists. Pot-grown plants are most likely to succeed indoors because their roots are not disturbed unduly. Except in Hawaii and Arizona, mail-order plants arrive with the soil ball

complete about the roots, the stems cut back to about two inches. They start growth almost immediately, and flower within six to eight weeks.

If you have miniature roses in the garden, you can root cuttings in early fall and force them into winter bloom indoors. Or you can dig the plants, pot them, and give them their necessary dormant rest before you bring them indoors for forcing. Sink the pots to the rim in soil, in the cold frame or in some spot protected from severe winter weather. When the temperature dips low, mulch with salt hay, straw, evergreen branches, or the like. In late December or January, after six weeks or more of dormancy, lift the pots and bring the plants indoors. Prune back the leafless stems and water sparingly until new leaf buds appear.

After they have flowered indoors all winter and spring, I always plant my miniature roses out in the garden and let them resume their natural outdoor growth cycle. I may root cuttings, or I may bring others indoors the following fall; but I have never tried to force a plant a second time without letting it live at least one year in the garden first. I have heard that some growers (probably city dwellers or others who have no outdoor garden facilities) simply let the plants rest outside in summer—on a shaded ledge or in a window box with moist peat—prune them severely in fall, and grow them again. I haven't heard how many years a plant will take this unnatural treatment or how much it suffers from missing its cool fall nap.

Soil

A fairly heavy potting soil packed quite firmly in the pot seems to help keep the plants small without sacrificing foliage or flower. One expert recommends a mixture of two parts garden soil, two parts humus, one part moderately coarse sand, with a light sprinkling of superphosphate or bone meal. When I use my ready-prepared potting soil, I always add sand, and sometimes some humusy soil dug from immediately under the leaf mold in the woods.

Potting

I've used both clay and plastic pots with equal success, always of the shape with the greatest depth, as miniature roses are naturally deep-rooted. Each pot has the usual layer of rocks or pebbles in the bottom for drainage.

Small, newly purchased plants usually start off in three-inch pots, are shifted to four-inch pots before they become severely root-bound. Some larger varieties may take larger sizes.

Sun

This is one of the three important cultural requirements. Miniature roses must have sun if they are to bloom. A minimum of three hours is sometimes set, but I should think this amount would be applicable only to midsummer or to mild climates. In winter the plants need all the sunshine they can possibly get.

Temperature and Humidity

Second in importance is a cool 65 degrees or even much lower (maximum, 70 degrees), and third is the humidity which keeps the plants at their best. Leaves curl and dry, buds and flowers drop when the air is hot and dry. Miniature roses should not be set anywhere near a heater or radiator of any kind. Unless the air in the growing area is really cool and moist, set the pots on moist gravel or make some other provision for increasing humidity, as outlined on pages 76–77. It even helps to cover the plants with a tent of plastic every night, and let them emerge only for the day.

Watering

Keep the soil always moist, never soggy and muddy, never dry and caked. As a humidifier and refresher, mist the foliage as often as you can.

Fertilizing

A balanced soluble house-plant fertilizer (never one with high nitrogen content) can be fed in half-strength solution every three weeks beginning about three weeks after a freshly potted plant begins active growth. Or you can use any special rose food according to directions and at half the strength recommended on the package. The idea is to encourage the plant to grow and flower, but not stuff it with so much nutrition that it gallops gaily up to nondwarf size with leaves only.

Pruning and Grooming

I seldom prune miniature roses indoors except to cut off cleanly any stems that may have been accidentally broken or that may grow unattractively long or misshapen. I do try to douse the plants in slightly sudsy water, to clean the foliage, every few weeks; and I pick off faded flowers promptly. Actually, instead of being in continual bloom, these plants usually flower for a few weeks and then take a short rest before they send up buds again.

Insects and Disease

Again, preventive spraying is all I've ever done. I use my handy house-plant aerosol bomb almost every week. If disaster should strike, I'd probably use the special rose spray or dust I use on the regular garden roses.

OUTDOOR PLANTING AND CARE

In all except mild or warmer climates, bare-root plants bought by mail from nurseries or garden-rose specialists should be planted in early spring, when vigorous root action and growth are beginning. Potted plants or any that come complete with a soil ball around the roots can be planted almost any time the garden soil is not frozen. But in sections where winters are severe, I think spring planting is always safest. In fact, in Connecticut I like to give new plants a longer growing season their first year by starting them a few weeks early, in pots, indoors or in the greenhouse.

Location

Plant miniature roses where they will get at least half a day of summer sun (a full day is best) but where it is possible to keep the soil suitably moist. Avoid low, muggy pockets where air does not circulate freely or where water can collect and make the soil muddy. Good drainage is vital. In cold areas, select a spot sheltered from icy winds by a wall or low shrubs.

Soil

To help keep them dwarf, miniature roses need a fairly heavy soil, but not, of course, too clay-like. Dig down at least ten inches, to prepare for the deep-growing roots, and improve the soil you remove with whatever is

needed before you replace it around the plant. Clay-type soils will need the addition of sand, for drainage, and leaf mold needs rotted or dry cow manure or other humus to lighten the texture. Sandy soils need humus to help hold moisture. In even average fertile soil, miniatures appreciate an extra ration of humus at planting time.

The ideal soil for miniature roses will pack firmly around the roots, yet won't cake and crack in the sun. It drains perfectly, so excess water does not stand around the roots, particularly in winter. Yet it holds enough moisture so that the roots don't dry out so fast you can't keep up with the watering job. Soil should also test neutral or slightly acid (pH 6.0), never extremely acid. In acid-soil areas, apply a light sprinkling of horticultural lime each winter.

Planting

Plant miniature roses about a half-inch deeper than they were before, and far enough apart so that they will have room to spread as wide as they will be tall. Crowded plants have no individual beauty, but they are prime targets for mildew. If the weather turns sharply cold or dry and windy after planting, protect the plants by mounding up soil around the stems. Remove it gradually as spring days grow balmy.

Watering

Miniature roses suffer seriously from drought, and will drop their buds and flowers after only a few days of hot, dry weather. Keep the soil constantly moist, and spray or mist the foliage once or twice a day. A mulch of pebbles, a mixture of half soil and half peat (peat alone packs down too heavily), or something similarly porous, will help keep the soil cool and moist.

Fertilizing

Light feedings of organic fertilizers such as bone meal or cow manure once a year, in early summer, are usually recommended. Or supply small amounts of a balanced garden fertilizer, or special rose food, in late spring and again in early July. A weak solution of liquid manure is also good and can be fed about twice as often.

Pruning and Training

In early spring, when new growth is first beginning to show on bush types, shorten all stems severely. I usually cut mine back to a uniform four inches. And of course, cut out cleanly any dead or mutilated branches. Make every cut just above a new shoot or leaf bud. Otherwise, pruning is limited to keeping the bushes shapely, removing faded flowers, and occasionally thinning the growth of old plants to admit air to the center.

Miniature climbers bloom on last year's wood. They can be cut back to six inches when first planted, but are otherwise not pruned except to control ungainly canes or remove dead ones. Train the climbing canes into an attractive, open pattern as they grow, by tying them to the arch, fence, or other support they are to climb on.

Miniature tree roses can be cut back to a small but symmetrical head in spring and all dead wood should be removed. To keep them shapely, prune as needed during the growing season.

Insects and Disease

I protect my miniature roses, as I do the others, with an all-purpose rose spray or dust applied first when leaves begin to unfold, and repeated every week or ten days until the plants go dormant. Occasionally, during a long spell of hot, humid weather, I see signs that mildew threatens. If the all-purpose spray contains a fungicide (which most of them do), I use it immediately. If not, I may resort to dusting sulfur (which does mar the beauty of the flowers) or whatever sterilant is on hand.

Winter Protection

What you do to protect miniature roses in winter, or whether you do anything at all, depends not only on your climate but also on the health of the plants. If they've been growing well, they'll take lots of abuse; if they're weak and ailing, their chances of survival are reduced.

Sometimes, a flower pot inverted over the leafless stems is all that's needed. Or you may mulch with salt hay or evergreen boughs. In Connecticut we mound up soil so it covers the first three or four inches of the stems, and remove it gradually in spring. It is most important to make

sure that water does not stand around the roots in winter, next most important that alternate freezing and thawing don't heave the roots out of the ground and break them.

If they are likely to be whipped by wind or covered with ice, the canes of climbing varieties are removed from their support and laid flat on the ground, where they can be covered with either soil or mulch. Since tree roses are inclined to be touchy, we wrap ours in burlap, with an extra layer or two around the graft and crown.

PROPAGATING MINIATURE ROSES

I've grown many miniature roses from seed, and had a lot of fun doing it. They usually germinate in about three weeks (best temperature about 60 degrees), quickly send out tiniest true rose leaves, and are ready for transplanting into small pots in another two or three weeks. I usually pinch the tip growth at least once, when the plant is about five inches tall. The flowers can appear within three months after sowing.

Of course, seedling plants are not named varieties. Most of them, in fact, have small single flowers in pale shades of pink or white. Your chances of double, more brightly colored flowers increase if you can get seeds of a good strain.

For new plants of named varieties, take cuttings in August or September—three-or four-inch pieces of healthy wood produced in the current season. If the stem can be pulled off gently with a sliver of the main stem still attached (a heel), rooting may be faster and is surer. Dipping the cut ends in hormone rooting powder is also helpful.

Make the moist propagating material firm around the base of the cutting, and make sure the air is kept humid in the propagating box or plastic tent, or invert a glass jar over the cutting. New growth is the signal that roots have formed and the cutting is ready for potting. These plants, too, will grow more compact and bushy if the tips are pinched out when the stems are about five inches tall.

NAMED VARIETIES OF MINIATURE ROSES

Although new varieties of this popular plant are constantly being introduced, and most likely will have a wide appeal eventually, it may be

some time before they appear in plant and seed catalogues. In compiling this modest list I have thought chiefly of what is available at the moment, miniature roses I have either grown, seen in friends' gardens, or admired vicariously on the printed pages of magazines, books, and booklets. If you are interested, I am sure these bushes are readily available. If I am old-hat and you feel avant-garde, there are many persons propagating new varieties. Talk to some of them, or try it yourself, a most gratifying hobby:

'Baby Bunting'—A delightful, small rose with red flowers of a deep, startling shade. This variety is an inch or so taller than some, but many of my friends think it ideal in that they like to make miniature flower arrangements and appreciate slightly longer stems. Among its other charms, the rose is most delightfully fragrant.

'Baby Crimson'—I've never had this one grow taller than six inches. Since I have maternal instinct for the wee ones, I love it. Under the right conditions it will bear tiny crimson flowers and exquisite buds up until frost time.

'Bo-Peep'—This one has a charm in its name, and is one of the more popular miniature roses. It has double pink blooms, forms a bush with a neat conformation, and is another favorite with those who make miniature arrangements. You needn't worry about cutting its blooms. It is always growing more.

'Cinderella'—This one fits beautifully into the legend about the girl with the glass slipper. It has dainty white blooms touched with pink and is in the true tradition of rose shapeliness. Seemingly, it loves to bloom.

'Granada'—In some listings I find the name spelled 'Granata.' Regardless of the spelling, I love the bushes I have grown, for the lovely, semidouble, red flowers which the bush bears so profusely. A tiny vase filled with them makes one wish to build a doll house in which to display it.

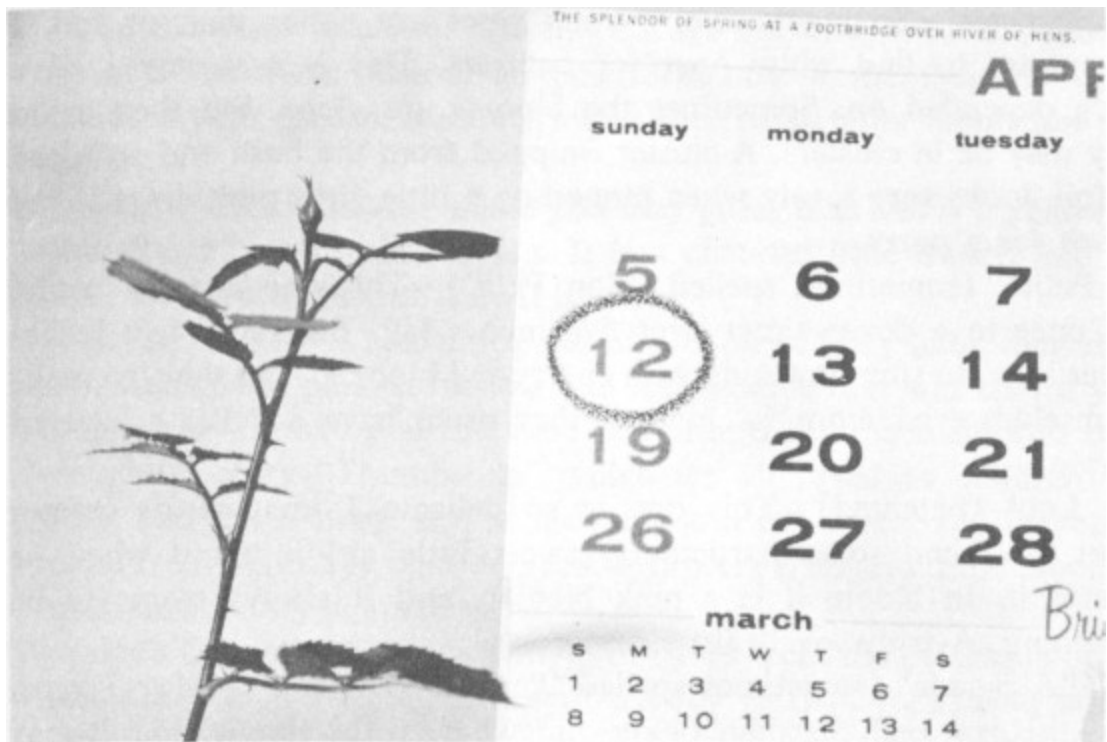


Days in the life of a miniature rose:

a. Leafed out



b. A growing bush



c. First bud



d. Full bloom

‘Little Princess’—So many of my small roses are either pink or red, I always try to find white ones for contrast. This is a variety I have often depended on. Sometimes the blooms are alone, but then again they may be in clusters. A cluster, snipped from the bush and wrapped in foil, looks very lovely when pinned on a little girl’s pink dress as she leaves for a party.

‘Mon Petite’ (sometimes spelled ‘Mon Petit’)—This one is truly petite, not once in a dozen times over five inches tall. But those five inches never seem to stop flowering with cherry-red blooms. And then, to make themselves even more fascinating, they often have a delicate haze of purple.

‘Patty Lou’ (patented)—This one is so delicate I imagine its creator must have had some particularly sweet little girl in mind when he named it. In bloom it is a pink bicolor, and it always seems to be blooming. A truly lovely little rose.

‘Perle d’Alcanada’ (sometimes spelled ‘Perle d’Alconada’)—May I warn you, this is a real charmer. Being somewhat on the stately side, it may grow to nine inches. It makes a neat and most

attractive bush and then comes forth with pink blooms that slowly change to a white pearl-like effect.

‘Pixie Gold’—This is another dainty one, a yellow miniature with a lovely soft color to add to its beauty. It is really a miniature, and would consider itself a giant if it topped five inches. It has an attractive bush, but that is only part of it, the blooms follow a perfect pattern from the time they are buds until they are full-blown. It follows all rose traditions.

‘Red Imp’—Many persons consider this beauty the most perfect of all miniature roses. Certainly, with its deep-red blooms, it is one of the best known. Such a feeling of affection must be deserved. I think the first miniature I ever owned was a ‘Red Imp.’ If for no other reason, that would make me love it.

‘Rosa Oakington Ruby’—Some years back the English Royal Horticultural Society thought so much of this rose they gave it the Award of Merit, and well they might. The blooms are a rich ruby-carmine, are double, and are around all summer. In planting this variety, remember it tends to be an inch or so taller than the wee ones.

‘Rosata’—I love this one for its fragrant, pinkish flowers, which also have a touch of salmon. The blooms are delightful when made into corsages or miniature bouquets.

‘Rouletti’—This is a great favorite in rock gardens, and in edgings around beds of big roses. It is a true “shorty” and seldom exceeds five inches in height. But the buds, rose-pink, are so exquisite one wishes to put them into a setting for a ring to be worn on the finger.

‘Scarlet Gem’—This is what is known as a newcomer among miniature roses. But it has so much charm, I know it will be called an old favorite in the years to come. The flowers are an orange-scarlet, and fairly cover a handsome bush of nice conformation. Remember when setting it out in your garden, it may grow an inch or so taller than some of the others.

‘Sunbeam’—From the very name you may guess that this is a yellow rose, literally, a beam from the sun. It is a cheerful little dwarf, and has a tea-rose type of bloom. You’ll love it.

‘Sweet Fairy’—This is something out of a book of fairy tales, delicate and fanciful. It has pinkish blooms and a fragrance that will charm you.

‘Thumbelina’—Looking at this rose will bring back the memories of that delightful story “Thumbelina” which we all loved as children. As a rose, and not a story, it is semidouble and has lovely red flowers that open from delicately pointed buds. It flowers freely, and the blooms are most enchanting in small vases.

‘Wayside’s Garnet’—As I first bought this rose from the catalogue of Wayside Gardens, I’ll let them describe it for you: “... a neat, compact small plant with many perfect, fully-double, garnet-red little flowers. It is a prolific bloomer and a good grower ... much like Oakington Ruby, which is one of its parents. It probably has the brightest, darkest and deepest red to be found among miniature roses.”

‘Yellow Miniature’—This is a charming little rose, yellow, with a cheering tint. Many consider it to be the most attractive of all yellow miniatures. I will not argue with them. I’ve always been enchanted with it.

MINIATURE SINK GARDENS

Take the concept of dish gardens and model landscapes, but execute it with miniature garden plants. Take the outdoor plants of bonsai, but don't dwarf them unduly or train them into unusual shapes. There you have the mixture that makes up these specialized miniature gardens, called "sink" or "trough" gardens for the old-fashioned stone sinks and horse troughs they were planted in when the fad first swept England, some thirty years ago.

Now, the old sinks and troughs are practically nonexistent, and the name is anachronistic. But I have been totally unable to dream up anything better. "Sink garden" is a specific title for a composition of plants or a landscape scene in small scale, planted in a sturdy, sink-like container, grown outdoors and used in limited ways to decorate the garden and grounds. No other phrase seems to define it.

My interest was originally aroused by the books of Anne Ashberry, England's sink-garden specialist, and by the warm affection she has for her specialty. But it was not until I began to work with miniature plants in our Connecticut gardens—and to find out what a great variety is available—that I was inspired to plant a sink garden of my own. Originally, I was intrigued; soon, I was fascinated; now, I'm an addict. With the flimsiest excuse I'd have so many of them it would look as if our grounds had broken out with measles.

These sink gardens are not for big, burly gardeners who like cabbage roses and gaudy shrubs. They're for connoisseurs who appreciate the minuscule perfection of a tiny plant, more effectively displayed at eye level. They're for those who grow alpenes and other difficult plants and find them less finicky under these controlled conditions. They're for gardeners who can't, or don't want to, squat in the hot sun for hours, weeding or transplanting; who want the pleasure of creating gardens, but take the accompanying chores in small doses. And sink gardens are for people, like me, who simply find irresistible charm in the miniature.

If our grounds were spacious, I'd find a place where I could have a collection of sink gardens, set up on pedestals and arranged in neat rows, so

I could move easily from one to the next with the watering can. But they're probably much more ornamental and distinctive if used the way the few we have now are.

Instead of a sundial at a break in the shrubbery border, we have a sink garden set on a two-foot column of mellowed brick. Two narrow gardens outline the corner of the small patio by the front entrance. A small sink garden enlivens a shelf beside the door to the lath house. There's one at the end of an old stone bench.

Or you can display one of these gardens against the wall at the end of a garden walk; as a centerpiece on the lawn or terrace; on top of a low wall or at the edge of a balcony; in place of an inanimate statue or urn. If possible, let the background be light and not bright-colored; neutral shades show off the plantings to best advantage.

CONTAINERS

Picturesque old sinks are obviously not available to us, and any horse troughs I've seen have been much too monstrous. Miss Ashberry casts her own containers of concrete (its porosity is excellent for plants), and we can do the same.

Sometimes I think the sinks and troughs look a little heavy in relation to the plantings. Certainly they *are* heavy, and almost impossible to move, when filled with soil. But I've found a goodly number of acceptable substitutes. First, of course, I shopped my favorite junk yard and found the round concrete planter and the wash-tub lid that served as containers for my first sink gardens. I also saw possibilities in a big old butter tub that could be cut down, and in a leaky birdbath.

Some of today's building tiles are perfectly beautiful and, if shallow enough, could be fitted with a metal or wooden bottom. They come in all sizes, shapes, and colors. Thick, old wood is another possibility. I'm thinking of some weathered planks we found at the seashore last summer; they'd make a handsome and sturdy container for a wind-blown, woodsy garden.



Miniature garden of dwarf evergreens, *Cyclamen neapolitanum*,
and tiny trumpet narcissi not yet blooming

In designing or selecting a container, you have few rules to go by. It should be strong and weather-resistant, of course, because it is to be placed outdoors. It must have drainage holes in the bottom, so fallen rain won't stand in it. It should be deep enough (six to eight inches) to give small trees and plants root-room. And artistically, it should be in harmony and proportion with the garden to be planted in it, not as a feature in itself, but as a subordinate element in the picture.

Unless a sink garden is to be placed on top of a wall or some other existing support, it will probably need a base to hold it two or three feet off

the ground. This can be made of cement blocks, rustic brick, tile, or concrete, according to the design of the container itself.

PLANTS FOR SINK GARDENS

Unlike bonsai, these plants are not to *be* dwarfed, they *are* dwarf by nature. Miniature perennials, such as *Calceolaria biflora*, never top two inches, nor do some of the tiny narcissus species. Some trees have never been known to grow taller than six inches. And if you can't find trees that are small at maturity, you can find many that grow so slowly they'll stay in proper scale (even without pruning) for five years or more. There are miniature garden plants of all habits and shapes—stiffly erect, tufted, bush-like, sprawling, creeping, hanging, climbing—and even pinhead-size water plants for tiny pools.

Actually, there are miniature plants in every horticultural category—annuals, biennials, perennials, bulbs, shrubs, trees, aquatic plants, and wildlings—and most of them are suitable for sink gardens. You have only to select those that are in scale and sympathy with your design, and that are culturally compatible, one with the other. You can grow many of them from seeds or cuttings. Miniature perennials are available in widest variety from growers of alpine and rockery plants. Trees and shrubs can be bought by mail from suppliers of small plants for bonsai work. Native and aquatic plants are plentiful from mail-order wild-flower houses.

ACCESSORIES

The worst thing you can do with one of these little outdoor gardens is to clutter it up with little artificial props such as benches, bridges, and old oaken buckets. At all costs, avoid the cute and the trite. Practice moderation and the utmost restraint.

One prop—a hand-carved well-head, an alabaster birdbath, a lichen-covered rock—is usually plenty for any one garden. If it is handsome in its own right, the whole garden may be designed to set it off. If it's a supporting element, play it down and let the plants stand out in the picture.

The same is true of streams, pools, walks, walls, and other miniature landscape constructions. They're pretty and they're fun to make; but just one too many can spoil a garden.

Naturally, any accessories and props to be used in a sink garden should be sturdy and weather-resistant. And as in any other miniature composition, proportion and scale are terribly important.

THEMES AND DESIGNS

Many of the principles and suggestions for dish gardens and model landscapes in Chapter 3 are equally applicable to sink gardens. The design needs, first, a basic idea or theme. Will the garden be formal, or informal and woodsy, or simply an artistic arrangement of living plants with or without a piece of tree stump or rock? Should it be built around an important accessory, or will one plant or a group of plants be the center of interest? Does the style of the container suggest the style of the garden to go in it?

Since a sink garden is usually planned to have some permanency, it is particularly important to plan the design in every possible detail and, if at all possible, to put the plan on paper—and in proper scale. You can tell, before it's too late, whether a tree will be too large, a fence too high or prominent, a grouping of plants too far off balance.

When you plan the planting, keep proportion and perspective clearly in mind. If the design is to have formal balance, arrange pairs of trees, clipped hedges, straight walks, and other elements with geometric precision. If the effect is to be informal, make sure the center of interest is off-center, with a large airy area or low planting to balance it at the other side.

In crowded plantings the beauty of the form of individual plants is lost. Be sure to space them so that they have room to grow without becoming entangled with their neighbors. To blend the garden with its container, plan to have a creeper or trailer dangling over the edge.

Artistic plant compositions are arranged, like dish gardens, with outstanding accent plants, low growers often around the base, usually arranged naturally at the base of a rock or around a piece of log or stump. Colors and textures of flowers and foliage are contrasted and blended as they are in arrangements of cut flowers. Setting the plants in the empty container and rearranging them until the best effect is achieved may save shifting them about during planting.

All kinds of landscape designs can be re-created, in miniature, in sink gardens. And the scenes can change naturally with the seasons of the year. One of my informal gardens has a basic arrangement of rocks, small evergreens, and ground cover. In spring, miniature narcissus species bloom; in summer, tiny annuals such as *Ionopsidium acaule* and perennials such as *Erodium chamaedryoides roseum*; in fall, small cyclamen species.

Woodsy wild gardens can also have basic, permanent plantings—seedling evergreens, moss, foliage plants such as small ferns, rattlesnake plantain, and pipsissewa—through which spring-blooming squirrel corn, hepatica, and spring beauty can push up their flowers.

One of the most effective formal-garden designs makes good use of miniature roses as a flowering hedge in front of a high wall at the back, or as twin specimens on each side of an arch. Other formal gardens adapt the designs of the Victorian age, or the Colonial gardens of Williamsburg.

Someday I want to try an Oriental garden featuring a bonsai-style dwarf tree and planted sparsely, in the Japanese manner, with tiniest shrubs and perennials and a ground cover of fine moss or sand, and perhaps a curved bridge over a still stream.

With a suitable container you could do an outdoor desert garden. Many miniature desert plants are hardy or semihardy and would live through the winter with some protection. There are many other possible themes, and many types of plants and containers with which to carry them out.

PLANTING AND CARE

Unless you can control watering (which means keeping the garden out of the rain), make sure that the container has plenty of small holes in the bottom for drainage. And for extra insurance that drainage will be perfect, start out with a layer of pebbles or sand. A covering of burlap or sheet moss will keep soil from sifting down into it.

Soil should be light and porous, capable of holding some moisture but not too much. The standard recipe of one-third garden loam, one-third humus, and one-third sharp sand is a good basic mixture to start with. Add extra sand if the plants are succulent-like, extra humus for woodsy plants, a sprinkling of lime for plants that dislike acid soil. A slow-acting organic

fertilizer such as bone meal can be mixed in, but in very small amounts. Run the mixture through a coarse sieve, to remove stones and debris.

As you place the plants, firm the soil gently around the roots. Don't fill the container so full that the soil is level with the rim; leave an inch or so to hold water while it seeps down to the roots below. Place the ground-cover plants, and those to dangle over the edge, last. Some gardens are finished with a thin mulch of stone chips or sand, some with a carpet of moss.

Location

A sink garden planted in a real trough or sink is a mighty heavy thing, once it's filled with soil and planted; and so may be many others. If you can place the empty container in its permanent spot and plant it there, you may save someone an aching back.

These gardens are meant to grow out in the open air, but not where searing sun and hot dry winds can dry the soil too fast and burn the plants. If the plants are all of the type that need sunlight, give them only the dappled shade of a high-branched tree or the windbreak and slight noonday shade of a low wall. Woodland plants and others that like shade can be grown in more protected spots. Naturally, the two types are not combined successfully in the same garden.

Don't place sink gardens where they will receive the drip from eaves or an overhanging tree. Don't set them tight up against a wall. Newly planted gardens need some special protection—a cheesecloth tent or newspaper on a temporary frame overhead—until plants are well settled in their new home.

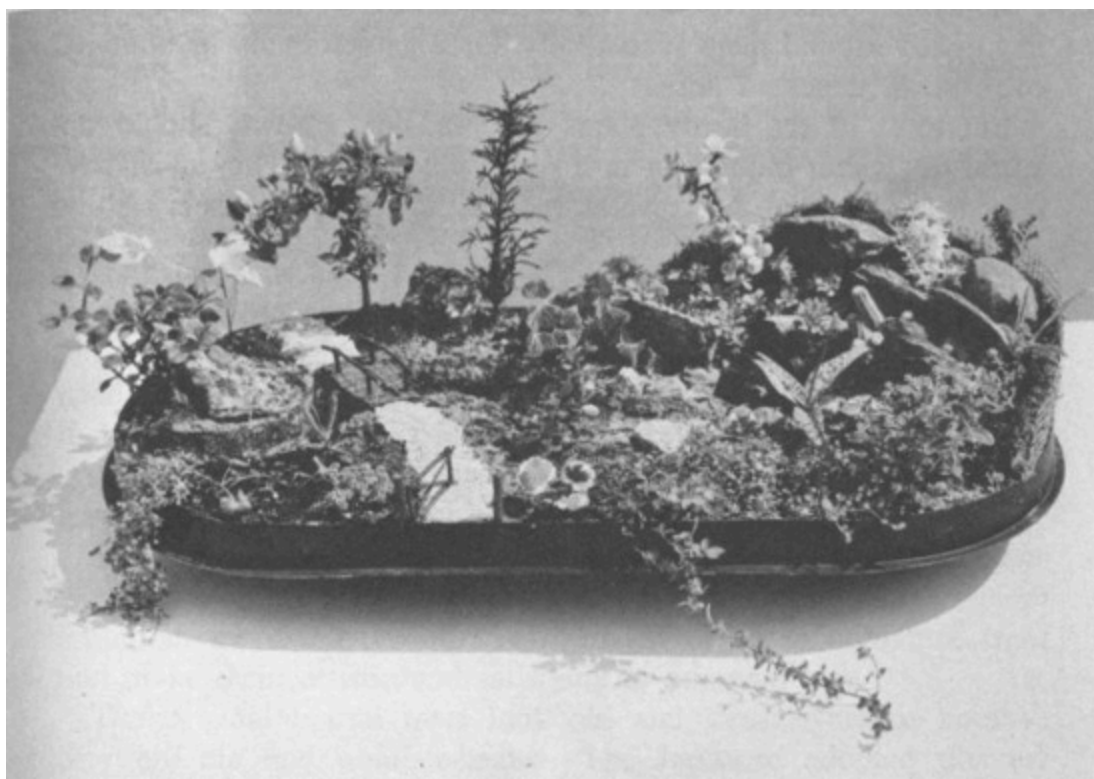
Watering

A safe general rule is never to let the soil dry out all the way through, never to water so much that it is soggy and sour. For most plants, you can scratch into the soil surface with your fingers. If it feels moist, don't water; if it feels dry, do. However, succulent plants should be grown drier, boggy plants more constantly moist. Frequency of watering depends upon type of plant, size and type of container, the soil, the weather—depends, in fact, upon how often each individual sink garden needs water.

Fertilizing

If a fertile soil mixture is used in the first place, and particularly if it is enriched with a slow-acting fertilizer such as bone meal, most gardens will not need extra feeding for many months after planting, often not for the first year. The point is to give the plants just enough food to keep them healthy, not enough to make them grow out of proportion to the garden.

If you see signs of malnutrition—few, small leaves with poor color; failure to bud and flower; sickly, stunted growth—feed quickly but lightly. A weak solution of organic food such as fish emulsion or liquid manure is usually recommended. Established gardens can take this light feeding once in spring when active growth begins, and once or twice during the early summer, without outgrowing their bounds.



Rock garden set in an old wash-boiler lid

Pruning and Grooming

Pick off all faded flowers promptly, so the plants will not exhaust themselves by setting seed. Remove any dried or fallen foliage so it will not

rot and invite disease. Pinch the growing tips of plants that threaten to grow too tall and lanky. Shear hedge plants regularly and nip back creepers that spread out too far and strangle other plants. Refresh and renew any mulch or moss carpet as needed. In a garden so small, the least imperfection seems magnified.

Insects and Disease

Once a week, all summer long, my sink gardens get a quick treatment from an all-purpose aerosol bomb, used according to label directions. So far, with one exception (the mysterious plague of “inchworms” we had in the spring of 1961), this has kept insects and disease at a safe distance.

Winter Care

In mild or warm climates, sink gardens should not need any special protection in winter. But in Connecticut, the deep-freeze is so long and severe, I move my gardens to the cold frame. To make sure that the soil does not freeze and crack the container, I sometimes sink it to the rim in the soil. I’ve also packed salt hay tightly around them successfully. Or a garden could be wintered over on an unheated porch.

But most of the hardy plants used in sink gardens should not spend the winter indoors or in a warm greenhouse. They must have a cool rest period for several months to complete their natural growth cycle.

MINIATURE PLANTS, BONSAI-STYLE

Only in the true Oriental bonsai do art and horticulture combine in such an extreme state of perfection—and in miniature to boot. Paintings may be as magnificent, but they're inanimate. Ancient trees of the forest may have equal artistic virtue, but they're not shaped by the hand of man. Living bonsai trees, sometimes centuries old, become masterpieces because, says Claude Chidamian, "they're planted in philosophy, shaped by art, grown with love."

If that sounds as if I am awed by bonsai—I am. I would never have the talent and patience to prune and shape, trim and train, in minute detail year after year, so that every branch, twig, and tiny needle or leaf would be perfectly placed and proportioned. Even if I were an artist, I doubt that I could create the illusion of grandeur in minuscule scale. Nor would I ever dare assume the responsibility for caring for these priceless, age-old plants.

But that doesn't mean that bonsai is beyond me, or any other gardener who admires it. Without committing the sacrilege of inept imitation, we can have our own version of these miniature trees and make them artistic and satisfying in our own way.

The original bonsai trees look old and weather-beaten because they *are* old and weather-beaten. The Japanese adopted this art from the Chinese many centuries ago. Our trees in bonsai-style are not likely to have that venerable age, but they can have character. They can have the lines of trees that have held a precarious footing on the side of a rocky slope, have been bent by the wind or twisted by mighty storms.

Because every part of it is in perfect proportion to every other part, a fine bonsai tree creates an illusion of tremendous size—as if you were looking through the wrong end of a telescope to a giant more than a hundred years old. Our dwarfed trees can be perfectly proportioned and create the same illusion. Although there is no substitute for true antiquity, our dwarfed trees can be artistic in their own way without pretending to be ancient.

By making some concessions (without desecrating the art) we can take suitable trees and turn them into “Orientalized” garden ornaments, and do it in one year, not ten. If the pruning and training is done with care and artistry, the result will be a bonsai which is a distinguished ornament and particularly appropriate for contemporary architecture, and also for landscape architecture.

I have seen a bonsai of Sargent’s juniper set beside a garden pool, its twisting branches swaying out and over the water, and reflected in it. Twin (but not identical) bonsai trees are startlingly effective; for example, one on each side at the top of a set of formal steps. Bonsai can be used as a center of interest on a patio or terrace to accent an entrance, on top of low walls, or against the wall at the end of a path.

Last summer, in our wild garden, my husband dammed up a tiny stream at a point where it began to run down a short but rather steep and shaded slope. This created a small pool from which the water trickles over the dam and drops onto a series of rock ledges below. We planted the banks on both sides with ferns, wild ginger, bloodroot, trillium, and other wildlings. But something was needed at the top, some small tree or shrub that would integrate the dam into the picture and would be in harmony with the woody surroundings.

We considered all the dwarf, shade-tolerant evergreens our local nurseries had to offer, but nothing seemed just right. We scoured our woods, but the only low-growing trees (which are mighty few in our area) were too straight and erect. The mountain laurels and other shrubs with interesting lines would eventually grow too large. So we decided to try what, for us, is an experiment.

We found a white pine less than two feet tall with a suggestion of the irregular shape we had been looking for. We lifted it carefully, took it home, root-pruned it, and planted it in the best bonsai tradition, in a sturdy box just large enough to hold the roots but leaving a little room to spare around the edge. Then we took the tree to the top of the dam and planted it by sinking the box in the soil. After some weeks, when the pine showed no sign of ill effects from being moved, and was making new growth, we shaped and pruned it, and then wired it, bonsai-style, along the lines of the tree we had been hunting for.



Streptocarpus—this variety is the delightful little Weismoor hybrid.

With sensible care and winter protection, pruning and root-pruning when it threatens to grow too large, and training in the way we think it should grow, the little pine will, we hope, mature into a gnarled gnome in proportion and harmony with its woodland setting. Of course, we plan to provide a new box at root-pruning time before the old one can rot and set the dwarfed roots free to roam the soil around it.

INDOOR BONSAI, HARDY OR SEMIHARDY

The real Japanese bonsai is an outdoor inhabitant, usually one of a collection that is brought indoors for display purposes and for only a few

days at a time. But by using tender plants that want more warmth, growers are beginning to create bonsai that can be used decoratively indoors the year round. In a shadow box, on a coffee table or special stand, even as a table centerpiece they should always be alone and with no accessories to detract from their unique style.

Last winter I thoroughly enjoyed starting a collection of indoor miniatures, frank copies of Japanese *name* bonsai, and even the wee fingertip *shinto* type. These are a special challenge because to preserve proper proportion, leaves and needles must be extra small, and pruning and training are particularly crucial. Water is applied with an eyedropper; fertilizer is administered in microscopic amounts. To provide protective humidity and warmth, and to help keep the small amount of soil in tiny containers from drying out, we rigged up plastic-covered quarters on a window sill, with a layer of moist vermiculite. Our tray was also improvised from a double layer of heavyweight aluminum foil. Now, all but the youngest, and the very smallest, miniatures are strong enough to grow on glass shelves outside the plastic tent.

Indoor plants, bonsai-style, are more than just tender seedlings, or cuttings, kept small in small containers. They are patterned after true bonsai, with interesting character, artistic lines, and perfect proportion. The mechanics of pruning and training are very much the same. But because they are not hardy outdoor growers that resent the hot dry air of a house, and because they don't need annual dormancy, they are fascinating indoor ornaments to be lived with and enjoyed the year round.



Different types of bonsai trees and containers



Bonsai in citrus

Once we became intrigued with the concept of “indoor bonsai” we found so many house and greenhouse plants with picturesque prospects that I fear we will never get to try them all. I have seedlings and cuttings of all sorts, even including those from a breakfast orange and a pomegranate out of the fruit bowl. And, of course, small plants sold by mail-order suppliers are just the right size to begin the process of dwarfing and shaping.

(In the list at the end of Chapter 6, plants suitable for indoor bonsai use are indicated.)

OUTDOOR PLANTS FOR DWARFING, BONSAI-STYLE

With proper care, any woody plant—any tree, shrub, or vine with persisting trunk or stems—can be grown indefinitely in a pot. With some

skill in pruning branches and roots, it can be permanently dwarfed. And with imagination and artistry, it can be trained to re-create in miniature one of the majestic pictures of nature. However, some plants are more amenable to rigorous dwarfing than others; some adapt more willingly to growing in containers; and some are by nature more suitable in habit and appearance.

The easiest plants to dwarf are those that are naturally small, or slow growing, and those with small leaves or needles, flowers or fruit. Proportion is the most important factor. Every element—leaf, twig, branch, trunk, root, container—must be in harmony and balance with all others.

It's not impossible to use larger-leaved plants. It's just a little more difficult. Long needles can be cut shorter, for example, but they must be kept the proper length. Large leaves can be thinned to relieve any feeling of heaviness, and each leaf can be used to represent a branch. With some types of deciduous trees—say, maples—the leaves that come out first in the spring can be pinched off. The leaves that come out to replace them will be smaller.

Almost equally important is the “character” of a dwarfed plant—its irregular or fluid lines, illusion of age, unusual aspect of bark or twig that make it dramatic and vibrant. Any form of art can be dull if it has nothing except perfect proportion to offer. With good proportion, plus intriguing line and design, it becomes interesting.

In the original bonsai the artist transplanted a tree he found growing in the wild and carefully conserved the misshapen lines made by buffeting weather, or he very carefully copied, or re-created, a tree he had seen holding a precarious footing high on a rocky ridge and perhaps dipping down into a windy gorge. These shapes and forms are now the basis for specific classes of bonsai which we can borrow or adapt. So the “character” of a plant may suggest that it be trained as if it were growing out at a right angle from a rocky slope with its roots covering a stone; as a grove, to weep or cascade; or as a gigantic, single-trunked forest monarch with pitted, weather-worn bark.

Or perhaps, lightning might have split the trunk, leaving part of it jagged and dead. The two trunks may have become entangled with each other. Branches may have been blown in one direction so long that they bend that way permanently.

Types of Plants

Any woody plant whose parts are in proper proportion for dwarfing—and particularly, any that shows promise of interesting lines or “character”—is a good prospect for bonsai. The coniferous evergreens are most popular, because they hold their foliage all year, and because small-needled types are comparatively plentiful. Small-leaved deciduous trees can be at their most attractive best with spring’s budding new growth, summer’s airy foliage, fall’s brilliant color, or with the silhouette of a naked trunk in winter.

With flowering trees and shrubs the choice of varieties narrows even more. Large flowers with brilliant color and overpowering fragrance may destroy balance and proportion, and detract from the beauty of the plant itself. More delicate plants are more likely to enhance the picture. For obvious reasons, plants that bear fruits and berries in the proper scale are the hardest to find, and culturally the most difficult.

(For plants that are suitable for outdoor bonsai, see list at the end of Chapter 14.)

Sources

Bonsai becomes a reality faster, of course, when you start with a fairly mature plant. The most fruitful source is the selection of three-year-old trees and shrubs at your local nursery. Those growing in gallon-sized cans, or other containers, naturally accept pruning and transplanting with the least setback. They should be healthy plants, not overgrown or neglected, that have been regularly root-pruned and transplanted through their infancy. It is perfectly safe to buy them pot-bound if that condition has not persisted for several seasons, with the outer roots all dead as a result.

In selecting a specimen, first examine varieties that are by nature slow-growing and have leaves, or needles, in perfect scale. Be selective, pass up the symmetrical specimens that are best for ordinary landscaping purposes, and look for that one plant in a thousand, the one with interesting “character”—peculiarity of shape, irregular branches, low horizontal growth, stocky or twisted trunk. The most ideal plant has its largest branch near the base, and has no regular or opposite branches to be pruned away. Next to above-ground development in importance, is the below-ground root

system. To be ideal, the root system should be a compact, shallow mass rather than one long taproot with a few offshoots. To make it even more ideal, the stoutest roots should be growing near the surface.

Younger, smaller bonsai plants are readily available from mail-order suppliers who specialize in them. Here, make your selection according to variety, and then train your plants as you grow them. But beware of cheap “bargain offers.” I speak with this advice because of the experience of some of my friends. They were too intrigued by Sunday-newspaper ads. Buy only from reliable growers who have invested time and care in developing healthy plants and, better yet, have labeled them true to name. For those who are more interested in the finished work of art than in the growing and creating of it, some florists and nurseries advertise mature, or nearly mature, bonsai. Again may I say, “Beware.” The supplier’s reliability is even more important.

Sometimes you can find precious bonsai-type shrubs or trees growing in the wild. Small seedlings of hemlock, ash, birch, maple, and some elms, if they are dug very carefully—and at just about any time of the year—can be used as miniature bonsai, or they can be planted in the garden for a few years until they are larger. Usually, the taproot has to be cut back rather severely to encourage the development of a spreading mass of smaller roots. Start your regular pruning as soon as the plant recovers from transplanting shock.

The larger native plants should be taken when they are dormant, in late fall or the earliest spring. In searching, look for those that have been naturally dwarfed and misshapen by misfortune. In digging, make sure the soil is so moist that much of it will cling to the roots. Keep the roots moist and well covered against drying sun and winds until the transplant is safely in the soil again. For any such wildlings, take along enough of the surrounding soil to fill the container into which they are going. This will make them feel at home in their new place in the garden bed.

Many growers now propagate their own bonsai plants; thus they can control shape, root-spread, line, and design from the very beginning. Almost all of the propagating methods outlined in Chapter 10 are useful here, some especially so. Cuttings of all kinds will produce stout-trunked plants much faster than seeds. Plants such as willow, holly, ginkgo, and ivy

can be started from stems as thick as an inch or so in diameter. With cuttings you can have quantities of new plants from one parent, all with the same variegations or other characteristics.

Either ground or air layering can give you plants that are larger than those obtained from cuttings. By selecting a certain branch you can be surer of getting the characteristics and shape you want. If your bonsai is to be well balanced, select a well-proportioned branch with close twiggy growth. By layering you can also correct an unattractive bonsai, growing a new one from a plant that has become too tall and gangly, or one that has badly formed roots.

Grafting for bonsai plants is not widely practiced because too often it leaves a visible scar or some other sign of artificiality, and also because there is the danger of undesirable sprouts shooting up from the roots. Plants from a graft are generally weaker than cuttings or layers which have their own roots. If your plant happens to be a valuable one, you can often improve its shape by grafting on new branches where they will do the most good; or good branches can be grafted onto picturesque, gnarled roots. I have never made a serious effort to graft in such cases, but I have seen some good results from the work of other growers.

Growing bonsai plants from seed requires infinite patience—as a warning, the process takes years. But seedlings, once you have them going, and with the healthiest of root systems, will live indefinitely. Seeds of bonsai-type plants are available from several growers. Should they need special preparation, such as nicking or stratifying, the seed packets should say so.

When seedlings have several true leaves and are ready for transplanting, cut back the strong taproot (should there be one) by at least one-third. This will encourage root-branching. When potting seedlings, spread the side roots so they will develop evenly near the soil surface. Pruning and training can begin while the plants are still quite young. Plant the stem on a slant. Pinch new tip growth for development of side branches. Tie straight trunk stems to a bamboo cane, or perhaps just a sliver; or otherwise guide the young shrub, or tree, toward the lines you have in mind for its mature effect.

In no other art form is it more obvious that a subordinate element such as a container can make or mar the perfection of a picture. For instance, with a painting the frame can have small faults without lessening the impression of a masterpiece. But bonsai is so stark that a slight imperfection can become a glaring error. So, although the container is merely a supporting feature, it becomes only a little less important than the center of interest, the plant. It must harmonize with the plant, reflect and supplement its beauty, and not detract from its leading role.

Imported Japanese bonsai containers of the traditional type, now readily available, are usually shallow bowls or dishes of glazed (or unglazed) ceramic. They are never glazed on the inside and are usually sold in sets of three matched units of graduated size. Contemporary American artists and manufacturers are also turning out pans and trays made of dull metals, tile, and wood. The wood may be either of a natural finish or artistically weathered. For outdoor bonsai, wood must necessarily be treated for durability and weather resistance.

For indoor plantings, containers can very often be improvised. I've used bronze ash trays, wooden salad bowls, shapely plastic dishes (shallow), and odds and ends of pottery—anything of the right motif in which it is possible to bore, drill, or chip drainage holes on the bottom. Many of our modern ceramics are in complete harmony with bonsai. But you must have those drainage holes, otherwise your watering problems are compounded.

The ideal bonsai containers are seldom ornate; really they shouldn't be. They should have the grace and elegance of "expensive simplicity," which doesn't mean they are expensive. They just look as though they were. Colors are subdued, not bright or showy. As a rule to follow, darker and somber shades are used for evergreens, lighter shades for flowering plants, the specific choice depending on the color of the bloom. In shape and form, bonsai containers are simple and graceful and are selected to set off the shape or lines of the plants. Erect trunks often take shallow, rectangular containers. Hanging or weeping lines call for round containers with more depth. Square or oval containers are used for extremely delicate, graceful subjects.

Container size, of course, depends on the plants themselves. As a general rule, the smallest and most shallow container that is culturally practical, and

in good proportion, is the best. In true bonsai containers, diameters range from two to twenty-five inches; depths, from one to ten inches. The accepted rule for good proportions allows the plant to occupy 80 per cent of the picture, the container 20 per cent. Should you have very small plants the ratio is slightly changed—60 per cent for plant, 40 per cent for the container. However, these figures should not be considered as hard and fast, but as a guide when purchasing containers. Your eye may tell you what is better suited for your particular plant or shrub.

PLANTING AND CARE

Rather specific cultural practices have evolved for dwarfing hardy garden plants, trees, and shrubs, and growing them in small containers. Tender house and greenhouse plants, dwarfed and grown as indoor bonsai, are also potted, pruned, trained, and watered as outlined in this chapter. Otherwise, they are grown like the window-garden plants in Chapter 1.

Root-pruning

This is often the first thing you do for a plant that is to be grown bonsai-style. It is a procedure that is repeated regularly if your plant is to have a long life. Dwarf trees and shrubs are root-pruned so they will fit their small containers, or to make room for fresh soil when they are repotted, or to keep the root system in balance with the growth above soil that is being restricted. Pruning also keeps the roots compact, near the surface of the soil, and vigorously young. Removing old, woody roots encourages the growth of fine new ones.

When seedlings, rooted cuttings, and small newly purchased or collected plants are made ready for their first bonsai containers, they are not immediately root-pruned in a severe sense. Any roots that are dead should be trimmed off and long taproots should be cut back at least one-third. Otherwise, it's a matter of trimming off the root ball with as little disturbance as possible, just enough to fit the container.

After they have been established, plants are root-pruned when they are repotted. When roots are crowded and completely cover the soil in a close mesh, it's time to repot and root-prune. For some plants this may come once in a year, for others once in five years.

Hold the base of the trunk in one hand—your left hand if you are a right-hander. Use a dull-pointed pencil (the Japanese use a chopstick) and loosen the soil around the outside. Pick away about one-third of it if the plant is established, somewhat more on younger plants. When you have finished, the soil ball should be, roughly speaking, one-fourth smaller than its container. Then, with sharp scissors, cut away all the loose root ends which you have removed from the soil. This is rather drastic surgery, although not like removing an arm and a leg of a gardener, because the plant or shrub has the happy faculty of growing new roots. However, it will need special care and protection until it is back on its feet again. Bonsai plants are root-pruned and returned to the same container year after year. None of this making each container one size larger each time the plant is moved into a new house. That's for house plants per se.

Soil

Potting soil for dwarfed trees and shrubs is particularly important. There is so little of it in small containers. In general, it should provide good drainage and aeration while also holding a certain amount of moisture. On the specific side, soil contents and textures should vary to meet the individual plant's needs.

If your plant happens to have been dug locally, take along a supply of local soil. If you happen to have purloined the plant from a neighbor's woods, purloin a little soil. It's no more of a sin to have stolen a sheep than half a sheep. In our neighborhood it is a standing joke about how many plants, flat paving stones, etc., we swipe from each other. If you have been honest enough to have bought your plant from a nursery, ask their advice on the soil. If they are not smart, although most of them are, get the reference book down off the shelf and find out whether the plant craves a mixture that is sandy, rich in humus, acid or alkaline, fine or coarse. You have a baby on your hands. Treat it right and it will award you with adulthood in bonsai. Neglect it and it will curl up and die. Mix your soil as you would a baby's formula. Remember, babies cry when the formula is faulty; plants can't. They silently pass away.

Potting and Repotting

I am frequently asked how often an established tree or shrub should be repotted. It depends, first, on how fast it grows. A willow may need this attention twice a year, a conifer once in five years. A general rule is to repot flowering and fruiting plants about once a year, most deciduous varieties every other year, evergreens every three to five years. But don't follow that schedule too literally. If a tree begins to wither or look weak, if its color turns sickly and it shows no sign of growth, if its roots are so packed they hump up the soil, root-pruning and repotting are often the "shot in the arm" that can save it.

The most favorable repotting seasons vary with types of plants and the climate. Again, it's a good idea to consult some authority or reference book. In general, and in most areas, it is safe to repot evergreens, deciduous foliage varieties, and summer-or fall-flowering types in early spring before new growth begins. As for spring-flowering plants, repot immediately after flowering; for fruiting types, in early autumn before cold weather sets in.

As a side light, it is possible to use watertight containers minus drainage holes. That is rather desirable when a plant is to be displayed on finely finished furniture in the house. The container should be a shade larger than usual so that, in potting, a corner area or space along the side can be left empty and the soil sloped down toward it. If there is any standing water it will show up in this section. Just up-end the container and drain off the excess water. In the process make sure that the plant is not also drained off or disturbed. But that you will know from plain common sense.

If a pot has drainage holes, I always make sure they are partially covered with crockery or aluminum window screen so the soil will not sift down and clog the holes. To insure even better drainage, I screen the first layer so that it is coarse and granular, put a little soil on top, and then set the plant in place. In oblong or oval containers, the trunk is usually one-third of the distance from the end and a little behind the center line. Of course, that depends on the shape of the containers. In round or square containers, the plant is usually centered.

When planting, fill in the soil gradually. Use a pencil, or if you have the true Oriental feeling, a chopstick, to tuck the soil in tightly around the roots. Whatever you use, be sure to eliminate any pockets of air. Roots don't like it. As to how tightly you pack it, that depends on the type of plant and the

texture of the soil. If your soil is built up around the edges and sloped toward the center, you will have a depression that will hold water until it can seep down to the roots below. Unless you have some particular plan or design, finish off the surface with a ground cover of moss, or perhaps a layer of gravel. Moss is not only an added attraction but functions as a mulch and delays the evaporation of moisture from the soil underneath.

After repotting, soak the soil thoroughly, and soak the container, which is probably dried out. Mist the foliage and remove any debris such as fallen leaves and petals. Then keep the plant sheltered for several weeks until it is re-established. Remember, it has had what a human being would regard as a major operation.

Some of the most appealing bonsai plants are grown with their roots around a rock of interesting and harmonious size, shape, color, and texture. Soft, porous tufa is especially good for this purpose because you can cut and shape it easily. You can hollow out cavities and crevices, just the place for roots to grow. Any rock should have a stable base. Please don't let it tip or wobble.

For this sort of planting, select a healthy plant in vigorous growth with roots long enough to reach down the side of the stone. Prune away any roots that are too short or dead, and remove the soil from the others. Set the base of the trunk in place on top of the rock and arrange the roots so that the tips reach down and can be buried in the soil in the container. Both roots and plant may need to be anchored in place for a while; that is, until they can stand on their own.

Such plants naturally need special watering and protection for the first few months, perhaps even for a year or so. You can help them along by rubbing soil into the roots after they are fastened in place, or covering them temporarily with a thin layer of moist moss. Should you do any repotting, be careful not to loosen the roots on the rock.

Recently, in the home of one of my Redding neighbors, I saw a bonsai arrangement that pretty much follows what I have been describing above, except that it goes it one better. This gardener, in the search for the "right" rock, had walked miles along the stone fences that divide fields, woods, and properties in our area. Eventually she found a beautifully weathered specimen with a large pocket in the top. This she filled with a rich,

moisture-holding soil mixture, encasing some of the roots of her bonsai in it. Then she set the rock in water in a shallow container to help keep it moist. Someday I am going to defy the snakes that are said to lurk in the cracks and crevices of our rock walls and see if I can't find something to approximate what my neighbor discovered. I hope it will have a few lichens on it. That would be a crowning touch of age and antiquity.

Shaping and Pruning

These are the techniques, says one author, that “make bonsai culture an art.” That statement alone is sufficient reason to invalidate just about all of the specific rules. But there are others—the wide variations among plants and types of plants, plus the personal element, the variances in personal tastes and degrees of artistry.

Instead of trying to cover bonsai pruning and wiring in complete detail (there are many excellent books on the subject), I'd rather speak from personal experience and set down the basic principles as I see them and have used them. By following these principles—not word for word, or too literally, but with your own creative ingenuity and imagination—I feel you can shape a true bonsai plant, or adaptations in the bonsai manner.

First, let's define the objective—a miniature tree or shrub with every part in perfect scale, the line and design of trunk and branches clearly outlined and not forced or distorted but naturally picturesque. If you start with a very small plant, a seedling, or a rooted cutting, the choice of shape and design is mostly up to you. But more mature plants almost always suggest their own form. A trunk may be slightly twisted, a branch slightly bent, the leaves or needles heavier in one area than another. You simply carry on the illusion in as natural a manner as possible.

In pruning, the first step is to remove all weak or dead wood, and any unwanted branches that cross unattractively or perhaps go off in the wrong direction. Then, you decide whether more branches should be pruned away—to reveal the basic form of the tree in general or the trunk in particular; to lighten the over-all effect; to help create the desired symmetrical or asymmetrical shape. If the plant has been root-pruned, the top should be pruned proportionally. Top growth and roots should always be kept in balance. Growing tips are pinched or cut back to encourage the

development of side shoots, or merely to keep the plant in dwarf proportions.

From here on, pruning depends on the plant and the planned design. Slow-growing types may need trimming only once a year; others need constant attention. Any undesirable or excess growth is best removed while the plant is still young so the operation won't leave an unsightly scar. Needles that are too long can be cut down to proper size from time to time. Leaves that are too large and heavy can be severely thinned. Because fruit that is too plentiful can weaken a dwarfed tree severely, some of it should be removed.

At repotting time you may find a few roots immediately under the soil surface that have enlarged to such a good size they can be exposed above the soil. If possible, spread them out slightly so they form a swelling base for the trunk. This is fun, but takes a bit of doing. You don't want to end up with something that resembles an unanchored telegraph pole with leaves at the top.

Stems, branches, and trunks are trained by wiring them and then bending them along the lines you desire. Wiring is best done in the spring. New growth is just starting and the older woody parts are more supple and pliant. If the soil is left dry a few days before wiring, the wood will be even easier to work with. You can use ordinary galvanized wire. If you happen to have only copper wire, try annealing it. Hold it in a fire and then plunge it into cold water. The weight and strength of the wire will depend on the size of the trunk or branch with which you are working. Be careful not to get a wire that is so stiff you can't bend it easily.

As to the actual process, first fasten one end of the wire. Perhaps you wrap it around the trunk, or better still, stick it deeply into the soil. Next coil it rather loosely around the branch that is being trained and fasten it again at the end. Now, using both hands, bend the wired branch in the direction you want it to go. Be firm, but be gentle and cautious. Ease up if the branch shows signs of breaking or being injured. Don't hurry the job. If necessary, only bend it a little the first time. Bend it a little more the next week, and the weeks after that. If you are tempted to rush, stop and think of the bonsai creations you have seen on display at flowers shows. Usually you will have seen a card that says the creation is fifty or more years old.

Be especially careful with old hardened growth or plants with tender bark that is easily bruised or broken. Most important, once you start to bend a branch, don't change your mind and try to bend it back the original way. It will almost surely die. Plan before you act.

Depending on the type and age of the plant, wires may be left in place for six months, a year, or even more. However, watch carefully for signs that the plant is being choked or disfigured due to heavy growth.

There are also some easy ways to train parts of plants without wiring them. Branches can be pulled down to a horizontal line by looping soft cord around them and hitching it to the container. They can be made to hang down, or weep, by hanging weights at the tips. If you want to straighten a trunk, tie it with something strong, like raffia, to a straight bamboo stake. To narrow a wide fork between two branches, pull them together with raffia. To widen a fork, prop it apart with a light wooden wedge.

Location

Whether you have a collection of bonsai growing on tables or benches, or perhaps just a few plants, in summer give them outdoor growing quarters where there is a free circulation of air. Full sunlight is good except during the hottest weeks or months. Be careful to protect them against hot, drying winds and burning sun, which they cannot stand. Being in small containers, excessive heat or dryness is dangerous. We had some old bamboo shades that once enclosed our porch which were good protection. We also had some lath screens which came in handy. Lacking either, hang up old sheets or sections of burlap. You can help by keeping it moist with the garden hose.

Watering

Many factors determine how often your plants should be watered—age, type, size, how recently they were repotted, size of root system, and the usual climatic conditions such as temperature and humidity. About the only concrete thing I can say is that your plants will resent neglect, and will show it. Newly potted plants should be kept moist constantly until they begin to make new growth or show other signs that they have recovered from pruning and transplanting. For plants that are established, the ideal is

limited moisture in the soil. Don't keep it so wet that rank growth is encouraged. And please beware of rot. That is one of the most evil of evils when you are too generous with the watering can. At the other extreme, the soil shouldn't be so dry that the plant wilts beyond recovery. You, as a grower, will have to determine this for yourself. I do it by feeling the soil in the container. If it feels moist to the fingers, no water is needed. Let it feel dry and it probably needs a drink. When you do water, be thorough. Make sure the soil is so completely moistened that the excess water runs out the drainage hole in the bottom of the container.

Fertilizing

There is a definite, delicate balance between too much and too little fertilizer for a bonsai plant. Too much food and it grows too large and is loaded with large foliage, flowers, and fruit. If you feed it too sparingly, it will suffer from malnutrition. I most certainly wish I could give you an exact rule to use. I fear no one can. Requirements differ for different plants. All I can offer is a few basic principles. The rest is up to you.

For the first few months after you have repotted a plant in fresh soil, withhold all fertilizer. Don't fertilize a plant that is weak or sick or approaching dormancy, and don't fertilize when the soil in the pot is dry. Be alert to fading leaf color, reluctant growth, and all similar signs that a plant is suffering for want of nutrients. These signs may be most apparent during the spring season of most active growth.

Organic fertilizers such as bone meal, liquid manure, or fish emulsion are usually recommended, and should be used sparingly in weak solutions. The purpose of this feeding is to keep the plant healthy but still small, and not to encourage larger growth.

Insects and Disease

My bonsai plants are much too precious to take chances with any sort of infection or insect infestation. As a preventative, I use an all-purpose aerosol bomb regularly, according to directions on the label. Constantly I keep a sharp eye for any signs of trouble. Thank goodness, up to now I have had no serious threats, but should they come along, there is a remedy I have used on other plants. I would dip them, container and all, in an appropriate

solution, thus making sure it would penetrate every crevice and cover every surface. For outdoor bonsai and the problems most likely to beset plants while in their summer quarters, I have elsewhere discussed insects and diseases of trees and shrubs. Those general principles also apply to bonsai.

Winter Care

In areas where freezing temperatures are the rule or, as here in Connecticut, where temperatures are much lower, dwarfed potted trees and shrubs should spend the winter where the soil in the small containers will not freeze. We have a tight cold frame where we plunge the pots into the soil, then cover everything with straw and salt hay. Those who happen to have an unheated porch, one that is glassed in, but where the temperature does not go below freezing, have a good winter quarters. But, be careful, don't let the soil in the containers dry out completely. Keep a watchful eye.

After that warning, may I offer another. Please don't coddle plants by keeping them warm in the house or greenhouse. A cool period of complete rest is often very beneficial, or even a touch of frost.

MINIATURE GARDENS IN THE LANDSCAPE

A picturesque surprise on a gently sloping, sunny bank could be a wind-blown mugho pine beside an outcropping rock that's lightly draped by a sprawling cotoneaster, with colorful sempervivums clustered at the base. Or in the light shade at the base of a clump of white-barked birches, there could be a group of dwarf rhododendrons displayed against dark, humus-rich soil or a pine-needle mulch. On a patio or terrace, it could be a contemporary grouping of low, fluffy juniper, perennial lavender, and water-polished rocks.

These are miniature gardens—not complete landscaping plans for small properties, but appealing plantings that often pop up in unexpected places and bring beauty into otherwise unusable or undecorated areas. They're gardens because, by definition, they are groups of plants that achieve an effect a single plant could not create alone. They're miniature gardens because they occupy small space, and because most of their plants are of miniature proportions. Properly designed and executed, they're equally effective in large grounds or small, with contemporary suburban homes, in informal settings, and even on estates with traditionally formal landscapes.

There's a special enchantment in these miniature gardens. The eye is attracted by their modesty and restraint, and by the utterly natural way they seem to suit the scene. Each plant is seen intimately, in close-up; its character is revealed in each small detail. And in these gardens the creator can express his individuality so easily; seldom does a design even resemble the one next door.



Miniature white poppies featured in a tiny garden beside an outcropping of rock

Part of the appeal, of course, lies in the charm of miniature plants—tiny annuals and perennials, small or slow-growing evergreen and deciduous trees and shrubs, available in amazingly wide variety if you take the trouble to find them. There are also frequent opportunities to use inert objects with special effect. Small gardens can be designed as settings for works of art or products of the hand-craftsman—ceramic bowls and urns, authentic wrought-iron grilles, wood carvings, statuary, sundials, pools, even fountains and waterfalls. Or a garden may be designed with a background of well-placed rock, a tree stump, or a piece of driftwood. A bench, arch, gate, antique hitching post, or well-house may inspire a miniature planting.

Inanimate ground covers such as gravel are often a definite part of the design.

In some ways miniature gardens are easier to design than, for example, standard items such as flower beds and foundation plantings. It's easier to achieve originality. Mistakes are usually small and easily corrected. On the other hand, really good design is more critical than in large plantings where space can swallow errors and provide lucky effects. In miniature, even a minor defect shows up immediately, and may be a major calamity.

DESIGNING MINIATURE GARDENS

The first and basic requisite is an idea the garden is to express, a theme for the picture it is to create. The objective may be to embellish some neglected nook, disguise an unattractive corner, feature an unusual plant or art object, soften the lines of a small pool and blend it with its surroundings. Once the goal is set, it is pursued without deviation. For example, a featured plant is kept dominant—not necessarily in size, but always in visual importance—and everything else is subordinate. A pool planting does not become so elaborate that the beauty of mirrored reflections or rippling water is lost.

In both conception and execution the design for a miniature garden should be in harmony with its surroundings. Nearby buildings may call for certain harmonious lines and proportions. Land contours, and constructions such as walls and steps, may dictate size and shape. The architecture of a house and its landscape has a style that should not be violated.

Our Connecticut landscape demands informal or naturalistic design in no uncertain terms. Straight lines and formal geometric shapes would be not only out of place, but practically impossible to achieve. The land's slopes and rises call for beds with flowering curves. Points of interest such as massive lichen-trimmed boulders, gnarled old trees, or a winding stream are sublimely situated by nature's unerring instinct for what looks right. We've merely cleaned them up and made the most of them.

For not-too-modern houses built on regularly shaped, level lots, some sort of formal design is easier to achieve and much more suitable. In the traditional style, elements of equal size and weight balance each other. There is strict adherence to a predetermined pattern. Identical beds may

make a formal dooryard garden, for example; matching groups of plants may ornament opposite sides of a gateway.

For houses of contemporary architectural style there are gardens of contemporary design, often featuring paved areas and patterns with distinct angles and curves. Plantings are based on the tone, texture, and form of the plants themselves. The object of interest is off-center, balanced by a larger area of subdued importance.

Miniature gardens should be not only in harmony with the style of the surroundings, but also in proper proportion. A birdbath with miniature roses planted underneath can look lost in the center of a spacious lawn, but could be in correct scale for a niche or arch in a brick wall.

Simplicity is extremely important. The smaller the space, the faster it will take on a spotty, cluttered look when crammed with too many different plants. The smaller the space and the plants, the more care and thought should be given to combining various types, forms, textures, and colors for best effect.

In addition to aesthetic principles, there are practical aspects to designing miniature gardens. Any site should be checked for cultural and environmental conditions that favor healthy plant growth. Is a wall so high it throws too much shade for sun-loving plants? Is an area too exposed in winter for questionably hardy plants? Is the spot so low that water collects and stands in the soil, making it suitable only for bog plants?

Think of the work of maintenance, too. Use ground covers instead of lawn in areas so small you can't move a mower around. Avoid fast-growing plants that need constant trimming and pruning. Don't use plants that require a lot of protective spraying or dusting unless you have time to keep up with the job. A healthy dwarf barberry is more attractive than a neglected, sickly rose.

MINIATURE BEDS, BORDERS, AND ISLANDS

There are a number of ways to use small-scale versions of the familiar mixed-flower border. Annuals and perennials can be selected for continuity or succession of bloom, and arranged so that colors harmonize, textures contrast, and carefully placed taller spikes or spires provide accent and relieve monotony. In the background there may be dwarf evergreens or

flowering shrubs, a low fence or wall, the side of a tool house or other small building.

Use miniature flower borders to add interest to tiresome areas such as long, narrow strips between the house and boundary line of the lot, or along a service walk. Use them at the edge or in corners of small lawn areas—never scattered through it—or at the base of low foundation plantings. Use them as visual space dividers between driveway and front yard or entrance to the house.

Miniature beds are also effective at one side of a breezeway, at the edge of a terrace or patio, or beside the back door. If they are raised slightly, perhaps the height of just one brick, the small plants are nearer to eye-level and their full beauty is more clearly revealed. I've also seen a tiny cutting garden, by the back fence, that was as colorful and pretty as any border.

Whenever there are walks, there are spots for miniature beds—in the diamond-shaped points where two walks join; as little islands along the curves; in the angles where corners turn. Or use small formal beds for dooryard plantings.

The intimate dooryard garden, by the way, is enjoying a revival. Instead of advice to keep flowering plants away from the front of the house, we're encouraged to plant little gardens that can be seen from the picture window and also will make the house more attractive to passers-by. These don't replace foundation plantings and front-yard landscaping, but supplement them in a small way. And the small gardens help relieve the monotonous sameness of many modern house fronts.

All kinds of corners are obvious sites for ornamental treatment with tiny gardens—architectural corners between front entrance and house wall, between one wing of the house and another; the back corner of a lot where, perhaps, a small tree is a focal point or background or where, if you're fortunate, a small stream winds its way across your property.

On almost any grounds there are natural nooks that seem to be made for miniature gardens up and down the sides of informal steps, at the top or base of low walls, in a patch of soil at the foot of a high-branched tree.

Landscape or architectural focal points can often be enhanced by little gardens—gates, bay windows, sundials, the mailbox, a birdbath, or a well.

Little pocket-handkerchief gardens are built right into patios and terraces. Little creepers planted in crevices between paving stones make a miniature garden of their own.

MINIATURE ROCK AND WALL GARDENS

Anyone who can live and garden in a place like our part of Connecticut without going overboard for rocks and rock-loving plants has more self-control than I. My bewitchment began one Sunday afternoon when, the weekend's weeding done, I strolled up toward the small wild-flower plantings in "the point." Just beyond the garage, before the trees began to cut off the sun, I stopped to glare at an ugly, erupting mound of soil and stones that had frustrated our every attempt to make this area more attractive.

I was pondering the monstrosity, when I noticed an intriguing detail. A partly submerged rock poking its head out of one side of the mound had the same color and patina as a snub-nosed piece sticking out the other side; and the lines of stratification were on the same slant. Could these be opposite ends of one continuous rock formation?

Five wheelbarrow-loads of rocky debris later, we had uncovered a beautiful boulder, perfectly placed so that its lines led gracefully into twin trunks of a wind-blown birch. It was a boulder with ancient age and character, artistically chiseled by the elements. And it said "rock garden" as plainly as any landscape plan.

This five-foot-long discovery may not be miniature; and the naturalistic kind of planting it inspired is neither practical nor suitable for many home grounds. But it was my introduction to rock plants, and to how effectively rocks and plants combine; and it led me to create and watch for rock gardens that *are* miniature, practical, and suitable for all kinds of grounds. Sometimes I think these are the most enchanting little gardens of all.

If they are to look like anything better than a pile of stones, miniature rock and wall gardens are not composed of miniature rocks. They simply contain fewer, perhaps somewhat smaller, rocks than average gardens; and they occupy far less space. These small plant-and-rock compositions brighten nooks and corners where other plantings would seem out of place.

By their nature, most rockery plants are miniature in height, foliage, and flower. Alone or combined with miniature bulbs, the smaller perennials and annuals, and the tiniest shrubs and trees, they invite close-up enjoyment of their daintiness in these small plantings. Rocks and rock-loving plants have a perfectly natural way of blending and contrasting, each bringing out the best qualities in the other.

Although they're of different design and construction, rock gardens and dry walls are usually planted with the same or similar plants, and given similar care. Paths, pavements, and garden steps are close kin. But pools and water plants, even though they are often included in rock gardens, are a separate subject, and are covered in Chapter 12.

ROCK-AND WALL-GARDEN DESIGN

"Cemetery," "penitentiary rock pile," "collection of geological specimens"—these and other unflattering phrases have been used to describe what a rock garden should not be. This eliminates the artificial rock-studded circle in the center of a lawn, rocky slopes with nothing to back them up, isolated paths and steps that have nowhere to go, stone-stuffed piles of loam with no conceivable excuse for being where they are.

What a rock garden should be, is an integral part of the landscape, completely at home and in harmony with its surroundings, and having a sound reason for its existence. Except for a few functional affairs such as walls that retain terraces and steps that climb banks, their purpose is mainly ornamental. Neither rocks nor plants are featured at the expense of the other. Both work together to create the picture—the plants to add brightness, vitality, and life; the rocks to help provide favorable growing conditions and a naturally suitable background for their display.

Designs for these gardens can be, as for others in the landscape, formal or informal. The fixed geometry of walks, walls, and raised beds in formal pattern are seen less frequently. But this is a type of rock garden that might be placed advantageously in even the flattest, barest suburban lot where naturalistic plantings and rocks look out of place. If you long for a rock garden, try something like this in either classic or contemporary style.

Informal rock gardens must, above all else, look convincingly natural, as if the glaciers might have placed them. This is equally important to a cluster

of bulbs at the base of a single, half-sunken boulder; the arrangement of rocks and plants on a small slope; or the series of rocks that line a winding path. Builders of ambitious rock gardens are often advised to study the nearby countryside and use it as a guide.

Designs for Naturally Uneven, Rocky Land

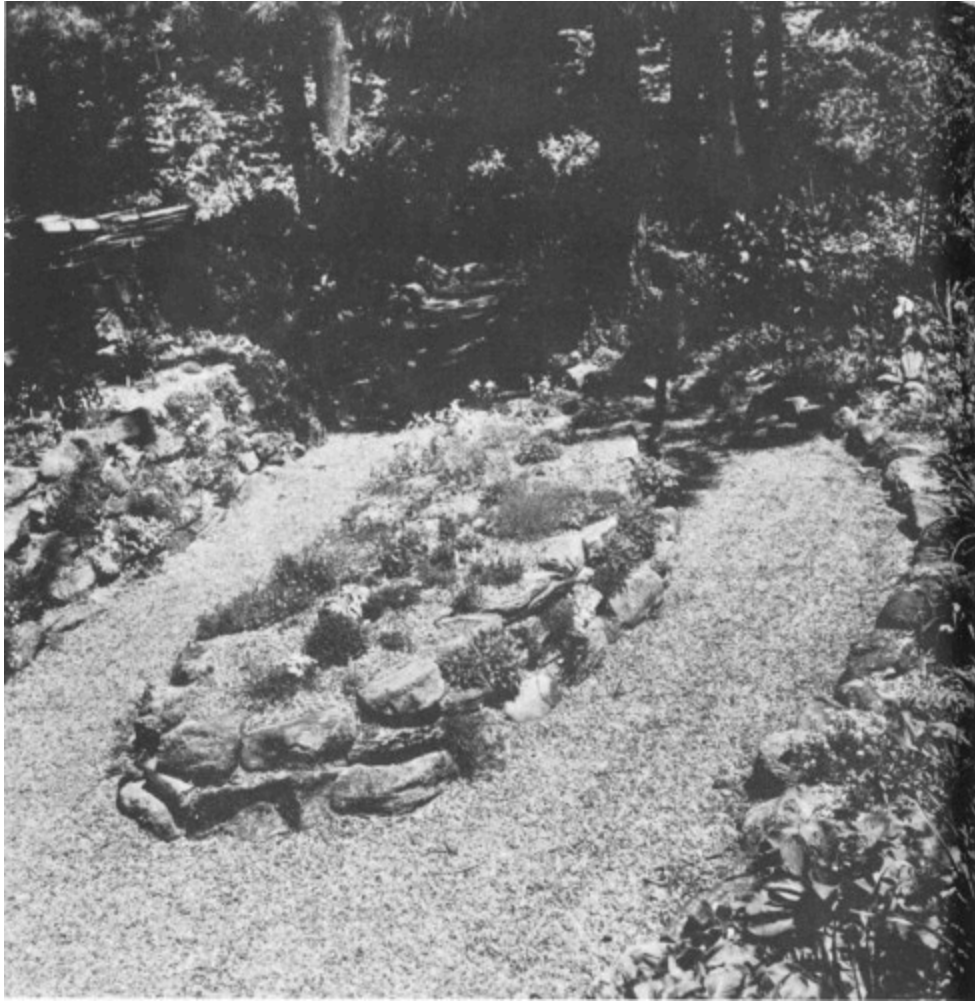
This type of terrain makes a natural setting for rock and wall gardens of all sizes, shapes, and contours. Generally, the design is informal and naturalistic. I don't know whether it's really easier to work out such designs in miniature, or whether I merely think it's easier because that's the kind of land I have to work with. So far, we've never disagreed with Nature about the placement of a rock outcropping, for example. The few we've uncovered and cleaned up have almost told us what and how to plant.

A small existing slope or bank is a perfect site for a composition of rocks and rockery plants, or, if you prefer, a retaining wall between the two levels with plants grouped at the bottom or top, or planted horizontally in the cracks. Or consider a series of ledges that create small, irregular terraces.

A slope or grade may be the perfect place for informal stone steps, with, of course, a path leading up to them and away. Colonies of small plants look well on either side, with, perhaps, very miniature evergreens at both sides at the top. Cracks in the steps and a walk can be planted with small carpeters.

If you possess a small stream, gulch, or ravine, by all means consider a little rock garden along either side. If you're fond of a gnarled old stump, see what a few rocks and plants might do at its foot. Where your driveway curves, try a small rock garden. Before you fill in and flatten out a small depression, or level off a small knoll, find out how it would look as a small "island" rock garden.

Corners like those formed by an old-fashioned "stoop" of a country cottage sometimes make effective frames for miniature rock gardens. But in general, informal designs usually fit best out in the grounds, away from angular architectural lines.



Raised flower bed copied from an old English garden

Designs for Level, Rockless Grounds

Here there are fewer existing, desirable locations for rock and wall gardens. But for the sake of enjoying the dainty plants, and of marking your garden with originality, you may want to create a site. It is possible to dig out a “natural” depression or build up a slope, if it’s skillfully designed—and if you can get rocks that don’t look foreign and far away from home. Or dig down a foot or two to make a formal or semiformal sunken garden, the beds raised and held in place by a dry wall with crevices to plant in, and flagstone walks.

Corners sometimes offer possibilities—a far corner, where there is a background of trees, fence, or wall; or a front corner where the lawn slopes

slightly down to the sidewalk. Or there may be a place near the patio for a tiny pool in a rocky setting.

A birdbath, sundial, or shrine that's a focal point in a small garden can sometimes be embellished with a small rock garden at the base. So can the edge or corner of a raised patio. Terrace pavements with cushions of thyme rooted in the crevices are enriched with color, fragrance, and texture. The lines of stark steps are softened and the appearance mellowed by crowded clumps of fat sempervivums. Wherever you can find reason for a rock wall, or even a row of rocks laid horizontally and only a few inches high, you can plant it with rockery plants at top, bottom, or in between. In the artificial settings of many suburban lots, semiformal walls separating two levels of lawn or supporting the edge of the family living area are very suitable substitutes for rock gardens.

MINIATURE ROCK AND WALL PLANTS

One of the main virtues of rock and wall gardens is their individuality—they don't look like other gardens, or even like each other. But commonplace plants such as bedding petunias can cancel this distinctiveness in a minute. If you use annuals at all, get less ordinary varieties and use them sparingly, for temporary color in a bare spot or over the dying foliage of spring-flowering bulbs.

The plants that look best with rocks are those that grow naturally among them. Hundreds of rock-loving plants are available, and more hundreds of alpine plants from rugged mountain heights. Only the easier alpine plants are included here, the most adaptable to more luxuriant climates and soil, the least likely to pine for their rigorous, high-altitude homes. Tricky types from above the timber line are left to the dedicated collector.



Saxifraga seedlings—a natural rock-loving plant

Nurseries and catalogues of rock-garden specialists are so full of distinctive and delightful miniature plants that my first reaction was blissful delirium. And after I acquired as many enticing items as I could, my second thought was: These gems are too little known and grown. So I began to sort out some that might flourish in our small perennial borders. With favorable conditions of drainage and air circulation, a number of rockery plants have already made themselves at home in other, rockless gardens.

Most miniature bulbs are effective in rock gardens, and some miniature perennials—not ubiquitous types, and not those that spread voraciously by runners. Miniature shrubs and trees are indispensable in rock gardens of all sizes—to give variety, contrast, and substance; to act as accents; to create boundaries or backgrounds. Use them with restraint. Place spreading, wind-

swept types at the top, bushy shapes down lower, upright exclamation points at the very bottom. Make sure they are in scale with the rocks, the plants, and the garden as a whole.

Select all kinds of rock and wall plants for their effectiveness of form, texture, and color in the complete design. And this repetition is important and unavoidable. Select varieties that naturally like, or will adapt to, the cultural conditions like sun and soil type you have or can provide.

MINIATURE ROCK GARDENS

For setting a few rocks into the side of a handkerchief-sized section of slope, or building many more into a garden in a good-sized corner, there are a few guiding principles that will help make sure you are happy with the result. Most important, have your design as complete as possible before you start to build. A sketch drawn to scale is a valuable visualizing aid. Then, make your construction solid and substantial, and favorable to plant growth and health.

If you have a choice of exposure, face a sloping rock garden toward the north or northeast, so it will get enough summer sun but not so much that it dries and cooks. Shrubs and trees in the background should not create dense shade in the garden, nor drip moisture on the plants and steal nourishment from their roots. Drainage is extremely important in selecting a site. If the subsoil is packed and clay-like, you can either replace it, make it more porous, or select another spot.

What kind of rocks to use and where to get them depends on the part of the country you live in. Anything that's indigenous to an area is first choice, always. Limestone, sandstone, fieldstone, porous tufa—native rocks “belong” as no artificial or imported rock ever can. If your land is not blessed (or cursed) by plenty of rocks, you can sometimes collect them near rivers, mountains, or abandoned quarries, or where a new road is being blasted through. Whether you have them, collect them, or buy them, all the rocks in one garden (or on the same property) should be of the same general type. Varicolored geological specimens are seldom attractive.

Consider each rock's characteristic color, texture, size, and shape before you decide where to place it, or whether to place it at all. Even in the tiniest rock gardens, tiny rocks look ridiculously salt-and-peppery. Start with rocks

of substantial size, in keeping with the size and contour of the garden; use smaller ones where they fill in effectively.

Naturally worn and weathered rocks look more natural than freshly chiseled surfaces; irregular, angular forms are better than uniformity. Rocks with stratified layers suggest ledges and cliffs. Glacial boulders or “hardheads” are not good in formal patterns. Use them naturalistically, in varying sizes, arranging them with studied carelessness as if they’d been scattered haphazardly and then left.

Avoid overcrowding with so many rocks that the plants are overshadowed by the debris. Place rocks as they might lie naturally, partly buried in the soil, the tops smoothed over by the elements, and so solid that they don’t wobble when you step on them.

Construction

Perfect drainage, so that water cannot stand around the roots and crowns, has been called “the most important need for all alpine plants.” The selected location (raised or sunken) and composition of subsoil and topsoil can either contribute to good drainage or deter it. In small gardens it’s not too formidable a job to install a system of drainage tiles, if necessary, or to add gravelly sand to subsoil to make it more porous. Easy insurance in any rock garden is to start with several inches of rather coarse gravel or rubble as a bottom drainage layer.

Any topsoil that is removed during construction should be saved carefully. If you can pile it in the center of a tarpaulin or piece of strong burlap, it can’t be scattered and lost; and you can also mix it with any additives with pleasurable ease. Simply pile on the peat, leaf mold, sand, or other ingredients; then, holding opposite corners (you may need a helper to hold one corner of a large piece), pull it back over the soil pile and fold it over the other end. Repeat this, from opposite ends, until the materials are thoroughly mixed.

The first rocks are usually set at the lowest part of the garden, partly sunk in soil or otherwise made extra-sturdy and secure. Build up from that point, arranging rocks with their broadest surfaces at the bottom, all lines of stratification horizontal or on the same slant. Each rock should slant slightly

inward and downward, to catch moisture and encourage it to run into the soil in the garden.

Provide plenty of soil for spreading roots, and a minimum depth of eighteen inches or more. Plants in shallow pockets may be stunted and will certainly dry out too fast during droughts. Whether or not you plant as you go depends partly on your agility. Don't crush the first plants under your feet or knees as you set rocks and plants above them. It's often faster and easier to spread out roots and make soil firm around them before the next rock is set in place.

Planting

Set plants as deep as they were planted before; no deeper (danger of crown rot) and not too shallow (they may dry out). Tuck them firmly in cool, moist, rock-shaded soil. Keep the soil constantly moist until firm new growth shows that the roots are established and growing.

Rock-garden plants are seldom suitable for planting in neat rows or geometric patterns. Tiny, slow-growing types are arranged in groups of three or more; one or two spreaders per planting spot will do. Don't crowd these plants. Give them room to develop and for air to circulate around them, and to display their charms at their best. Provide private quarters for the smallest gems, away from hearty growers that might overpower them. Keep color combinations and contrasting textures in mind. Set trailers where they can droop over stones. Place those that like moisture low in the garden; their roots keep cool in the shade of rocks.

As a finishing touch, after the rock garden is built, planted, and thoroughly watered, cover all soil surfaces with a mulch of fine gravel or stone chips. Work it up to each crown, under prostrate stems. Aesthetically, this gives a neat, well-groomed look and ties plants, rocks, and setting into one coherent picture. Functionally, the chips absorb heat and keep soil cool and moist, lift small, low leaves out of mud and water, discourage slugs, protect crowns against rot, and firmly discourage the intrusion of weeds.

MINIATURE WALL GARDENS

Any small section of low wall—brick, cement, stone, or concrete—can be the background for a miniature garden. But it is in a dry wall, built of

stones without cement, that many small rockery plants best display their jewel-like perfection. Planted horizontally, with their roots in pockets between the rocks, they make a vertical rock garden of glowing colors and fascinating foliage textures.

For small, low walls, and those in somewhat formal settings, thin-layered rock such as slate or shale is usually preferred. Irregularly shaped fieldstone or other rocks of native origin are used in more natural, informal designs. Any lines of stratification are, of course, laid horizontal and parallel to each other. Another good rule to go by is: The smaller the wall, the smaller the rocks.

Construction

Low walls (two feet high or less) usually do not need a foundation layer sunk below the frost line. They can be started at, or just under, ground level. But they must be set solidly on a firm, wide base—two feet wide, for example, for a wall two and a half feet high. Set the largest stones and complete the lower layer first. Tilt each rock slightly backward and downward so soil won't wash out of pockets, water will run back to plant roots, and the heaving pressure of frost won't thrust the rocks out of place in winter. A guide line of string or wire moved up with each new layer will help you keep the construction level.

The wall should slant slightly in toward the bank behind it, and usually grows more narrow as it goes up. Each layer is made absolutely solid before the next is applied. Set each stone so that it is completely steady, and rests on parts of two stones beneath it. This distributes weight evenly, avoids crushing plant roots, and prevents unattractive and impractical long, vertical cracks created when joints coincide. Use small, flattish rocks to fill chinks under large stones and make them steady.

Behind the wall, as you build, firm in a goodly layer (one foot thick is usually recommended) of topsoil that has been enriched and aerated to promote drainage. Roots will reach back here for water and food. As you ram soil into pockets, make sure it reaches all the way back to this layer. Unless all the soil is made very firm, it may settle and the stones may slide out or into the bank.

At the top you can leave some pockets for plants, or plant them just behind the wall. Cover cracks at the top with flat stones, so water won't wash out soil.



A rock, a rose, and a miniature rose.

Planting

If you can remember not to damage plants at the bottom as you're working above them, set the plants in each layer as you build. It's easier then to make sure the roots are firmly embedded well back into the soil. Plant sparsely; leave plenty of room for growth and spread, and plenty of uncovered wall surface to contribute pattern and texture to the picture. Space the plants at irregular intervals, never in straight lines either horizontal or vertical. Water thoroughly, making sure the layer of soil behind the wall is completely moist; and keep it moist at least throughout the first growing season.

PATHS AND PAVING

Many little rock plants will make soft cushions and carpets, with their roots in crevices between pieces of paving, and also in stone steps. They're very low-growing and very hardy, and usually spring back resiliently when stepped on. It's important that acid-loving varieties not be planted near cement, which sometimes neutralizes or alkalizes the soil. And, of course, the situation should provide the amount of sunshine and moisture the plants need.

The simplest kind of informal path or paving for an informal patio consists of flagstones or slates set in the ground, the top level flush with the surface of the soil. Plants are spaced irregularly in cracks between the stones. I've seen one path of old millstones laid this way, the plants rooted in the center hole.

Bricks create a more formal pattern, either laid on a bed of sand or set into a foundation of concrete over gravel. The popular "crazy pavings" can be odd pieces of either brick or stone arranged in any conceivable pattern and laid in mortar. In any such solid constructions, planting pockets should be scooped out before the concrete sets hard; or a chunk of wood can be inserted to hold the space open.

Give the carpeting plants as much good soil as you can, and keep them moist until they are growing well. Occasional trimming will stop them from growing straggly, and keep them neat, compact, and in a pretty pattern.

PLANTING AND CARE

Planting times vary according to climate and season of dormancy. In warm areas it's almost always late fall. Where winters are rugged, it can be during brief semidormancy in early September, but it's safest in early spring when severe frosts are no longer likely and favorable weather blesses plants at the height of their annual growing power. In temperate sections spring-flowering varieties can be planted from mid-August through October, others in spring.



Lemon thyme and ivy showing results of being trimmed and kept neat

Mr. and Mrs. Alex O'Hare

Strong, healthy, pot-grown plants can be planted throughout the season in almost all areas. They transplant easily, with little disturbance and shock. In fact, whenever we acquire plants during the fall, we keep them in their pots or pot them up, and protect them in the cold frame during the winter. One of my favorite and most knowledgeable rock gardeners tells me that he always pots tender or difficult plants, and keeps them potted until their roots are fully developed, before he sets them out in the garden, no matter what the season.

Soil

Providing the proper soil is vitally important to rock-garden plants, and particularly to alpine. They're naturally dwarfed by the rough conditions in their native mountain areas. But when they move down to our comparatively lush climates and longer growing seasons, they respond to a too-rich diet by growing too tall and rangy, with too few flowers, and they lose their charm. They do need some nourishment, of course. And they are extremely sensitive to soil textures and drainage conditions. Heavy, soggy soil causes them to rot quickly. Soil that lacks organic matter dries out too fast.

Because of climatic and other variations, I hesitate to set down any "ideal soil mix" for these plants. Instead, I suggest a basic formula that can and should be varied according to the aridity or rainfall in your area, the size and location of your rock or wall garden, the composition of your garden soil, the availability of some ingredients, and the personal preferences of the plant varieties you want to grow. Here it is:

Three parts fairly friable, fertile loam.

Two parts humus—peat, leaf mold, compost—for lightness and fertility.

Two parts stone chips or crushed rock—perhaps the most important ingredient, to keep soil open and cool.

One part sharp builders' sand, or the gravelly sand used in mixing concrete, to lighten heavy soil and promote drainage.

If your garden loam is sandy or gravelly, of course, you would increase the proportion of humus. If it's rich in organic matter, you would decrease the quantity of humus. Very sticky, heavy clay soils are best replaced.

Rock and alpine plants are also particularly sensitive to the acidity or alkalinity of soil. This is measured by the pH scale, ranging from very acid pH 4.0, through 7.0 at neutral, to very alkaline 9.0. Rock-garden specialists usually indicate the preferred pH for the plants they sell.

By using a soil-test kit or, better still, sending a sample to your state agricultural service to be tested, you can easily find out whether the soil in your garden is acid or alkaline, and to what degree. Then, to make acid soil more alkaline for plants that need it, mix in small quantities of agricultural

lime—the amount depends on the degree of acidity and on the plant itself. Don't overdo it. Too much lime can do as much harm as good. Alkaline soils are not quite so readily converted to acid, but you can use oak-leaf compost (not completely rotted), acid peat, or pine "duff" (rotted pine needles). Or you can be satisfied with the tremendous number of rockery plants that really prefer alkaline soil. These preferences vary, not only between different plant families, but also between individual members of the same family.

For our small gardens, I've found it not too tiresome to provide soils of several types. For specially precious plants, I often mix special, small batches of soil. Then I clean out the planting pocket, make sure that it is tightly closed in with rocks so water won't wash this soil out or other soil in, and refill it with the preferred mixture.

Watering

Except during rare droughts in spring, when the plants are in active growth, we seldom have to water our rock and wall garden. Our summer rains are usually sufficient for their needs. But whenever or wherever there are long weeks of extreme heat and dry winds in summer, water may be needed. Use a fine spray gently, so that the small plants won't be washed loose. And let the water soak in long enough so the entire area is moist in depth.

Fertilizing

Some rock plants can take and use fertilizer, in spring; but most alpenes are better off without it. For them, we mix a little leaf mold with the gravel mulch when we renew it. For the more ordinary types, we provide a very light feeding of whatever balanced fertilizer we have on hand.

Grooming

In rock gardens (and particularly in very small rock gardens) even slight untidiness or imperfection shows up all too clearly. But I consider it a pleasure to weed and care for the little plants a few minutes every weekend. Faded flowers are snipped off. Weeds are removed with determination, but with gentle protection for any small, weak plants. Spreading shoots that

grow long and out of line are cut off, or sometimes uprooted, and removed. Louise Beebe Wilder says it's an art to keep a rock garden in balance, with the proper proportion of plants against stones. And she is an artist with hers.

In spring, even before the last of the winter mulch is removed, we look for cracks in the soil and for plants that have been heaved out by frost, and press them firmly back into place so their roots can take hold again. We also fill any washed-out pockets with fresh soil mix, and top-dress any low areas. Any rotted leaves and stems and other debris are removed; and the entire garden gets a thin fresh layer of stone-chip mulch.

In fall there's the general cleanup operation—weeding, pressing in loose roots and packing them with soil, trimming back rambunctious growers, replenishing the mulch of stone chips if needed.

Winter Protection

If you can count on a continuous cover of snow over your rock garden all winter, there's little you need to do for these hardy little plants. But we can't; we have the danger of alternate freezing and thawing that can do so much damage. So we provide special protection.

In winter, after the ground is frozen at least an inch deep, we cover the complete garden with salt-marsh hay. (Some types of Fiberglas, and wood excelsior from packing cases, are also suitable, reusable materials). This light, airy blanket keeps plants cold and dark, yet gives them air to breathe. If it threatens to blow away, we hold it in place with a few light evergreen branches.

PROPAGATING ROCK-GARDEN PLANTS

These are, of course, perennials and with slight variations can be propagated like other perennials. (See pages [255–59](#).)

Seeds

Some need a pre-germination cold and some don't; some germinate best at cool temperatures, some at warm; some need light, others need dark. Check seed packet, catalogue, or reliable reference book for specific information for each type of seed.

We start seeds of delicate varieties in the greenhouse, where we can hover over them, in late winter; others may be sown in the cold frame in March or April, so the plants will be as mature as possible before their first winter.

Stem Cuttings

For the smaller plants, these can be as short as three inches or less. Try to take snappy new growth, neither weak and watery nor hard and tough. Sharp sand is a suitable rooting medium for many types.

MINIATURE POOLS AND WATER PLANTS

If you can work out a way to use water in your landscape, by all means do. Whether it's a dime-sized tub the birds dip into, or a full-scale formal pool in a tiled patio, any garden with water attracts immediate attention, gives quick pleasure, becomes an unquestionable center of interest. Actually, the excitement water adds to any landscape scene is the antithesis of its calm, cooling, serene effect. I don't know of any other feature that, with a few plants and a minimum of care, makes a garden spot at once so dramatic, artistic, and restful. And if the water moves—ripples through a stream, drips over stones, falls from one level to another—soothing sound is added to the other assets.

Natural brooks, streams, and pools usually require little designing beyond bringing out the best of their inherent beauty. But there's also not a suburban lot for which some kind of artificial water garden can't be designed, and few grounds that are not thereby enhanced. The only requisite is that the design be in harmony and scale with its setting; that it be conceived, located, and constructed with imagination and skill.

The fact that pools are inevitably natural focal points makes their faults more obvious and more difficult to correct. The most you can do, once a pool is in the ground, is to soften or otherwise improve its outline with coping, rocks, or plants. If there's too much cement, you can't hide it with aquatic plants without covering, too, the desirable reflections on the water. If it's too small, you can hardly make it larger. If the shape is uninteresting, you can't change it very easily. If it's in the wrong spot or faced the wrong way, you can't move it.

On the other hand, the charm of water itself is so strong that it is comparatively easy to create a design that looks just right, particularly if simplicity is the keynote in both pool and planting. Simple shapes are safer and more effective than intricate ones; few plants, well placed, are better than many; water spraying from a concealed pipe is often more desirable than a formal fountain or statuary.

Depending on architecture and terrain, garden pools may be of formal design—a perfectly regular shape; semiformal—the shape balanced but not usually regular, often in severe contemporary style; or informal, completely asymmetrical, and integrated so naturally in the landscape the pool looks as if it has always been there. Each type of design can be executed in so many different ways that, even for tiny pools, there's good reason to study photographs and articles in gardening and homemaking magazines. There's also inspiration in the designs in books about water gardens, plus detailed information on various types of construction that I've had to omit. I've built miniature rock gardens but never a pool. Someone who fathoms the mysteries of water levels and understands the intricacies of cement work can give much more reliable advice than I.

Formal Design.

Small square, rectangular, circular, or oval pools can be extremely effective in almost any kind of landscape except one so rocky and untamed as ours. They're usually best in level ground, and most harmonious with borders and other plantings of fairly formal pattern. And, of course, they're constructed of materials that give a formal effect, such as concrete, tile, or brick.

A raised pool in the center of a small terrace, with a foot-high wall of tile or brick, can be scaled down from classic designs of other eras. A wide coping on the wall provides a place to set potted plants, or to seat yourself to feed the goldfish and dabble your fingers in the cool water.

A quadrant-shaped pool in the corner where garden walls meet can be either raised or level with the ground. Its water might reflect the image of a saint in a shrine above it, or might catch the constant stream from a lovely fountain.

At one end of a narrow garden a rectangular or oval pool becomes a striking focal point. A path may lead to a bench on a far side, between the pool and a background of shrubs.

At the end of a garden path a fountain may spout from a mortared wall into a projecting pool set at convenient dipping height from the ground, for filling watering cans. The planting underneath can be permanent, or an arrangement of potted plants.

And why not a tiny, formal pool in the center of a miniature rose garden? Or under a piece of traditional garden statuary? Or simply in the center of a garden path that splits to make room for it? If it's of proper proportion, a formal pool can even be set in the center of a garden, sometimes even in the center of a small lawn area.

Semiformal Design

Here's where the popular kidney-shaped pools are most useful. They're neither formal nor informal, but especially attractive with contemporary architecture. Planting is usually sparse but dramatic, making extreme use of contrasting colors and textures, and of unusual lines. Construction materials are simple to keep them subordinate in interest to the over-all design.

A refreshing sight outside a picture window is a small pool recessed in the patio floor. A recirculating pump sends up a spray of water from a pipe that ends just below the surface of the pool. These pumps make it possible for fish and plants to coexist with fountains because they reuse water in the pool. Fresh water from a spring, a stream, or the pipes that supply the house is too cold; and it may lack the small organisms on which fish feed.

Pools of contemporary design can also be placed in a corner of the property, with a tasteful grouping of shrubbery behind; on the bottom level of a series of terraces; in a depressed spot in the lawn; in the curve of a path; or at one side of a breezeway. They can be illuminated dramatically with the new underwater lights available in many sizes and styles.

Informal, Naturalistic Design

These are either adaptations of existing streams or pools, or designs of uneven form deliberately created to look as if they had been there all the time. They're best suited, of course, to naturalistic landscapes where slopes, rocks, and hidden springs give them a reason for existence. The water-holding basin may be of any material, as long as it is not noticeable. Any edgings, copings, or nearby trimmings should be just as natural as the pools.

A rock garden on a slope is a perfect setting for a pool with rocks that jut out over the water and, if possible, a rock or two coming up through the water from below. Or two small pools, one above the other, can be joined by a miniature waterfall. If there's no water supply on the spot, use a

recirculating pump. But be careful that the waterfall doesn't resemble man-made steps. It should be as craftily haphazard as if the elements had worn it into the rocks.

A dripstone is another delightfully musical device—an overhanging rock from which water drips down onto the surface of the pool. The sound has more resonance when there is an empty chamber behind the dripping water.

A meandering brook can be widened to make a pool. Or a completely artificial brook can be constructed of cement, like a pool, with drainage and overflow pipes and similar appurtenances. This isn't easy to execute effectively, but it can be completely intriguing. It might be wise to try to recreate part of a real stream in the nearby countryside.

One small boulder half-sunk in the ground can inspire an irregular, shallow pool that makes a natural birdbath. A miniature shrub may back up the boulder; rock-garden plants may grow at its base.

Many nooks and corners in the naturalistic landscape spontaneously suggest the creation of a tiny pool, and become exciting little garden surprises. These can often be so simply made that there's not even a drainage hole in the bottom. Scooping out the water, as needed, is no chore at all.

CONSTRUCTING MINIATURE POOLS

There are as many materials and methods for building pools as there are sizes and shapes. The depth depends on the requirements of the aquatic plants to be grown, and so does the location in sun or shade. Pygmy water lilies, for example, need all available sun, and at least ten inches of water—four inches for the container that holds soil and roots, eight inches above the crowns or growing points. Other miniature water plants are satisfied with much less.

One vitally important construction detail is that the sides of even the smallest tub garden be perfectly level, parallel with the water surface. Otherwise, there will be an unattractive expanse of bare wall at one side—one of those awkward errors that quickly catches the eye and spoils the whole effect.

Some pools are simply sunk flush with the surrounding lawn or other surface, without edging or coping of any kind. This can be messy and sometimes disastrous to plants when surface water collects and floods the pool during heavy rains. Extending the rim an inch or so above the surrounding area usually avoids this problem, but then the rim should be concealed with a coping of some sort—bricks and tiles in formal pools, well-arranged rocks, gravel, or flagstone paving in informal design.

Tubs and Other Containers

Attractive but relatively impermanent miniature pools can be improvised or specially planned with many kinds of containers—metal or wooden tubs made and sold for the purpose; half-barrels or half-kegs; discarded tanks, kettles, dishpans; even discarded sinks and laundry tubs. These last can be eyesores, we found out, if their ignominious origins are not kept carefully secret. Rubber-base paint is the only way we found to conceal the pristine white porcelain, and it was tricky to use so that it wouldn't peel off. The rim at the top of the sink is an unpleasant giveaway that must be covered with something such as overhanging rocks. Finding a plug or stopper that fit the drainage hole wasn't the least of our problems.

For the welfare of both aquatic plants and wildlife such as fish and scavengers, some containers need special treatment before they can be used, and some should not be used at all. Wooden kegs and barrels that have held oily substances such as gasoline, roofing compound, and wood preservatives can't be cleaned sufficiently to be safe. Copper poisons fish; so do paints with oil or lead bases. Steel, lead, iron, and other metals should be protected against rust and corrosion with sound coats of rubber-base paint.

Containers freshly made of new wood (except white cedar) can be unhealthy for fish. Let them stand out in the weather for the winter, or season the wood by filling the container with water, slaking a chunk of lime in it for a few days, and rinsing well before using.

Prefabricated Pools

These are now available in metal or plastics of a sturdiness that varies with the cost, in a wide range of sizes and shapes. Many of them make

lovely miniature pools which, because they're small, are not necessarily calamities if they don't last forever. Most have been designed after consultation with experts on fish and water lilies. And most are simply installed by digging a proper hole and setting them in place.







Steps in constructing a no-cost pool out of materials on hand:

- a. An old kitchen sink
- b. Placing rocks that were a nuisance anyhow
- c. Finished pool with goldfish, and an interested cat

Permanent Pools

Permanency increases, of course, with sound construction. Concrete laid on a gravel or cinder base and strengthened with wire-mesh reinforcement is usually considered best. But the new concrete must be cured and the alkalinity of its lime neutralized before the pool can be safely used for plants or fish. Leaving the pool open to the weather for the winter will usually accomplish this purpose. Or the pool can be drained and refilled several times, then thoroughly cleansed with household vinegar, as in the “quick cure” recommended by G. L. Thomas, Jr., of Three Springs Fisheries.

Bricks, cement blocks, tile, puddled concrete—for each type of material there are detailed procedures to be found in readily available and up-to-date

reference books. Prefabricated pools can be found in the catalogues of most water-lily specialists.

CARE OF POOLS

Miniature pools are easy to scrub in the spring and to keep clean through the summer season. Floating dead leaves or other organic debris that can give off poisonous gas are skimmed off with your hands. Discolored or dying leaves of lilies and other plants can be cut away cleanly. Plants can be thinned out by removing superfluous leaves or pulling out portions of weedy varieties. The pool is kept nearly filled with fresh water—but without adding so much at one time that the water is dangerously chilled.

Permanganate of potash will control slimy, green algae. To one gallon of water add about two teaspoons of permanganate; let it stand until there is little residue at the bottom. For every gallon of water in the pool, add one teaspoon of this saturated solution whenever algae become unsightly. Don't make the solution too strong. It might kill the fish.

Insects such as the black aphids common on water lilies can be sprayed off with the garden hose to make a juicy meal for the fish. Spraying pool plants with insecticides or fungicides is dangerous unless you use a preparation made specifically for pools, and use it strictly according to package directions.

In winter, most tubs and other containers should probably be lifted and stored indoors. I should think this would lengthen the useful life of small prefabricated pools, too. Small permanent pools should be drained, so ice will not form and crack the construction. This means that the pool must remain empty; either the drain must be kept open or a roof of canvas or boards should be fitted over the top.

FISH AND SCAVENGERS

Goldfish earn their upkeep—a pinch or two of food a week—by gobbling up mosquito larvae the minute they start wriggling. In pools where the mud at the bottom does not freeze, they'll winter safely out of doors.

Several kinds of snails, tadpoles, and other scavengers offered by specialists consume impurities in the water, including algae. Frogs are just for fun. They'll often make themselves at home in a pool without any

invitation. A grandpappy bullfrog lived in our tiny sinkpool all last summer, retreating to a dark cave between the rocks when the dogs came down for a drink.

It seemed to me the epitome of modern efficiency when I first learned, as a city-dweller, that full-grown frogs could be bought by mail. That purchase made us the most popular family on the block. But we couldn't offer them comfortable winter quarters in the city; and the last I saw of the frogs, in early fall, they were hopping down the gutter of Fortieth Avenue toward the sewer drain.

MINIATURE AQUATIC AND POOLSIDE PLANTS

Obviously, the location of a pool (in sun or shade, warm climate or cool), its size (some plants would smother a miniature pool in a few weeks), and its design should be considered in selecting plants to grow in or with it. In fact, very tiny pools carefully placed to catch an artistic reflection may be better without any plant embellishment.

A frequent error is overplanting, with the water surface covered by a confusion of foliage and flowers, the perimeter a jungle. In small pools one pygmy lily is plenty; it needs clear, open water to set it off.

For pools of all designs—formal, contemporary, informal—plants are seldom arranged in neat rows, groups that repeat themselves regularly, or matching masses in perfect symmetry. The closest to formal balance might be adding accent and height by placing the tallest plant in the center of a formal pool. More often, it's effective in one corner or at one side.

Few plants offer such striking variation in the forms available to create interesting patterns and dramatic contrast. There are flat, leathery leaf pads and slim, spiky swords; glorious china-like cup flowers and fluffy plumes; modest creepers and bold elephant leaves. Seldom does a flower-arrangement artist have such a wealth of exciting material with which to make her prize-winning compositions.

Miniature Water Lilies

Of the two general types of water lilies—tropical and hardy—the tropicals are least likely to be in scale with miniature pools. The leaves and flowers are larger, the leaves spreading out wider and the flowers standing

out higher. The tropicals need lots of sun and warmth, can be shocked into dormancy if the water chills, and are usually treated like annuals and planted fresh, each summer. Some can be propagated from viviparous plantlets that grow on the leaves.

Among hardy water lilies there is a selection of pygmies with four-inch leaves and two-inch flowers in many colors—white, yellow, pink, red, and bicolors, some with attractively marbled foliage. Given generous sunshine, warm quiet water, and good nourishment, they will flower from July to frost, and can be held over the winter either outdoors or in.

In our area we set out hardy water lilies in late April or early May. If a late cold snap occurs when the plants arrive by mail, we keep them moist until it's over. Since the roots are hardy only if they do not freeze, and since it's difficult to keep small pools from freezing, each lily is planted in its own box, which is set on small stones or blocks so it is raised off the floor of the pool. Or plant in soil at the bottom of a tub-pool that will be taken up in fall. Lay the rhizome horizontal, with its growing tip out of the soil. Cover the surface with a web of clean sand, to keep the soil from muddying the water. If the plant is leafed out and the leaves don't reach the surface of the pool, they'll lengthen their stems in a few days.

For best growth and flowering, a pygmy lily needs at least a third of a bushel of soil and about eight inches of water over its crown. Soil can be heavy and clay-like, or average garden loam. If possible, add one-fourth the quantity of well-rotted cow manure—no other kind. Experts recommend strongly against the use of other-than-cow manures, swamp muck, leaf mold, peat moss, sand, or lime.

Lacking cow manure, you can get fertilizers prepared specially for water lilies, or you can use commercial garden fertilizers like those with 5-10-5 analysis. Supply nourishment at planting time, and once again a month or so later. Wrap balls of rotted cow manure, or handfuls of commercial fertilizer, in thin paper (a paper napkin will do) and thrust them down into the soil around the roots. Water lilies have voracious appetites, and even the pygmies eat heartily. On a starvation diet, they'll stop growing and flowering.

In winter, store the rhizomes in an unheated cellar or garage where they will keep cool but not freeze. Don't let them dry out completely. Keep the

soil in the container barely moist; or if the rhizome is not in soil, wrap it in burlap that can be kept moist.

Hardy water lilies are propagated by planting divisions of the roots, with each section having at least one growing point.

In the wide selection of available varieties, the following pygmies are suitable for most miniature pools.

Nymphaea Nymphaeaceae Pygmy Water Lily

adorata minor—Pint-sized variety of the native pond lily with fragrant, dainty, white three-inch flowers in generous profusion; leaves lined with red beneath. ‘Helen Fowler’ is a fragrant pink-flowered variety.

aurora—An old hybrid, more dwarf than miniature, but suitable for small pools. Changeable flowers open soft yellow, darken to rusty orange in the second stage, then to deep red in the third stage. The small leaves are mottled with wine-red lines.

‘Jo Ann Pring’—True pygmy with three-inch leaves, dusty-pink flowers lighter in the center.

‘Patricia’—One of the few small tropicals with crimson flowers, brown-metal buds. Young plants are borne on the leaves.

tetragona (pygmaea)—Smallest of all, with long-lasting two-inch white flowers with a fragrance like China tea, four-inch soft green leaves. Easy to grow, even in shallow water. The variety *alba*, or white pygmy, seems the same to me as the variety *helvola*, an old hybrid usually listed as yellow pygmy, and is even smaller than the species, with brown-blotched leaves and glowing yellow, star-shaped flowers.

‘Royal Purple’—A new red-purple, tropical lily.

Floating Plants

Another advantage of miniature pools is that any of the attractive surface plants that multiply too fast can be easily scooped out as often as you wish. And they are unusually attractive in forms, colors, and textures. Feeding from nutrients in solution in the water, they make shade for fish, and their dangling roots provide a safe place for fish to spawn. Simply drop the plants in the water and let them grow. Or, if you want them to raise a family, put some soil in a shallow spot where the roots can anchor. Most are annuals, to be bought each year.

Oxygenating Plants

These are aquatics that grow down in the water and help keep it sweet and clean. Bunches of some can be simply dropped in the pool. But all will

grow better and save you the trouble of replacing them if their roots are in soil, in pots set on the pool floor. Although they're mostly perennials, we prefer to start each season with a fresh supply. They're available in variety at pet stores or by mail from lily specialists. Since they're neither true miniature plants nor as decorative as they are functional, separate descriptions are not included here.

Other Aquatic Plants

With the water lilies and other aquatics, these are the only plants that can grow with their roots standing in water—some in deeper water than others. With few exceptions, this does not mean sour, stagnant water. Even in bogs there is some circulation. Adding chunks of charcoal that absorbs impurities will often help keep the water fresh and healthful.

Some of these plants are hardy, some are not; some need their roots in soil; some can do without it; some are better growing in the water of the pool; some in the boggy soil beside it.

Earlier in this chapter I said I had never built a pool. I meant that I had never done it personally and worried about water levels and the general engineering. But we have had several pools, little more than puddles, which my husband constructed. Small though they were, they gave us the chance to enjoy water lilies and bog plants. It was then that I began to realize the delights and magnitude of this sort of gardening. Someday I hope to have a pond of some size—a spot for lilies, frogs, and goldfish and all of those things G. L. Thomas, Jr., writes about so charmingly in his book, *Garden Pools, Water-Lilies and Goldfish*. I also get great pleasure out of the catalogues several suppliers send out. They are most delightful reading. Read that book, and the catalogues, and I know you will be converted to water-gardening.

MINIATURE WOODLAND GARDENS AND PLANTS

For ten years we lived within the boundaries of New York City in a big, old house that occupied more than a third of the squared, bare eighty-by-hundred-foot lot. Now we have a smaller house, and the gardenable land is completely surrounded by woodlands where native plants grow as they please. In both places we've had small woodland gardens; and no matter how pleasurable our other plantings, we've loved these best.

If your idea of a pretty garden centers around statuesque delphiniums, precise rows of roses, bold splashes of boisterous color, you may call insipid what we find enchanting. But if your senses respond to the fresh, sharp aroma of moist woods soil, the whispering of trees, the patterns of cool shadows, the shy delicacy of the spring beauty, or the gnome-like pomposity of a Jack-in-the-pulpit, be sure to have at least a tiny woodland garden, whether you have to create a spot for it or have it naturally. Once established, it gives more refreshment for less care than any other garden I know.

NATURALISTIC DESIGN

Here, of course, there is no place for classic or contemporary formal design. The single purpose is to create, or re-create, a small section of the woods with the shade, humus, and other environmental conditions woodsy plants can't live without. The essence of the tiniest planting—even a colony of trillium at the foot of a spreading evergreen—is its naturalness. It doesn't look planted, it belongs; it's been there all the time. And it looks natural, of course, because it's been planned that way.

Planning a woodland garden for the informal landscape is simplified by the fact that you begin with what you have—a shaded spot (preferably shaded by trees that drop needles or leaves, to enrich the soils) where hot, dry summer winds can't parch the plants. There may be a natural stream, rocks, a slope, a hollow; or a place to put a rustic bench, rotting log, small pool—a focal point harmonious with the naturalistic theme. Or the size, shape, or contour of the garden may make it the object of interest without additional accent.

I know of one design that began with pruning a stunted sapling to reveal its bonsai-like lines, another that featured the trunk of a rheumatic, old, lightning-shattered evergreen. A soggy, low spot can be planted as a bog and backed up by rocks and shrubs. A brook can be coaxed to make space for a small, hummocky island. The narrow, meandering, stepping-stone path through the sharp point of our land invites you to take a walk in the woods.

Originality has challenging freedom in creating a woodland garden for level, regularly shaped grounds. You may replace the ubiquitous pachysandra in the needle-carpeted shade of a tall pine tree with native plants. Or use one of those difficult areas such as the angle of a hedge, wall, foundation or shaded patio; an unused corner of the property; the space underneath an overhanging bay window.

We used a useless area, about six feet wide and less than thirty feet long, between our city house and the property line, shaded not only by a few straggling pines but also by the house next door. There was a path, a pool, and a little bench. This was the last place I visited before I left for work in the morning, and the first when I came home at night. Somehow, the flower face of a bloodroot with a shawl of unrolling leaf around its shoulders had added piquancy in the city, where building banished the wild flowers a hundred years ago. The bloom of the first lady-slipper was a noteworthy triumph; the green of the ferns seemed specially fresh and mossy; the blue of the bottle gentians was almost unbelievable.

CLEARING THE SITE

Even when we planned that metropolitan oasis, there were a few existing features we carefully kept—a valiant, five-leaved woodbine to train over the rail fence that kept out the trampling feet of the neighborhood small fry; a scrubby shrub I never managed to identify, which had voluntarily masked an ugly cellar window; a few precious patches of moss.



A wild garden in New York City

In Connecticut, of course, the area was almost completely overgrown, mostly by poison ivy. After that was cleared away, undesirable undergrowth was removed, leaving a rock here, a dogwood or azalea there. Baring the ground let us see more clearly what we had to work with.



A wild flower grows in New York City: bloodroot

When the ground is cleared, the designer makes a final decision about permanent features such as steps, path, pool, bog, bench. You might make a mental note of the place where an old tree stump might be artfully installed, when you find one, or where a dead branch might fall and make a background for some small plants. Of course, these are all kept carefully in scale and harmony. A path should be narrow and not sharply defined, perhaps just a series of barely matched flat stones. Steps are not conspicuous, but made to look as natural as possible.

SOIL

The next is the most laborious but most vitally important part of the procedure—supplying suitable soil. Woods plants grow naturally in soil so rich in humus that it will hold many times its weight in water. Season after season the trees drop leaves, needles, twigs, and branches that decay and maintain the high level of humus, which contains micro-organisms that help sustain the plants. In our area and in most other woodsy sections, forest soil is very acid. But the loam in a cleared area adjoining the forest may contain

little or no humus, and may not be nearly acid enough for the native plants growing a hundred feet away.

When a miniature woodland garden is planted in a wild, woodsy spot, the soil is probably suitable, and the trees will probably keep it that way. Then, you simply use plants that thrive naturally where the amount of available light, and the composition of the soil, are the same or similar. Otherwise, you need to prepare the soil thoroughly and deeply—a minimum of eight inches, and up to eighteen inches for small native shrubs.

You'll need incredible quantities of leaf mold, very old manure, or composted leaves and grass. Over humus-poor soil, spread a layer of at least six inches, and mix it in by digging the soil and turning it over several times. At the same time, remove matted roots, stones, and other unwanted obstructions to planting. For a very small garden, you may find it simpler to remove all the soil to a depth of twelve to eighteen inches, and replace it completely with humus and leaf mold brought in from the woods or purchased for the purpose. Leaf mold, by the way, is the layer of partially decayed leaves immediately under the top layer of whole leaves; humus, in the next layer down, is more completely decayed. Don't buy humus from door-to-door salesmen offering "big bargains." This is a racket the police and Better Business Bureau are trying to extinguish. Buy only from a nurseryman you can trust.

While you're conditioning soil, make sure there is good drainage. Only bog plants will grow with their roots in muck or standing water. You may need an initial layer of fine gravel or coarse sand. Raising the surface of the garden will also help.

If you think that the importance of suitable humus-rich soil is either exaggerated or overrated, think also of this: With the type of light and humus (usually acid) in which they grow naturally, woodland plants will settle down contentedly in your garden and establish families and colonies that increase every year. If you don't provide it, they'll get homesick and pine away. Not only that, but once they've made themselves at home, they'll require practically no care. They don't want to be cultivated, fertilized, weeded, watered, mulched, or given special protection in winter, as long as there are trees that drop leaves. The most you might have to do

will be making sure the leaves aren't blown away, or, if there are no trees, lugging in leaves from somewhere else.

PLANTING

Although plants are chosen for any garden according to light, soil, moisture, and other cultural requirements, woodland plants permit less leeway than most others. This is a matter of ecology, defined as “the total impact of the environment and the plant's accommodation to it.” Some delicate plants are, of course, less adaptable than other more rugged species. Check catalogues, reference books, and other sources to learn all you can about each plant before you plant it.

A quick word about shade. There is full shade, or deep shade, created all year by evergreens, walls, or buildings. And there is woodland shade created only in summer by deciduous trees. The hill near our house where rattlesnake plantain, pipsissewa, and pink lady-slippers grow is dark and cool in summer, but bright in spring, fall, and winter when the trees are bare of leaves. This is an important distinction.

In the interest of conservation—American wild flowers are threatened not only by bulldozers but also by thoughtless, criminally careless humans—private woodland gardens often become the home for plants dug from the wild. Last fall, one of my dearest friends methodically moved dozens of maidenhair ferns into a tiny bit of woods on her property to save them from extinction when their fronds were being picked in bunches to fill out bouquets. But this is conservation only when the plants are transplantable (a number of the most precious species are not) and are moved to quarters with growing conditions to their liking. Otherwise, they might as well die where they are.

In digging woodland plants, always dig deep and take as much of the surrounding soil as possible, and disturb the roots as little as you can. Protect the transplants against drying sun and wind until they are set in their new homes. If you can dig them with the roots intact, you can take most plants at any time during the growing season. The safest times are immediately after flowering or during fall dormancy.

Unless it is a matter of conservation, I never dig woodland plants. I buy them (they're surprisingly inexpensive) from a local nursery or from several

mail-order specialists. The nursery plants are pot-grown, and even fussy types transplant without loss. By mail they arrive bare-rooted—in early fall for all but the summer and fall varieties—but crisp and lively in their packing of moist sphagnum moss. Sometimes they even start to send out new roots and sprouts en route. If I can, I get mail-order plants into the ground before dusk on the day they arrive. Soil around transplants is kept moist and mulched with leaves until they take hold.

Years ago some suppliers collected the plants they sold, and some sources of slow-spreading types were completely devastated. Today, I believe, specialists grow their stocks of woodland plants, either under glass or in woodsy nurseries. They're doing more to preserve these native treasures than to annihilate them.

The best of all sources is by propagation—seeds, cuttings, division of wild plants—because you're not only increasing the population, but you're also starting with plants that from babyhood are accustomed to your growing conditions and don't have to make difficult adjustments. Even though it may be slow, this is the only method for a number of nontransplantable species.

If the garden contains shrubs that will serve as background for small, shy woodland plants, that's fine. If it doesn't, you may want to plant some (these, too, are available by mail), because few native woods plants are spectacular enough to make a big display of their own. And because woods plants are modest, they're best planted in colonies a half dozen or more of one species, not in mixed colors or varieties. Set the groups in the foreground, where they're easy to see. Allow plenty of space between groups for natural increase without crowding. Arrange more striking, tall, spiky plants here and there to add the interest of accent, just as you would in a conventional flower bed.

Care after planting includes the expected careful watering, and keeping the plants moist and shielded from heat and wind until they are growing on their own. For their first winter, you may want to supplement the natural mulch of leaves with salt hay or something similarly light and airy, held in place by chicken wire or branches. This mulch must be removed extra early for early-flowering species. By their second season the plants should be ready to be watered by rain, and fed and protected by trees, without your

help. Don't fiddle with them, pull off leaves or seed pods, or move them about unless you must.

PROPAGATING WOODLAND PLANTS

These are, of course, hardy perennials. Except for seeds, propagating methods are much the same as for cultivated types. It is important only that, if possible, you find out what is the most effective (often, the only) way each plant can be reproduced. If you can't track down this information, try several ways at once. In general, those with masses of fibrous roots can be divided immediately after flowering. For upright and branching types, you can usually root stem cuttings in a frame or propagating box. If everything else fails, try layering. You can't lose anything by it.

Seeds are planted the minute they are ripe, in a cold frame—or in flats that can be set in the cold frame—in a finely sifted mixture of equal parts of loam, woods leaf mold, and coarse sand. For varieties adapted to extremely acid soil, use half as much coarse sand as Michigan peat, without soil. For added insurance, acidify the planting mix with leaf mold or peat, or neutralize it with lime, until the *pH* is somewhere near that of the soil the plant grows naturally in.

Set the flats in shade and keep the soil moist, and leave the seeds to their own devices, summer and winter, until they germinate. Some species take two months, some take two years. After germination they are handled exactly like other perennial seedlings, except that the only safe fertilizer is very weak liquid manure at very infrequent intervals. Tender types should be grown in pots until they are fully mature, before they're set out in the garden.

PLANTS FOR WOODLAND GARDENS

Aquilegia canadensis *Ranunculaceae* Columbine

Although the native Eastern columbine can grow two feet tall, it seems to stay closer to six inches in my woodland garden and on the banks that line our Pine Tree Road. The clover-like leaves, and characteristic columbine flowers with yellow sepals and knobbed red spurs, are scaled down proportionately.

CARE. Poor, dry soil, acid (strong to neutral). Fertilizer promotes growth that is too rank. Full sun to three-quarter shade.

PROPAGATION. Seed.

SPECIAL USES. Woodland planting, bank planting.

Arisaema triphyllum *Araceae* Jack-in-the-Pulpit

The pulpit stands eight inches tall or more, curving up and over the stiff preacher-like spadix in late spring. The divided leaves also exceed true miniature size. But prim Jack spreads himself slowly enough to allow him entry to all but the very tiniest woodland garden; and his comical dignity is not to be dispensed with.

CARE. Humus-rich soil, moist. Shade or light shade.

PROPAGATION. Tuberous roots.

SPECIAL USES. Wild gardens.

Asarum *Aristolochiaceae* Wild Ginger

In the Northern part of the country this is a deciduous plant, while in much of the South it retains its kidney-shaped leaves. Its purplish flowers are somewhat hidden under the foliage. There are three general types.

canadense—Eastern wild ginger.

caudatum—Western wild ginger.

shuttleworthi—Mottled wild ginger (native of Virginia and South).

CARE. Rich soil, moist. Shade or partial shade.

PROPAGATION. Division.

SPECIAL USES. Wild gardens, ground cover.

Asplenium *Polypodiaceae* Spleenwort (Fern)

Lilliputian, native evergreen fern with deeply cut fronds.

platyneuron—ebony sweetheart—Eight-inch feathery fronds with brownish-purple stems in tidy circles. (Needs acid soil.)

trichomanes—maidenhair spleenwort—Thick, closely clustered three-to five-inch fronds with crowds of little one-inch leaflets on black ribs. Plant it sideways, its roots in a rock crevice, with

leaf mold and a touch of lime.

CARE. Soil acid or neutral according to variety. Leaf mold. Shade.

PROPAGATION. Division.

SPECIAL USES. Terrariums, rock gardens, wild gardens.

Camptosorus rhizophyllus *Polypodiaceae* Walking Fern

Small native evergreen fern, hardy from Canada to Georgia. Its name comes from the fact that the fronds are pointed on the ends and root on contact with the soil. It prefers to creep around limy rocks.

CARE. Lime-bearing soil, shade, dry. Hardy.

PROPAGATION. Root division.

SPECIAL USES. Terrariums, rock gardens, wild gardens.

Claytonia virginica *Portulacaceae* Spring Beauty

When the leaves of the forest trees are barely ready to break out in early spring, this lovely little plant is already in full bloom down on the ground. It makes colonies and even mats of precious white flowers which are tinged and streaked with pink and scarcely a half-inch across. But before the trees are in full leaf the spring beauty has gone back to sleep and disappeared, flowers, five-inch grass-like leaves, and all.

CARE. Woodsy soil, moist, shady or semishady.

PROPAGATION. Tubers.

SPECIAL USES. Rock gardens, bog gardens, swamp planting.

Clintonia *Liliaceae*

Woodsy plants spreading out into colonies by underground rhizomes. The leaves are rather broad, and thin and glassy. The little lily-like flowers are held atop stems above. The berries that follow are blue, blue-black, or black.

borealis—broadlily—Nodding, greenish-yellow flowers on seven-inch stems in May. Berries are blue, a half-inch long.

umbellata—Leaves lightly fuzzy; purple-spotted white flowers in clusters; shiny black berries.

uniflora—Western native with one precious flower per stem (one and a half inches across), berries bright blue.

CARE. Rich, humusy, acid soil. Shade or partial shade. Moist.

PROPAGATION. Division of roots in spring.

SPECIAL USES. Rock gardens, bog planting.

Convallaria majalis *Liliaceae* Lily of the Valley

This fragrant ground cover is a favorite for under trees and other shady spots. Perhaps it spreads too fast for very small gardens. The oval, lily-like leaves are lasting, but not evergreen. The dangling flowers, all on the same side of the stem, are like scalloped petticoats. Unless you need a ground cover, try one of the fancier and less robust varieties, such as *rosea*, with light-pink flowers that fade to near-white in the sun. Or try *flore plens* with double white hoop-skirts.

CARE. Garden soil, shade, moist, hardy. Persists for years and doesn't have to be touched. If you want to increase your planting, dig and thin every three years.

PROPAGATION. Division of small bulbs.

SPECIAL USES. Ground cover, forcing in February and March.

Coptis trifolia *Ranunculaceae* Goldthread

This tiny bog plant, only four inches high, has three-way clover-like leaves and tiny white waxy flowers. It creeps and spreads by means of thready yellow roots.

CARE. Humusy, peaty, sandy, and decidedly acid soil. Plant needs a cool, moist location and will not survive warmth and dryness.

SPECIAL USES. Bog planting.

Cornus canadensis *Cornaceae* Bunchberry, Creeping Dogwood

This is a real dogwood, but a very miniature one. The stems standing about six inches high are encircled by a whorl of rich green leaves. Above are the up-facing flowers (in May), like elves wearing white ruffs. The

clusters of red berries are as gay as Christmas, in August. After taking a year or so to get settled, the plants spread out into colonies or clumps.

CARE. Rich woods soil, highly acid, damp, but with drainage. Shade, cool location. Takes several years to get established. Needn't be disturbed once settled. Can be transplanted if large chunks of sod are lifted with it.

PROPAGATION. Separate clumps when dormant. Seeds from ripe berries will germinate the following spring.

SPECIAL USES. Bog planting, ground cover.

Cryptogramma crispa *Polypodiaceae* Parsley Fern

This is a small alpine fern with clusters of six-inch evergreen fronds so finely cut and frilled they could garnish a standing rib roast in place of parsley. Don't overlook this one when next planning a terrarium.

CARE. Peaty, dry soil. Hardy.

PROPAGATION. Division.

SPECIAL USES. Rock gardens, terrariums.

Cypripedium *Orchidaceae* Lady-slipper, Moccasin Flower

This is the fabulous earth-growing orchid that is native to many parts of the country, with culture varying accordingly. The broad, long fresh leaves fold around the stem at the base. The flowers have a lip inflated into a pouch and are in various colors.

acaule—Northeastern native with dusty-pink flowers. Requires dry, sandy, woodsy, intensely acid soil, full shade. Difficult to transplant unless taken with a large amount of soil. Attempt it only after flowering, or in October. Don't cover crown more than a half-inch. No good method of propagation is known.

calceolus pubescens—Fuzzy yellow lady-slipper from moist, wooded hillsides of the North and mid-South. Not difficult to transplant and easy to grow. Likes dappled shade and slightly acid, humusy soil.

candidum—Small white flowers marked and veined with brown and purple. From limy, marly bogs. Needs moisture and shade.

CARE. Soil, moisture, and shade as described above.

PROPAGATION. Division when dormant.

SPECIAL USES. Rock and wild gardens. Can force.

Dalibarda repens *Rosaceae* Dewdrop, False Violet

Precious little creeper, pretending to be a violet, with two-inch heart-shaped leaves on long stems and five-petaled, half-inch white flowers in July. A native of the Northeastern section.

CARE. A sheltered spot. Rich woodsy soil.

PROPAGATION. Seeds and division.

SPECIAL USES. Woodland and rock gardens.

Epigaea repens *Ericaceae* Trailing Arbutus

Flat evergreen creeper with overlapping hairy stems and bright-green oval leaves. In early spring it has white or pink phlox-like flowers, not very large but interestingly fragrant. Although it is usually cultivated in shady, protected spots, it can thrive in hot dry areas.

CARE. Humusy and very acid soil, sandy, with leaf mold. Good drainage. Moist. Don't dig wild plants which are difficult to transplant. Use pot-grown plants.

PROPAGATION. Cuttings. Seed if strictly fresh.

SPECIAL USES. Wild gardens, rock planting.

Erythronium *Liliaceae* Adder's-Tongue, Trout Lily, Dogtooth Violet

A very large clan of bulbous wild flowers native to many parts of the country. All species have the marks of family resemblance—two leathery basal leaves, often beautifully marbled, and miniature lily flowers nodding from the tops of central stems in spring. Height, usually six inches.

albidum—Leaves may be matted or not. In the East flowers are white, in April.

americanum—Best-known Yankee with marbled leaves and bright-yellow flowers in April.

californicum—fawn lily—Mottled leaves, creamy or pale-yellow flowers. (From California, of course.)

dens-canis—dogtooth violet—European emigrant with green leaves splotched with brick red; flowers are rosy purple. Also available in white and other colors.

grandiflorum—Giant of the family, plain leaves, golden flowers in clusters. The variety *parviflorum* has the most midget flowers. Western.

hendersoni—Mottled leaves, dark-centered purple flowers. Western.

revolutum—Western with brown-blotched leaves, purple-tinged white or lavender flowers. The variety *johnsoni* has rosy-lilac flowers, white in the center.

tuolumnense—California with unmarked chartreuse leaves, large deep-yellow flowers.

CARE. Woodsy soil, not especially acid, with leaf mold. Drainage. Moist, never hot and dry. Winter mulch.

PROPAGATION. Seeds, offsets (from underground stolons).

SPECIAL USES. Rock gardens, woodland gardens, forcing.

Goodyera pubescens *Orchidaceae* Rattlesnake Plantain

This terrestrial herb is a close relative of the most glamorous of orchids and bears no resemblance to a reptile or the weed that is part of its common name. On the woodland floor you will come upon it as a flat cluster of durable, oval, deep-green leaves with gleaming-white veins that cross and recross in a delicate pattern. Perhaps you will find it in bloom with tiny whitish flowers on tall spikes.

CARE. Woodsy acid soil, with humus. Slightly damp. Cool. Shade. North side of a glacial ridge is ideal.

PROPAGATION. If possible, buy section of root from dealer who specializes in woodland plants and seeds. Pick sparingly in the woods, by breaking off section of rhizome with rootlets and plant. New plant will appear in about one month.

SPECIAL USES. Terrariums, woodland gardens.

Hepatica *Ranunculaceae* Liverleaf

Six-inch evergreen plants that shed their old leathery leaves for new after the blue-lavender (or pinkish-white) flowers fade and fall in May.

acutiloba—Sharp-lobed hepatica, distinguished by the points on the lobes of its leaves. (Needs alkaline soil.)

americana (*triloba*)—Round-leaved hepatica with dainty flowers on stems upholstered with silky hairs. Leaves maroon in winter

CARE. Not too acid soil. Full shade. Drainage. Not difficult to transplant but needs time to get established. Plants in the woods best dug in the fall.

PROPAGATION. Seeds, division of roots.

SPECIAL USES. Woodland gardens, colonies.

Houstonia *Rubiaceae* Bluets, Quaker Lady

This is an enchanting wildling with grass-like leaves and flowers less than a half-inch across in the shape of four-pointed stars. It is a very profuse bloomer throughout the spring and early summer.

caerulea—Northeastern favorite with leaves in tufts about an inch high. Flowers are blue or lilac, often fading to white in the sun. Seeds itself willingly.

purpurea—Larger than miniature, and native from Maryland to Georgia. It makes a soft evergreen foliage mat, and in July has purple flowers, several per stem.

serpyllifolia—creeping bluets—The stems stretch out to ten inches or more with little half-inch teardrop leaves. Sky-blue flowers.

CARE. Loamy, acid soil with peat. Moist. Part shade.

PROPAGATION. Division. Some varieties are self-seeding.

SPECIAL USES. Rock and woodland gardens.

Lemna minor *Lemnaceae* Duckweed

This tiny one is often called the “smallest known flowering plant.” A floating plant that has quarter-inch leaves (actually the body or “thallus”), it has hair-like rootlets and flowers so small one needs a magnifying glass to see them. In the fall the plants sink to the bottom of the pond, to rise again in the spring.

CARE. No attention is needed.

PROPAGATION. None necessary. Increases naturally.

SPECIAL USES. Dish-garden pools, rock-garden pools, fish food.

Lycopodium *Lycopodiaceae* Club Moss

In prehistoric times this was a towering tree; now it is a moss-like poor relation of the ferns, bearing needle-like leaves and spores instead of flowers.

clavatum—running pine—Slowly but steadily the stems creep over the woodland floor, sending up four-inch stems at lax intervals, each with several spikes packed tightly with needle-like leaves.

complanatum—ground cedar—The stems trail faster and more sturdily, the erect branches dividing and spreading out to a lacy green fan.

lucidulum—shining club moss—Trailing stems turn upright at the tips and are covered completely with glassy, dark-green “needles.”

obscurum—ground pine—Christmas trees in miniature about six inches tall and popping up from stems that creep about underground.

CARE. Rich, woodsy soil, not especially acid. Shade, moist.

PROPAGATION. Cuttings. Division of roots at joints.

SPECIAL USES. Terrariums, ground cover.

***Mitchella repens* Rubiaceae** Partridgeberry

This native creeper bears the popular partridgeberries widely used to fill “Partridge Bowls” at Christmastime. It has glassy green leaves which are close to the ground with the berries beneath them. In the springtime it has fragrant, fuzzy white flowers. (Note to those who hunt the berries in the woods: pick from the tallest plants and do not tear up the roots.)

CARE. Rich woodsy soil, not especially acid. Drainage. Shade. Cut back to encourage branching. Transplants easily.

PROPAGATION. Divide rooting stems.

SPECIAL USES. Rock gardens, terrariums, woodland gardens.

***Mitella* Saxifragaceae** Bishop’s Cap

For shady rock and wild gardens here is a saxifrage-like plant with mounds of heart-shaped leaves and spires of finely fringed flowers on tall stems.

breweri—Pacific Coast native with deep-green leaves and whiskery yellow-green flowers (in May and June).

caulescens—Also from the Pacific Coast. More dainty, and more difficult.

diphylla—East Coast native with eight-inch flower spikes on taller stems. White flowers (in April and May).

CARE. Rich woodsy soil. Drainage, shade.

PROPAGATION. Seeds.

SPECIAL USES. Rock and wild gardens.



Garden in the landscape in the finest tradition—it features *Azalea macrantha* *basaminæflora*, *Juniperus radicans*, and *Calluna* County Wicklow. (Atlantic Avenue Nursery, Inc.)

***Podophyllum peltatum* Berberidaceae** May Apple

The May apple, a native of much of the country, is actually too large for most small wild gardens, but find a place for it if possible. In spring it is fascinating to watch the stumpy stems push up through the ground and the foliage slowly unfold to wide flat leaves. Then come gold-centered white flowers and finally the “apple” in chartreuse and nearly two inches long. (It is edible only when fully ripe.)

CARE. Rich woodsy soil. Shade, or partial shade.

PROPAGATION. Root division.

SPECIAL USES. Wild-garden planting.

(Additional plant listings suitable for woodland gardens can be found in Chapter 15.)

MINIATURE TREES AND SHRUBS

There's a *chamaecyparis* that forms a fluffy, green two-inch ball by the time it's seven years old; a cone-shaped Norway spruce that rarely grows more than twelve inches tall; a one-foot rhododendron with deep-crimson trumpet flowers; a nonfruiting viburnum that makes a two-foot globe tightly packed with ivy-like leaves. There are junipers that form plush carpets; ericaceous evergreens with neat mounds of twinkling flowers over shiny teardrop leaves; deciduous shrubs with golden blooms, seeds in silky pea pods, and green stems that look leafy the year round.

That is just a tantalizing hint of the fantastic variety of little trees and shrubs, and how they can steal the show in the garden. We have three magnificent pines on Pine Tree Road (it may have been named for them) but our guests are more likely to comment about the starry cushions of *Leiophyllum buxifolium* beside the front walk. We dug and lugged tons of rock to make beds for our favorite roses, but the small edging plants draw more attention. When the rock garden is a mass of flamboyant spring color, we'll be asked the name of a heather that's not even in flower, or the juniper (*Juniperus procumbens nana*) two inches high that spreads like velvet over a rock.

There are miniatures among all types of trees and shrubs—deciduous and evergreen, broad-leaved and needle-leaved. Those described in this chapter are almost all three feet tall or less at maturity, or are so extremely slow-growing they seldom top that height in twenty years. In just a few cases, slightly taller varieties are included, which can be kept to three feet or less with a little pruning. Some low types are omitted because they spread too rambunctiously to be called miniatures or to be suitable for small gardens.

These limitations automatically exclude the small, flowering trees and shrubs, and dwarf fruit trees, that are part of the over-all landscape plan for small grounds. They're not really miniature garden items, but background features around which miniature gardens are planned.

We've thought of a dozen different ways to use the really midget trees and shrubs, and I imagine there must be dozens more. We have a colony of tiny rhododendrons blooming at the base of a boulder, and another near an old stump at the edge of the woods. Several sprawly evergreens and some precious ericaceous gems adorn the rock garden and spots near the front terrace.

There are plenty of miniature shrubs of varying heights, foliage, colors and textures, and flowering times to compose a small shrubbery border, with tiny bulbs to plant along the edge. Small trees and shrubs can be used as background for mixed borders of small annuals and perennials; evergreens can make a setting for miniature rose gardens. Almost any small pool, wall, raised bed, or set of steps provides a place to plant the right tree or shrub. And, of course, these are perfect plants for sink and trough gardens, bonsai work, or even for indoor forcing in the greenhouse.

If you can't find the varieties you want at your local nursery, you can order them by mail from reliable suppliers. Either way, plants that have been transplanted regularly by the grower develop a compact system of fibrous roots rather than a few long straggly tap-like roots, will take hold faster and transplant easier, will fit into your garden quicker. All that will make you a happier gardener.

Don't be surprised if miniature trees and shrubs cost more than the regular varieties of the same name. The little fellows take much longer to reach that size, and require more costly care. Many of them are difficult to propagate, some even have to be grown from seed.

Your first thought in selecting a miniature tree or shrub, of course, is the decorative effect it is planned to achieve—whether it should be formally upright and symmetrical, or irregularly shaped and naturalistic; whether it is to act as an accent or specimen, or as a background or blender. Consider the texture of the foliage, the time and manner of flowering, whether the colors will fight with others in the garden. Plant deciduous types where they won't be an eyesore in winter, or select one with an attractive winter habit and appearance. Make sure the plant and all its parts are in scale with the setting and with other plants.

Once you decide that a variety will look right in your garden, make sure it will also grow right. A plant that needs full sun will not flower in a shady

spot. Acid-loving plants will not flourish in alkaline soil. Those that like their roots kept cool and moist will wither where it is hot and dry. Watch for unfavorable factors such as high walls, drip from overhanging eaves, low, muggy spots where there is little circulation of fresh air. If the plant requires a lot of pruning, training, and spraying, make sure you have the time to take care of its well-being. A healthy, easy-to-care-for barberry is infinitely more ornamental than a glamorous shrub struggling to stay alive.

HARDINESS

The main factor that determines whether a plant will live through the winter in your garden (and it is a factor you can't control) is minimum winter temperature. This is the basis for the newest (1960) Plant Hardiness Zone Map issued by the U. S. Department of Agriculture. Other uncontrollable factors include: How long the low temperature will last; how early and suddenly a freak freeze occurs; whether or not winter provides a beneficial blanket of continuous snow; whether there are punishing gales with driving sleet and encrusting ice.

But even in severe areas, or with plants of questionable hardiness, there are some ways we can increase the odds against winter-kill. Selection of suitable varieties, careful planting in favorable locations (never in low "pockets" where frost settles, or where water collects around roots under eaves or around walls), careful fertilizing, winter protection, prevention of attacks by insects and disease—all of the cultural principles that promote healthy, thriving growth, will help to increase winter hardiness. In other words, "The combined effects of all (climatic and cultural) factors determine the true plant adaptability."

Actually, true hardiness means more than mere winter survival. A tree or shrub is hardy when it can live through many years to maturity, and when it not only survives but develops normally and completes its annual growth cycle. For example, in some sections, very late or very early frosts will kill flower buds on some varieties. Trees and shrubs that require a period of winter dormancy are not hardy in Florida, where there is no protracted cold period. In arid areas where summers are extremely hot, dry, and windy, hardiness may depend on how much time you have to spare for special care.

If you would avoid disappointment, especially with rather expensive miniature trees and shrubs, buy the strongest, healthiest plants of the

varieties most suitable to your climate, and do all you can to satisfy each individual's cultural needs. Thus, unfavorable weather has two strikes against it from the start.

PLANTING AND CARE

No matter how careful we are, fall planting is seldom successful for us in Connecticut. But in spring planting it seems as if we can't miss. Once we were given a dozen ornamental shrubs—named varieties of *virburnum*, lilac, and two of the supposedly tricky *caryopteris*—whose bare roots had been out of the ground for several weeks. The *caryopteris* and one invincible lilac even flowered that year!

In early spring, just before they awaken from dormancy (or a little later when new growth and buds are starting to show), woody plants have more vitality than at any other time of the year. Vigorous new roots grow faster. And the plants have a full growing season to get established before winter sets in. We start planting the minute the soil dries enough to be workable, sometimes as early as March. Of course, spring-flowering varieties are safest if they are container-grown or balled and burlapped.

Early fall planting is fine for milder climates—from the time the season's growth has matured on evergreens, or when deciduous varieties begin to drop their leaves, until the ground freezes. Except for container-grown plants, winter planting is usually recommended for the South, when the plants are as nearly dormant as possible.

Bare-root plants (risky for evergreens) should be kept moist until they are put into the ground. Thoroughly soak the roots, and even the stems, in a bucket of water. The roots and soil of balled and burlapped plants should also be kept moist. Either way, protect the transplants against wind and sun until they're set out.

Dig a planting hole wide enough for the roots to spread out in without bending or crowding, with an extra six inches of loosened soil in the bottom for them to penetrate. This is the time to check on drainage, one of the most important factors in plant survival. Run some water in the hole—enough to fill it. If the water seeps away fairly fast, the drainage is probably good. If it remains for any length of time, you'd better dig much deeper and remove the subsoil. Fill in with sharp sand or fine gravel for about six inches.

Enrich or condition the topsoil that you dug out, before you replace it around the roots. Set trees and shrubs at the same level they were growing in the nursery, never more than an inch or so deeper. The burlap around a soil ball doesn't need to be loosened. It will rot away rather quickly. Fill in around the roots until the hole is three-fourths full; flood with water to settle soil in any air pockets; let it drain; and finish filling the hole until it is just level with the surface, but don't mound it up. A raised ridge around the edge of the hole will form a saucer to hold water until it runs down to the roots. Water again, slowly and thoroughly, so that the soil in the hole and some of the surrounding area is thoroughly saturated.

A newly planted tree or shrub is likely to wilt in sun or dry wind unless you provide some temporary shelter or shading. This is rather simple to do. An upturned basket over a small shrub, or a screen of light cheesecloth, or an old window curtain, or a piece of burlap suspended on stakes will suffice.

Soil

What you do to enrich or condition topsoil depends on the existing soil, and on each plant's individual preferences. Add sharp sand to heavy, clay-like soil to improve the drainage. If you have extra-sandy soil, add moisture-holding peat or leaf mold. Almost any soil is improved by mixing in liberal amounts of organic matter such as leaf mold or well-rotted manure. We seldom add fertilizer for miniature trees and shrubs.

If your soil is alkaline, and you are planting acid-loving varieties, use liberal quantities of well-rotted cow manure, acid peat, or woodsy soil from under pine, beech, or oak trees. To neutralize, or alkalinize, acid soil, use horticultural lime.

Watering and Mulching

Transplanted trees and shrubs should be watered with extra care during their first growing season. Soil should be kept moist constantly, but not muddy, to the full depth of the roots. Don't rely on light rains that moisten only the surface. During hot, dry periods, spray the foliage with the garden hose as often as possible. A light, airy mulch—buckwheat hulls, crushed

sugar cane, or something similar—will help keep the soil from drying out and keep it cooler, too.

Established trees and shrubs are kept moist during the spring and summer growing seasons. But in August and September, when the year's new growth is maturing and hardening for the winter, less watering is needed. Resume watering in October, and keep it up until the ground freezes. It is particularly vital that evergreens—both needle-leaved and broad-leaved—should never go into winter with dry soil about the roots.

Fertilizing

Feeding miniature plants too heavily sometimes can cause them to outgrow miniature size. Actually, they don't need a rich diet. An annual top-dressing of well-rotted, or dried, cow manure, or a light sprinkling of balanced commercial fertilizer, in early spring is usually sufficient. Fertilizing in late summer or early fall dangerously promotes soft new growth that is susceptible to winter damage.

Pruning

Most miniature trees and shrubs are best when allowed to keep their natural habits and shapes, and pruned as little as possible. Of course, damaged or broken branches should be cut away at planting time, or at the end of the winter. Unwanted suckers, and any growth that is out of line, ungainly, or unattractive can be removed at any time. Any other pruning should be done in earliest spring before new leaves appear. There is one exception. Spring-flowering varieties that bloom on last year's wood are pruned immediately after flowering. Fall pruning of any type can stimulate new growth that may winter-kill.

Some evergreens, for example the mugho pine, are encouraged to branch and keep more compact by breaking off half of the partially matured "candles" at the ends of the branches. Evergreen or deciduous shrubs sometimes are grown in formal shapes and, like hedges, trimmed regularly in spring and summer.

Some summer-and fall-flowering shrubs, referred to as "cut-back" or "die-back" types, are perfectly root-hardy although the tops are likely to be partly, or completely, killed by winter. But they make new growth each

spring on which normal flowers and fruit are produced. Some of these may grow too tall and awkward when the branches are killed back only part way. To keep them small and shapely, cut all stems back to mere stubs in earliest spring.

Insects and Diseases

We've never had to hover over our garden constantly with duster and spray gun, and I hope we never will. It's a dull, tiresome, unrewarding occupation—confusing and frustrating, too. There are so many different kinds of pests, with so many different habits, appetites, and life cycles, infesting so many different types of plants in so many different ways, that only an expert such as the county agricultural agent, or “plant-doctor” Cynthia Westcott, can keep them straight. And even Miss Westcott asks, “Is this spray necessary?”

Don't misunderstand. We do have pests and diseases on our garden plants; and we do fight them; and we've been known to mutter about the injustice of it all. But we try not to let them take all of the joy out of summer gardening.

Two or three times during the growing season we fill the tank of the small pressure sprayer with a solution of “all-purpose” garden spray. Everything in the garden gets the treatment, including shrubs and trees we can reach without a tall ladder.

We also wage annual war against a few familiar enemies as soon as they appear. In February we cut off twigs encircled by the brownish egg bands of tent caterpillars. When the nests appear in spring we wipe them out with rags or crumpled paper and spray the surrounding area with DDT. When cankerworms are all over the place (as they were in the spring of 1961), DDT protects the foliage of small plants, particularly our precious miniature trees and shrubs. We discovered we couldn't possibly cope with the gigantic shade trees of the woods.

We have very few Japanese beetles since we started grub-proofing all cultivated land. Any time after the ground thaws in spring, and through October, we spray or dust with chlordane. Five pounds of 5 per cent chlordane dust will treat a thousand square feet, can be bought for about two dollars, and can be applied in an hour with a good-sized duster. This

treatment is effective for three years. The few beetles that start working over the roses in midsummer quickly succumb to DDT.

The infrequent appearances of aphids are met with a dousing of nicotine sulfate solution, and the same treatment is used for some kinds of scale, although others require dormant oil spray. During one very hot summer, mites yellowed some of our evergreens. Aramite, used faithfully according to label directions, routed them. We do keep either sulfur or Karathane on hand to combat mildew on plants such as roses where it really matters. For any more complicated or unidentifiable problems, we rely on the advice of our friendly county agent and his staff.

Winter Protection

Newly transplanted or very young trees and shrubs—or any of questionable hardiness—need special protection against winter severities. Our favorite method for small specimens is to pack salt hay or leaves around them loosely and put an upturned basket on top. Somewhat taller shrubs may be encircled with evergreen branches, the ends being stuck in the ground and tied together over the tops. For groupings of shrubs, we put a burlap barrier on the windward side, especially for tender types, and sometimes make a tent of sorts. Evergreens planted where the late winter sun might burn them need shade of some sort—a lathed frame, snow fence, or the like.

PROPAGATING MINIATURE TREES AND SHRUBS

This is one of my favorite addictions. I can't resist planting any seeds that come my way, can't bear to discard an evergreen or a broken branch of cotoneaster, can't resist the impulse to root cuttings. As a result, our rather small nursery bed is closely populated with small, young shrubs and trees of all kinds. I doubt that we'll ever find space to plant them once they're grown; and I imagine I could be rather popular could I ever bring myself to give them away.

Propagating trees and shrubs is predominantly a trial-and-error proposition because each variety has its own preferred method of reproducing itself. Some are difficult to raise from seeds, some won't propagate any other way; some root readily from cuttings, some take three years; some are amenable to layering, others are not. For me this guesswork

is a good thing. If *all* my seeds and cuttings took root and grew into bushes and trees, there'd be no challenge, no fun, and no place to put them.

We always keep a propagating box handy—a shallow box or flat with sides and top of glass or polyethylene. In summer it rests under a tree; in winter it may be on an unheated porch, in the cold frame, or in the greenhouse. The propagating medium may be Pelonex or perlite; a three-way mixture of peat, vermiculite, and perlite; half-and-half sterilized leaf mold and sharp sand (peat packs too hard for me); or clean sharp sand alone. The medium is kept constantly moist but not soggy and is never allowed to dry out. Cuttings are inserted in rows, and removed for potting as soon as the roots are an inch or so long. Sometimes seed is sown in part of the box, or we may rig up a second box when we have a goodly number to plant. The glass, or plastic, sides and top preserve moisture and keep the air humid.

Seeds

Most types of shrubs and trees are rather slow to grow to maturity from seed. And seeds of miniatures produce all-miniature seedlings only when the seed-bearing parents are natural species with natural miniature characteristics. Small varieties of larger species, “sports” or mutations, can't always be counted on to produce small offspring.

Seeds vary in the time they take to germinate (three weeks to three years, depending on the plant) and in germination requirements. Some need warmth, some cool temperatures, some darkness, some light. And some need a period of dormancy or “after-ripening” before they are planted. In a process of “stratification” seeds are stored in moist sand for several months, usually at about 40 degrees or slightly less. However, some types (cedar, mugho pine) need to be frozen; a few others are stratified in warmth. Some seeds that you buy have already been stratified. The package will tell you so.

Fleshy seeds, such as chestnuts, are usually planted as soon as they ripen and fall. When seeds are covered with a fruity coat, like holly berries, for example, the berry should be soaked in water for a few days until the seeds can be squeezed free of the pulp. Hard-coated seeds need to be gently cracked, or nicked, to hasten germination, and can be sown outdoors in the

fall to germinate the following spring, or even the spring after that. Most winged seeds of conifers need to be stratified (remove the wings) in moist sand all winter at 35 to 40 degrees and are planted in the spring. Fine seeds, as for rhododendrons, can be harvested from the split pods in fall and stored cool and dry until spring.

These are, of course, generalizations. When I am in doubt about how to handle seeds, and if my reference books don't give me the information I need, I plant them several different ways. It is surprising how often they'll germinate, regardless of method.

There is, of course, one inviolable rule about planting seeds of any sort—the planting medium should never dry out from the moment of planting, through germination and up to transplanting time.

Softwood Cuttings

Take the ends of this year's growth when it is half ripe, in summer, or longer sections that can be cut into pieces with at least two buds or "eyes." Make the cut just a little below the node. Or on slow-rooting plants, pull off the cutting with a sliver of the main stem (a "heel") attached to the rooting end. Remove the leaves at the base, and dip the cut end in one of the hormone rooting powders if you wish. Check the formula on the label to make sure you have the correct one for that type of cutting, and follow the directions precisely. Insert cuttings in rows in the propagating box and firm the medium around the stem ends.

Almost all deciduous trees and shrubs can be propagated from summer softwood cuttings, and so can evergreens—both the needle-leaved and the broad-leaved varieties. Needle-leaved evergreens can also be produced from similar cuttings taken in late fall, but they seem to take longer to make roots. With mild bottom heat in the greenhouse, however, the process is fast enough so the cuttings are rooted before time to set them out in the nursery in the spring.

Hardwood Cuttings

When deciduous trees and shrubs have dropped their leaves, and this year's growth has had additional ripening from a few sharp frosts, take cuttings of the ends of branches that seem fairly thick and sturdy. Cut off

any immature tips, and trim the cuttings (of miniatures) to four to six inches long. Tie a dozen or so into a bundle with covered wire, or some similar tie that will neither rot nor injure the bark. Bury the bundles (some growers bury them standing, some in a horizontal position) so they are completely covered with soil in a cold frame, or use a box that can be kept where the temperature will be cold but not freezing. In spring, unbundle the cuttings and root them like any others in a propagating box, greenhouse, frame, or nursery bed.

Grafting

There are several different methods of attaching a stem or branch of one tree or shrub to the roots of another until the two parts grow together as one plant. So far, I've willingly left this field to the professionals, or to amateurs who are seriously interested and make a thorough study of it. I haven't found the patience or free time for it.

Layering

Here is a foolproof method that guarantees a healthy new plant almost every time. Although some root faster than others, there's hardly a tree or shrub with long or low-hanging branches that can't be layered. Of course, the safety lies in the fact that the new plant is part of, and supported by, the parent plant until it is well rooted and strongly established.

In spring, when the plant is growing vigorously, select a low-hanging branch that can be pulled down to the ground. At that spot nick the bark, or slit the branch, to encourage rooting. Pin it down with a forked twig, a hairpin, or a clothespin and cover the tip with a little soil. When roots have formed and the tip is growing lustily, cut the branch between the new plant and the parent, then pot or transplant as you wish.

Air Layering

Plants that do not have a low branch are air-layered by nicking a stem or branch three or four inches from the tip, covering the nick with a ball of moist sphagnum moss, and wrapping the ball tightly in polyethylene, which is tied tightly at both ends. Hormone rooting powder can be applied to the nick with a small brush if you wish. Make sure the moss doesn't dry out.

When you see several good-sized roots, cut the stem below the ball, remove the plastic, and transplant or pot as you wish. Rooting speed varies with different plants. Some take only six weeks; some, several years.

Air layering can be done in spring, using ripened wood of last year's growth, or in summer with new growth that is not yet woody and hard.

MINIATURE TREES AND SHRUBS

***Abelia* Caprifoliaceae**

Ornamental shrub, persistent or deciduous, often half evergreen, spreading. Leaves simple, opposite, nearly stalkless. Flowers white or pink tubular, persistent purplish sepals. Native of Asia Minor and Mexico.

CARE. Semisun, or some protection. Well-drained soil with leaf mold. Amenable to pruning. Cool greenhouse. Hardy in zones 4 and 5.

PROPAGATION. Greenwood cuttings in summer, ripe wood cuttings in autumn, layering in spring, rarely from seed.

SPECIAL USES. Foundation planting, hedges, ground cover, slopes, and as specimens; use as an evergreen in mild climates.

***Abeliophyllum distichum* Oleaceae** White Forsythia, Korean Abelia

Deciduous ornamental shrub two to three inches high. White flowers in dense clusters in May and June or in February in mild climates. Leaves like the abelia, branches arching. Close relative to the forsythia; flowers smaller but more of them. Deep-purple flower buds form in the fall.

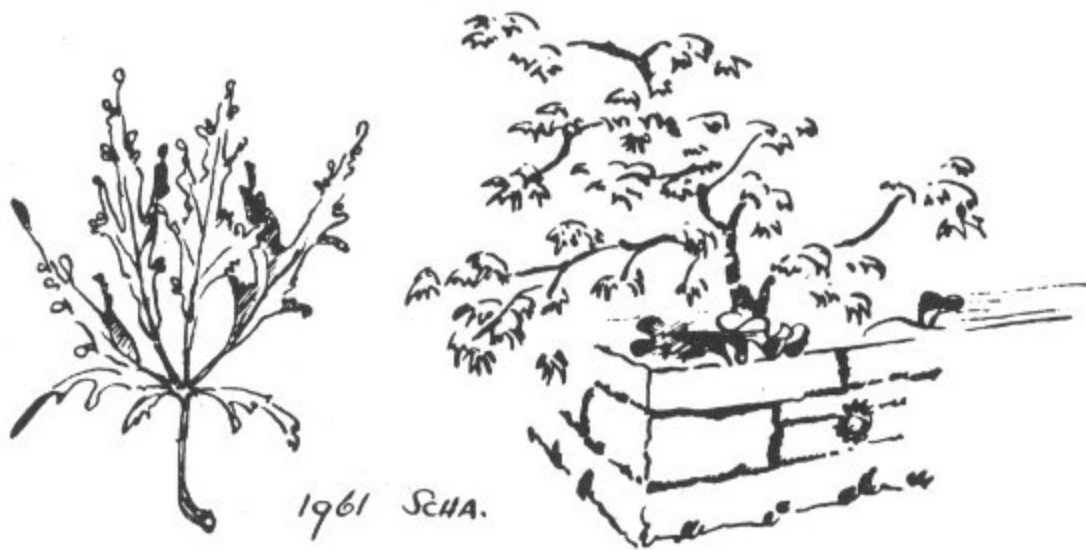
CARE. Sun, drainage, moist soil, hardy in zones 4 and 5. Too severe winter weather may kill the buds for spring. Needs some cover.

PROPAGATION. Same as for abelia.

SPECIAL USES. Same as for abelia.



Foliage details of popular miniature trees and shrubs
Chamaecyparis pisifera filifera aurea



Acer palmatum dissectum

Abies *Pinaceae* Fir

Pyramidal evergreen trees with whorls and graceful branches. Cones are erect; leaves flat, narrow, whitish on the underside with two lines, and shiny green above. It is sometimes difficult to distinguish from spruce, picea. (Spruce sheds leaves when dry, and its cones hang.) Bark is smooth and thin when young, thick and furrowed when old. Dwarf varieties are *A. balsamea nana*, *A. grandis nana*, *A. procera (nobilis) glauca prostrata*, two feet high.

CARE. Cool, humid. Hardy in the North.

PROPAGATION. Cuttings, grafting, seed.

SPECIAL USES. Bonsai.

Andromeda *Ericaceae* Bog Rosemary

Low, dwarf evergreen shrubs with narrow leaves, urn or bell-shaped pink flowers in May. Dwarf varieties are *A. glaucophylla*, less than a foot tall with leathery green leaves lightly frosted beneath, stems erect; and *A. polifolia*, a root-creeping species, the leaf margins often rolled.

CARE. Cool, very acid soil, peaty, deep acid leaf mold. Prefers bog garden; if among rocks, prepare soil to conform to above. Sun or part shade.

Very hardy.

PROPAGATION. Seeds, cuttings (mature wood under glass), layers, division.

SPECIAL USES. Rock and bog gardens.

Azalea (Rhododendron) *Ericaceae*

The azalea differs from the rhododendron only in technical points. The size varies from tree-like to almost prostrate. It flowers from April to June and is the most vivid of all shrubs. Colors pink, yellow, salmon, purple, white, red, and orchid. Some are fragrant.

CARE. Acid soil, sun. Moist, unsuited for areas where there is less than twenty-five inches of rainfall per year. Water after flowering when plant is making new shoots. Reduce watering in summer. Prune to check irregular growth after blooming season. Remove old blooms. Fertilize after flowering. Dust soil with sulfur.

PROPAGATION. Seeds; layering, in air and in the soil; cuttings. Balled and burlapped plants can be set out almost any time.

SPECIAL USES. Rock gardens, foundation planting, bonsai.

Betula nana *Betulaceae* Birch

Deciduous shrub, spreading and branching to two feet, leaves to one-half inch long. When young the leaves are sticky and fuzzy on the underside. A native of Alaska. Flowers, catkins.

CARE. Moist sandy soil.

PROPAGATION. Seeds (stratify at once), layering, green-wood cuttings under glass.

SPECIAL USES. Rock gardens, trough gardens.

Bruckenthalia spiculifolia *Ericaceae* Balkan Heath

Evergreen, heath-like shrub ten inches high. Small pink bell flowers in summer. Gives the effect of billowy daintiness.

CARE. Drainage and not soggy wet. Hardy, but mulch in zone 4 with salt hay in winter. Sun; soil gritty and acid.

PROPAGATION. Seeds and cuttings.

SPECIAL USES. Rock and trough gardens.

Buxus *Buxaceae* Box, Boxwood

Evergreen shrub, very small with clusters of small flowers. Slow-growing; in fact, boxwood I set out two years ago seems hardly to have changed in size. But we await the possibilities with anticipation and patience. It is a versatile shrub.

CARE. Little or no care seems necessary. It grows in just about any soil not sandy or rocky. It has shallow roots. We protect it in winter with a light mulch. Plant in spring or late summer; use little or no fertilizer. It accepts being pruned or sheared.

PROPAGATION. Hardwood cuttings, suckers, division, layering.

SPECIAL USES. Edging, foundation planting, bonsai, hedges, pots, and dish gardens.

Calluna vulgaris *Ericaceae* Heather, Ling

Low evergreen shrub, clusters of pinkish bell flowers in August and September, leaves very small.

CARE. Soil light, peaty, sandy, with acid leaf mold. Full sun best but light shade is acceptable. Drainage is necessary in winter but prevent dehydration by dry winds. Hardy.

PROPAGATION. Cuttings, layering, division.

SPECIAL USES. Rock gardens, foundation planting, edging, containers.

Caryopteris incana *Verbenaceae* Blue Spirea

Deciduous shrub, gray-green toothed leaves. Shrub grows to two or three feet and forms a ball the same width; firmly packed with fuzzy, fringed flowers in late summer, of blue or purplish tints.

CARE. Full sun, light soil, the sandier the better. Tolerates drought. Not entirely hardy and may be killed-back but will bloom anyhow. Prune severely in spring, making a more compact plant.

PROPAGATION. Cuttings of mature wood in August (root under glass), layering, seeds.

SPECIAL USES. Borders.

Cassiope lycopodioides *Ericaceae*

Low evergreen from two to three inches high and formed in a fat dome of the same width. Needle-like leaves are tiny and overlap like shingles. Has white bell flowers in spring.

CARE. Moist and cool, part shade, soil peaty or sandy and acid. Suffers in long hot summers or drought. Native of Siberia and Alaska.

PROPAGATION. Cuttings from mature wood in August (root under glass), layering, seeds.

SPECIAL USES. Rock and trough gardens.

Chaenomeles (Cydonia) *Rosaceae* Japanese Quince, Flowering Quince

Deciduous or semideciduous shrub with alternate toothed leaves of shiny green tone. Blooms in spring before leaves appear, in shades of white or pink. Later forms hard, quince-like fruit that is fragrant.

CARE. Almost any soil, but if fertile more flowers; drainage; partial shade produces more flowers. Top-dress in spring with garden fertilizer. Can be pruned for hedges (not too much).

PROPAGATION. Seeds (plant in spring and stratify), root cuttings, cuttings from ripe wood (root under glass), layering, grafting.

SPECIAL USES. Foundation planting, specimens, hedges (do not prune as much as privet), bonsai.

Chamaecyparis (Retinospora) *Cupressaceae* False Cypress, White Cedar

Actually this is an evergreen that grows to a hundred feet, but it also comes in a wide variety of two-foot dwarfs. Basically the tree is pyramidal and has leaves more like scales than needles. They are very dense and tight, on drooping branches. Some of the dwarf species are *C. obtusa coespitosa* (tennis ball), *C. lawsoniana*, *C. compacta*, *C. compacta glauca*, *C. nana compressa*.

CARE. Acid soil; needs moisture, due to shallow roots. Fertilize, prune some to keep shapely, root-prune for rock gardens.

PROPAGATION. Seeds from the small cones.

SPECIAL USES. Bonsai, rock gardens, specimens.

***Chamaedaphne calyculata* Ericaceae** Leatherleaf

Evergreen shrub that is a native bog dweller and therefore good for locations unsuitable for many plants. The evergreen two-inch leaves look rusty underneath and are dull brown in winter; branches are sparse. In spring it has dangling clusters of little white flowers. One-foot variety nana effective in moist part of rock gardens.

CARE. Moist location, acid soil with peat.

PROPAGATION. Seeds, cuttings from ripe wood (rooted under glass), layering.

SPECIAL USES. Bog gardens; dwarf variety for rock gardens.

***Cotoneaster* Rosaceae**

This is a small shrub; some varieties are deciduous, some persistent. Has small pink or white flowers in the spring. In fall has bright-red berries. Deciduous leaves are colored before they drop.

CARE. Sun, but will accept some shade. Cool and moist but never waterlogged. (Subject to red spider if hot and dry.) Use lime to produce a slightly alkaline soil; use loamy soil. Needs circulation of air and drainage. Prune only to remove dead wood or broken branches. Doesn't transplant readily. Plant in permanent location.

PROPAGATION. Seeds, sown when ripe or stratified by layers in autumn; cuttings of young wood in late summer (root under glass).

SPECIAL USES. Bonsai, rock walls, hedges, foundation planting.

***Cryptomeria japonica* Taxodiaceae**

This is a Japanese evergreen tree that grows to towering heights but has several dwarf varieties: lobbi, pygmaea, and nana. It has shreddy red-brown

bark and distinctive root formations above ground, making it very appealing for bonsai work. Blue-green leaves.



Cotoneaster humifusa and *juniperus sabina tamariscifolia*



Cotoneaster humifusa with flowers and berry

CARE. Protect during hot and dry summer, and from low temperatures. It is best in warmer climates. Enriched soil is not necessary but produces a more pleasing tree.

PROPAGATION. Seeds, cuttings, grafting. Seeds are stratified, and do not always come true. Cuttings are more satisfactory.

SPECIAL USES. Bonsai.

Daboecia cantabrica *Ericaceae* Irish Bell Heather

An evergreen shrub, upright and dwarf (ten to eighteen inches). The leaves are dark green but fuzzy white underneath. Purple, drooping-urn flowers, in clusters, last all summer.

CARE. Soil gritty, sandy, and peaty. Protect in winter and be careful about drainage. Mulch with salt hay or evergreen leaves.

PROPAGATION. Seeds, cuttings from ripe wood under glass.

SPECIAL USES. Rock gardens.

Daphne *Thymelaeaceae*

Some of these little shrubs are evergreen, some are not. Some of them grow very low. All of the several varieties bloom in early spring, often on bare wood. Colors vary from white to pink. Most varieties are fragrant.

CARE. Plant in a cool, moist location in light soil, drained and slightly sandy. Protect in winter with straw or salt hay.

PROPAGATION. Seeds are sown at once or are stratified; softwood cuttings after flowering; layering; hardwood cuttings in fall.

SPECIAL USES. Foundation planting, hedges, border planting, rock gardens, greenhouse forcing, ground cover, bonsai.

Deutzia *Saxifragaceae*

This is a heavy flowering shrub with a compact, plump shape. The flowers, resembling small hoop skirts or bells, cover the plant almost completely. Colors vary with varieties, white, pink, and pink with red touches.

CARE. Will grow and flower in some shade. Accepts most any soil. Needs drainage. Most varieties are hardy.

PROPAGATION. Both green-wood and hardwood cuttings, layers, seeds in spring.

SPECIAL USES. Foundation planting, hedges, greenhouse forcing.

Euonymus *Celastraceae* Burning Bush

Sizes can vary from dwarf shrubs to medium-size trees. Sometimes it is evergreen, but is more likely to be deciduous. Simple, opposite leaves; clusters of small flowers in spring. Has showy fruit which turns red in fall and opens to drop the seeds. Has brilliant red coloring in autumn.

CARE. Grows in ordinary soil and does well in shade (bright sun makes for better colors in fall). Most varieties are hardy.

PROPAGATION. Seeds (sown in spring), layering, hardwood cuttings.

SPECIAL USES. Bonsai, pots, dish gardens, rock gardens, hedges.

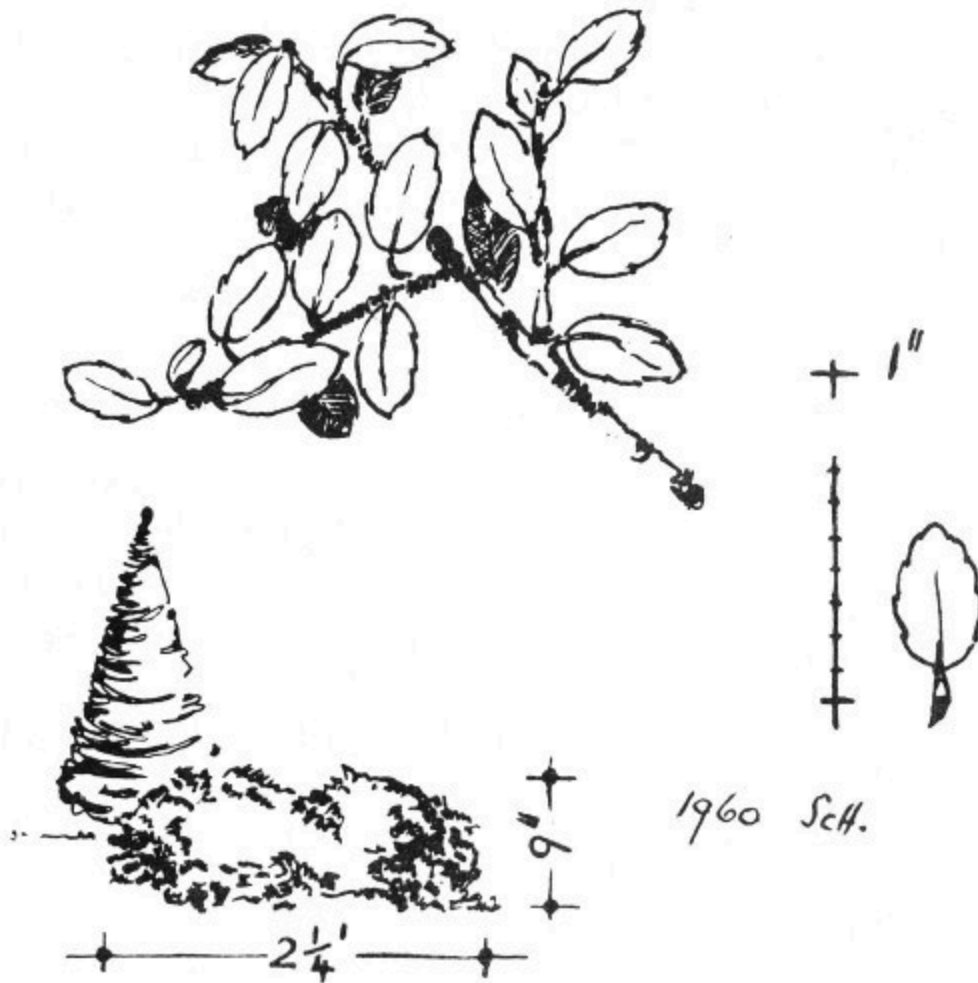
Fothergilla *Hamamelidaceae* Bottle Brush

Native American shrub, deciduous with alternate coarse-toothed leaves, hazel-like in appearance and brilliant in fall colors. During the flowering season blooms of a lustrous white resemble shaggy dust mops. It is slow-growing and adapts itself to bonsai treatment.

CARE. Seemingly, it requires little pampering. It likes moist situations. It is hardy and can be pruned to size and shape.

PROPAGATION. Seeds, layers (may take two years to root), and root cuttings.

SPECIAL USES. Foundation planting, bonsai.



Ilex crenata helleri ca. 20–25 year old spec. with *Abies glauca conica*

***Ilex crenata helleri* Aquifoliaceae Holly**

Holly may be a tree or a shrub and is sometimes an evergreen. The waxy green leaves are known to anyone who has ever seen a Christmas wreath. The red berries are almost as striking. Most varieties grow slowly.

CARE. Most like rich well-drained spots. They need acid soil or the leaves turn yellow. They are hard to transplant and it should only be tried in the spring before growth starts. The deciduous varieties are easier to move, but are not so hardy. If you are buying a tree, get nursery stock that has been freshly dug. Mulch, but not near the trunks in winter because of mice. If you desire the red berries, spray with hormones. Prune yearly for shape, in winter.

PROPAGATION. Seeds (stratify, slow to germinate), cuttings of young ripe wood (root under glass).

SPECIAL USES. Hedges, rock gardens, containers, foundation planting, bonsai.

Juniperus *Cypressaceae* Juniper

Evergreen trees and shrubs with needles or scale leaves. Can be tall trees or prostrate shrubs that hug the ground. Foliage is a beautiful blue-green.



A beautiful specimen of *Juniperus procumbens nana* with pansies in a corner planting

CARE. Moderately moist loamy soil. Some (*J. communis*) thrive in dry spots amid rocks. Most are hardy. Many varieties are exceptionally hardy

and accept sun or shade, city dust, fumes, and smoke. All like drainage and wide spacing.

PROPAGATION. Cuttings, seeds, layering.

SPECIAL USES. Rock gardens, foundation planting, containers, bonsai.

***Picea abies* Pinaceae Spruce**

Pyramidal, evergreen, coniferous trees native to the cooler parts of the country. They have four-sided leaves that fall easily, leaving cones that usually droop. Although many varieties are tall and majestic, there are some of the most shapely dwarfs in the family. They grow so slowly they are desirable for small landscapes. Among the dwarfs are: *clanbrasiliana*, *gregoryana*, *maxwelli*, *nana*, *nidformis*. The latter, with its dark-green needles and flat open top, is also known as the bird's nest spruce. It is less expensive because it grows very easily from cuttings.

CARE. Hardy, tolerates more wet than firs or pines, and is at home in most soils.

PROPAGATION. Seeds and cuttings.

SPECIAL USES. Foundation planting, hedges, bonsai.

***Pinus* Pinaceae Pine**

Among the many familiar pines of ornamental interest there are several dwarf forms for limited landscapes. All bear needle-leaves in typical bundles of two, three, or five according to type. Here are some possibilities: *P. cembra chlorocarpa*, *P. mugo (montana) mughus*, and a variety of the white pine, *P. strobus nana*.

CARE. Light, dry, sandy soil. Plant where sheltered; for example, a northern slope. Dry summer winds and unseasonable, warm winter sun will turn the leaves brown. In transplanting trees, remember that the roots do not stand exposure to air. Take the biggest possible chunk of dirt with them. In buying, get stock that is balled in burlap.

PROPAGATION. Seeds and grafting.

SPECIAL USES. Bonsai, containers, rock gardens, foundation planting.

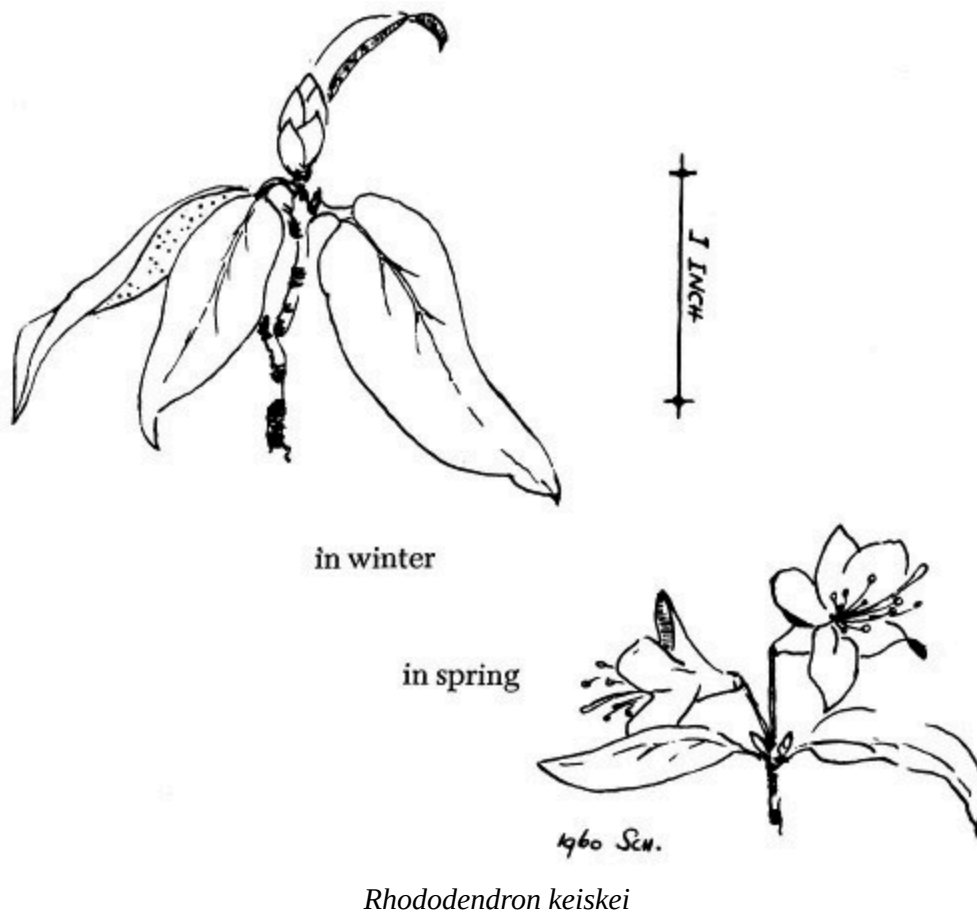
Rhododendron *Ericaceae*

Some of the most charming of our truly miniature shrubs, far too infrequently known and grown. *Rhododendron* species and hybrids may be under a foot tall at maturity and have leaves as small as a little fingernail. The only thing out of proportion is the size of their flowers. Here are some of them: *cantabile*, *chryseum*, *cinnabarinum*, *cuneatum*, *didymum*, *fastigiatum*, *ferrugineum*, *flavidum* (*primulinum*), *glaucum* (*glaucophyllum*), *hypolepidotum*, *impeditum*, *kotschyi*, *radinum*, *riparium*, *russatum*, *williamsianum*.

CARE. Humus-rich soil, sandy and of open texture. Mulch for winter season and water before freezing weather. Withered leaves in the dead of winter mean lack of moisture as well as cold weather.

PROPAGATION. Seeds, softwood tip cuttings (in June).

SPECIAL USES. Bonsai, rock gardens, edging, foundation planting.



Rhododendron keiskei

Sciadopitys verticillata *Taxodiaceae* Umbrella Pine

The five-inch needles of this slow-growing Japanese evergreen spread out from new tip growth like the ribs of a half-open umbrella, giving it an airy distinction. The tree, a squat pyramid, with short branches and a stubby main trunk, is ideal for bonsai cultivation.

CARE. Grows well in most any soil but does better in one that has been enriched. As it is slow-growing, there is slight danger that it will outgrow its dwarf proportions. Transplant only when young and do not expose the roots. Protect from severe winter conditions.

PROPAGATION. Seeds.

SPECIAL USES. Bonsai, rock gardens, specimens.

Spiraea *Rosaceae* Spirea

This popular deciduous shrub varies as to size, with most varieties far out of the miniature classification. However, there are some short ones, with *S. bullata (crispifolia)* an out-and-out midget of twelve to fifteen inches. It has rose-colored flowers in midsummer. *S. arguta (compacta)* is slightly taller. In May it has garlands of fragrant white flowers. *S. bumalda*, with deep-pink flowers in early summer, grows to about two and a half feet. Most others are considerably taller.

CARE. Almost any soil is acceptable providing it is not too heavy and has drainage. However, plenty of moisture is essential. Plenty of sun is necessary for good flowering. Pruning also promotes flowers, but do it after the blooms have fallen. Transplanting is easy, even with bare roots. Fertilize occasionally.

PROPAGATION. Seeds (ripe or stratified), cuttings of green or hard wood, layers.

SPECIAL USES. Borders, rock gardens, hedges, foundations.

Taxus *Taxaceae* Yew

This evergreen shrub is very valuable to those who garden in smaller-scale plots and grounds. There are a number of extremely low and slow-growing varieties available at most nurseries and in several different forms. All have the typical yew masses of glassy, evergreen needle-leaves and satiny red, berry-like fruit. All accept shearing, even need it to preserve their symmetry and small size. It is often difficult to differentiate between the many varieties when they are young, so be sure to check them when buying.

baccata—English yew—A tall tree but available in these small forms: *adpressa*, a broad, dense bush with half-inch needles in thick masses; *repandens*, a flattish, low, and spreading form with drooping branches, two to four feet tall.

canadensis—American or Canadian yew—Three to four feet tall and for planting in the shade. Suffers from unusually warm winter sun. Its variety *stricta* is neat and upright and stays under two feet in height. The green needles are yellow-tinged when it is young.

cuspidata—Japanese yew—This variety is recommended for Northeastern planting. The slow-growing variety *densa* is plump and rounded and twice as wide as its four-foot mature height. Another variety, *nana (brevifolia)*, is considerably larger, horizontal, and spreading in habit.

cuspidata nana—This species slightly different, has a deeply fissured trunk with red berry fruit and very dark evergreen leaves. It is highly regarded as a miniature and excellent for bonsai work.

stricta (fastigata, hibernica)—An upright column that tops dwarf proportions sooner than most others.

CARE. Slightly acid soil. Fertilize frequently in early spring and early summer. Good drainage. Don't plant under rain gutters. Top-dress with old manure. Easily transplanted. Shear just before spring growth starts, to control size and shape. Winter sun may burn needles. Comparatively free of disease.

PROPAGATION. Seeds (stratified), cuttings.

SPECIAL USES. Foundation planting, specimens, hedges, rock and wild gardens, tub planting, bonsai.

Teucrium chamaedrys *Labiatae* Germander

This is an aromatic Old World shrublet that is adaptable to many uses in the garden, whether formal or informal. Most varieties are less than a foot in height. It flowers in late summer. Many gardeners raise it for its decorative value and snip off the blooms. It is hardier than boxwood and less costly.

CARE. It grows in almost any soil but requires good drainage and full sun. Prune the top and sides twice a year for formal effect. Give winter protection with evergreen boughs.

PROPAGATION. Seeds (good but slow), division, cuttings.

SPECIAL USES. Beds, edging, rock gardens, miniature hedges, foundations.

Thuja *Pinaceae* Arborvitae

Certain species of the "tree of life" are majestic monsters. But there are a number of shrubby varieties of delicate value in small plantings. All have characteristic scale-like leaves arranged along fan-shaped branches, making fluffy sprays. But shapes vary from balls to pyramids. Foliage may be light, or blue-green, or even golden. Smaller forms are:

occidentalis—This species has two small varieties, *compacta* (slow-growing, dense pyramid) and *ellwangeriana* (a low round mound-shape).

minima—A very small and a slow-growing ball.

pumila—'Little Gem'—Dense-leaved and globe-shaped.

pygmaea—A two-foot globe or ball.

CARE. Cool, moist location. Rich soil and leaf mold. Either full sun or half shade. City soot harmful.

PROPAGATION. Same as for conifers.

SPECIAL USES. Foundation planting, rock gardens, tubs and other containers, specimens.

Tsuga canadensis *Pinaceae* Hemlock

This graceful evergreen conifer has slender horizontal branches with small leaves and small cones. Not all nurseries have it in its few dwarf forms. Best known of these small varieties are the conical compacta, and the completely prostrate, creeping prostrata.

CARE. Slightly acid soil, organic fertilizer, shade-tolerant, hardy (but protect from cold winds), can be pruned.

PROPAGATION. Seeds (stratified), cuttings.

SPECIAL USES. Foundation planting, hedges, specimens, rock gardens, bonsai.

Viburnum *Caprifoliaceae* Snowball, European Cranberry Bush

An ornamental shrub. Most varieties are deciduous, and are valued for their heavy flower clusters, fragrance, autumnal color, and attractive fruit. Sizes and colors of flowers vary.

carlesi—Korean snowball—Fragrant pink buds that open as white flowers about the same time the foliage appears.

davidi—About three feet tall with handsome crinkled leaves sharply creased at the veins. Small clusters of white flowers in June; blue fruit in the fall.

nanum—A real dwarf popular for low hedges and rock gardens. It is strictly ornamental, with no flowers or fruit.

CARE. Any soil is acceptable. Moisture required. Hardy.

PROPAGATION. Stratified seeds, cuttings (hardwood or green wood under glass), layers, grafting.

SPECIAL USES. Rock gardens, hedges, specimens.

MINIATURE PERENNIALS AND ROCK PLANTS

Our odd piece of Connecticut countryside is shaped rather like the blade of a paring knife. We cleared the point to make a wild-flower garden. At the blunt end, on one side, a tree-lined bank dips down to the road. On the other, there is an old, low rock wall that's not particularly beautiful, nor is it suitable for dry-wall planting. When we finally whipped the central lawn into shape, we realized that a flowering border would be nice against the wall—a low, narrow border with colorful bloom all summer long.

To be in proportion and harmony with the setting, the border had to be completely in miniature—following accepted principles for selecting and arranging plants, but executed in small scale. Finding the proper plants has been a challenge; arranging them has been a trial-and-error proposition, because many varieties are entirely new to us. But last summer, the little border began to look the way we had visualized it—taller specimens in the background, ground-hugging cushions facing them down in front, small groups of straight spires for accent at strategic intervals. It will probably take another season or two to perfect the blending and contrast of flowers and foliage, and to get the most colorful and constant succession of bloom.

This is not, of course, our first acquaintance with miniature perennials; and it couldn't possibly be our last. They combine beautifully with small shrubs, show off to advantage in small rock gardens, add an artistic finishing touch to garden pools, birdbaths, and other ornaments. They're at their very best in outdoor planters and raised beds on patios and terraces, and in ribbon edgings along walks and walls.

Although they're precious in flower, many varieties—candytuft, lavender, sempervivum, thyme, plumbago, to name a few—are equally decorative without bloom. This is a valued advantage when planting space is limited. Gardeners with limited time use perennials lavishly because, once established, they're mostly permanent, and they require a minimum of care. Somehow, I sympathize with a writer who found more pleasure in spring's tufts of fresh new growth than in the full bloom of summer. Nostalgically, one welcomes the return of an old friend; practically, one

knows it will tend to its business of growing and flowering with very little personal attention.

This leads us to the question, “How perennial *is* a perennial?” and to the obvious answer, “It depends....” To begin with, it should live at least three years in your garden. After that, some varieties need to be lifted and divided into smaller plants with fresh new vigor; some may need to be replaced completely; some few may grow on indefinitely without renewal, or may replace themselves with seedlings that are often of doubtful value.

In the language of the catalogues, a “hardy” perennial is one that can withstand fairly severe winter cold, and may not be hardy in Southern gardens where it is not frozen into dormancy. Most hardy perennials are “herbaceous”—the stems usually die back to the ground each winter; some few have stems that may or may not be somewhat woody and persistent. A “tender” perennial is likely to be killed by cold weather, and is grown as an annual in the North, if it is grown at all.

The life cycle of a biennial extends over two years. It is grown from seed, may or may not need special protection to carry it over a cold winter, produces flowers and seeds the second year, then dies. Some biennials sow themselves with such faithfulness they’re regarded as perennial. The comparatively few biennials among flowering plants are grown much like perennials; miniature varieties are covered in this chapter.

Originally, I was firmly resolved to admit as miniatures only those perennials that grow less than eight inches high. But I couldn’t resist stretching the limits an inch or so to allow for a ten-inch flower stem above a small, flat rosette of leaves; and I found that the height limitation allowed free entry to some rambunctious spreaders that could quickly ruin a miniature garden. So I finally settled on biennials and perennials that are miniature in general appearance and habit, without strict measurements, and with flowers and foliage in suitable proportion for miniature gardens.

For the first year or so, locating some of the small perennials we wanted was rather like a treasure hunt; we never knew where we’d find a clue. Occasionally, we’d spy a prize on a visit to a local nursery. Frequently an advertisement or a tip from another gardener would give us the name of a promising mail-order supplier. Those who specialized in rock-garden plants proved to be particularly fruitful prospects. Often, we started our plants

from seeds and discovered later that mature plants were available, had we known it. Now, we have quite a list of sources for miniature perennials which we happily share with you, beginning on page [259](#).

In selecting suitable varieties, decorative effect is naturally the first consideration—size, habit, and blooming season; color and texture of flowers and foliage; how the plants fit into your design and planting plan. Equally important is hardiness; not only ability to survive a cold winter, but also adaptability to other climatic conditions. Some perennials simply will not grow where summers are hot and dry; others will live but can't mature enough to flower where nights are cool or growing seasons are short.

Like other types of plants, perennials should be selected according to their cultural needs, and should be planted only where they can get the amount of sunlight and moisture they need, and where the soil is suitable or can be made so. An acid-loving plant in neutral or alkaline soil is a poor, short-lived thing; a moisture-lover withers pitifully where it's dry as a desert most of the time; a "hardy" perennial won't be hardy unless its environment is to its liking.

PLANTING AND CARE

Unless it's utterly impossible, we do all our planting in the spring, so there is plenty of time for plants to develop vigorous root systems before a mean Connecticut winter comes along. This is particularly important to shallow-rooted perennials, and to any that are planted in windy, exposed sites. But for us, it also includes spring-flowering varieties. In spring we've planted dormant roots with or without mere nubbins of new growth, and plants that were fully leaved out and bursting with buds, with very little loss. But fall planting has proved to be a gamble; and if we're forced into it, we're more likely to find room in the cold frame for the new arrivals until spring.

Of course, this is not usually necessary in more temperate sections where plants set out in late summer and early fall have plenty of time to become acclimated before heavy frost. And it can be all wrong for Southern gardens, where planting during deepest dormancy is often recommended.

Even if you're planting only three little pinks in a nook the size of a lady's handkerchief, soil should be dug deeply and be suitably conditioned

or enriched, drainage should be made perfect, roots should be spread out carefully, and watering should be as thorough as if you were planting a priceless miniature tree or shrub. To prevent a serious setback from wilting, put up some sort of temporary protection against sun and wind—an encircling screen of leafy branches, even a tent of newspaper or old sheeting.

Soil

There are few soils that can't be improved by the addition of organic matter such as leaf mold, manure, or compost; and few plants that won't grow better for it. It helps hold moisture in sandy soil, improves texture and increases drainage in clay-like soil, enriches ordinary garden loam. Mix it thoroughly with the soil before you place the plants. Also before planting, add lime to acid soil for plants that need it, acid peat for acid-loving varieties if your soil is neutral or alkaline. Above all, don't plant anything (except for a few bog inhabitants) where water stands in puddles. Poor drainage has killed more perennials than Old Man Winter himself.

Watering and Mulching

Most miniature perennials naturally have rather shallow root systems, so they suffer quickly from drought. Don't let them dry out seriously (particularly after transplanting); water often and deeply enough that the soil below the roots is wet. Thorough watering in late fall, just before the ground freezes, often makes the difference between winter-survival and winter-kill.

Mulching with any available, airy material—buckwheat hulls or crushed sugar cane, for example—will help keep soil moist and will also discourage germination of weed seeds, thereby saving you a bit of work.

Fertilizing

Most miniature plants don't like or need a heavy diet. A light top-dressing of balanced fertilizer, in early spring, is usually enough to nourish them without forcing soft and straggly growth. Reluctant bloomers may need a small ration of superphosphate or a booster drink of liquid-manure "tea" as flowering time approaches.

Pruning and Grooming

To keep them compact and attractive, some varieties should have growing tips pinched out once or twice in spring and early summer; others may have a few long, straggly stems to be removed; some carpeting types should be sheared off after flowering. Always pick off faded flowers unless you have reason to want the seeds.

In fall, when foliage is frostbitten, cut back the old stems and take them away, along with any fallen leaves or other debris, to be burned. Many diseases and insects winter over in decaying vegetable matter.

Insects and Disease

Our trusty duster or sprayer, filled with an all-purpose insecticide-fungicide formula, gives all our garden plants a preventive treatment several times a season. So far, this has been enough to keep problems and pests away. For some special infestations or epidemics, we keep a few specific remedies on hand—sulfur for powdery mildew, and Aramite for mites, for example.

Winter Protection

In our area we never know whether winter will bless us with a constant covering of snow, or the ground will be bare and exposed to alternate freezing and thawing that “heaves” tender roots up out of their beds. After the ground is frozen two or three inches deep, we cover sleeping perennials with a light blanket of salt hay or, sometimes, evergreen boughs. Crowns that stay evergreen are surrounded by a collar of sharp sand. Questionably hardy varieties are lifted and moved to the cold frame.

In spring, as the weather begins to warm up, we remove the protective mulch a bit at a time. There’s a fine line between taking it off prematurely, thus exposing new growth to a late freeze, and leaving it so long that the soil underneath gets soggy and the roots rot. But, rot can be fatal, and frozen tips of new growth are not, so we remove the winter covering as fast as we dare.

One item in our Connecticut landscape that's completely out of scale with its surroundings is the monstrous cold frame near the back boundary line. The cement-block wall goes down below the frost line, and up high enough to make room for winter storage of fairly good-sized plants. The discarded storm windows are hinged across the back and completely removable in summer, when they are replaced by light wooden frames of the same size with laths nailed a lath-width apart. The construction slants toward the south, to make the most of all winter sunshine; light shading is necessary in summer to protect tender seedlings and rooting cuttings.

The cold frame serves dozens of purposes and has more than paid for itself with plants it has protected or produced. When we plant perennial seeds in the cold frame, we throw a piece of burlap across the top and keep it moist until they germinate. Tender seedlings spend their first winter within its walls, and so do newly rooted cuttings. Questionably hardy perennials or any that we acquire in fall are held over until spring. Every year, it seems as if we take more out of the cold frame than we put into it!

Other, smaller, portable devices are equally useful for all kinds of summer propagating. Low square or rectangular wooden frames can be set over an area of prepared soil and the top covered with glass or polyethylene to keep the soil from drying out. An empty fruit crate from the grocer can be equipped with a glass or plastic top. A few cuttings can be rooted in soil in some shady spot with a clear glass jar inverted over them. There are many devices that keep soil moist and air humid while seeds germinate or cuttings root. How large or elaborate yours should be depends on how much propagating you want to do.

Seeds

Many hard-to-find miniature perennials can be easily grown from a twenty-cent packet of seed. You can also harvest seeds from your own plants; but only natural species will "come true." Complicated hybrids will have unpredictable offspring, most of them not particularly desirable.

We plant seeds of most biennials and perennials in June, when the soil is comfortably warm and the seedlings will have the whole summer to grow large and lusty. Some of our own seeds that ripen in midsummer are planted as soon as we can harvest them; those that mature later are usually stored on

a cool shelf in the cellar in plastic boxes or little pill bottles that keep them dry.

Soil for the seed bed is sifted to remove pebbles and debris, and mixed with equal quantities of sharp sand and peat or sieved leaf mold. To prevent “damp-off,” a fungus that chokes off stems at the soil line, soil should be sterilized if at all possible. Small quantities may be baked in the oven. Special easy-to-use fungicides are also available; follow label directions.



Rocks, water, and planting—an effective combination of all the elements of a rock garden. (*Mr. and Mrs. Norman Cherry*)

Most seeds are covered to the depth of their diameter; very fine seeds are merely firmed down into the soil. Seeds with hard coats may be gently nicked with a knife or soaked in water for a day or so. Some seeds, like primroses, need to be treated as if they had wintered outdoors before they

will germinate. You can put them in moist sand in a small, tight container and alternate freezing in the ice compartment and thawing in the warmer part of the refrigerator for several days. Or you can plant these seeds in a small box or flat of their own and leave them outdoors, in a protected spot, where winter will supply its own natural conditions.



Cold frame planted and ready for top made from discarded storm windows

Some seeds germinate faster in the dark, some with light; some like cooler temperatures than others; some come up in seven days, some take months or even a year. Seed packets usually supply pertinent information.

Seeds will not germinate in dry soil, or if allowed to dry out even temporarily during the critical period. To avoid washing out fine seeds, water gently with a fine mist, or set the flat or other container in water up to

the level of the soil inside. When the soil surface looks shiny and moist, remove the container and set it aside to drain.

When the first “true” leaves are of fair size, transplant the seedlings to peat pots, or flats, or rows in a prepared bed or cold frame. Shade against sun and wind until they resume growth.

Stem Cuttings

Many perennials can be propagated easily and in quantity by rooting softwood cuttings like those of trees and shrubs. Tip cuttings three or four inches long are usually best, with all the flower buds and the lower leaves removed. Some types, such as dianthus and lavender, root faster and more surely if the cutting is taken with a heel.

For spring-flowering varieties, take stem cuttings after flowering has finished and up until midsummer; for those that bloom later, take cuttings in May or June.

Division

This is a wonderful way not only to increase perennials, but also to keep them young and vigorous. How often you divide them depends on each plant’s individual performance. When growth is so thick it seems to be choking itself, when there’s little healthy new growth and a poor show of bloom, when a clump or crown becomes tough and hard in the center, it’s usually time to refresh the plant by division. Some plants need it almost every other year, some not for many years.

How you divide a plant depends upon how it grows. If there is a central crown of leaves, it can be cut cleanly into two or three sections, each with its own share of roots. Or there may be new, young crowns clustered around its edge that can be pulled or cut off. If the plant simply has a crowded colony of stems and fibrous roots, you can often pull it apart gently into several pieces. Or if it’s the type that sends out rooting runners, you can sever these and replant them. Generally, the old, tired center of the plant is discarded.

In cold climates, even spring-flowering perennials are most safely divided very early in the season, when new growth is beginning. Cool,

moist spring weather favors rapid recovery from the operation, and there is plenty of time for the new plants to mature before fall. Summer-flowering and fall-flowering types are also divided in earliest spring. In humid climates the spring-bloomers can be divided in August or early September.

Layering

Almost any perennial with low or low-hanging branches can be propagated by pinning a stem to the soil, several inches from the growing tip. When new growth appears, cut off the rooted stem and transplant it. This propagating method may not produce great numbers of new plants at one time; but it is surely one of the easiest and least troublesome practices—and particularly safe because the new plant is supported by the parent until it is well rooted.

I've been intrigued with one good gardener's method of creating a low hedge from one plant of dwarf lavender. She layered one stem at each side of the original plant. When these two new plants were fairly mature (but not cut away from the old plant), she layered one stem of each. By repeating the process, and locating each new layer in a straight line with the last one, she can extend her hedge as far as she wishes and plant it as she goes, along any lines.

ROCK GARDEN PERENNIALS

Achillea Compositae Yarrow, Hilfoil

Resembling the field yarrow, but dainty, mottled, and tufted. The leaves, some finely cut and ferny, wear thick wool coats. The saucy flowers are in heads, some flat-topped and not unlike daisies.

ageratifolia aizoon (*Anthemis aizoon*)—Six-inch mat of silver-haired, uncut leaves topped with bright white flowers in May and June.

clavennae—Tufts of hoary, fine-cut leaves less than a foot high, tight round white flower heads three-fourths of an inch across in May and June.

tomentosa—Best-known little species, quickly makes a thick mat of semi-evergreen gray-green leaves six inches tall. It has sunny yellow flowers from June to midsummer. The variety *aurea* has more golden flowers and blooms earlier; *nana* is a true midget and makes a thick carpet studded with many white flowers. Five-inch 'Moonlight' has flowers of sulfur yellow, and greener foliage.

umbellata—Four-inch mounds of fuzzy, silvery, ferny leaves. It has cream-white flowers in June and is evergreen in temperate areas.

CARE. Any soil, even sandy or poor. Dryish to moist. Sun or very light shade. Easy to grow.

PROPAGATION. Division in spring or fall, seeds (flowers the second year).

SPECIAL USES. Rock plantings, dry-wall planting, edging, ground cover, pavements.

Aethionema Cruciferae Stone-Cress, Persian Candytuft

This is a heathery little shrub-like perennial with slim leaves and steely-blue needles, cheerful clusters of flowers at the stem tips in May and June. It is often compared to a minuscule daphne.

armenum—Neat tufts of short, sharp leaves and petite pink flowers in June. It is never over four inches high.

iberideum—Low, crowded, and cushiony-minute with gray-green leaves and large white flowers in short clusters. This one is earlier than other varieties, blooming even in April if the weather is favorable. (Needs gritty soil with some lime.)

schistosum—By comparison this one is a giant growing to eight or ten inches, with two-inch leaves and fragrant pink flowers.

CARE. Soil light, sandy, well-drained, and gravelly; dry; sun. Hardy in southern New England. Give protection in severe winters.

PROPAGATION. Cuttings in summer, division and seeds in spring.

SPECIAL USES. Rock gardens, dry-wall planting, edging, borders.

Ajuga Labiatae Bugle

This is a low, creeping or spreading perennial with pinwheels of rather coarse leaves and small blue flowers which come in May and June. It is of miniature height and should only be planted where you can control it, or want it to spread. Some of the better-known varieties are:

genevensis—Oval toothed leaves, light green. Flowers deep blue.

pyramidalis—Less likely to spread and stays neat and small. Flowers blue and slightly larger than those of other varieties.

reptans—Also known as carpet bugle, it is nearly supine, with stems that root as they creep. Flowers blue or purplish. Variety *alba* has white flowers; *variegata*, foliage marked with cream and pink. *Rubra* pink flowers.

‘Silver Beauty’—Compact and ground-hugging. White markings on leaves give a silvery appearance. Blue flowers.

CARE. Ordinary soil, even poor; sun or part shade; dry. Plant in fall or spring six inches apart. Easy to grow.

PROPAGATION. Seeds in spring, division.

SPECIAL USES. Rock gardens, dry-wall planting, ground cover, edging.

Alyssum *Cruciferae* Madwort

Plushy, mat-forming, like a silver-gray rug with flecks of white or golden flowers in spring and summer. The grayish leaves are small, the flowers in thick clusters. Popular forms are:

alpestre (*serpyllifolium*)—Three-inch gray carpet with bright-yellow flowers through June.

montanum—Fragrant gold flowers in June; about eight inches high.

mulfenianum—Smaller (four-inch) version with balls of yellow flowers from May to June.

saxatile—basket of gold—Blooms in May; nine inches high; spreads thickly. Variety *compactum* is more compact and shorter; about eight inches. Variety *citrinum* has lemon-yellow flowers.

spinosum—Dense growth, shrubby and spiny, about eight inches high. White flowers sometimes tinged with pink, in June and July.

CARE. Any ordinary soil with drainage. Sunny, open location. Plant in spring or fall six to eight inches apart.

PROPAGATION. Seeds in fall, division of roots in spring.

SPECIAL USES. Rock gardens, borders, dry walls, pavements, edging, carpet for bulbs, ground cover.

Anemone *Ranunculaceae* Windflower

One of the earliest to bloom in the spring, it has lacy leaves and colorful flowering saucers not made up of petals but of sepals (leaves that encircle the flowers at the base).

apennina—Tuberous species six or seven inches high with deeply cut leaves; arrives in March with bright, sky-blue flowers. Variety *alba* has white flowers; *purpurea*, rich lavender-rose.

blanda—Resembles *apennina* but is slightly larger and has darker flowers.

nemorosa—European wood anemone—Similar to American forms. Variety *alleni* has lavender flowers. Variety *rosea*, pink.

palmata—Six-inch dwarf with whorls of leathery leaves, golden-white flowers in May or June. The flowers of the variety *albida* are heavenly white with gold centers.

pulsatilla—pasque flower—A more robust alpine growing to eight inches tall. Hardy even in Maine rock gardens. Flowers are purple-blue bells that come early and last as long as the weather is cool. Its finely cut leaves are softened with silky hairs. Other varieties are *alba*, white; *rubra*, plum red; and *camla*, silvery lilac surrounded by white.

CARE. Rich sandy loam with leaf mold. Part shade for some varieties.

PROPAGATION. Seed in late fall, root division in early spring.

SPECIAL USES. Wild gardens, rock gardens, forcing, flower beds.

Antennaria Compositae Pussy-Toes

This is a furry little creeper with flat, silvery, spoon-shaped leaves and bristly flowers sometimes dried as everlastings.

dioica—Fast-creeping carpet with pink-tipped flowers in spring.

microphylla (parvifolia)—Western native, slower growing, with white flowers.

neodioica—Eastern native that makes a low mat of silver-hairy leaves and sends up pink-tipped white flowers in early summer.

CARE. Sandy, even poor, soil; full sun; dry. Cut off leaves after flowering to make neater plants.

PROPAGATION. Seeds and division.

SPECIAL USES. Rock walls and pavement.

Aquilegia Ranunculaceae Columbine

Hardy little perennials with two-or three-part leaves like clover but often cut or indented on the edge, and drooping or noddling flowers with flaring sepals and spurs of different sizes and shapes.

akitensis—Six-inch Japanese doll with stemless blue-green leaves and large lavender-blue flowers with yellow at the center. June flowers.

discolor—Alpine with glowing blue flowers, white inside and frosty-green leaves. Five inches tall.

flabellata nana-alba—fan columbine—Three divided leaves like blue-green fans, lustrous white flowers in May.

jonesi—Diminutive, two to three inches tall. Small leaves in small mounds, flowers upturned and deep blue in June.

saximontana—Alpine with two-inch tufts of crisp leaves; aquamarine flowers with yellowish sepals on four-inch stems in April.

CARE. Soil light and sandy, with leaf mold; drainage; shade or semishade. Hardy. Plant in fall.

PROPAGATION. Seeds in the spring (flowers the following year), division in the spring.

SPECIAL USES. Rock and wall planting, wild-flower gardens, beds.

Arabis Cruciferae Rock Cress

Mat-forming perennials with blankets of hairy leaves under spreads of flowers rather like stocks or candytuft. They flower in spring or summer and are often fragrant.

albida—Unless the form is marked “compact” the species may top ten inches. Leaves green and woolly, flowers white (in April and May). Variety *flora plena* is about six inches tall with double white flowers; *rosea*, orchid pink and single; *variegata*, white-marked leaves. ‘Pink Charm’ is single with bright-pink flowers.

blepharophylla—Clusters of deep-green leaves with eyelashes on the edge; pink-purple flowers in April. Height, four inches.

procurrens—Matting plant with creeping stolons, white flowers in May. Dwarf variety, *stari*, spreads slowly and stays under four inches.

CARE. Medium, even poor, soil; warm and sandy. Not too moist. Sun or light shade. After flowering cut back straggly stems.

PROPAGATION. Division (in spring or fall), seeds, cuttings.

SPECIAL USES. Rock gardens, wall planting, ground cover, edging.

Aster alpinus Compositae Rock Aster

Small leaves mat closely to make a mound four inches high. The flat lilac daisy-like flowers come in June and July on eight-inch stems. The variety *superbus* is slightly larger; *albus* has white flowers. ‘Goliath’ is a variety with flowers ranging from yellow-centered lavender to near-purple. ‘Spring Charm’ is a midget with lavender-pinkish flowers.

CARE. Any good soil, full sun.

PROPAGATION. Seeds in the spring (flowers the next year).

SPECIAL USES. Rock gardens, borders, edging.

Bellis Compositae Daisy

Miniature daisies of cheerful charm, some varieties barely six inches tall.

perennis—English daisy—Nests of green leaves sprinkled in spring and summer with flowers of white, rose, or red. Several new strains are now being offered in singles, doubles (some have quilted petals), and varied colors.

rotundifolia caerulea—alpine daisy—Soft-blue flowers all summer long.

CARE. Fertile soil; moist, but good drainage; sun or semisun. A cool summer produces larger flowers. Not suitable for hot climates.

PROPAGATION. Seeds (flowers following year), division in cool weather.

SPECIAL USES. Rock gardens, edging, ground cover.

Campanula *Campanulaceae* Harebell, Bell Flower

These are the beloved bell flowers which should be included in all gardens. Included on page [264](#) are varieties that are easy to get and grow.

CARE. Light loamy soil, slightly on limy side; drainage. Sun, except dwarfs, which take light shade. Remove dead flowers. Shelter with evergreen boughs in severe winter.

PROPAGATION. Division in early spring or fall, seeds (plant as soon as ripe), cuttings.

SPECIAL USES. Rock gardens, walls, borders, edging, pots, and baskets.

Cerastium *Caryophyllaceae*

Perennial creepers and carpeters with slim silk-hairy leaves and showers of white flowers with an over-all effect of soft fuzzy wool. Maximum height six inches, but most plants less.

alpinum lanatum—Smallest downy leaves, white flowers in clusters in June and July.

bierbersteini—Longer leaves, flowers one month earlier.

tomentosum—snow in summer—This, the most popular species; quickly forms a large mat like a silver carpet under starry white flowers in May and June. It is said it will grow in pure sand.

CARE. Ordinary soil, sun, dry. Easy to grow.

PROPAGATION. Seeds (plant in fall or spring), division, cuttings after flowering.

SPECIAL USES. Rock gardens, edging, flower beds, ground cover.

VARIETIES OF HAREBELL, BELL FLOWERS

	<i>Leaves</i>	<i>Flowers</i>	<i>Blooming season</i>
<i>carpatica</i>	mats 4" tall	blue saucers	June to Oct.
<i>carpatica alba</i>	mats 4" tall	white	June to Oct.
<i>carpatica</i> 'Wedgwood'	cushions	violet-blue cups	June to Oct.
<i>carpatica</i> 'White Wedgwood'	cushions	white	June to Oct.
<i>carpatica turbinata</i>	low mass	large, flat, violet	June to Aug.
<i>cochlearifolia</i>	shell-shaped mass	blue bells	May and June
<i>cochlearifolia alba</i>	shell-shaped mass	white	May and June
<i>cochlearifolia</i> 'Mironda'	shell-shaped mass	silver-blue	June to Aug.
<i>collina</i>	fuzzy clumps	purple bells	June
<i>elatines</i>	dense 3" mass	stars, white center	June and July
<i>elatines alba plena</i>		double white 'Star of Bethlehem'	for baskets and pots
<i>garganica</i>	may be woolly	blue	June and July
<i>portenschlagiana</i> (<i>muralis</i>)	roundish, toothed crinkled tufts	violet bells	June and July
<i>saxifraga</i>	broadened	upturned violet bells	spring

Ceratostigma plumbaginoides *Plumbaginaceae* Leadwort

Shrubby little perennial "bushes" about eight inches high and spreading into clumps twice as wide, with hairy-edged leaves which are green with bronze overtones. It turns to a brickish color in autumn. In August the plants cover themselves with blue phlox-like flowers and continue to bloom until heavy frost.

CARE. Most any soil with drainage. Will grow in sand. Full sun best but semishade is acceptable. Don't plant in moist location. In winter give protection.

PROPAGATION. Seeds in the spring, division at time of new growth in spring, cuttings.

SPECIAL USES. Borders, edging, rock gardens, ground cover.

Coreopsis auriculata nana *Compositae*

Neat little plant with tufts of divided, or lobed, leaves; golden-orange daisy-like flowers on three-to six-inch stems, from June through August.

CARE. Light sandy soil. Shade in extremely hot climate. Cut faded flowers.

PROPAGATION. Sow seeds in fall for flowering next year; division of clumps.

SPECIAL USES. Rock gardens, edging.

Corydalis *Fumariaceae* Fumitory

Ferny-leaved herbaceous perennials with spurred flowers like bleeding heart but more colorful and blooming in the spring. Some types are tuberous-rooted, or partly so, but are grown more like other perennials.

cava (*bulbosa*)—Tuberous type, about eight inches tall, with gaudy clusters of purple flowers in April.

cheilanthifolia—Perennial with thickened roots, finely dissected eight-inch leaves, sprays of yellow flowers from May to frost. Requires shade from hot sun. Striking for its foliage alone.

halleri (*densifolia*)—March bloomer, tuberous. Not many leaves but plenty of clusters of rosy, or purple, flowers; six to eight inches.

lutea—Eight-inch clump-makers with lacy blue-green leaves. Yellow flowers appear in June and repeat later. Likes to keep cool in the shade, or have its roots under a rock. Won't grow in extreme heat.

nobilis—Tuberous type, upright to eight inches, with leaflets both wedge-shaped and deep-toothed. It may have as many as twenty spurred white flowers per cluster, tipped with yellow and spotted with purple.

rupestris—Ferny-leaved eight-inch perennial with short-spurred yellow flowers. Nontuberous.

CARE. Any garden soil with drainage. Partial or light shade, or sun. Plant tuberous types in fall.

PROPAGATION. Division, seeds.

SPECIAL USES. Borders, rock gardens, wall plantings, edging.

Dianthus *Caryophyllaceae* Pink

There are some very special treasures for every garden in this big family of spicy-scented plants with flowers that seem to have been fringed with pinking shears. There are impermanent perennials that flower the first year like annuals, biennials like sweet William that often self-sow, longer lived types that tuft or spread out to make low flowering mounds or mats, elusive

alpines for the rock-garden connoisseur, pixies with tiny half-inch flowers, and great garden carnations. Every year new hybrids make the list longer.

Make your selection, of course, according to size, scent, growing habits, and the color scheme of your little garden. For the miniature rock garden, the rose-colored species *D. alpinus* stays under three inches; *D. neglectus* combines pink and buff, and there are many more. The Allwoodi hybrid varieties are sturdy, lasting, and flower freely for most of the summer. The cheddar pinks (*D. casius* varieties) are clumpy; the maiden pinks make mats; the cottage or grass pinks are tufted; sweet William (*D. barbatas*) is one of the clusterheads; the “annuals” (usually forms of *D. chinensis*) will flower early when seed is started indoors, repeat the performance if cut back after the first flowering, and may even live to bloom once more the second year.

For the most miniature, look for these newcomers: double pink, three-inch ‘Pixie’; rose-red ‘Tiny Rubies’; four-inch ‘Wee Willie’ (sweet William’s son), not reliably perennial but behaves beautifully as an annual; deep-red ‘Little Joe.’ One seed house offers a perennial ‘Midget Double’ sweet William under four inches high and like a button-size carnation.

CARE. Warm sandy soil, not too rich. Lime, dryish, drainage. Full sun, cool temperature. Cut off faded flowers.

PROPAGATION. Division, layers, cuttings, seeds.

SPECIAL USES. Flower beds, rock gardens, edging.

Geranium *Geraniaceae* Cranesbill

This is not the popular potted plant (pelargonium) or the wild geranium of the Eastern woods (*G. maculatum*). These are lacy-leaved perennials with a delicate look but a tough constitution. Here are some of the smallest available types, flowering mostly in the spring but often repeating in spurts throughout the summer.

farreri—A little three-inch doll from China, the many-fingered leaves making a low nest under the astonishingly large, black-anthered, blush-pink-on-white flowers.

macrorrhizum—The species is largish and makes a buxom six-inch bush with light sprays of pink flowers. The foliage is aromatic and turns beautifully bronzy in fall.

pylzwianum—Eager spreader, two to three inches high, with finely divided leaves and rosy flowers in May and June.

CARE. Gravelly soil with drainage, dryish. Sun or light shade. Mostly hardy.

PROPAGATION. Seeds; root division in spring or fall.

SPECIAL USES. Border and dry-wall planting.

Gypsophila *Caryophyllaceae* Baby's Breath

Here is something for every rock garden, the beginner's or the connoisseur's. Over a soft cushion of tiny leaves the foamy flowers are heart-tuggingly romantic and delicate. The plants are hardy, eagerly branching and spreading, and veiled in bloom for most of the summer.

cerastioides—Four-inch creeper with gray-velvet mouse-ear leaves, sprays of white flowers with pink veins. Soil should be alkaline, sandy or gritty, and perfectly drained.

muralis—An unusual "annual" about six inches high, with narrow leaves and warm rosy flowers. Looks and grows best when crowded.

repens—Midget trailer with slim, silvery blue-green leaves and white or pinkish flowers only a few inches above them. Available varieties include, white-flowering alba, compact double pink bodgeri, and single rose-pink rosea.

CARE. Not too rich soil, dryish, with drainage. Full sun or light shade. Usually hardy. Don't thin too much.

PROPAGATION. Seeds (sow where you want the plant), division.

SPECIAL USES. Rock gardens, wall plantings, pavements, ground cover.

Iberis *Cruciferae* Perennial Candytuft

Extremely decorative and useful evergreen with durable, fresh-looking foliage and clusters of sparkling white or pale-tinted flowers in spring. The plants spread, but not enough to be troublesome in small gardens.

saxatalis—Miniature even among miniatures, with two-inch clumps of hairy-edged leaves and flat clusters of white flowers in May. Needs gravelly, limy soil.

sempervirens—The better-known named varieties are offspring of this stalwart, nearly shrub-like perennial with blunt-tipped narrow leaves and flowers forming an umbrella-shaped cluster on six-to eight-inch stems. One plant can spread out to several feet across. Among the dwarfs are 'Autumn Snow,' seven inches high and often blooming again in September; 'Little Gem,' compact mounds four to six inches high; 'Purity,' the purest of white; and 'Snowflake,' a little later (June) and a little larger (eight inches).

Iris *Iridaceae*

Basically, iris are of two types—bulbous and rhizomatous—each with miniature species, varieties, and hybrids. Leaves may be broad and flat, or slim and grass-like. Stems may bear one or several flowers, the earliest in March and the latest in June. Natural colors include white, yellow, and all shades of lavender and purple; hybrids widen the range to many other tints.

There are many miniature species, some with several varieties, and also hybrids, of their own. By far the largest group is the ‘Dwarf Bearded Iris,’ so classified by the American Iris Society according to the height of the flower stalk, and further divided into two series: miniature dwarfs (four to ten inches) and standard dwarfs, Lilliputs (ten to fifteen inches). These are rhizomatous, of course, like the tall bearded giants of the garden, and bloom midway between the earlier species and the big ones.

bakeriana—Bulbous, about six inches high, with round leaves and early, fragrant, blue-purple flowers.

cristata—Crested iris with rhizomes and spreading by stolons to make wide mats. About six inches high with one or two white-crested, lilac, fragrant flowers in mid-May. There is an all-white variety. It likes light shade or dappled shade and cool moisture for its roots.

flavissima—Rhizomatous, spreading by stolons to make open clumps under six inches high. It has narrow leaves and several bearded, brown-marked yellow flowers per stem when it blooms in April and May. It needs gritty or sandy soil, only slightly acid.

histroides—Bulbous, about four inches high, with pale-blue flowers which appear early before the leaves. Various colors available.

innominata—Rhizomatous West Coast dwarf with long grassy leaves and flowers varying between yellow and orange, and between lavender and purple. Blooms in May and June. Takes light shade and neutral, or even slightly acid, soil.

orchioides—Bulbous, with sword-shaped leaves to nine inches long; has several purple-blotched yellow flowers per stem.

reticulata—Bulb covered with netted membrane. Reedy stems about eight inches long disappearing in June. Crested flowers on six-inch stems with a fragrance like fresh violets. Gold-marked on deep purple. Blooms very early in March, even in snow.

tenuifolia—Rhizomatous, about six inches high, blue-purple flowers in June.

vartani—Bulbous, flowering in late winter in mild areas, stemless slate-blue or violet flowers.

voina—Native rhizomatous species which spreads by stolons; four to six inches tall. Has tufts of narrow leaves and beardless flowers which are lilac, or velvet marked yellow, on three-inch stems; blooms in May. Provide light shade and acid soil.

CARE. For bulbous type: sandy or gritty soil with humus; drainage; sun, except the hottest. For dwarf bearded type: loamy alkaline soil, and drainage. Moist. More sun for best flowering.

PROPAGATION. Bulbous: divide bulbs in summer and replant in summer. Rhizomatous: divide in summer.

SPECIAL USES. Rock gardens, forcing, flower beds.

Lavandula officinalis *Labiata* Lavender

Here is one that deserves a place in every garden, and for many reasons: Its aromatic foliage and flowers, its nostalgic charm, and the pleasant sight of it against evergreen with its gray-felt leaves and dark-lavender flowers. It usually stops growing just under the height of one foot and can be clipped lower if you wish. The leaves are lacy, slim, and dainty, especially with their pure-white fur when young.

Among the varieties available are the Munstead strain, dwarf, plump, and bearing pink flowers; 'Hidcate Blue,' very hardy; compacta (nana compacta), earlier flowering and only a few inches high; and atropurpurea, with dark purple flowers.

CARE. Poor soil so plants will not grow too vigorously and winter-kill. Drainage, sun, winter protection. Dryish. Lime for maximum fragrance. Prune almost any time; mulch.

PROPAGATION. Seeds, division, cuttings in late fall or early spring on this year's growth with heel of older wood.

SPECIAL USES. Edging, rock gardens, ground cover.

Myosotis *Boraginaceae* Forget-Me-Not

Sentiment, compelling as it is, is not the only reason for having this delightful little plant in one's rock garden. Small though it is, it has its own quiet charm in its delicate foliage and transparent cloud of tiny blue flowers. The intensity of color of those blooms catches your eye and draws you to it. You look at it and understand the reason for its name. Here are a few members of the family:

scorpioides (palustris)—Light, low, and sprawling; light-blue flowers touched with gold at the center during the spring. (They last much longer if you have the dwarf variety *semperflorens*.) The fact that these perennials are often listed as "half-hardy annuals" indicates their need for winter protection, but they will self-seed. (Water-lily specialists offer an aquatic variety that grows in bogs.)

sylvatica (alpestris)—Self-sowing biennial (also offered as an annual) that has blue flowers with touches of pink, sometimes entirely pink. May flowers.

CARE. Ordinary soil, cool, moist, part shade.

PROPAGATION. Seeds (flowers the second year), division of clumps in late summer.

SPECIAL USES. Rock gardens, edge of pools.

***Nepeta hederacea (glechoma)* Labiatae** Ground Ivy

This is a ground cover, one that can get out of hand unless checked, but it has its uses in shady spots where other covers do not flourish. About the best-liked variety is *variegata*, which has round leaves neatly scalloped on the edge and embroidered with white. In the spring it has spires of blue flowers.

CARE. Ordinary or sandy soil. Drainage, either sun or shade. Cut off faded flowers.

PROPAGATION. Seeds, division.

SPECIAL USES. Ground cover, containers, baskets, house and greenhouse plantings, rock gardens.

Oenothera Onagraceae Evening Primrose, Sundrop

The sundrops bloom by day; the evening primroses stay open at night and play host to the moths of the garden. There is a miniature of each type for the small garden. Each has the common characteristic of large yellow poppy-like flowers which are short-lived but are quickly replaced by others for most of the summer.

missouriensis—Evening primrose from the West, six inches high. The leaves are long and oval, covered with ash-gray hairs. The nocturnal flowers, poppy-like, can measure six inches across and are yellow.

perennis (pumila)—sundrop—This is the daytime bloomer, often with one-inch blooms on a plant only three inches tall. The silk-hairy leaves make a lush-looking mound.

CARE. Medium sandy soil, dry, with drainage (heavy dampness is fatal). Lime and manure. Sun. Remove seed pods to promote more flowers. Plant has long taproot which must not be broken when transplanting.

PROPAGATION. Seeds, division (in early fall or spring), cuttings (in late summer).

SPECIAL USES. Rock gardens and wall plantings.

Phlox *Polemoniaceae*

There are so many varieties of this furiously spring-flowering plant, and all with many admirers, that the trick is to find the ones which appeal to you personally. You will be guided by the colors you want and the amount of space available. Many of them, after the blooming season, make lush, plushy, green foliage mats.

amoena (procumbens)—Neat, not rampant, cushions of two-inch oval leaves with flat clusters of fragrant pinkish, or purplish, flowers on six-inch stems in May and June.

bifida—sand phlox—A native of the Midwest with low, tufted, and slightly hairy foliage and starry blue flowers (early, in April and May).

divaricata (canadensis)—wild sweet William—This slow creeper is a little tall—about twelve inches—but may fit your planting scheme. It has two-inch oval leaves and clusters of lavender-blue flowers at daffodil time.

douglasi—alpine phlox—Three-inch Western native with half-inch knife-like evergreen leaves and half-inch white or pastel flowers (in June).

rigida—sandhill phlox—Western alpine, similar to *douglasi* except for thinner, more sharply pointed leaves and white flowers blending to blue and purple. The more acid the soil, the deeper the colors.

stolonifera—creeping phlox—Partly or fully evergreen creeper with four-inch stems, oval, hairy leaves, and full clusters of light or dark lavender flowers in May and June.

subulata—ground moss, or mountain pink—In April and May the six-inch mats of evergreen leaves are almost invisible under the profuse cover of one-inch, fragrant flowers. They may be purple, purplish-pink, or white, and some varieties are red or blue.

CARE. Fertile soil, acid or alkaline according to the variety. Moist. Most are hardy, again according to the variety.

PROPAGATION. Seeds (do not always come true), cuttings in July and August, division of roots.

SPECIAL USES. Rock gardens, edging, ground cover.

Primula *Primulaceae* Primrose Official “Sections,” by common cultural requirements.

Auricula Section. *P. auricula* and other European alpine and their hybrids; usually evergreen; flower in early spring. Stony, drained, rich loamy soil lightened by leaf mold and sand; will stand more

sun than most, but not hot midday sun; likes its roots under stones; mulch with stone chips.

Candelabra Section from Asia. Flowers in tiers, one whorl above another, open-faced; most tall with big leaves. Needs moist soil, part shade. Flowers in late spring and early summer. Disappear over winter.

Capitatae Section from the Himalayas. Biennial here, perennial where snow covers all winter and on the West Coast. Flowers late (July). Flat rosettes of delicately toothed leaves, twelve-inch silver stems; flat silvery heads of slightly drooping flowers. Disappears over winter.

Cortusoides Section from Asia. Woodsy-rich humusy soil. Decorative leaves usually soft hairy. Flowers in spring. Disappears over winter.

Denticulata Section. Small group from the Himalayas. Easy perennial. Be careful about dampness in winter. Among the earliest to flower in spring.

Farinosae Section. Huge group from many lands. Moisture, drainage, part shade. Stand more sun if moist. Small bud, usually white at surface, shows in winter.

Nivalis Section from Asia. Biennial here. Long narrow leaves. Flowers in tiers closer than candelabra and more drooping. Cool, rich, moist soil, part shade. Flowers in early spring; disappears in winter.

Sikkimensis Section from Asia. Belled primulas, with hanging heads of bell-shaped flowers. Biennial here; perennial in both colder and warmer climates. Most tall with heads of fragrant flowers held well above large toothed leaves. Flowers in late spring; disappears in winter.

Vernales Section includes all the easiest-to-grow and best-known primulas. Requires part shade, rich soil, moisture; should never be completely dry. Evergreen. Includes: *P. vulgaris (acaulis)*, single flower per stem; *P. polyantha* with many varieties, and stalked umbrella of flowers above the leaves; *P. juliae* hybrids. Among *P. polyantha* there are some dwarf hybrids (not miniature).

CARE. Some shade for hottest part of the day. Woodsy soil rich in humus, neutral or slightly acid. Mulch with manure if possible, around and between plants; don't cover crown. Keep moist. Provide winter protection of salt hay, evergreen boughs, or excelsior, which results in gradual thawing of the ground around the plants in spring, and also prevents premature blooming during unseasonably warm winter days. Fertilize in spring, and in summer for next year's flowers. No nitrogen in summer fertilizer. Watch plants for signs of slugs and red spider. Divide when plants get too crowded, preferably after flowering. Hearty seeders.

PROPAGATION. Seeds when ripe (others must be frozen first and thawed), division after flowering.

SPECIAL USES. Carpet under shrubs and under early-flowering trees. Naturalize by streams and beside pools. Wall planting, rock gardens, borders, wild gardens.

Saxifraga *Saxifragaceae* Rockfoil

There are so many species and hybrids of this popular plant it is difficult to know all of them. Generally, they are characterized by flattish clusters of leaves around the crown, and airy sprays of flowers on top of taller stems. But from this point the variations take many forms and sizes. The leaves may be the size of a nickel or five times that size; flowers may be minute, or large and open-faced; there may be creeping stems or runners, or not.

For most gardeners it is not necessary to know how botanists classify these plants. If you plan to delve deeply into rock gardening there are a number of excellent books on rock plants, some of which are listed in the appendix.

Following are a few of the better-known, and easier grown, saxifraga, including several that have a place in flower borders and other garden spots.

aizoon—Arctic native with rosettes of evergreen, stiff, gray-blue leaves edged with a thin “crust” of limy deposit, and purple-spotted white flowers on tall stems (in May or June). There are a number of varieties, including yellow-flowering *lutea*, pink *rosea*, and tiny *baldensis* with leaf clusters only a quarter-inch across. (Drainage, moist, with northwest exposure. Shade from noon sun; provide limy soil with leaf mold.

cuscutaeformis—Thick, hairy, roundish leaves tinged with copper and marked with a network of white veins; spreads by reddish runners that produce new plants at their tips.

decipiens rosacea—Mossy mats of finely cut leaves and large cupped white flowers on six-inch stems (in May and June). Drainage; gritty sandy soil with leaf mold and humus. Moist, shade from sun. Also pink-and red-flowering varieties.

sarmentosa—strawberry begonia—Favorite pot and basket plant, hardy in nearly every garden. Spreads by strawberry-like runners with plantlets at the ends. Makes mats of round, white-veined leaves and tall airy sprays of white or purplish flowers in June. Requires shade, and moist acid soil.

umbrosa—London pride—Makes a carpet of three-inch-high rosettes of leathery leaves; spires of pink flowers in early summer. Needs moist, rich soil and shade or semishade. The small-scale variety *primuloides* is daintier.

virginiensis—Slender, oval three-inch leaves in flat clusters, green tinged, but bronze in the fall; small white flowers on ten-inch stems in May. Native of our Northeast. Nice in wild gardens.

CARE. Soil varies with the species (lime for the encrusted ones, which also tolerate more sun). Semishade, moist, but perfect drainage. A hot dry climate is bad for the alpiners. Mossy varieties need heavier shade, richer soil, and are best moist.

PROPAGATION. Divisions of sets, cuttings, seeds (most varieties).

SPECIAL USES. Baskets, pots, dry-wall planting, containers, borders.

Silene *Caryophyllaceae* Champion, Catchfly

The perennials of this genus include some of our most colorful native plants (the flaming fire pink *S. virginica*, for example), all related to the garden pinks but each one having its own characteristics.

acaulis—cushion pink, moss campion—Dainty two-inch mossy tufts of needle-like leaves slowly increasing to clumps and having half-inch purple-red flowers all summer. (Provide gritty, limed soil.)

caroliniana (*pennsylvanica*)—wild pink—Sticky-fuzzy plant with slim four-inch evergreen leaves and pink flowers on six-inch stems (April-June). Native to Northeastern woods. Good in rocky spots. Has deep taproot so handle with care. Dig deep when dormant.

maritima—Sea campion—Slim, gray-blue leaves on trailing stems, making a mat four inches high with clusters of white flowers in June.

saxifraga—Tufts of knife-shaped leaves under summer flowers which are white inside, tinted green or red outside, on six-inch stems.

schafra—moss campion—Spreading mounds of silky-hairy leaves covered in August with rose or purple flowers on six-inch stems.

CARE. Good soil, gritty for some varieties with lime for some. Top-dress yearly with mixture of soil and cow manure. Drainage.

PROPAGATION. Seeds, division, cuttings (of young growth).

SPECIAL USES. Rock planting, borders, walls, wild gardens.

Thymus *Labiatae* Thyme

These are actually tiny shrubs with tangling, woody stems and masses of little leaves that are evergreen in all but the coldest and most exposed locations. Some creep flatly; some make three-inch mounds. You can choose from varying fragrances, leaf textures and colors, and colors of early summer flowers. In fact, there is a much wider variety available than most gardeners realize.

herba-barona—caraway thyme—Ground-hugging creeper with fresh green caraway-scented leaves, and tight clusters of rose-purple flowers. It does not seed and is propagated by division of roots. Hardy.

lanicaulis—Four-inch carpet with half-inch leaves slim as needles at the base and broadening higher on the short stems. Stems feature a woolly coat of hairs as long as they are thick. Tight clover-like flowers are rosy pink.

nitidus—French thyme—Six-to eight-inch shrub with shiny, oval, gray-green leaves, and lavender flowers. Older clumps resent being disturbed, so start young plants from cuttings.

serpyllum—mother-of-thyme, creeping thyme—Perennially popular stem-rooting creeper, less than four inches high with half-inch oval green leaves and purple flowers loved by bees. There are many varieties: white-flowered *albus*, silver-marked *argenteus*, gold variegated *aureus*, larger, red-flowering *coccineus*, bright-pink *roseus*, robust, romping, red-flowering *splendens*. Almost in a class by themselves—and often listed as separate species—are *lanuginosus*, woolly thyme (completely covered with silvery fuzz), and the variety *vulgaris* (*citriodorus*), lemon thyme with citrus-scented leaves.

vulgaris—common thyme—Kitchen herb, spreads by underground stems. The erect stalks are covered with fine white hair and set with half-inch oval leaves; there are deep-lilac flowers in May and June. It spreads so fast it has been accused of being a weed.

CARE. Ordinary soil, or rich and sandy loam (gritty for some plants). Drainage. Fairly sunny. Protect lightly where winters are severe.

PROPAGATION. Division, layers in spring, cuttings in early summer, seeds (not all varieties have seed, and seeds do not always come true).

SPECIAL USES. Pavements (plants not damaged), border planting, herb gardens, ground cover, edging, dry-wall planting, cover for bulbs.

Veronica *Scrophulariaceae* Speedwell

Most of the smaller veronicas are matmakers; some are clumpy, and a few are even woody or shrub-like. Also there is a fairly wide variation in the foliage. But the flowers are always arranged in spires; a few are white, purple, or pinkish, but mainly there is a wide choice of true blues from baby blue to royal blue. Here are a few varieties:

allioni—Carpet of little leathery leaves, violet flowers in July.

armena—Ferny, furry tufts of finely cut leaves on trailing stems, five-inch spikes of vivid blue flowers (from May to July). Provide drainage, filtered sun, or half shade.

filiformis—An idiosyncratic creeper that delights some gardeners and drives others crazy. It has been described as “pernicious” and a weed. Actually it’s rather pretty, with small rounded leaves and galaxies of starry-blue flowers in late spring. Grows in either sun or light shade.

fruticans (*saxatilis*)—Woody stems tightly clothed in dark evergreen leaves making a shrubby plant under six inches. In June and July sapphire flowers open their eyes with pinkish lids.

fruticulosa—Shrubby and semi-evergreen, about four inches high with half-inch leaves and pinkish-blue flowers in June.

gentianoides—This species has clusters of gleaming leaves close to the ground. And in June the flowers are very large, a luscious Wedgwood blue. But there are alarming reports that it has topped twenty inches in height. To play safe, try the miniature version *nana*, which has the same characteristics but is much less than half that high.

repens—This is a precious pet, a massy creeper with tiny teardrop leaves. It sparkles in May or June with baby-blue flowers. It is all of two inches tall at maturity.

spicata nana—The midget reproduction of the better-known species, slowly swelling to low, leafy clumps and topped with three-inch spikes of blue-purple flowers in July. A white-flowering form has smaller leaves but taller flower stems.

CARE. Any good soil. Moist but with drainage. Sun or light shade depending on the variety (see notations in plant descriptions). Cut back faded flowers. Trailers start best if plants were grown in pots. Divide occasionally.

PROPAGATION. Seeds in spring, division in fall or spring, cuttings early in the summer.

SPECIAL USES. Rock gardens, ground cover, dry-wall planting, pavements, steps, stepping stones.

Viola *Violaceae* Viola, Violet

Thanks to the viola, gardens of all sorts are richer and more pleasant places in which to work, visit, or spend a pleasant afternoon. Here are some of the viola's gifts to gardening:

cornuta—Tufted pansies.

odorata—sweet violets—Fragrant violets, once strictly a specialty of florist shops but now widely grown in home gardens.

rosina—Pink flowers with intense fragrance.

semperflorens—Mats of fresh green leaves with purple flowers that last much of the summer.

tricolor—Commonly known as Johnny-jump-up, an all-summer-long bloomer with small purple and gold flowers.

tricolor hortensis—Pansies which grow in many colors and color combinations.

CARE. Rich garden soil, partial shade, moisture, light winter protection.

PROPAGATION. Seeds (sown in flats and exposed to freezing weather over winter for spring germination), division, layering of runners of those varieties that produce them.

SPECIAL USES. Cut flowers, beds, edging, rock gardens.

MINIATURE ANNUALS

Even if every inch of our gardens were permanently planted, I'd probably still start some annuals every spring. In those depressing, letdown weeks of January after the holidays nothing brightens my life like leafing through the new seed catalogues. In the dark days of March when late-winter blusters snarl at spring and threaten to hold it off forever, my perky indoor seedlings tell me that it is just around a bend in the road. My traditional birthday treat, in early June, is a day all my own when I get my favorite flower beds ready for the summer. I absolutely refuse to fry an egg, wash a dish, or lift a dustcloth. Before dark on that wonderful day I see that every seedling is planted in moist warm soil.

Actually, I can't imagine our grounds being so completely planted that there isn't a place for at least a few annuals—no bare spots in the perennial border, no shrub or tree around which to set flowering clumps, no window boxes or planters in need of decoration. And if ever there is a spring when I don't ask my husband to spade up “just one more spot” for a few more annuals, I'll eat my garden gloves finger by finger.

Because they cost so little, and because they last only one season, you can plant annuals with careless abandon. You can experiment with bizarre color schemes. If they don't work out, try something different next year. If you are saving a special spot for a special shrub or bush, and you feel it is too expensive for the moment, or you can't find a specimen that is exactly what you want, let annuals fill in until you have that particular plant. If you are creating a new garden and are not quite sure of the design, test it with annuals before you make it permanent.

As with the miniature perennials, the annuals included here have been selected not only because they are small, but also because they are in suitably modest scale for use in miniature gardens of many sorts. Descriptions of the varieties should be helpful in selecting the truly miniature annuals at local nurseries and garden centers. Seeds are available from various mail-order suppliers listed in the Appendix.

Like all other garden plants, miniature annuals should be selected according to such cultural conditions as the amount of moisture and sunlight and the type of soil. If these requirements are not clearly stated on the seed packets or in the seed catalogue from which you ordered, check for details in a good reference book rather than risk disappointing failure.

GROWING ANNUALS FROM SEED

When and where you sow seeds of annuals depends on whether the particular varieties grow best in coolness or warmth, and how long the growing season has to be for maturity and flowering. This is the basis of catalogue listings that classify annuals as hardy, half-hardy, or tender. Hardy annuals germinate best when air and soil are cool and make their best growth before hot weather sets in. Seeds are sometimes sown out of doors in autumn to germinate in early spring, or they may be sown in the garden the moment the frost is out of the ground.

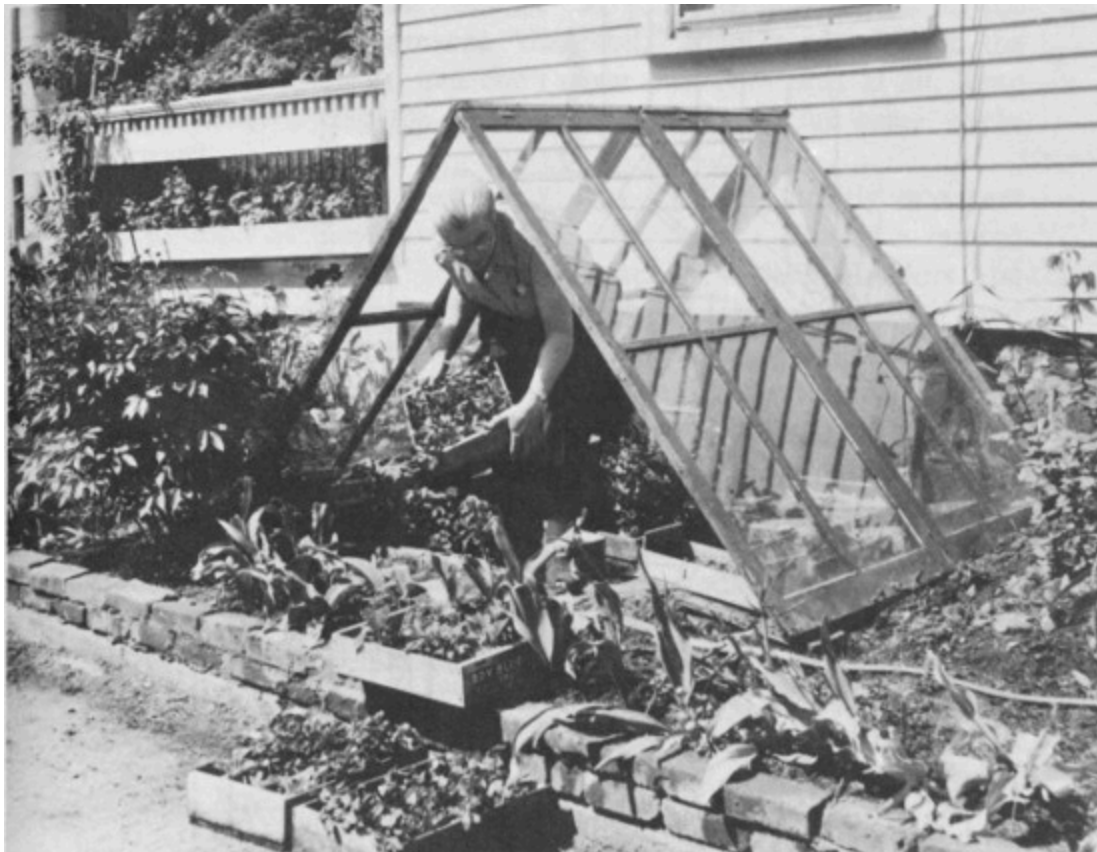
Half-hardy seeds are planted outdoors where the plants are to grow, or you can give them a head start either indoors or in a hotbed or a cold frame. Plant about six weeks before the time to set them outdoors. Tender perennials grown in cold areas as annuals follow the same schedule.

Seeds of tender annuals should not be planted outdoors until all danger of frost has passed and the soil has warmed up considerably. But many tender varieties, like petunias, are slow-growing and need plenty of time to reach flowering size. These should be started indoors no later than March.

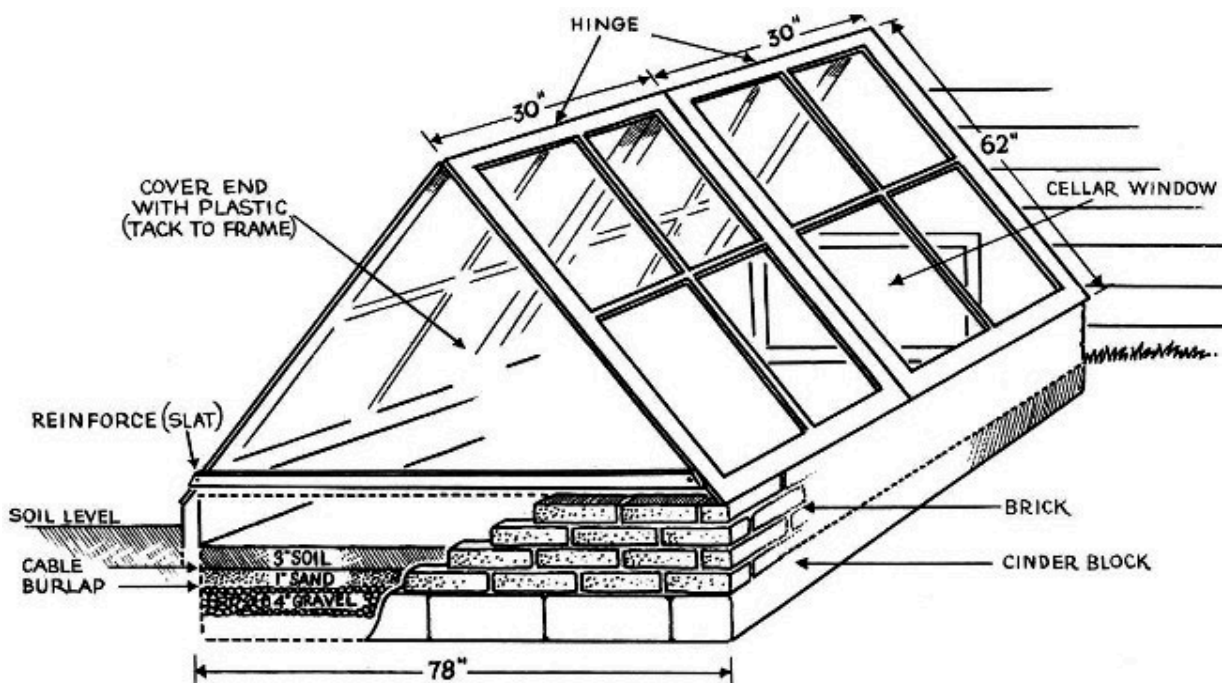
Unless you have a greenhouse or hotbed, it's risky to start seeds too early indoors. Poor light and high temperatures will cause the seedlings to grow limp and leggy and often to topple over toward the source of the light. At transplanting time they may be so weak they will be slow getting established; thus, little time is saved after all.

But you can save several weeks by starting seeds early in a sunny window (just not too early), and you can start them even earlier in a cool greenhouse, a hotbed, or a cold frame. When we lived on Long Island we rigged up a workable substitute outside a cellar window by using discarded storm windows. In fact, since it had a brick foundation that reached below the frost line and had a heating coil in the soil, we grew many cool greenhouse plants in it during the winter. The open cellar window provided

enough heat for flats of annual seedlings we propagated in it in the spring. Once these seedlings were transplanted to the garden, we dismantled our temporary greenhouse, filled the spot with topsoil, and made a flower bed. I've seen several small greenhouses like the one we improvised on sale commercially, one, of metal and glass, selling for less than ten dollars.



Improvised greenhouse costing \$00.00 which saved author, shown in photograph, at least a hundred dollars per annum in the cost of annuals



Greenhouse construction diagram

Indoors or out, seeds germinate best in clean, sifted soil that is light and porous and drains perfectly. Add organic matter such as peat or humus to help hold moisture, and sand to aerate and insure drainage. If possible, sterilize the soil with heat or one of the special fungicides, to prevent "damp-off." Plant seeds only to the depth of their diameter, and don't cover dust-fine seeds at all. Keep the soil moist during the germination period and for some weeks afterward.

For several years we've transplanted seedlings directly into peat pots as soon as they have their first few leaves. This is a tedious job, but how it saves backache and heartache later. Indoors you can transplant while seated comfortably at a table or bench, and while listening to soft music on the radio. When the seedlings have flourished in the peat pots and it is time to put them outdoors, there is much less time spent bent double on one's knees. And since the roots aren't disturbed, the plants seldom suffer from transplanting shock and seldom fail to thrive.

Plants grown indoors should be hardened before they're put out of doors for good. Every day expose them to fresh outdoor air for an increasing

length of time, until they are accustomed to the cooler temperatures, particularly at night.

SUMMER CARE

Annuals take the same kind of care you give perennials, except that most of them require less of it. Few need, or even thrive in, especially fertile soil. Fertilize lightly, if at all. Many annuals get along with less moisture and mulching than perennials. And of course, none need be prepared for winter. During their one big season, simply pinch or prune those that grow straggly, and remove fading flowers before they set seeds. That usually prolongs the blooming season. After frost strikes, burn the dead tops, which helps control the spread of insects and disease.

As you study miniatures you will realize how many of the old regular annuals of the garden, plants that usually top two or three feet, can also be had in five-to eight-inch sizes. You have but to look through your seed catalogues. The tiny ones are usually listed at the end of each category, though in the affections of miniature-lovers they should come first and take precedence over the “new giant-size,” “bigger-than-ever” varieties.

In searching for miniatures I’ve culled through my large collection of catalogues from Burpee to Vaughn and have compiled a list of small annuals I have either grown, or expect to grow the coming season. Not all of them are compact versions of their larger relatives. Many by nature are on the diminutive side. So many of them are old friends, let’s call them by their first names—the ones by which they are listed in most seed booklets. The plants I have in mind are mostly eight inches in height or less, though some of the longer-stemmed charmers may slip in simply because they are too lovely to be left out. We’ll give colors for those who have color schemes in mind for their gardens.

Ageratum

This annual, a great garden favorite, is also known as the floss flower. In general it is a profuse bloomer that is fine for borders, edgings, rock gardens, small flower beds, and even as a house plant. It comes in several colors but usually in the same sizes, two to six inches.

‘Blue Ball’—This name is especially descriptive. The plants shape up as round as a ball and then cover themselves with dark-blue flowers. Who can expect more of one so small?

‘Blue Mink’—My friends who write seed catalogues, and so see many beautiful plants, rave about this one as being true blue, the real ageratum color. Its blooms, pretty as they are, grow surprisingly large for a bush so small.

‘Imperial White Dwarf’—For its contrast I love this one, a pattern of white surrounded by the blues, pinks, and reds of other annuals.

‘Midget Blue’—This is my real pet, a really tiny one of uniform shape, never over three inches tall and with the most delicate blue ageratum flowers.

Alyssum (Sweet)

This is just about the nicest thing that can happen to your garden be it annual, miniature, or otherwise. Although some members of this family may reach twelve inches, most of them run from three to five inches.

‘Carpet of Snow’—Much like ‘Royal Carpet’ for spreading, but has white flowers instead of purple.

‘Little Gem’—This one is about six inches tall, but atones for its height by being a more profuse bloomer, with small white flowers.

‘Pink Heather’—The blooms open in a delicate pink and slowly darken. It has vigor and scent, and seems to want to bloom constantly.

‘Rosie O’Day’—The pride and joy of the family, and the winner of an award comparable to being chosen Miss America. The blooms are a deep pink, even more so when the weather is a bit cool. It is quick to bloom, easy to grow, and nicely fragrant.

‘Royal Carpet’—This one is only about two inches high. With its purple flowers it fills in vacant spots most beautifully. A single plant may spread out ten inches.

‘Violet King’—Somewhat shorter than ‘Violet Queen,’ but has a deeper color and more compact shape.

‘Violet Queen’—Averages about five inches, and is as its name says a rich violet color. Delightfully scented, it is wonderful for edging, borders, and flower beds.

Alyssum has long been one of my garden favorites, whether for a ground cover, for rock gardens, as an edging for walks and beds, or just for its own sweet alyssum self.

Balsam

In my concept of miniature plants I had always thought of balsam as a towering giant of at least two feet or more. Then one night while going through seed catalogues I discovered an entire column listing ‘Tom Thumb Double Flowered Dwarf,’ ‘Tom Thumb Purple,’ ‘Tom Thumb Scarlet,’ ‘Tom Thumb Shell Pink,’ ‘Tom Thumb Pure White,’ ‘Tom Thumb Rose.’

According to growing directions, Balsam likes a rich soil, well-drained but slightly moist, and full sun.

Coreopsis

It was love at first sight when I saw a clump of this delightful annual. I was even more delighted to learn that the plant can be grown as a miniature. In the smaller form the cute little fellows spread out into low mounds of six or eight inches and cover themselves with blooms that last all summer. As an edging, for rock gardens, and as a cover for otherwise bare spots, they are truly charming.

‘Golden Ray’—Yellow with crimson touches.

‘Petite Purple’—Rich purple covering the small mounds.

‘Tiger Star’—Rich-crimson-striped and splashed with yellow. Plants stay at about six to eight inches and become mounds of color.

Cornflowers

Here is another of the taller plants which fortunately has some small relatives. There are:

‘Jubilee Gem’—Has a vivid blue flower on a compact bush.

‘Lilac Lady’—Purplish blooms on a neat bush.

‘Little Snow Man’—Pure-white blooms.

‘Rose Gem’—Red blooms, double.

Most of these plants will stay under eight inches. They are ideal for rock gardens.

Marigolds

Marigolds are very well represented in the gallery of miniatures. Loving the large ones as I do, I love the tiny ones even more. In one of my favorite catalogues I find an entire group with “Petite” as a first name:

‘Petite Gold’—About seven inches tall and golden yellow.

‘Petite Harmony’—Mahogany in tone, with a gold crest.

‘Petite Orange’—About the same size, only tending to be more orange in color.

‘Petite Yellow’—Another variation on the popular marigold color. In another catalogue I find the small ones called “Pygmies”; all of them are just a little short of nine inches in height.

‘Fireglow’—Mahogany-scarlet with gold centers.

‘Helen Chapman’—Rich gold in color with flecks of red.

‘Primrose Pygmy’—Primrose-yellow flowers with a red base.

‘Yellow Pygmy’—Lemon-yellow pompon-type flowers.

There are other miniature marigolds, from the double dwarf French part of the family, within my idea of how small a small one should be.

‘Brownie Scout’—Golden little flowers with a base flecked with red.

‘Butterball’—Canary-yellow blooms and plenty of them until frost time.

‘Fireball’—A few inches taller than some varieties, but with flame-like markings flashing up through the otherwise orange petals.

‘Gold Laced’—Basically mahogany in color but with each petal edged in gold.

‘Rusty Red’—The brightest of red, extra-large flowers for a small plant.

Nasturtiums

I love this plant so much I hate to leave it out of my listing even though it might be a few inches taller than my other miniatures. I always am careful to plant it toward the rear of flower beds where it will not overpower the smaller plants. There are several dwarfs, but the one I know best is ‘Cherry Rose.’

Nemophila

This is a very beautiful little dwarf, each plant spreading out into a shapely clump, and covering itself with pretty cup-shaped blossoms. When sown in the spring it blooms all summer. When sown in the autumn it is an early spring-bloomer. It is at its best when used in a semishaded spot for a ground cover, for a border, or in a rock garden. Its more usual name is baby blue eyes.

‘Insignis Blue’—The best-known variety; covers itself with lovely, cup-shaped, sky-blue flowers with white centers.

Petunias

Petunias, being favorite annuals, belong in just about every garden, but it is difficult to find the small ones for the miniature garden. But they are available despite the fact that most hybridizers try to develop a larger and larger plant. Here is a list of smaller ones; few of them are taller than eight inches.

‘Blue Star’—Velvety blue with a sharp white star.

‘Bright Eyes’—Rosy-pink blooms with white throats.

‘Heavenly Blue’—Light sky-blue, a heavenly color.

‘Igloo’—Compact and pure white.

‘Rosy Morn’—Brilliant rose with white throat.

‘Salmon Gem’—Bright salmon color.

‘Twinkles’—Brilliant rose-starred white.

Phlox

This is another garden favorite that may give you problems when you try to find seed for the small ones. Again, the tall varieties are being developed, but concessions have been made to those who like the diminutive. I find five varieties listed as the ‘New Dwarf Compact.’

‘Cinnabar Red’—Scarlet vermillion.

‘Fireball’—Rich fiery crimson scarlet.

‘Isabellina’—Primrose yellow.

‘Pink’—Bright chamois-pink.

‘Snowball’—Flowers of purest white.

Another source lists:

‘Dwarf Globe’—A plump-shaped plant with flowers of various colors.

nana compacta—Not six inches high and a constant bloomer with large radiant flowers.

‘Twinkle’—Various colors with fluted, fringed petals.

Portulaca

Seemingly this is an insignificant little plant, though I don’t know how one so charming could be classified as such. I had never raised it until the editor of one of my favorite magazines asked me to make some photographs of it. Of course, I had to grow it first. I planted a bed and waited for results. They were not long in coming. Plants barely out of the ground started blooming a few days later, and started spreading about the same time. The blooms, when single, were cup-shaped; the doubles looked like roses. The plants had a sunny exposure, a sandy soil, not much nutriment, and an inordinate attraction for our dogs (like catnip for cats). One day I took a photograph of my husband’s Labrador-retriever pup looking over a short picket fence at my portulaca. I promptly put in a higher fence, but the puppy grew faster than the fence. Eventually, I got my

photographs, but it was a dog-gone struggle. If you do not have dogs, or if your dogs are not portulaca-minded, I offer you a listing of a few varieties.

‘Alba’—Pure white.

‘Aurea’—Lovely yellow.

‘Coccinea’—Scarlet.

‘Red Foundling’—A little larger than some of the others, but semidouble, carmine-red, and most lovely.

‘Rosea’—Rose.

‘Salmon’—Salmon-colored and delightful.

Snapdragons

I didn’t think it possible that the lovely tall snapdragons, the ones that are waist high or taller, could be reduced to a mere six or eight inches. But it is possible. I offer two varieties in substantiation.

‘Miniature Magic Carpet’—Four to six inches. Colors are mixed and very dainty. As a carpet plant it is a creeper when given a chance.

‘Tom Thumb’—A plant that varies from six to eight inches tall, very free-flowering, good for window boxes, rock gardens, porch boxes, and flower pots for the window. Colors are in all of the brilliant shades.

Sweet Peas

One seed company came right to the point when listing the small varieties of sweet peas, calling them ‘Little Sweethearts,’ which they are. Eight varieties were listed in one booklet.

‘Boy Blue’—Clear mid-blue and delightful.

‘Caprice’—Red-maroon.

‘Coquette’—A rose-red variety.

‘Heidi’—Another red bloomer.

Those are varieties we have grown. Others which I hope to meet are ‘Fantasy-Face,’ ‘Pollyanna,’ ‘Sinbad,’ and ‘Snow-White.’

Another source lists ‘Cupid,’ ‘Little Sweetheart,’ and ‘Color Carpet.’ These three are bushy, not viny, make six-inch mounds of small-scale pea-type leaves, are slightly fragrant, and are usually sold in mixtures of colors.

Verbena

Actually this is a perennial, but it is usually grown as an annual. It is rampant both as a bloomer and as a creeper. The flowers are tubular with flat heads and show up in many colors—pink, lavender, scarlet, white. Our sources list *Verbena bipinnatifida*, the most hardy; and *V. hortensis*, many colors with white eyes, and a low compact shape. Another, ‘Dwarf Compact’ is white.

Zinnias

In running down my alphabet of annuals, from alyssum to zinnias, I find it a bit difficult to fit the latter into what I consider the miniature category, eight inches or thereabout. But zinnias are so effective in a garden I’ll forgive them if they are a few inches above eight. I’ll plant them in the back rows of my beds and pinch out the tops if they get too rambunctious. In the Lilliput class, those delightful little plants which cover themselves with the pompon type of bloom, I offer the following list.

‘Canary Yellow’—Clear yellow.

‘Peach Blossom’—Soft light pink.

‘Rose Gem’—Deep clear rose pink.

‘Scarlet Gem’—Bright fiery scarlet.

‘White Gem’—Pearl-like white.

Sorry I can’t offer more varieties, but all growers seem to be devoted to growing giants in the zinnia class. I happen to love the little ones, those cute pompons that look so charming in the buttonhole of a man’s jacket.

MINIATURE GARDEN BULBS

Ring-around-a-rosying at the foot of a gray birch, twenty dazzling-white spring snowflakes nodding at the turn of a path, a company of bright-eyed kaufmanniana tulips; perching lightly in a rocky crevice, a flock of tiny canary daffodils that seem ready to wing away.... There's something about the miniature bulbs that's more fauna than flora, more fairy tale than real. Each one has its own elfin character and a happy-go-lucky way of stamping your garden "personally yours."

In the rock garden, at the edge of a border, with evergreen ground covers, clustered wherever there's a stone or stump to back them up, used by the hundreds as drifts in the lawn, miniature bulbs more than anything else give you that spring feeling. Potted and forced into flower indoors, they're winsome window-garden items, delicate decorations for coffee table or what-not, delightful dish-garden inhabitants. They're indispensable for sink gardens, precious in the small greenhouse.

Perhaps part of their charm lies in their unfamiliarity. So few are well known and widely grown. So many more are easy to find, easy to grow, and especially easy to pay for. And they're such beguiling plants, I don't see how any garden could be without them, miniature or not.

All of the bulbs included here hold their flowers no more than six inches high. They're not small versions of better-known, larger plants, but mostly completely different, with their own individual characteristics—and with bulbs, foliage, and flowers in miniature proportions. The word *bulb* is used in its generalized sense, and includes true bulbs, corms, and tubers. To be botanically accurate, a few plants such as tuberous-rooted dahlias, rhizomatous iris, and bulbous erythroniums should also be included; but to be consistent in such matters as culture and decorative use, they're grouped with plants that grow similarly, in the chapters on annuals, perennials, and wild flowers.

Some of these miniature bulbs are available from nurseries and other local suppliers. But the majority, and particularly the most unusual, are offered by mail by bulb importers and specialist growers. Reading some of

the catalogues is like taking a quick trip to exotic parts of the world. And for me, ordering no more than I can find time to plant and care for is a severe exercise in self-restraint. There are always a few more that would look well in some special spot, others that are intriguing simply because I've never grown or seen them.

Most bulb catalogues give specific cultural recommendations that are helpful in selecting varieties that will adapt to your climate, and to the sun, soil, and moisture in the spot where you want to plant them. In general, bulbs are either hardy or not; they will survive a deep winter freeze, or they must be lifted and stored before the ground freezes at all. There are some natural borderline exceptions, like many fascinating varieties native to the West that are not so touchy about cold as they are about other climatic and cultural conditions.

GROWING MINIATURE BULBS OUTDOORS

Tender, summer-flowering bulbs are planted in late spring, when soil is warm and danger of severe frost has passed. Hardy bulbs are usually planted in late summer and early fall, when foliage has ripened and died back and the plants are in deepest dormancy. This includes fall-flowering types like the colchicums. The earlier bulbs can be planted, the stronger the root systems they can develop before winter, and the stronger their flowering during the first season.

Make sure the selected site has perfect drainage. Bulbs rot quickly when water stands around their roots. Dig generously, to about eight inches deep; enrich the soil with organic matter such as leaf mold or compost; increase aeration and drainage in sticky, clay-like soils with sharp sand; add a light sprinkling of bone meal or superphosphate, if fertility is low. Since few bulbs like very acid soil, lime is a "must" except where the soil tests so extremely alkaline that the addition of organic matter does not make it acid.

An average measure for depth of planting is twice the diameter of the bulb in cooler climates, an inch or so deeper in areas like southern Virginia to provide protection against summer heat. For quick effect, plant about a dozen bulbs in a group; six bulbs more widely spaced will usually increase and give the same effect in several years.

Watering

Most bulbs need moisture before, during, and after flowering, when foliage is green and growing or ripening. They're better off on the dry side during dormancy. This is a perfect setup for most hardy types, because they need the least watering in summer, when droughts are most common. But it does increase the urgency for perfect drainage for some of the Western species that can't bear moisture in winter.

Fertilizing

Except for an early-spring top-dressing of leaf-mold compost, we seldom feed our little bulbs. Occasionally, some healthy specimens may get a puny, undernourished look that calls for sprinkling bone meal or superphosphate over the soil and scratching it in. Or we may water with manure "tea" during or after flowering time.

If soil is properly prepared at planting time, supplemental feeding should not be necessary for most types for several years. By that time some bulbs have multiplied so enthusiastically they should be lifted, separated, and reset in freshly mixed soil.

Problems and Pests

If you plant bulbs where they don't stand in soggy mud or water, you'll have little loss from rot. But if you're in a suburban or rural area inhabited by cute chipmunks, squirrels, or mice, you won't want to plant juicy morsels such as tulips, crocus, and eranthis just to feed the animals. I've never had the time or patience to plant bulbs in wire cages. Poison baits can be dangerous when you have children or pets. An effective safety measure is to put the bait inside a clean milk bottle and cover it with a heap of straw or leaves, with evergreen branches to hold the heap in place.

Winter Protection

Because the root systems may not be completely matured, newly planted bulbs should be mulched with a light, airy covering of something like salt hay during their first winter. After that, the necessity for protection varies with hardiness and with climate. Sometimes these mulches hold in more moisture than is good for bulbs. If alternate freezing and thawing should

heave the shallow roots up through cracks in the soil, press them back gently but firmly and cover with soil again.

PROPAGATING MINIATURE BULBS INDOORS

Whether they're to be exhibited in competition at a flower show, or displayed for close-up enjoyment in the living room, uniform perfection is the objective. First, then, buy the finest-quality bulbs available—the most plump, firm, healthy bulbs you can find. If the catalogue listing includes “recommended for forcing” or “specially treated for forcing,” so much the better.

Time for potting, and length of time required for rooting, vary with the varieties. We're usually potting up bulbs for forcing through most of October, occasionally into the first week of November. Our favorites are potted at two-week intervals, so we'll have a continuous show of flowers in winter and early spring.

We prefer clay pots because of their porosity, and shallow pots because most miniature bulbs are shallow-rooted. A generous layer of pebbles or broken crocks goes in the bottom of every pot. The potting soil is a house-plant mixture with a teaspoon of lime mixed in for a six-inch bulb pot. Bulbs are set so their tips are barely covered with soil; then pots are set in water up to the rims to make sure both soil and pots are thoroughly moist.

Before they can be forced into flower, bulbs need time to grow sound root systems. They need to be stored cool, and in the dark so that premature top growth won't be encouraged.

Our potted bulbs are pre-rooted in the cold frame, the pots sunk in the soil, each pot covered by another pot turned upside down over it. Soil is added to cover the top pots; and after severe weather sets in, we top with at least six inches of salt hay.

In lieu of a cold frame, a fairly deep trench, with a layer of pebbles or sand for drainage, can be used in the same way, with twice as much salt hay on top or even more. Some growers set the pots on shelves in cool, dark corners of the basement, but our cellars have always been much too warm. Soil should be kept moist, but not wet, during the rooting period.

When the pot is completely filled with roots, and a few stick out the drainage hole in the bottom, the potted bulbs are moved to a cool, shady spot in the house or greenhouse for just a few days. Then they are brought into the sun for growth and flowering. For the best and most lasting flowers, try to provide temperatures between 55 and 60 degrees by day, five degrees or more cooler by night.

The window of an unheated bedroom or attic, or an enclosed porch, may be the only place where temperatures are low enough for healthiest growth. These are good places to grow the plants until they open their flowers, then they can be brought into warmer quarters for a flowering display.

If you are trying to force bulbs into flower at some specific time, and if the buds aren't swelling as fast as they should, warmer temperatures and a booster feeding of liquid manure should speed up the process. Cooler temperatures will help retard flowering.

Keep the soil always moist before, during, and after flowering. When the leaves begin to yellow, gradually hold back on watering and keep it barely moist until time to set the bulbs out in the garden. Forcing bulbs two years in succession is not usually successful.

MINIATURE BULB PLANTS

Chionodoxa *Liliaceae* Glory-of-the-Snow

Early April flowers that disregard unfavorable weather. Leaves are narrow and grassy and disappear after the flowering period. Flowers are open-faced with short tubes.

gigantea—Largest flowers, pastel lavender-blue. Strong stems that don't fall over in bad weather.

luciliae—Five or more flowers per five-inch stem, bright sky-blue with white centers. The variety *alba* is white; *rosea*, pink.

sardensis—Like *luciliae*, but deeper blue without the white eye.

CARE. Gritty soil. Moisture. Mulch with manure in the fall. Light and sun.

Hardy and easy to grow. Makes a ground carpet.

PROPAGATION. Seeds (seeds itself except for some varieties which increase below ground), offshoots.

SPECIAL USES. Ground cover, rock gardens, forcing.

Colchicum *Liliaceae* Autumn Crocus, Meadow Saffron

Usually called crocus, this fall bloomer actually is a lily.

autumnale—Crocus-like flowers four inches across from September to frost. Leaves two inches wide in spring. Usually has purple flowers. There are several varieties: album, white; roseum, pink and often double; ‘The Giant,’ large, late, lilac and white (best-known variety).

bornmuelleri—Generally considered the finest species with five-inch-wide flowers, rosy-lilac turning to purple and the tube white.

luteum—Rare spring-flowering species, flowers yellow and smaller, leaves slim and less conspicuous.

speciosum—Large leaves, and flowers with pink or white throats. The white variety album is also large and is often considered the most beautiful of hardy bulbous plants.

CARE. Soil loamy, slightly acid. Sun or partial shade. Plant in colonies in August, two to three inches deep. Let leaves mature.

PROPAGATION. Separate bulbs. Division of corms. Seeds.

SPECIAL USES. Borders, rock gardens, forcing.

Crocus *Iridaceae*

There are three general kinds of these grassy-leaved plants with flaring flowers of several colors. The large flowering “Dutch” hybrids flower in the spring; the usual spring-flowering varieties come earlier and last longer; other species flower in the fall.

biflorous—Spring. Bright-purple flowers with darker stripes, yellowish throat.

medius—Fall. Bright-orchid flowers with very pale throat.

sativus—Fall. Pastel-lilac flowers accented with scarlet stamens.

seiberi—Spring. Yellow-throated lavender flowers.

speciosus—Fall. Light blue with fiery orange stigmas.

tomasinianus—Spring. Palest pinkish-blue.

versicolor—Spring. Lilac with purple streaks, or, in the variety *picturatus*, white with purple feathering.

zonatus—Fall. Rosy lavender with orange inside.

CARE. Soil, sandy and drained. Not much fertilizer. Full sun. Hardy, but winter mulch is often helpful. Divide seldom and replant at once. Be on alert for rodents.

PROPAGATION. Offsets, seeds.

SPECIAL USES. Forcing, rock gardens, colonization.

Cyclamen *Primulaceae*

These are the country cousins of the florist's cyclamen but smaller in size. Their substantial, heart-shaped leaves are often beautifully marked in marbled patterns and are in a circle near the ground. Generally they are evergreen through the winter but drop before the flowers appear. Each blossom hangs from its own stem with turned-back tubes. Some varieties blossom in summer and fall, some (indoors or in suitable climates) flower in winter or very early spring. A number of varieties are available.

atkinsi—Crimson flowers in January, foliage marbled. The variety *album* has white flowers; *roseum* is pink.

cilicicum—Purple-throated pale-pink flowers on five-inch stems (in September), small marbled leaves.

coum—Blooms in February and March; choice of red, white, or pink. Dark-green foliage.

europaeum—Fragrant red flowers in late summer. Leaves marbled.

libanoticum—Large flowers with dainty pink blending into white, in early spring. Foliage marbled.

neapolitanum—Exquisite silver-zoned leaves follow the rosy-pink flowers that push their buds through the soil in September. The variety *album* is white.

repandum—Red-purple flowers with wide-flaring petals, in April.

CARE. Humusy soil, with drainage. Must never dry out, but roots must never stand in water. Humidity in greenhouse. Outdoors, shade. Small amount of lime. Once established, small amounts of rotted manure.

PROPAGATION. Seeds, and division (plant with top of tuber at surface; may be dormant first year after planting).

SPECIAL USES. Rock and woodland gardens, forcing.

Eranthis *Ranunculaceae* Winter Aconite

Often during a mild January or February this little flower blooms with its butter-yellow cup on a stem only four inches above the ground. The finely-cut leaves form a collar below the blooms.

'*Cuinia*'—Slightly later and faintly fragrant.

hyemalis—Long-stemmed ferny leaves with clear yellow flowers. The variety *cilicica* has deeper-colored flowers and brownish leaves.

'*Tubergeni*'—A hybrid with larger golden flowers.

CARE. Soil woodsy, not too acid. (Plant is dormant in summer.)

PROPAGATION. Division of roots.

SPECIAL USES. Rock gardens, pockets atop walls, forcing.

Galanthus nivalis *Amaryllidaceae* Snowdrop

An early-flowering plant that often has blooms sticking up through March snow. The narrow leaves form a base for six-inch stems with fragrant white bell-shaped flowers.

Several varieties are available: flore-pleno, double; maximus, larger and less frail-looking flowers; viride-apice, white petals tipped with green.

CARE. Soil heavy, moist, and cool. Considerable shade, but only opens fully in the sun. Top-dress with manure in the fall. Hardy.

PROPAGATION. In fall plant bulbs three to four inches deep, near deciduous trees or shrubs. Most effective in groups. Spreads by self-seeding.

SPECIAL USES. Forcing, rock and wild gardens.

Gladiolus *Iridaceae*

In a small woodland garden only the miniature gladiolus is suitable; they are little fellows two and a half inches in height or less. A number of these small ones are available. Named varieties are:

‘Glee,’ ‘Gnome,’ ‘Impresario,’ ‘Little Egypt,’ ‘Little Fawn,’ ‘Little Red Star,’ ‘Little Spice,’ ‘Periwinkle,’ ‘Pinkie Dots,’ ‘Pint Size,’ ‘Whirlybird,’ ‘Red Button,’ ‘Red Dot,’ ‘Red Tweedle,’ ‘Tom Thumb,’ ‘Wee Red.’ As a point of information, the term “small” applies to plants from two and a half to three and a half inches high.

CARE. Good garden soil. If manure is to be used, it should be applied the year previous and worked into the soil; it should not come into contact with the corms. Drainage. Sun.

PROPAGATION. Plant corms or bulbs.

SPECIAL USES. Rock gardens, beds.

Hyacinthus *Liliaceae* Hyacinth

The large formal hyacinths of the regular garden have a number of relatives that are small, delicate, and graceful, including two midget species:

amethystinus—Early-flowering Spanish species with typically flat leaves and nodding blue bell flowers on spikes seldom over five inches. There is also a white variety, *albus*.

ciliatus (azureus)—Slightly smaller with stiff three-inch spikes, blue-green leaves often hairy on the edges, and blue flowers. The variety *albus* is white.

CARE. Rich fertile soil. Drainage. Mulch if no leaves fall on plants. Sun.

PROPAGATION. Division, seeds.

SPECIAL USES. Forcing, rock gardens.

Leucojum *Amaryllidaceae* Snowflake

The long leaves are about a half-inch wide and in a basal circle from which the stems stretch up to hold nodding bell flowers marked on the edge with bright-green dots.

aestivum—summer snowflake—Flowers in May. About nine inches tall, it may be a little large for really small gardens.

vernum—spring snowflake—One fragrant bell flower to each six-inch stem, in April.

CARE. Soil rich and sandy, drainage, protect from hot sun.

PROPAGATION. Detach bulblets after leaves wither. Self-seeds. Increases rather fast. Plant bulblets three to four inches deep.

SPECIAL USES. Rock gardens.

Muscari *Liliaceae* Grape Hyacinth

These slim-leaved plants come in a wide variety with a selection of flowering times from early March through May and have clusters of flowers that look bright and grape-like or soft and plummy.

armeniaceum—Violet-blue flowers on six-inch stems. The variety *cantab* blooms in mid-spring, clear blue and sweet-scented. ‘Early Giant’ is larger and earlier. True blue.

botryoides album—Heavenly white and fragrant. The most compact and suitable for smaller gardens, or forcing.

comosum monstrosum (plumosum)—feather or tassel hyacinth—Lavender feather dusters in late May.

moschatum flavum—Out-of-the-ordinary purple flowers turning to yellowish in March and April.
One of the smaller species.

pacemosum—Fragrant, deep-blue flowers on eight-inch stems.

parodoxum—Flowers of blue so dark they're nearly black.

tubergianum—A newly introduced species, with the top flowers turning lighter, and the lower ones darker. (Mid-April.)

CARE. Soil deep, rich, and sandy. Drainage. Sun or light shade.

PROPAGATION. Offsets, seeds.

SPECIAL USES. Forcing, rock gardens.

Narcissus *Amaryllidaceae* Daffodil

It is very unlikely that the average gardener realizes how many varieties in the narcissus world are under six inches in height at maturity. One catalogue lists over two hundred names of little varieties for little gardens. Obviously, only a few can be mentioned here.

bulbocodium—hoop-skirt daffodil—The “trumpet” is drawn in a bit at the mouth, making it more like a balloon than a trumpet. Six smallish petals twist and turn shyly behind. All varieties are under six inches tall. Among the smallest are three-inch, scented, white-flowering monophyllus (for forcing only); obesus, with a fat yellow trumpet; tunifolium, smaller, brighter, and later than the species; and scented vulgaris nivalis, the smallest of all at two to three inches and very early. The variety conspicuus is the easiest to grow and about four inches tall.

cyclamineus—The nodding trumpets are long and slim, the petals fold straight back around the stem, giving a feeling of motion. The species is all yellow and about four inches high. Among the smallest hybrids are ‘Milky-White,’ ‘Yindee,’ ‘Minicycla,’ deep yellow ‘Jetage,’ and ‘Flute.’

jonquilla—Like the familiar jonquils, these are usually scented and have small, flattish cups with a neat ruff of flat petals around the neck. The variety atlanticus is pure white and four inches high; the variety juncifolius is three inches high and has several scented yellow flowers per stem. ‘Peachblossom’ is similar and one inch taller.

tazetta—Similar to jonquils, but the cups are even smaller and flatter. They are mostly early flowering and have several blooms to a stem. The species is twelve inches tall, but the variety lacticolor (canaliculatas) is only four inches, with yellow cup and white petals that flare back. ‘Halenzy’ combines pale yellow and white. ‘Shrew’ is the tiniest.

triandrus—angel’s tears—Several drooping flowers per stem, the center cup-shaped, the petals turned back and twisted rather like a tiny fuchsia flower. Among the varieties, albus is all white, and three to four inches high; concolor is all gold; ‘Frosty Morn’ is white; ‘Poppet’ has yellow petals and a white cap.

trumpet—Miniature replicas of the familiar larger types, the flaring trumpet standing out from a collar of six petals. In some types the petals are flat, in others wavy or twisted. Among these, asturiensis (minimus) is the smallest of all daffodils; minar (nanus) is slightly larger, two to three inches tall; ‘Wee Bee’ is lemon yellow and five inches tall. ‘Little Beauty’ has two shades of yellow; ‘Sneezy’ is all yellow, and four inches high.

CARE. Average soil (acid or alkaline), not too heavy, not too light. Drainage. Fertilize little and use only bone meal. Mulch with stone chips to prevent flowers being spattered with mud during rain.

PROPAGATION. Bulbs. Plant in August if possible. Plant in groups because of small size. Allow space for increase. Plant smallest bulbs one and a half inches deep; large bulbs, the depth of bulb itself. Don't disturb unless crowded.

SPECIAL USES. Forcing, rock gardens, containers.

Oxalis Oxalidaceae

Bulbous or tuberous shamrock-leaved perennials with five-petaled, buttercup-shaped flowers that close in the dark, most of them much hardier than realized and all suitable for indoor or greenhouse planting.

adenophylla—Blue-green leaflets, clusters of pink flowers with deeper pink veins (May and June). Three inches tall. Hardy.

enneaphylla—Tuberous-rooted perennial with gray-green leaves and large white-purple veined flowers; two inches. Spring bloomer.

lasiandra—More fragile species, bulbous with tight clusters of deep-rose flowers. Dig and store in fall, or provide protection. Sun.

rubra (crassipes)—Not bulbous but with thickened roots. Pink flowers in clusters above the leaves.

variabilis rubra—Dense mats of fresh green leaves under deep-pink flowers, some two inches across. Hardy as *rubra*.

CARE. Average soil. Dig and store bulbs of some species.

PROPAGATION. Division of roots, bulbs.

SPECIAL USES. House plants, rock planting, containers.

Sternbergia lutea Amaryllidaceae Winter Daffodil

Popularly called a daffodil and compared to a crocus, but these small amaryllids beat both of those plants for durability and distinction. They have up-facing butter-yellow flowers like squat, flaring, two-inch vases. They bloom in mid-September with the flowers lasting some weeks. The narrow leathery leaves often last through the winter but die back by spring. Fragrant.

CARE. Heavy soil, barely acid. Dry, drainage. Sun.

PROPAGATION. Bulbs.

SPECIAL USES. Forcing, rock gardens.

Tulipa *Liliaceae* Tulip

These are typical tulips with flowers that are up-facing bells or saucers, but are varied in shape, colors, and markings of the petals. They keep tightly closed after dark and on dark days but open wide in the sun. The foliage may be wide or grassy, clean green or mottled. Included here are only the smallest available species.

batalini—Twisted, narrow leaves and soft creamy-yellow flowers (in April).

biflora—Only two or three slim, tapered leaves around a four-inch flower stem bearing from one to three long, striped buds which open to flat, white, star-shaped flowers with golden centers. Often blooms in March. Variety *turkestanica* has wider leaves and up to ten flowers per stem.

dasytemon (tarda)—Polished blue-green leaves fan out flat under three-inch stems with bouquets (three to five) of yellow-centered white flowers. Some have pointed petals striped lengthwise with green on the outside. (Mid-April.)

eichleri—Typically broad leaves, stems six to eight inches, fiery-red flowers with gold and blue-black centers and gold stripes outside.

kaufmanniana—water-lily tulip—The leaves are very broad, short, and sharply pointed; the stems are sturdy and five inches long. The flowers are very large, creamy yellow marked with carmine (in April). There are many named varieties with flowers in shades and combinations of gold, scarlet, cream, and even salmon pink.

kolpakowskiana—Long, blue-green leaves waved on the edges, yellow flowers blotched with brick red on the outside (in April).

linifolia—Narrow, wavy leaves and six-inch stems topped with satiny scarlet flowers, marked with blue-black at the heart (in April).

patens (persica)—Slim, arching leaves, stems with one to three flowers, yellow, darker outside, fragrant (in May).

puchella—Early, even before the daffodils. Slim leaves. Flowers an odd shade of lavender and tinged red. Slightly larger.

stellata chrysantha (originally considered a species, *chrysantha*)—Curved-down blue-green leaves, six-inch stems; cherry-flushed yellow flowers opening to flat stars (in April).

EPILOGUE

There is a touch of sadness to any book that is published posthumously. It means that an author who has devoted years to acquiring a skill with words and their ways, one who has something to say, will write no more.

This book, *All About Miniature Plants and Gardens, Indoors and Out*, is the work of Bernice Gaines Brilmayer. It was written in the last months of her life and was practically completed except for the mechanics of preparing it for publication. Although she didn't realize the seriousness of her illness, she asked me to do the final work on the manuscript.

Her dedication, "For my family, again and again. The future is coming closer," is typical of the magnificent woman she was. She lived for the days when we would be free of the commercial world, when we could write, dream, plant, and write. It was to be for, and with, the family she loved so well. I've never heard the philosophy of her existence expressed so well as by an old fishing friend, Captain Bill Stiles of Stone Harbor, New Jersey. Captain Bill, of the *Skipper*, is not one who professes to write phrases and words, but he does know people. On the long ride out to the fishing grounds off the Jersey coast Bill used to observe her. When the anchor was down, Bill would give the order to "drop lines, Folks." He noted that she was always the last to get her line in the water. Her family came first and she waited until they were set.

If ever I do a biography of Bernice, the title will be "The Last to Get Her Line Down."

The family to which she dedicated her books, has a dedication to her:

TO BERNICE—A GALLANT WOMAN
1913–1962
IN LOVING MEMORY
THE FAMILY YOU LOVED SO MUCH

Robert Brilmayer
Redding, Connecticut, 1963

BIBLIOGRAPHY

Agricultural Research Service, U. S. Department of Agriculture, PLANT HARDINESS ZONE MAP. Washington, D.C.: Government Printing Office, 1960.

Ashberry, Anne, MINIATURE GARDENS. New York: Van Nostrand, 1952.

Ashberry, Anne, MINIATURE TREES AND SHRUBS. London: Nicholas Kaye, Ltd., 1958.

Bailey, L. H., and Ethel Zoe, HORTUS SECOND. New York: Macmillan, 1941.

Beale, James H., THE EVERGREENS. New York: Doubleday, 1960.

Brooklyn Botanic Garden, handbooks on many subjects—DWARF TREES AND SHRUBS, DWARFED PLANTS, FLOWERING SHRUBS, PROPAGATION, ROCK GARDENS, WHERE TO BUY TREES AND SHRUBS. New York: Brooklyn Botanic Garden, various dates.

Chidamian, Claude, BONSAI, MINIATURE TREES. New York: Van Nostrand, 1955.

Chidamian, Claude, CACTI AND OTHER SUCCULENTS. New York: Doubleday, 1958.

Clark, William H., GARDENING THE SMALL PLACE. Boston: Little, Brown, 1952.

Clarke, J. Harold, GETTING STARTED WITH RHODODENDRONS AND AZALEAS. New York: Doubleday, 1960.

Cumming, Robert W., and Lee, Robert E., CONTEMPORARY PERENNIALS. New York: Macmillan, 1960.

Foley, Daniel J., GROUND COVERS FOR EASIER GARDENING. New York: Chilton & Co., 1961.

Free, Montague, PLANT PROPAGATION IN PICTURES. New York: Doubleday, 1957.

Genders, Roy, GARDENING IN MINIATURE. London: Robert Hale, Ltd., 1958.

Genders, Roy, MINIATURE ROSES. London: Blanford Press, 1960.

Graf, Alfred Byrd, EXOTICA II. Rutherford, N.J.: Julius Roehrs, 1960.

Gray, Alec, MINIATURE DAFFODILS. London: W. H. & L. Collingridge, Ltd., 1955.

Ishimoto, Tatsuo, ART OF GROWING MINIATURE TREES, PLANTS AND LANDSCAPES. New York: Crown Publishers, 1956.

Klaber, Doretta, ROCK GARDEN PLANTS. New York: Henry Holt, 1959.

Moore, Harold E. Jr., AFRICAN VIOLETS, GLOXINIAS AND THEIR RELATIVES. New York: Macmillan, 1957.

Newsom, Samuel, DWARFED TREE MANUAL FOR WESTERNERS. Tokyo: Tokyo News Service, Ltd., 1960.

Noble, Mary, and Merkel, J. L., PLANTS INDOORS. New York: Van Nostrand, 1954.

Rickett, Harold William, BOTANY FOR GARDENERS. New York: Macmillan, 1957.

Rockwell, F. F., and Grayson, Esther C., COMPLETE BOOK OF BULBS. New York: Doubleday, 1953.

Taylor, Norman (ed.), ENCYCLOPEDIA OF GARDENING. Boston: Houghton Mifflin, 1961 (Revised Edition).

Taylor, Norman, WILD FLOWER GARDENING. New York: Van Nostrand, 1955.

Thomas, G. L. Jr., GARDEN POOLS, WATER-LILIES AND GOLDFISH. New York: Van Nostrand, 1958.

Van Melle, P. J., SHRUBS AND TREES FOR THE SMALL PLACE. New York: Scribner's, 1943.

Westcott, Cynthia, ARE YOU YOUR GARDEN'S WORST PEST? New York: Doubleday, 1961.

Westcott, Cynthia, GARDENER'S BUG BOOK. New York: Doubleday, 1956.

Wilder, Louise Beebe, ADVENTURES IN MY GARDEN AND ROCK GARDEN. New York: Doubleday, 1929.

Wilson, Helen Van Pelt, NEW PERENNIALS PREFERRED. New York: Van Nostrand, 1962.

Wyman, Donald, SHRUBS AND VINES FOR AMERICAN GARDENS. New York: Macmillan, 1958.

Wyman, Donald, TREES FOR AMERICAN GARDENS. New York: Macmillan, 1951.

WHERE TO BUY MINIATURE PLANTS AND SUPPLIES

Nothing is as frustrating to a gardener as to hear or read glowing descriptions of plants that appeal to him and then be unable to find them in catalogues or elsewhere. For the unusual and out-of-the-ordinary plants the answer is quite often a mail-order supplier. For some general types such as water lilies and woodland or rockery plants, the most fruitful source is the specialist, who quite naturally has the widest selection and the greatest knowledge of the plants he grows the most. We have increased our collection of miniature plants by mail order from many (but not all) of the following. While I have not bought from all of them, enough of my friends and colleagues have so that I am rather familiar with all. In writing for catalogues of any sort, remember they are expensive to produce and often you are asked to pay a nominal fee for them. I usually get so much information I am happy to pay a few cents. You can often save time and trouble by asking if there is such a fee.

ALBERTS & MERKEL BROS., INC., Box 537, Boynton Beach, Fla.—Orchids and other tropical plants.

ALLGROVE, ARTHUR EAMES, North Wilmington, Mass.—Terrarium and woodland plants and supplies.

ALPENGLOW GARDENS, 13328 Trans-Canada Highway, North Surrey, New Westminster, B.C., Canada—Alpines, perennials, trees, and shrubs.

AMERICAN PERENNIAL GARDENS, 6975 Dover St., Garden City, Mich.—Perennials.

ARENIUS, ARTHUR, 123 Western Drive, Longmeadow 6, Springfield, Mass.—Gladiolus.

ARMSTRONG NURSERIES, Ontario, Calif.—Roses, dwarf citrus.

ARNDT'S FLORAL GARDEN, Route 2, Box 336, Troutdale, Ore.—African violets and other gesneriads.

BARNHAVEN, Box 218, Gresham, Ore.—Primulas.

BARRINGTON GREENHOUSES, 860 Clements Bridge Rd., Barrington, N.J.—House and greenhouse plants.

BEAHM GARDENS, 2686 E. Paloma St. Pasadena 8, Calif.—Epiphyllums and other succulents.

BEE, MRS. T. C., Route 3, Box 120, Newnan, Ga.—African violets and other house plants.

BIT OF FLORIDA, INC., Box 3305, West Hollywood, Fla.—Dwarf citrus.

BRECK'S OF BOSTON, 250 Breck Bldg., Boston 10, Mass.—Seeds, bulbs, and supplies.

BRIMFIELD GARDENS NURSERY, 245 Brimfield Rd., Wethersfield, Conn.—Trees and shrubs for garden, bonsai.

BRISTOL NURSERIES, Bristol, Conn.—Chrysanthemums.

BUELL'S, Eastford, Conn.—African violets and other gesneriads.
 BURGESS SEED AND PLANT CO., Galesburg, Mich.—Seeds, bulbs, perennials, trees, and shrubs.
 BURNETT BROS., INC., 92 Chambers St., New York 7, N.Y.—Seeds, bulbs, supplies.
 BURPEE, W. ATLEE, CO., Philadelphia 32, Penna.—Seeds, bulbs, supplies.
 BURT'S, Federal Highway No. 1, North Jupiter, Fla.—House and greenhouse plants.
 CACTUS PETES, 5454 Valley Blvd., Los Angeles 32, Calif.—Epiphyllums and other succulents.
 CALIFORNIA NURSERY CO., Niles, Calif.—Trees and shrubs.
 CAROLINA BIOLOGICAL SUPPLY CO., Elon College, N.C.—Woodland, bog, and water plants.
 CARROLL GARDENS, Westminster, Md.—Bulbs, perennials, trees, and shrubs.
 CHAMPLAIN VIEW GARDENS, Burlington, Vt.—Gladiolus.
 CHAUTAUQUA FLOWERFIELD, Greenhurst, N.Y.—Bulbs, perennials.
 CHERRY, NORMAN J., CO., Box 608, Merrick, Long Island, N.Y.—Supplies for house and greenhouse.
 COCHLAN, JAMES J., Box 331, Saddle River, N.J.—Bonsai plants and containers.
 CONARD-PYLE CO., West Grove, Penna.—Roses.
 DAFFODIL MART, Nuttall Station, Gloucester, Va.—Narcissus.
 DE JAGER, P., AND SONS, INC., 188 Ashbury St., South Hamilton, Mass.—Bulbs.
 DELKINS BULBS, 4205 Hunts Point Rd., Bellevue, Wash.—Bulbs.
 DOLBOW, DOROTHY J., 149 W. Main St., Penns Grove, N.J.—Supplies for house and greenhouse.
 EASTERN NURSERIES, INC., Holliston, Mass.—Trees, shrubs, alpine.
 EDENWALD GARDENS, Vincennes, Ind.—Iris.
 EL SEGUNDO GERANIUM GARDENS, Box 289, Gardena, Calif.—Geraniums.
 EMERSON INDUSTRIES, INC., 132 Adams Ave., Hempstead, N.Y.—Window greenhouses, etc.
 EVANS, HERBERT O., Solon, Ohio—Gladiolus.
 EVERGLADES ENTERPRISES, Box 48-811, Miami, Fla.—Tropical plants.
 EVERGREEN HOUSE, Box 433, Los Altos, Calif.—Bonsai plants and containers.
 FANTASTIC GARDENS, 9550 S.W. 67th St., Miami, Fla.—Bromeliads and other tropical plants.
 FENNELL ORCHID CO., 26715 S.W. 157th Ave., Homestead, Fla.—Orchids.
 FIELD, HENRY, SEED AND NURSERY CO., Shenandoah, Iowa—Seeds, bulbs, perennials, aquatic plants, trees, shrubs, and house plants.
 FIORE, CHARLES, NURSERIES, INC., Prairie View, Ill.—Trees and shrubs.
 FRENCH, J. HOWARD, Baltimore Pike, Lima, Penna.—Seeds, bulbs, supplies.
 GARDENS OF THE BLUE RIDGE, Ashford, McDowell County, N.C.—Native woodland, bog, and water plants, trees and shrubs.
 GERRY'S GERANIUM GARDEN, 221 W. 223 St., Torrance, Calif.—Geraniums.

GIBBS, WINSEL, SEED AND NURSERY CO., 1955 W. Florence Ave., Los Angeles 47, Calif.—Tropical trees, shrubs, plants.

GIRARD NURSERIES, Geneva, Ohio—Trees and shrubs for garden, bonsai.

GLEN SAINT MARY NURSERY, Glen Saint Mary, Fla.—Trees and shrubs.

GRAY, ALEC, Treswithian Daffodil Farm, Camborne, Cornwall, England—Narcissus.

GREEN LEAF NURSERY, 2047 Middlefield Rd., Mountain View, Calif.—House and greenhouse plants.

GRIFFING NURSERIES, Beaumont, Texas—Trees and shrubs.

HAGA, F. M., & SON, 5033 Doris St., Charlotte 5, N.C.—African violets and other gesneriads.

HARRIS, JOSEPH, CO., INC., Rochester 11, N.Y.—Seeds, bulbs.

HARROLD'S, Box 29, Grants Pass, Ore.—House-plant and garden seeds, plants, roses.

HAV'ALOOK GARDENS, 10045 Grand River, Fowlersville, Mich.—House and greenhouse plants.

HOLLY HILL, Straight Path, Huntington, N.Y.—Holly, heath, heather.

HOUDYSHEL, CECIL, 1412 Third St., LaVerne, Calif.—House and greenhouse bulbs and plants.

HOUSE OF PLANTS, 26 S. Hotchkiss St., Binghamton, N.Y.—Gesneriads.

HOUSE PLANT CORNER, Box 810, Oxford, Md.—Supplies for house and greenhouse.

ILGENFRITZ, MARGARET, Monroe, Mich.—Orchids.

INTER-STATE NURSERIES, Hamburg, Iowa—Seeds, bulbs, perennials, trees, and shrubs.

JACK'S CACTUS GARDEN, 1707 W. Robindale St., West Covina, Calif.—Cacti and other succulents.

JACKSON & PERKINS CO., Newark, N.Y.—Roses, perennials.

JOHNSON CACTUS GARDENS, Paramount, Calif.—Cacti and other succulents.

JOYNER'S, 404 Fisher Drive, Ocala, Fla.—Caladiums.

KARTUZ, MICHAEL J., 92 Chestnut St., Wilmington, Mass.—House and greenhouse plants.

KELSEY NURSERY SERVICE, Highlands, N.J.—Bulbs, trees and shrubs, woodland plants.

LAGER AND HURRELL, INC., 427 Morris Ave., Summit, N.J.—Orchids.

LAKEMONT NURSERIES, 915 S. Lakemont Ave., Winter Park, Fla.—House and greenhouse plants.

LAMB NURSERIES, 101 E. Sharp Ave., Spokane 2, Wash.—Chrysanthemums.

LESLIE'S WILD FLOWER NURSERY, 30 Summer St., Methuen, Mass.—Woodland plants.

LOGEE'S GREENHOUSE, 55 North Street, Danielson, Conn.—House and greenhouse plants.

LOUNDBERRY GARDENS, Oakford, Ill.—Bulbs and woodland plants.

MANHATTAN GARDEN SUPPLY, 305 N. Sepulveda Blvd., Manhattan Beach, Calif.—Geraniums.

MARX, WALTER, GARDENS, Boring, Ore.—Bulbs, perennials, roses.

MAYFAIR NURSERIES, RFD 2, Nichols, N.Y.—Alpines, trees, and shrubs for gardens, bonsai.

MCCLEELAN, ROD, CO., 1450 El Camino Real, South San Francisco, Calif.—Orchids, bonsai containers.

MERRY GARDENS, P. O. Box 595, Camden, Maine—Rare house plants, geraniums, begonias, and a large selection of miniatures.

MILLER, HOLMES C., 280 W. Portola Ave., Los Altos, Calif.—Geraniums.

OAKHURST GARDENS, Box 444, Arcadia, Calif.—House and greenhouse plants.

OPPEN'S GREENHOUSE, 4330 Auburn Rd., Salem, Ore.—House and greenhouse plants.

PARK, GEORGE W., SEED CO., Greenwood, S.C.—Seeds, bulbs, supplies.

PEARCE SEED CO., Moorestown, N.J.—Seeds, bulbs, perennials, plants.

PILLEY'S GARDENS, Box 757, Valley Center, Calif.—Iris.

PINE-CROFT NURSERIES, Exeter, N.H.—Perennials.

PLANTERS, THE, 385 Bleecker St., N.Y.C.—House and greenhouse plants, bonsai.

PUTNEY NURSERY, Putney, Vt.—Chrysanthemums.

RARE PLANT CLUB, Route 1, Box 155, Mill Valley, Calif.—Bonsai plants and containers.

ROEHRS, JULIUS, CO., Rutherford, N.J.—House and greenhouse plants.

SCHERER, S., AND SONS, Northport, N.Y.—Aquatic plants and water lilies.

SCHULZ, PEGGIE, 7714 N. Fairfield Rd., Minneapolis 12, Minn.—House-plant seeds.

SHERWOOD NURSERY CO., Corbett, Ore.—Trees and shrubs for garden and bonsai.

SKY-CLEFT GARDENS, Camp Street Ext., Barre, Vt.—Alpines, perennials, woodland plants.

SLOCUM WATER GARDENS, Binghamton, N.Y.—Aquatic plants and water lilies.

SPRING HILL NURSERIES, Tipp City, Ohio—Perennials, roses, trees.

STARK BROS., Louisiana, Mo.—Trees and shrubs.

STARKER, CARL, GARDENS, Jennings Lodge, Ore.—Alpines, perennials, trees and shrubs, bonsai containers.

STEWART, FRED A., INC., 8606 E. Las Tunas Drive, San Gabriel, Calif.—Orchids.

SUMMIT GARDENS, Box 567, Glenn Ellyn, Ill.—Gladiolus.

TERRACE VIEW GARDENS, Greencastle, Ind.—House and greenhouse plants.

THOMPSON & MORGAN, LTD., Ipswich, England—Seeds.

THREE LAURELS, Madison County, Marshall, N.C.—Woodland plants; trees and shrubs.

THREE SPRINGS FISHERIES, Lilypons, Md.—Water lilies and aquatic plants of all sorts.

THURMAN'S GARDENS, Route 6, Spokane, Wash.—Alpines, perennials.

TINARI GARDENS AND GREENHOUSES, Bethayres, Penna.—African violets.

TRICKER, WILLIAM, INC., 196 Allendale Ave., Saddle River, N.J.—Water lilies and other aquatic plants.

TROPICAL PARADISE GREENHOUSE, 8825 W. 79th St., Overland Park, Kan.—House and greenhouse plants.

UPPER BANK NURSERIES, Ridley Creek Rd., Media, Penna.—Trees and shrubs.

VAN NESS WATER GARDENS, 2460 N. Euclid Ave., Upland, Calif.—Water lilies and other aquatic plants.

VAUGHAN'S SEED CO., 24 Vesey St., New York 7, N.Y.—Seeds, bulbs, perennials, trees, shrubs, aquatic plants.

VILLAGE HILL NURSERY, Williamsburg, Mass.—Geraniums, perennials.

WAKE ROBIN FARM, Route 1, Box 33, Home, Penna.—Woodland plants.

WARE, ROMAINE B., Canby, Ore.—Lilies and other bulbs.

WARREN'S NURSERY, 2200 Fifth St., Berkeley 10, Calif.—House and greenhouse plants.

WAYSIDE GARDENS, Mentor, Ohio—Annuals, bulbs, trees, shrubs, etc.

WESTON NURSERIES, INC., E. Main St., Hopkinton, Mass.—Perennials, trees, and shrubs.

WHITE FLOWER FARM, Litchfield, Conn.—Bulbs, perennials, trees, shrubs.

WILSON BROTHERS, Roachdale, Ind.—Geraniums.

WOOD, THOMAS M., Constantia, N.Y.—Woodland plants.

WOODLAND NURSERY, Hendricks Rd., RFD 1, Perkiomenville, Penna.—Trees and shrubs.

WYANT, MELVIN E., Johnny Cake Ridge, Route 84, Mentor, Ohio—Roses.

YOARS HOUSE PLANT NURSERY, Bunker Hill, Ind.—House and greenhouse plants.

INDEX

Abelia, [236](#)
Abeliophyllum distichum (White Forsythia, Korean Abelia), [236](#)
Abies (Fir), [238](#)
Abutilon hybridum savitzi (Flowering Maple), [93](#)
Abutilon megapotamicum, [64](#)
Achillea (Yarrow, Hilfoil), [259](#)
Achimenes, [93–94](#)
Acorus gramineus variegatus, [94](#)
Adder's-Tongue, [222](#), [288–89](#)
'Adeline' ('Improved Darling'), [96](#)
Adiantum bellum (Bermuda Maidenhair Fern), [94](#)
Aeonium caespitosum spathulatum, [102](#)
Aeschynanthus, [63](#)
Aethionema (Stone-Cress, Persian Candytuft), [260](#)
African boxwood, [124](#)
African violets (See: [Viola](#))
Agave victoriae-reginae ("century plant"), [102](#)
Ageratum (floss flower), [282](#)
Aichryson (Aeonium) domesticum variegatum, [102](#)
Ajuga (Bugle), [260](#)
'Alba,' [285](#)
'Albatross,' [101](#)
All About Begonias, [78](#)
All About Vines and Hanging Plants, [36](#), [37](#)
Allophyton mexicanum (Mexican Foxglove), [94](#)
Aloe, [102](#)
Alpenglow Gardens, [13](#)
Alpine daisy, [263](#)
Alpine gardens, [187–98](#)
Alpine phlox, [271](#)
Alpine (rock) plants, [62](#), [150](#), [187–98](#), [273](#)
Alternanthera (Joseph's Coat), [94–95](#)
Aluminum plant (watermelon pilea), [130](#)

Alyssum (Madwort), [47](#), [138](#), [260–61](#)
Alyssum (Sweet), [282–83](#)
American Iris Society, [268](#)
American yew, [248](#)
Andromeda (Bog Rosemary), [238–39](#)
‘Andy,’ [96](#)
Anemone (Windflower), [261](#)
Angel’s tears, [296–97](#)
Annuals, [10](#), [38](#), [47](#), [138](#);
 miniature, [277–87](#);
 in rock and wall gardens, [187](#)
Antennaria (Pussy-Toes), [261](#)
Anthurium scherzerianum (Flamingo Flower), [95](#)
Aphids, [232](#)
Aptenia cordifolia (*Mesembryanthemum cordifolium*), [102](#)
Aquilegia (Columbine), [262](#)
Arabis (Rock Cress), [262](#)
Aramite, [232](#), [255](#)
Arborvitae (*Thuja*), [249](#)
Arisaema triphyllum (Jack-in-the-pulpit), [211](#), [219](#)
Arizona, [140](#)
Aroids, [51](#), [53](#), [63](#)
Artillery plant, [130](#)
Asarum (Wild Ginger), [160](#), [219](#)
Ash, [166](#)
Ashberry, Anne, [150](#)
Asia, [271](#), [272](#)
Asplenium (Spleenwort), [219](#)
Aster alpinus (Rock Aster), [262](#)
‘Astrid,’ [129](#)
Astrophytum (Star Cactus), [102](#)
‘Aurea,’ [285](#)
Autumn Crocus, [289](#), [292](#)
‘Autumn Snow,’ [267](#)
Azalea (*Rhododendron*), [177](#), [213](#), [226–27](#), [234](#), [239](#)

Babiana stricta, [95](#)

Baby blue eyes, [284](#)
'Baby Bunting,' [145](#)
'Baby Crimson,' [145](#)
'Baby Perfection,' [98](#)
'Baby Rainbow,' [100](#)
Baby Toes, [104](#)
Baby's Breath, [267](#)
Baby's Tears, [120–21](#)
Ball Cactus, [106](#)
Ballard, Ernesta, [13](#)
Balsam, [283](#)
Banana, dwarf, [63](#), [68](#)
Barberry, [228](#);
 dwarf, [180](#)
Basket of gold, [261](#)
Bayside, [23](#), [24](#)
Beech, [230](#)
Begonia: [19](#), [25](#), [28](#), [29](#), [37](#), [63](#), [95–101](#);
 fibrous-rooted, [95](#), [96–97](#);
 floriferous, [63](#);
 grown in artificial light, [24](#), [35](#);
 miniature (dwarf), [44](#), [53](#), [95–101](#);
 Strawberry (See: [Saxifraga](#));
 tuberous, [24](#), [95](#)
 (See also [Rex begonias](#), [Rhizomatous begonias](#), [Semperflorens begonias](#))
Bell Flower, [263–64](#)
Bellis (Daisy), [263](#)
Bermuda Maidenhair Fern, [94](#)
Bertolonia, [100](#)
Better Business Bureau, [215](#)
Betula nana (Birch), [239](#)
Billbergia, [114](#)
Birch tree: [177](#), [183](#), [239](#);
 as base for bromeliads, [69](#);
 in bonsai, [166](#)
Bird's nest, [246](#)

Bishop's Cap, [224–25](#)
'Black Falcon,' [98](#)
Blackleaf panamiga, [131](#)
'Black Vesuvius,' [126](#), [249](#)
Bloodroot, [160](#), [212](#), [214](#)
'Blue Ball,' [282](#)
'Blue Gem,' [95](#)
'Blue Mink,' [282](#)
Blue Oxalis, [125](#)
'Blue Star,' [285](#)
Bluets, [223](#)
Boea hygroskopica, [100](#)
Bog Rosemary, [238–39](#)
Bone meal (See [Superphosphate](#))
Bonsai: [159–76](#);
 character of the plant in, [164–65](#);
 in citrus, [163](#);
 indoor, [161–63](#);
 name, [161](#);
 plants for use in, [101](#), [110](#), [122](#), [124](#), [132](#), [134](#), [150](#), [160](#), [166](#), [227](#),
 [238–49](#);
 root-pruning in, [169–70](#);
 roses in, [138](#);
 shinto, [161](#)
'Bo-Peep,' [145](#)
Boston (Massachusetts), [64](#), [65](#);
 beans, [107](#)
Bottle Brush, [243](#)
Bouvardia longiflora humboldti, [101](#)
'Bow-Arriola,' [97](#), [99](#)
'Bow Chance,' [97](#), [99](#)
'Bow-Joe,' [97](#)
'Bow-Nigra,' [97](#), [99](#)
Box (See [Buxus](#); [Buxus microphylla japonica](#))
Boxing gloves, [106](#)
Boxwood, African, [124](#)
 (See also: [Buxus](#); [Buxus microphylla japonica](#))

‘Boy Blue,’ [286](#)
Bracts, [104](#), [136](#)
‘Bright Eyes,’ [285](#)
Broadlily, [220](#)
Bromeliad, [44](#), [45](#), [69–70](#), [114](#), [136](#);
 tree, [63](#), [69–70](#)
Brooklyn, [137](#)
‘Brownie Scout,’ [284](#)
Bruckenthalia spiculifolia (Balkan Heath), [239](#)
Bugle, [260](#)
Bulbs: [288–98](#);
 fertilizing, [290](#), [292](#);
 propagating miniature, indoors, [291–92](#);
 South African, [95](#);
 spring-flowering, [64](#);
 winter-flowering, [95](#)
Bunchberry, [221](#)
Bunny ears, [106](#)
Burning Bush, [243](#)
Burpee, [281](#)
‘Butterball,’ [284](#)
Button Fern, [128](#)
Buxus (Box, Boxwood), [46](#), [138](#), [239–40](#), [249](#)
Buxus microphylla japonica, [101](#)

Cactus: [49](#), [64–65](#), [78](#), [101–7](#);
 Christmas, [30](#);
 use in window gardens of, [19–21](#)
Caladium (Elephant Ears), [11](#), [107–9](#);
 bicolor varieties of, [107](#)
Calamondin, [113](#)
Calathea (*Maranta*), [109](#), [122](#)
Calceolarias, [38](#)
Calceolaria biflora, [153](#)
Callopsis volkensi, [109](#)
Calluna vulgaris (Heather, Ling) [226](#), [240](#)
Campanula (Harebell, Bell Flower), [263–64](#)

Campion, [273–74](#)
Camptosorus rhizophyllus (Walking Fern), [219](#)
Canada, [219](#)
Canadian yew, [248](#)
‘Canary Yellow,’ [287](#)
‘Candidum, Jr.,’ [107](#)
Candytuft, [252](#), [267](#);
 Persian, [260](#)
Cankerworms (See [“Inchworms”](#))
‘Caperata’ (‘Little Fantasy’), [129](#)
‘Caprice,’ [286](#)
Caraway thyme, [274](#)
Carex elegantissima (*Morrowi variegata*), [110](#)
Carissa grandiflora nana compacta (Natal Plum), [110](#)
Carnations, [266](#)
Carpet bugle, [260](#)
‘Carpet of Snow,’ [282](#)
Caryopteris, [229](#)
Caryopteris incana (Blue Spirea), [240](#)
Caryota urens (fishtail palm), [63](#)
Cassiope lycopodioides, [240](#)
Catchfly, [273–74](#)
Cedar, [234](#);
 White, [241](#)
“Century plant,” miniature, [102](#)
Cephalocereus senilis (Old Man Cactus), [102](#)
Ceramics, as plant containers, [39](#), [41](#)
Cerastium, [263](#)
Ceratostigma plumbaginoides (Leadwort), [265](#)
Ceropegia, [110](#)
Chaenomeles (*Cydonia*) (Japanese Quince, Flowering Quince), [240–41](#)
Chaenostoma fastigiatum (Little Stars), [111](#)
Chamaecereus silvestri (Peanut Cactus), [102](#)
Chamaecyparis, [226](#)
Chamaecyparis (*Retinospora*) (False Cypress, White Cedar), [141](#)
Chamaedaphne calyculata (Leatherleaf), [241](#)

Chamaedorea elegans bella, [111](#)
Chamaeranthemum, [111](#)
‘Chantilly Lace,’ [98–99](#)
Charcoal:
 for house plants, [78](#);
 use in terrariums of, [58–59](#)
‘Charm,’ [93](#)
Cheddar pink, [266](#)
Chenille plant, [103](#)
‘Cherry Rose,’ [284](#)
Chidamian, Claude, [159](#)
Chin Cactus, [104](#)
‘China Doll,’ [98–99](#)
China plant, [136](#)
Chinodoxa (Glory-of-the-Snow), [292](#)
Chipmunks, [290](#)
Chlordane, [232](#)
Chlorophytum bicheti, [111](#)
Chlorophytums, [90](#)
Christmas cheer, [107](#)
Chrysanthemums, [30](#);
 commercial, and photoperiodism, [23](#)
‘Cinderella,’ [145](#)
‘Cinnabar Red,’ [285](#)
Cissus striata (Miniature Grape Ivy), [111](#)
Citrus: [113](#);
 dwarf, [63](#), [68](#);
 evergreen, [117](#)
Claytonia virginica (Spring Beauty), [155](#), [211](#), [219–20](#)
‘Cleopatra,’ [97](#)
Clerodendrum thomsoniae, [63](#)
Climbers, [27](#), [47](#), [51](#), [70](#), [110](#);
 miniature, [18](#), [49](#), [111–12](#), [138](#), [143](#)
Clintonia, [220](#)
Club moss, [224](#)
‘Coccinea,’ [285](#)
Colchicum (Autumn Crocus, Meadow Saffron), [289](#), [292](#)

Coleus, [94](#)
'Color Carpet,' [286](#)
Columbine, [262](#)
Columnnea microphylla, [113](#)
Columnneas, [63](#)
Common thyme, [274](#)
Conifers (See [Evergreens](#))
Conophytum, [103](#)
Connecticut: [75](#), [142](#), [144](#), [150](#), [158](#), [176](#), [179](#), [183](#), [213](#), [251](#), [253](#),
[255](#);
 fall planting in, [229](#);
 Redding, [64](#), [81](#), [172](#)
Connecticut Power and Light Company, [35](#)
Conservatories, planted as gardens, [64](#)
Containers, gardens in, [38–52](#), [41](#), [49](#), [53–60](#)
Convallaria majalis (Lily of the Valley), [220](#)
Coptis trifolia (Goldthread), [220](#)
'Coquette,' [286](#)
Coral beads, [107](#)
Coreopsis, [283](#)
Coreopsis auriculata nana, [265](#)
Corms, [288](#);
 and fertilizing, [295](#)
Cornflowers, [283](#)
Cornus canadensis (Bunchberry, Creeping Dogwood), [221](#)
Corydalis (Fumitory), [265](#)
Coryphantha vivipara, [103](#)
Cotoneaster, [177](#), [233](#), [241](#)
Cottage (grass) pink, [266](#)
Cotton, [63](#)
Cranesbill (See: [Geranium](#))
Crassula, [103](#)
Creepers: [105](#), [111](#), [120](#), [129](#), [132](#), [134](#), [136](#), [182](#), [221](#), [224](#), [271](#), [274–](#)
[75](#), [286](#);
 in greenhouse, [70](#);
 in indoor garden, [19](#), [27](#), [49](#), [58](#), [60](#);
 in sink garden, [154](#)

(See also [Episcias](#); [Ivy Vines](#))
Creeping bluets, [223](#)
Creeping Charlie, [130](#)
Creeping Dogwood, [221](#)
Creeping fig, [27](#)
(See also: [Ficus](#))
Creeping phlox, [271](#)
Creeping thyme, [274](#)
Crocus, [290](#), [293](#)
Crown: division of plant, [90](#);
of thorns, dwarf (See: [Euphorbia](#));
Cactus (See: [Rebutia](#))
Cryptanthus (Earth Stars), [114](#)
Cryptbergia meadi, [114](#)
Cryptogramma crista (Parsley Fern), [221](#)
Cryptomeria japonica, [241](#)
'Cuinia,' [294](#)
'Cupid,' [286](#)
Cushion pink, [273](#)
Cuttings: [31](#), [73](#), [86](#), [88–89](#), [140](#), [198](#);
in bonsai, [166](#);
effect of red rays of spectrum on, [30](#);
hardwood, [235](#), [243](#), [250](#);
leaf, [89](#);
in propagation of perennials, [258](#);
rhizome, [89](#), [208](#);
root, [89](#);
softwood, [88](#), [234–35](#), [243](#)
Cyanotis, [114](#)
Cyclamen, [38](#), [155](#), [293–94](#);
neapolitanum, [152](#), [294](#)
Cymbalaria muralis (Kenilworth Ivy), [11](#), [40–41](#)
Cypripedium (Lady-slipper, Moccasin Flower), [212](#), [216](#), [221](#)

Daboecia cantabrica (Irish Bell Heather), [242](#)
Daffodils, Winter, [297–98](#)
(See also: [Narcissus](#))

Dahlias, [288](#)
'Dainty Spray,' [96](#)
Daisy, [263](#)
Dalibarda repens (Dewdrop, False Violet), [221–22](#)
"Damp-off," [256](#)
Daphne, [243](#), [260](#)
Davallias (rabbit's-foot ferns), [64](#)
Day-neutral plants, [30](#), [34](#)
Delphiniums, [211](#)
Desert Rose (See: [Trichodiadema densum](#))
'Dew Drop,' [100](#)
Dewdrop, [221–22](#)
Dianthus (Pink), [258](#), [266](#);
 barbatus (sweet William), [10](#), [266](#)
Dieffenbachias, [89](#)
Dionaea muscipula (Venus Fly Trap), [115–16](#)
Dish gardens: [38–39](#), [42–45](#);
 commercial, [51–52](#);
 plants for, [93–95](#), [101](#), [109–11](#), [124](#), [129–32](#), [134](#), [239–40](#), [243](#), [288](#);
 use of accessories in, [44](#)
Dogtooth Violet, [222](#), [288–89](#)
Dogwood, [213](#);
 Creeping (See: [Cornus canadensis](#))
'Dopey,' [126](#)
Double pink, [266](#)
Dracaena godseffiana, [115](#)
Drosera rotundifolia (Sundew), [115](#)
Duckweed, [223](#)
'Dwarf Bearded Iris,' [268](#)
'Dwarf Compact,' [286](#)
Dwarf crown of thorns, [104](#)
'Dwarf Globe,' [285](#)
'Dwarf Houghtoni,' [96](#)
Dwarf myrtle, [124](#)
Dyckia, [116](#)

'Early Giant,' [295](#)

Earth Stars, [114](#)
Earthworms, [60](#)
Easter Lily Cactus, [104–5](#)
Eastern wild ginger, [219](#)
Ebony sweetheart, [219](#)
Echeveria, [103](#)
Echinocereus melanocentrus (Hedgehog Cactus), [103](#)
Echinopsis (Easter Lily Cactus), [104–5](#)
Electricity: as source of light for plants, [23–37](#);
 use in greenhouse propagating of, [73](#), [86](#)
Elephant bush, [106](#)
Elephant Ears (See: [Caladium](#))
‘Elf,’ [126](#)
Elms, for bonsai, [166](#)
‘Emerald Ripple,’ [129](#)
England, [150](#)
English daisy, [263](#)
English Ivy, [11](#), [118–19](#)
English Royal Horticultural Society, [148](#)
English yew, [248](#)
Epigaea repens (Trailing Arbutus), [222](#)
Epiphyllum, [63](#)
Epiphytic plants, [44](#), [53](#), [62](#), [92](#);
 on bromeliad tree, [70](#)
Episcia dianthiflora, [116](#)
Episcias, [20](#), [37](#), [63](#), [70](#), [90](#), [116](#)
‘Epsilon,’ [126](#)
Eranthis (Winter Aconite), [290](#), [294](#)
Erodium chamaedryoides roseum, [155](#)
Erythronium (Adder’s-Tongue, Trout Lily, Dogtooth Violet), [222](#), [288–89](#)
Euonymous (Burning Bush), [243](#)
Euphorbia, [104](#)
Evening Primrose, [170](#)
Evergreens:
 coniferous, [165](#), [234](#), [246](#), [249](#);
 ericaceous, [226](#);

as mulch, [140](#), [144](#), [197](#), [233](#), [242](#), [249](#), [255](#), [272](#), [288](#), [292](#);
in rock garden, [49](#), [239](#);
in sink gardens, [155](#);
watering of, [231](#)
Exacum affine, [116](#)
Exotica II, [11](#)
Eyelash begonia, [97](#)

‘Fairyland,’ [126](#)
‘Fairy Tales,’ [126](#)
False Cypress, [241](#)
False Violet, [221–22](#)
Fan Columbine, [262](#)
‘Fantasy-Face,’ [286](#)
Faucaria (Tiger Jaws), [104](#)
Fawn lily, [222](#)
Feather hyacinth, [296](#)
Fenestraria (Baby Toes), [104](#)
Fern: [51](#), [54](#), [58](#), [66](#), [131](#), [133](#), [155](#), [212](#);
 Button, [128](#);
 maidenhair, [94](#), [216](#);
 Mexican tree, [63](#);
 rabbit’s-foot, [64](#);
 staghorn, [70](#);
 use in window gardens of, [20](#)
 (See also; [Asplenium](#))
Fertilizing:
 of greenhouse plants, [72](#);
 of house plants, [79–80](#);
 with liquid manure, [109](#), [123](#), [175](#), [218](#), [254](#), [290](#), [292](#);
 of miniature gardens in containers, [51](#), [156](#), [175](#), [208](#);
 with rose food, [141](#), [143](#);
 trees and shrubs, [230](#)
Fiberglas, as mulch, [197](#)
Ficus (Creeping Fig), [27](#), [117](#);
 pumila minima, [19](#), [117](#)
Fig:

creeping, [27](#);
dwarf ever-bearing, [68](#)
(See also: [Ficus](#))
'Fireball,' [284–85](#)
Firecracker Vine, [122](#)
Firefern, [124–25](#)
'Fireglow,' [284](#)
Fish emulsion, as fertilizer, [157](#), [175](#)
Fishtail palm, [63](#)
Fittonias, [117](#)
Fittonia verschaffelti (Nerve or Mosaic Plant), [117](#)
Flamingo Flower, [95](#)
'Fleurette,' [126](#)
Floribundas, [137](#)
'Florida Beauty,' [115](#)
Floss flower, [282](#)
Flower and Garden, [27](#), [61](#)
Flowering Maple, [93](#)
Flowering Quince, [240–41](#)
'Flute,' [296](#)
Forget-Me-Not, [269](#)
Formal gardens, [48](#), [64](#), [138–39](#), [180](#)
Fortunella (Kumquat), [117–18](#)
Fothergilla (Bottle Brush), [243](#)
Foxglove, Mexican, [94](#)
'Francois Cardinaux,' [93](#)
Free, Montague, [45](#)
French thyme, [274](#)
'Frosty Morn,' [297](#)
Fuchsia magellanica, [118](#)
Fuchsias, [63](#), [118](#);
 light requirement of, [75](#)
Fumitory, [265](#)
Fungicide, for roses, [143](#)

Galanthus nivalis (Snowdrop), [294](#)
Garden Pools, Water-Lilies and Goldfish, [210](#)

Gardens:

aquatic, [199](#), [207–10](#);

in the city, [213](#);

herb, [48](#);

Northeastern, [10](#);

Northern, [106](#);

Oriental, [48–49](#);

pests in, [83–85](#);

Southern, [109](#)

Gentian, [116](#);

bottle, [212](#)

Georgia, [219](#), [223](#)

Geranium (Cranesbill): [135](#), [266](#);

dwarf, [17](#), [87–88](#), [127](#);

miniature, [20](#);

temperature for greenhouse, [71](#)

(See also: [Pelargonium hortorum](#); [Saxifraga](#))

German Weismoor, [135](#)

Germander, [249](#)

Gesneriads: [20](#), [28](#), [63](#), [71](#), [96–97](#), [100](#), [134–35](#);

and artificial light intensity, [35](#), [37](#);

in Greenwich Village apartment, [23](#);

grown from seed, [86](#)

‘The Giant,’ [292](#)

Ginger, wild, [160](#), [219](#)

Ginkgo, in bonsai, [166](#)

Gladiolus, [294–95](#)

Glass: gardens in, [53–60](#);

as plant containers, [39](#), [100](#)

‘Glee,’ [294](#)

Glory bush (*Tibouchina semidecandra*), [64](#)

Glory-of-the-Snow, [292](#)

Gloxinia (Sinningia), [134–35](#)

‘Gnome,’ [294](#)

‘Goblin,’ [126](#)

Golden Easter Lily, [105](#)

‘Golden Hahni,’ [132](#)

Golden lace, [106](#)
'Golden Ray,' [283](#)
Goldfish, [36–37](#), [205–6](#)
'Goldheart,' [119](#)
'Gold Laced,' [284](#)
Goldthread, [220](#)
'Goliath,' [262](#)
Goodyera pubescens (Rattlesnake Plantain), [155](#), [216](#), [222–23](#)
Graf, A. B., [11](#)
'Granada,' [145](#)
'Granata,' [145](#)
Grape Hyacinth, [295–96](#)
Grass, in dish gardens, [47](#)
Greek myrtle, [124](#)
Greenhouse: [61–73](#);
 author's, [279](#);
 indoor, [27](#), [55](#);
 low-cost, [279–80](#);
 miniature, [53](#);
 pests in, [83–85](#);
 plants for, [136](#);
 propagating, [73](#), [86–90](#);
 rampant, [69](#);
 soil mixture, [71–72](#);
 window, [22](#), [138](#)
'Green Medora,' [97](#)
Gro-Lux lamps, [31](#), [35–36](#)
Gro-Master propagating box, [86](#)
Ground cedar, [224](#)
Ground Ivy, [270](#)
Ground moss, [271](#)
Ground pine, [224](#)
Gymnocalycium mihanovichi (Chin Cactus), [104](#)
Gypsophila (Baby's Breath), [267](#)

'Hahni,' [131](#)
'Halenzy,' [296](#)

Harebell, [263–64](#)
Hawaii, [140](#)
Haworthia, [104–5](#)
Hearts entangled, [110](#)
Heather, [226](#), [240](#)
‘Heavenly Blue,’ [285](#)
Hebe buxifolia variegata, [118](#)
Hedera helix (English Ivy), [11](#), [118–19](#)
Hedgehog Cactus, [103](#)
Hedges, in dish gardens, [46](#)
‘Heidi,’ [286](#)
‘Helen Chapman,’ [284](#)
‘Helen Fowler,’ [209](#)
Helxine soleioli (Baby’s Tears), [120–21](#)
Hemlock, [166](#), [249](#)
Henri, [124](#)
Hepatica (Liverleaf), [155](#), [223](#)
‘Hidcate Blue,’ [269](#)
Hilfoil, [259](#)
Himalayas, [271](#)
Holly, [166](#), [244](#)
Home decorating, and artificially lighted plants, [24–25](#)
Hoop-skirt daffodil, [296](#)
Hormone rooting preparations, [88–89](#), [234–36](#)
Hortus Second, [11](#)
House plants: [72](#), [74–136](#), [282](#);
 collecting, [18](#);
 culture of, described, [92–136](#);
 humidity for, [76–77](#);
 insects and disease and, [83–85](#);
 propagating of, [84](#), [86–90](#);
 repotting of, [79–80](#);
 temperatures for, [75–76](#);
 unusual, [91](#);
 watering of, [80–82](#)
Houstonia (Bluets, Quaker Lady), [223](#)
Hoya bella (Wax Plant), [120](#)

Humidity: gauge, [76](#);
for house plants, [76–77](#);
relative, [76](#)

Hyacinthus, [295](#)

Hybrids: [12](#);
fuchsia, [63](#);
man-made, [11](#);
Weismoor, [160–61](#)

Iberis (Perennial Candytuft), [252](#), [267](#)

‘Igloo,’ [285](#)

Ilex crenata helleri (Holly), [166](#), [244](#)

‘Imp,’ [126](#)

Impatiens, [19](#)

Impatiens repens, [120](#)

‘Imperial White Dwarf,’ [282](#)

‘Impresario,’ [294](#)

Inchworm plant, [105](#)

“Inchworms,” [158](#), [232](#)

Indian head, [106](#)

‘Insignis Blue,’ [284](#)

Institute de Biologia, [114](#)

Internodes, long, [75](#)

Ionopsidium acaule, [155](#)

Iris: [267–68](#);

bulbous, [267](#);

rhizomatous, [267](#), [288–89](#)

Irish Bell Heather, [242](#)

‘Isabellina,’ [285](#)

‘It,’ [100](#)

‘Itsy Bitsy,’ [119](#)

‘Ivalace,’ [119](#)

Ivy: [41](#), [194](#);

in bonsai, [166](#);

English (See: [Hedera helix](#));

grape, [11](#), [111](#);

Kenilworth (See: [Cymbalaria muralis](#));

poison, [213](#)

Jack-in-the-pulpit, [211](#), [219](#)

Japanese beetles, [137](#), [232](#)

Japanese Quince, [240–41](#)

Japanese yew, [248](#)

Jasmine, [63](#)

(See also: [Stephanotis floribunda](#))

Jelly beans, [107](#)

‘Jetage,’ [296](#)

Jewel plant, [107](#)

‘Jo Ann Pring,’ [209](#)

Johnny-jump-up, [276](#)

Joseph’s Coat, [94–95](#)

Joshua tree, little, [49](#), [107](#)

‘Jubilee,’ [119](#)

‘Jubilee Gem,’ [283](#)

Julius Roehrs Company, [13](#)

Juniper, [139](#), [177](#), [226](#), [245](#);

Sargent’s, [160](#)

(See also: [Juniperus](#))

Juniperus: [244–45](#);

communis, [139](#), [245](#);

procumbens nana, [226](#), [245](#)

Kalanchoe, [105](#)

Karathane, [232](#)

‘Kathy Diane,’ [98](#)

Kenilworth Ivy, [11](#), [40–41](#)

‘Kleiner Liebling’ (‘Little Darling’), [126](#)

Kleinia, [105](#)

Koellikeria erinoides, [120](#)

Korean snowballs, [250](#)

Kumquat, [117–18](#)

Lady-slipper (See: [Cypripedium](#))

Landscape, gardens in the, [177–82](#)

Lantana camara, [120](#)
Laurel, mountain, [160](#)
Lavandula officinalis (Lavender), [177](#), [252](#), [258](#), [269](#);
 dwarf, [138](#), [259](#)
Lavender (See: *Lavandula officinalis*)
Layering: [89–90](#), [166–67](#), [235–36](#), [259](#);
 air, [89](#), [166–67](#), [236](#);
 ground, [90](#), [166–67](#), [235–36](#), [259](#)
Leadwort, [265](#)
Leatherleaf, [241](#)
Leiophyllum buxifolium, [226](#)
Lemna minor (Duckweed), [223](#)
Lemon thyme, [274](#)
Leucojum (Snowflake), [295](#)
Leuconeura massageana, [123](#)
Levittown, [23](#), [27](#)
Light bulbs:
 distance from plants of, [32–33](#);
 fluorescent, [27](#), [31](#), [55](#);
 incandescent, [27](#), [31](#)
Lilac, [229](#)
 ‘Lilac Lady,’ [283](#)
Lilliputs, [268](#)
Lily of the Valley, [220](#)
Lime (citrus), [113](#)
Lime: [156](#), [208](#), [260](#), [289](#);
 horticultural, [78](#), [230](#);
 and humusy soil, [94](#), [218](#), [254](#)
Ling, [226](#), [240](#)
 ‘Little Beauty,’ [93](#), [297](#)
 ‘Little Darling,’ [126](#)
 ‘Little Egypt,’ [294](#)
 ‘Little Fantasy,’ [129](#)
 ‘Little Fawn,’ [294](#)
 ‘Little Gem,’ [96](#), [249](#), [267](#), [282](#)
 ‘Little Joe,’ [266](#)
Little Joshua tree, [107](#)

‘Little Princess,’ [148](#)
‘Little Rascal,’ [108](#)
‘Little Red Star,’ [294](#)
‘Little Snow Man,’ [283](#)
‘Little Spice,’ [294](#)
Little Stars, [111](#)
‘Little Sweethearts,’ [286](#)
Liverleaf, [155](#), [223](#)
Lobelia, [138](#)
Lobivia aurea (Golden Easter Lily), [105](#)
London pride, [273](#)
Long Island, [279](#)
Long-night plants, [30–31](#)
‘Lorraine Closson,’ [100](#)
‘Louise Closson,’ [100](#)
‘Lucille Closson,’ [100](#)
‘Lucy Closson,’ [100](#)
Lycopodium (Club Moss), [224](#)

Madwort (See: [Alyssum](#))
‘Magic Carpet,’ [132](#)
Maidenhair spleenwort, [219](#)
Maiden pink, [266](#)
Maine, [261](#)
Malpighia coccigera (Miniature Holly), [122](#)
Mammillaria (Pincushion Cactus), [105](#)
‘Mamorata’ (‘Silver Heart’), [129](#)
Mandarin orange, [113](#)
Manettia bicolor (Firecracker Vine), [122](#)
Manitoba, [103](#)
Mantha requieni, [63](#)
‘Maphil’ (‘Cleopatra’), [97](#)
Maple:
 in bonsai, [164](#), [166](#);
 flowering, [93](#)
Maranta, [53](#), [109](#), [122–23](#)
 (See also: [Calathea](#))

Marigolds, [283–84](#)
‘Maroon Beauty,’ [132](#)
Marumi kumquat, [117](#)
Maryland, [223](#);
 Beltsville, artificial light experiments, [23](#)
‘Masterpiece,’ [93](#)
May Apple, [225](#)
Meadow Saffron, [289](#), [292](#)
‘Medora,’ [97](#)
Medusa’s head, [104](#)
Merry Gardens, [13](#)
Mesembryanthemum cordifolium, [102](#)
Mexican Foxglove, [94](#)
Mexican snowball, [103](#)
Mexican tree fern, [63](#)
Mexico, [114](#)
Mice, [290](#)
‘Midget,’ [98](#)
‘Midget Blue,’ [282](#)
‘Midget Double,’ [266](#)
Mildew, [142](#)
‘Milky-White,’ [296](#)
Mimosa, [63](#)
“Ming urn,” [54](#)
‘Miniata,’ [93](#)
‘Miniature Magic Carpet,’ [286](#)
‘Minicycla,’ [296](#)
Mint, Corsican (*Mantha requieni*), [63](#)
‘Minx,’ [126](#)
‘Mischief,’ [126](#)
‘Misera,’ [93](#)
‘Miss Marveen,’ [108](#)
Mitchella repens (Partridgeberry), [53](#), [224](#)
Mitella (Bishop’s Cap), [224–25](#)
Mites, [37](#), [232](#)
Moccasin Flower (See: [Cypripedium](#))
‘Mon Petit,’ [148](#)

‘Mon Petite,’ [148](#)
Monstera, cut-leaf, [17](#)
Montreal Botanical Gardens, [120](#)
‘Moonlight,’ [259](#)
‘Morgan’s Pink,’ [103](#)
Morning glories, [24](#), [37](#)
Morrowi variegata, [110](#)
Mosaic Plant, [117](#)
Moss: [86–87](#), [155–56](#), [171–72](#), [208](#), [213](#);
 Club (See: [Lycopodium](#));
 Michigan peat, [78](#);
 sheet, [155](#);
 Spanish, [70](#);
 use in terrariums of, [58–59](#)
 (See also [Sphagnum moss](#))
Moss campion, [273](#)
Mother-of-thyme, [274](#)
Mottled wild ginger, [219](#)
Mountain laurel, [46](#)
Mountain pink, [271](#)
‘Mrs. Arno Nehrling,’ [108](#)
Mulching, [140](#), [144](#), [197](#), [233](#), [242](#), [249](#), [255](#), [272](#), [288](#), [293](#)
Munstead lavender, [269](#)
Muscari (Grape Hyacinth), [295–96](#)
Mutations, [11–12](#), [234](#)
Myosotis (Forget-Me-Not), [269](#)
Myrsine africana (African Boxwood), [124](#)
Myrtus communis (Greek Myrtle), [124](#)

Nagami kumquat, [118](#)
Narcissus (Daffodil): [17](#), [63](#), [288](#), [296–97](#);
 fertilizing, [297](#);
 miniature, [155](#);
 trumpet, [152](#), [297](#)
Nasturtiums, [284](#)
Natal Plum, [110](#)
‘Needlepoint,’ [119](#)

Nemophila (baby blue eyes), [284](#)
Nepeta hederacea (*glechoma*) (Ground Ivy), [270](#)
Nerve Plant, [117](#)
'New Dwarf Compact,' [285](#)
New England, [260](#)
New York City, [211](#), [213](#)
Nicotine sulfate, [232](#)
Night-neutral plants, [30](#), [34](#)
Notocactus (Ball Cactus), [106](#)
Nymphaea (Pygmy Water Lily), [209](#)

Oak, [230](#)
Oenothera (Evening Primrose, Sundrop), [270](#)
Offsets, [135](#);
 in propagating, [90](#)
Old lady, [106](#)
Old Man Cactus, [102](#)
Oleander, [63](#)
Olive, [63](#)
Oncidium pusillum, [12](#)
Opuntia, [106](#)
Orchids: [10](#);
 for bromeliad tree, [70](#);
 cattleya, [17](#);
 and electric lighting, [24](#), [37](#);
 miniature (dwarf), [11](#), [17](#), [44](#), [53](#), [63](#);
 in New York, [23](#);
 potting material for (See [Osmunda fiber](#))
'Oriental Music,' [98](#)
Osmanthus fragrans, [63](#)
Osmunda fiber, [44](#), [45](#), [70](#), [81](#), [92](#)
Otaheite orange, [113](#)
Overpotting, [79–80](#)
Oxalis, [63](#), [124–25](#)

Pachysandra, [212](#)
Pacific Northwest, plants shipped from, [91](#)

Painted lady, [103](#)
Panda plant, [105](#)
'Pansy,' [100](#)
Pansy, [245](#);
 hybrid, [46](#)
Parochetus communis (Shamrock Pea, Blue Oxalis), [125](#)
Parodia, [106](#)
Partridgeberry, [53](#), [224](#)
Parsley Fern, [221](#)
Pasque flower, [261](#)
Passiflora coccinea, [63](#)
'Patens Major,' [93](#)
Patient Lucy, [120](#)
'Patricia,' [209](#)
'Patty Lou,' [148](#)
'Peach Blossom,' [93](#), [287](#)
'Peachblossom,' [296](#)
'Peacock,' [100](#)
Peanut, [63](#)
Peanut Cactus, [102](#)
Peat: [86](#), [93](#), [140](#), [143](#), [196](#), [218](#), [230](#), [233](#), [254](#), [256](#), [258](#), [279](#), [281](#);
 Michigan, [218](#);
 moss, [115](#);
 orchid, [136](#)
Pelargonium hortorum (Geranium), [126–28](#)
Pellaea rotundifolia (Button Fern), [128](#)
Pellionia, [128](#)
Pelonex, [78](#), [86](#), [233](#)
Penwiper plant, [105](#)
Peperomia, [19–20](#), [58](#), [89](#), [128–29](#)
Perennials: [197](#), [209](#), [218](#);
 dwarf, [10](#), [132–33](#), [138](#), [251–76](#), [286](#);
 effectiveness in rock gardens of, [189](#)
'Periwinkle,' [294](#)
'Perky,' [126](#)
'Perle d'Alcanada' ('Perle d'Alconada'), [148](#)
Perlite, [47](#), [73](#), [77–78](#), [81](#);

in propagating, [86](#), [233](#)
Permanganate of potash, [206](#)
'Persian Brocade,' [98](#)
Persian Candytuft, [260](#)
Pests and house plants, [85](#)
'Petite Gold,' [284](#)
'Petite Harmony,' [284](#)
'Petite Orange,' [284](#)
'Petite Purple,' [283](#)
'Petite Yellow,' [284](#)
Petunias, [278](#), [284–85](#);
 bedding, [187](#)
Philadelphia, [118](#)
Philodendron, [63](#)
Phlox, [270–71](#), [285](#)
Photochrome, [24](#)
Photoperiodism, principle of, [23](#), [30](#)
Photosynthesis, [30](#)
Picea abies (Spruce), [246](#)
'Pied Piper,' [96](#)
Pilea, [129–31](#)
Pincushion Cactus, [105](#)
Pine: [212](#), [226](#), [230](#);
 "duff," [196](#);
 mugho, [177](#), [231](#), [234](#)
'Pink,' [285](#)
'Pink Charm,' [262](#)
'Pink Heather,' [282](#)
'Pinkie Dots,' [294](#)
'Pink Radiance,' [108](#)
'Pint Size,' [294](#)
Pinus (Pine), [246](#)
 (See also [Pine](#))
Pipsissewa, [155](#), [216](#)
Pitcher plant, [63](#)
'Pixie,' [129](#), [266](#)
'Pixie Gold,' [148](#)

“Plant rooms,” in contemporary homes, [64](#)
Plant sources, [58](#)
Plumbago, [252](#)
Podophyllum peltatum (May Apple), [225](#)
Poinsettias, [30](#), [104](#)
Poison ivy, [213](#)
‘Polaris,’ [126](#)
‘Pollyanna,’ [286](#)
Polyscias (Aralia) fruticosa, [131](#);
 ‘Elegans,’ [131](#);
 ‘Parsley,’ [131](#)
Polystichum tsus-simense, [131](#)
Pomegranate, tree, [68](#)
Ponderosa lemon, [113](#)
Pools, miniature: [66](#), [199–206](#), [223–24](#);
 charcoal as purifier of, [210](#);
 oxygenating plants for, [209](#)
‘Poppet,’ [297](#)
Poppy, miniature, [178](#)
Portulaca, [285–86](#)
Portulacaria afra variegata (Rainbow Bush), [106](#)
Powder puff, [106](#)
Prayer plant, [122](#)
‘Pride,’ [126–27](#)
‘Primrose Pygmy,’ [284](#)
Primroses (See: [Primula](#))
Primula (Primrose), [38](#), [256](#), [271–72](#)
‘Prince Valiant,’ [126](#)
Propagating: [197–98](#), [233–36](#);
 of cacti and succulents, [101](#);
 greenhouse, [73](#), [197](#);
 with the Gro-Master, [86](#);
 of house plants, [84](#), [86–90](#), [92](#);
 miniature roses, [144–45](#);
 of woodland plants, [217–18](#)
Pruning: [82](#), [231](#);
 of dish gardens, [43](#);

miniature roses, [141](#), [143](#);
of sink gardens, [157–58](#)
‘Purity,’ [267](#)
Pussy ears, [114](#)
Pussy-Toes, [261](#)
“Pygmies,” [284](#)
‘Pygmy,’ [127](#)
Pygmy Water Lily, [209](#)

Quaker Lady, [223](#)

Rabbit tracks, [122](#)
Rabbit’s-foot ferns, [64](#)
Rainbow Bush, [106](#)
‘Raspberry Parfait,’ [99](#)
Rattlesnake plantain (See: [Goodyera pubescens](#))
Rebutia (Crown Cactus), [106](#)
‘Red Berry,’ [100](#)
‘Red Button,’ [294](#)
‘Red Dot,’ [294](#)
‘Red Foundling,’ [286](#)
‘Red Imp,’ [148](#)
‘Red Tweedle,’ [294](#)
Resurrection plant, [134](#)
Rex begonias, [89](#);
miniature, [99–101](#)
Rhizomatous begonias, [89](#), [95](#), [97–99](#)
Rhizomes, [89](#), [97](#), [208](#)
Rhododendron, [246](#)
(See also [Azalea](#))
‘Robin Hood,’ [126–27](#)
Rock, as an accessory in gardens, [45](#), [47](#), [57](#), [66](#), [172](#), [177](#)
(See also [Rock gardens](#))
Rock Aster, [262](#)
Rock cress, [262](#)
Rock gardens: [183–98](#);
alpine and rock plants in, [187–98](#);

in containers, [49](#);
designs for, [184–87](#);
plants for, [119](#), [126](#), [187–91](#), [219–26](#), [239–50](#), [259–76](#), [282–83](#), [286](#),
[288](#), [292–95](#), [297–98](#)

Rockfoil (See: [Saxifraga](#))

Roots: cuttings from, [89](#);
division of, [90](#);
as plant holders, [45](#);
rotting of, [42](#), [81](#)

‘Rosa Oakington Ruby,’ [148](#)

Rosary vine, [110](#)

‘Rosata,’ [148](#)

‘Rosea,’ [286](#)

Rose gardens, [138–39](#), [227](#)

‘Rose Gem,’ [283](#), [287](#)

Roses: [137–49](#), [211](#), [226](#);
climbing, [139](#);
growing, under artificial light, [138](#);
rambler, [139](#);
in sink gardens, [138](#);
tea, [137](#), [139](#);
tree, [139](#), [143–44](#)

‘Rosie O’Day,’ [282](#)

‘Rosy Dawn,’ [126](#)

‘Rosy Morn,’ [285](#)

‘Rouletti,’ [148](#)

‘Royal Carpet,’ [282](#)

‘Royal Purple,’ [209](#)

‘Ruffles,’ [126](#)

Runners, in self-propagating, [90](#)

Running pine, [224](#)

‘Rusty Red,’ [284](#)

Sakrete, [47](#), [49](#)

‘Salmon,’ [286](#)

‘Salmon Comet,’ [126–27](#)

‘Salmon Gem,’ [285](#)

Sand:

builders', [78](#), [195](#);

dollar, [102](#);

sharp, [86](#), [195](#), [198](#), [233](#), [255–56](#)

Sandhill phlox, [271](#)

Sand phlox, [270](#)

Sansevieria (Snake Plant), [131–32](#)

Sarcococca ruscifolia (Sweet Box), [132](#)

'Saturn,' [126](#)

Saxifraga (Rockfoil): [272–73](#);

sarmentosa (Strawberry Begonia, Geranium), [90](#), [132–33](#), [273](#);

seedlings, [188](#)

Scale, treating, [232](#)

'Scarlet Gem,' [148](#), [287](#)

Schizocentron (*Heeria*, *Heterocentron*) *elegans* (Spanish Shawl), [132](#)

Schulz, Peggie, [13](#);

garden sold by, [87](#)

Sciadopitys verticillata (Umbrella Pine), [247](#)

Scotch moss, [134](#)

Sea campion, [274](#)

'Sea Gull,' [108](#)

Sedum, [107](#)

Seedlings:

blue rays of spectrum and, [30](#);

in bonsai, [167](#);

repotting of, [79](#), [281](#);

woodland, [218](#)

Seeds: [84](#), [86](#);

and bonsai, [167](#);

effect of red rays of spectrum on, [30](#);

house plants from, [86–87](#);

woodland, [218](#)

Selaginella, [58](#), [66](#)

Selaginella, [133–34](#)

Semperflorens ("wax") begonias, [73](#), [95–96](#)

Sempervivums, [177](#), [187](#), [252](#)

Serissa foetida (*japonica*), [134](#)

Shading, artificial, [70–71](#), [73](#)
Shamrock Pea, [125](#)
Sheet moss, [47](#)
‘Shell,’ [119](#)
Shining club moss, [224](#)
‘Shippy’s Garland,’ [63](#)
Short-night plants, [30–31](#)
‘Shrew,’ [296](#)
Shrubs: [101](#), [181](#), [212](#), [226–50](#);
 in bonsai, [169](#), [173](#);
 “cut-back,” [231](#);
 “die-back,” [231](#);
 tropical, [93](#)
Silene (Campion, Catchfly), [273–74](#)
‘Silver Beauty,’ [260](#)
‘Silver Hahni,’ [132](#)
‘Silver Heart,’ [129](#)
‘Silver Jewel,’ [99](#)
‘Silver Panamiga,’ [131](#)
‘Silver Star,’ [114](#)
‘Silver Tree,’ [131](#)
‘Sinbad,’ [286](#)
Sink gardens, [138](#), [150–58](#), [288](#)
Sinningia, [134–35](#)
Sinningia pusilla, [10](#), [40](#), [134–35](#)
“Slips” (See [Cuttings](#))
Slugs, [60](#), [117](#), [272](#)
‘Small Fortune,’ [128](#)
Snake Plant, [131–32](#)
Snapdragons, [286](#)
‘Sneezy,’ [128](#), [297](#)
‘Snowball,’ [285](#)
Snowball, [226](#), [229](#), [249](#)
‘Snowdrop,’ [96](#)
Snowdrop, [294](#)
‘Snowflake,’ [267](#)
Snowflake, [295](#)

Snow in summer, [263](#)
'Snow-White,' [128](#), [286](#)
Soil: [60](#), [74](#), [230](#);
 in greenhouses, [71–72](#), [92](#);
 humusy, [92](#), [214–16](#);
 mixture for miniature gardens, [50](#), [59](#), [77–79](#), [85–86](#), [92](#), [140](#), [142](#),
 [170](#);
 prepared, [78](#);
 "sour," [59](#)
South, winter planting in, [229](#)
Spanish Shawl, [132](#)
'Sparkle,' [128](#)
'Spaulding,' [97–98](#)
Speedwell, [275](#)
Sphagnum moss, [44–45](#), [86](#), [89](#), [92](#), [115–16](#), [133](#), [217](#), [236](#)
Spiraea (Spirea), [247–48](#)
Spleenwort, [219](#)
Spring beauty (See: [Claytonia virginica](#))
'Spring Charm,' [262](#)
Spring snowflake, [295](#)
'Sprite,' [128](#)
Spruce, [226](#), [246](#)
Squirrel corn, [155](#)
Squirrels, [290](#)
Star Cactus, [102](#)
Starch, in plant growth, [30](#)
Stephanotis floribunda, [63](#)
Sternbergia lutea (Winter Daffodil), [297–98](#)
Stilt-roots, [63](#)
Stolon, [90](#), [132](#), [222](#), [262](#), [268](#)
Stone-Cress, [260](#)
Strawberry Begonia (See: [Saxifraga](#))
Streptocarpus, [135](#)
Streptocarpus, [20](#)
String of hearts, [110](#)
Succulents: [49](#), [51](#), [64–65](#), [78](#), [81](#), [107–36](#);
 containers for, [42–43](#);

garden of, [68](#);
propagating of, [73](#), [89](#)
Suckers, [95](#), [116](#), [231](#), [240](#)
Sugar, in plant growth, [30](#)
Sulfur, [239](#), [255](#);
dusting, for roses, [143](#), [232](#)
Summer snowflake, [295](#)
‘Sunbeam,’ [149](#)
Sun cup, [106](#)
Sundew, [115](#)
Sundrop, [270](#)
Superphosphate (bone meal), [140](#), [143](#), [156–57](#), [175](#), [254](#), [289–90](#)
Sweat plant, [134](#)
Sweet Box, [132](#)
‘Sweet Fairy,’ [149](#)
Sweet Peas, [286](#)
Sweet William (See: [Dianthus](#))
Sylvania Electric Products, Inc., [31](#)

Tangerine (Mandarin orange), [113](#)
Tassel hyacinth, [296](#)
Taxus (Yew), [248](#)
‘Tear Drop,’ [119](#)
Teddy-bear plant, [114](#)
‘Telecurl,’ [119](#)
Temperatures:
for house plants, [75–76](#), [92](#);
for miniature bulbs, [291–92](#);
of plants in terrariums, [59](#)
Tennis ball, [241](#)
Terrariums: [53–54](#), [65](#);
cider jug as, [54–57](#);
commercial, [65](#);
cultural compatability in, [58](#);
plants for, [93–95](#), [97](#), [100](#), [101](#), [109–11](#), [115–17](#), [124](#), [128](#), [131](#), [134](#),
[219](#), [221](#), [223–24](#)
Teucrium chamaedrys (Germander), [249](#)

Texas, [103](#)
Thomas, G. L., Jr., [205](#), [210](#)
'Thousand Beauties,' [96](#)
Three Springs Fisheries, [205](#)
Thuja (Arborvitae), [249](#)
'Thumbelina,' [149](#)
Thyme, [187](#), [252](#);
 lemon, [194](#), [274](#)
 (See also: [Thymus](#))
Thymus (Thyme), [187](#), [252](#), [274–75](#)
Tibouchina semidecandra, [64](#)
Tiger Jaws, [104](#)
'Tiger Star,' [283](#)
Tillandsia, [136](#)
'Tiny Rubies,' [266](#)
'Tiny Tim,' [128](#)
Titanopsis, [107](#)
Tom Thumb Cactus, [106](#)
'Tom Thumb': [105](#), [286](#), [294](#);
 'Double Flowered Dwarf,' [283](#);
 'Pure White,' [283](#);
 'Purple,' [283](#);
 'Rose,' [283](#);
 'Scarlet,' [283](#);
 'Shell Pink,' [283](#)
Tradescantia (Spiderwort, Inch Plant, Wandering Jew), [114](#), [136](#)
Trailing Arbutus, [222](#)
Transplanting "shock," [50](#), [80](#), [166](#), [175](#), [281](#)
Trees: [226–50](#);
 in bonsai, [164–76](#);
 deciduous, [75](#), [165](#), [170](#), [216](#);
 miniature fruit, [68–69](#), [227](#);
 roots as plant holders, [45](#)
Trichodiadema densum (Desert Rose), [107](#)
Trillium, [160](#), [211](#)
Tropical gardens, [49](#), [62–65](#), [109](#), [115](#);
 "jungle," [26–27](#)

Tropical Gardens, [13](#)
Tropical Paradise Greenhouse, [98](#)
Trough gardens (See [Sink gardens](#))
Trout Lily, [222](#), [288–89](#)
Tsuga canadensis (Hemlock), [166](#), [249](#)
‘Tubergeni,’ [294](#)
Tubers, [288](#)
Tufa rock, [171](#), [189](#);
 as plant-holder, [44](#), [57](#)
Tulipa (Tulip), [288](#), [290](#), [298](#)
‘Twilight,’ [109](#)
‘Twinkle,’ [10](#), [128](#), [285](#)

Umbrella flower, [110](#)
Umbrella Pine, [247](#)
U. S. Department of Agriculture, [31](#)

‘Variegated Kleiner Liebling,’ [128](#)
Vaughn, [281](#)
Venus Fly Trap, [115–16](#)
Verbena, [122](#), [286](#)
Vermiculite: [77](#), [78](#), [81](#), [93](#);
 in indoor bonsai, [161](#);
 in indoor greenhouse, [22](#), [27](#);
 in propagating, [86](#), [233](#)
Veronica (Speedwell), [275](#)
Viburnum (Snowball, European Cranberry Bush), [226](#), [229](#), [249](#)
Vigoro, [116](#)
Vine, [63](#);
 rosary, [110](#)
 (See also: [Cissus striata](#); [Clerodendrum thomosoniae](#); [Passiflora coccinea](#); [Manettia bicolor](#))
Viola (Viola, Violet): [276](#);
 African, [20](#), [25](#), [51](#), [63](#), [90](#), [93](#)
‘Violacea Semi-Plena,’ [93](#)
‘Violet King,’ [283](#)
‘Violet Queen,’ [283](#)

Violets (See: [Viola](#))

‘Virbob,’ [98](#)

Virginia, [290](#)

Walking Fern, [219](#)

Wall gardens: [183–89](#), [191–98](#);

design of, [184–89](#);

planting and care of, [193–97](#)

Wandering Jew (See: [Tradescantia](#))

Wardian case, [53](#), [55](#), [77](#)

Watering of plants: [80–82](#), [91](#), [101](#), [156](#), [230](#);

excess, [42](#), [50](#), [59–60](#), [80–81](#), [180](#);

importance of drainage in, [190](#), [229–30](#);

with manure water, [60](#), [80](#), [109](#), [123](#);

in propagating, [86](#);

in terrariums, [59–60](#)

Water lilies, [205](#);

miniature, [202](#), [207–9](#)

Watermelon pilea, [130](#)

Water ‘N’ Watch garden, [87](#)

Water plants, [207–10](#);

floating, [209](#)

Wax Plant, [120](#)

‘Wayside’s Garnet,’ [149](#)

‘Wee Bee,’ [297](#)

‘Wee Red,’ [294](#)

‘Wee Willie’ (‘Sweet William’), [266](#)

‘Wendlinger,’ [109](#)

Westchester, [23](#)

Westcott, Cynthia, [232](#)

Western wild ginger, [219](#)

West Indies, [122](#)

‘Whirlybird,’ [294](#)

White Cedar, [241](#)

White Forsythia, [236](#)

‘White Gem,’ [287](#)

‘White Gossamer,’ [114](#)

White pygmy, [209](#)
'White Velvet,' [114](#)
Wilder, Louise Beebe, [197](#)
Wild-flower, mail order houses, [153](#)
Wild ginger, [160](#), [219](#)
Wild pink, [273](#)
Wild sweet William, [271](#)
Williamsburg, [48](#), [155](#)
Willow, [166](#)
Windex bottle, for misting plants, [77](#), [82](#)
Windflower, [261](#)
Window boxes, [20](#);
 plants for, [109](#), [286](#)
Window gardens, [18–22](#), [97](#), [117](#)
'Winkie,' [96](#)
Winter Aconite, [290](#), [294](#)
'Winter Jewel,' [96](#)
Woodbine, [111–12](#), [212](#)
Woodland gardens, [211–25](#)
Woolly thyme, [274](#)

Yarrow, [259](#)
'Yellow Miniature,' [149](#)
Yellow pygmy, [209](#)
Yew, [248](#)
'Yindee,' [296](#)

Zinnias, [286–87](#)

Transcriber's Notes:

1. Obvious printers', punctuation and spelling errors have been corrected silently.
2. Some hyphenated and non-hyphenated versions of the same words have been retained as in the original.

*** END OF THE PROJECT GUTENBERG EBOOK ALL ABOUT
MINIATURE PLANTS AND GARDENS INDOORS AND OUT ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.